COMPARISON-BASED DIAGNOSIS IN FAULT-TOLERANT, MULTIPROCESSOR SYSTEMS

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Comparison-Based Diagnosis in Fault-Tolerant, Multiprocessor Systems

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Abstract

In any system the toleration of faults in the individual components is achieved through redundancy. Redundancy can have many forms: spatial (modular) redundancy, temporal redundancy and redundancy of information. In a typical multiprocessor system of today the plurality of its processors can be exploited in two ways: (1) as redundancy for fault tolerance or (2) as combined computing power for improved performance. In some systems it is ideal to use the abundance of processors for both purposes.

The design of a fault-tolerant system must take into account the expense of the redundant elements. Expenses occur not only in the initial cost of the redundant components of the system, but also in the support needed to use these components or simply to have them be present in the system. Furthermore, the redundant components may not necessarily benefit the system when the faults that they were designed to tolerate have not occurred. A flexible design of a multiprocessor system allows the redundancy of the system to be used for fault tolerance when needed, and for improved performance when no faults have occurred.

System-level diagnosis is a technique for fault tolerance that strives to identify the faulty elements in a system. This is done by deduction, based on information in the form of results of tests applied to the elements. Once the faulty elements have been identified, the system is able to isolate them, ignore their output and to initiate a repair operation such that the reliability of the system can be maintained in the long run.

Comparison-based, system-level diagnosis is a variation of this method where the tests used for diagnosis are in fact the comparisons of the output produced by the identical elements for a certain operation (task). An advantage of this method is that the operation in question can be a something that constitutes real work performed by the system, rather than a specialized diagnostic task. In this way the normal operation of the system can be used for its own diagnosis, thus making the procedure more efficient.

Fault-tolerant techniques based on comparisons are not limited to system-level diagnosis. Modular redundancy (NMR) with majority voting, a well known technique, ultimately relies on the comparison of the outputs of the redundant modules for its ability to mask faults.

We present a new approach at combining the classical technique of modular redundancy and that of comparison-based, system-level diagnosis. Our goal is to utilize the plurality of processors in a multiprocessor system for both fault tolerance and improved performance.

First we propose a new model for comparison-based, system-level diagnosis, which is more suited to large, multiprocessor systems. Then, by characterizing a certain type of diagnosability under this new model, we classify system structures suitable for diagnosable systems with low number of comparisons. Finally, we propose a method for combining modular redundancy with comparison diagnosis under our model in order to obtain systems where inherent redundancy can be easily traded off between fault tolerance and performance.

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Comparison-Based Diagnosis in Fault-Tolerant, Multiprocessor Systems

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Résumé

Dans tout système, la tolérance aux pannes des composants individuels du système est assurée grâce à la redondance. La redondance peut prendre plusieurs formes : redondance spatiale (ou matérielle), redondance temporelle et redondance de l'information. Actuellement, dans les systèmes multi-processeurs typiques la présence de plusieurs processeurs peut être exploitée de deux façons différentes : (1) en tant que redondance pour assurer une tolérance aux pannes ou (2) comme une plus grande puissance de calcul dans le but d'augmenter les performances du système. Dans certains systèmes, il est idéal d'utiliser la pluralité des processeurs pour atteindre ces deux buts simultanément.

La conception de systèmes tolérants aux pannes doit tenir compte des dépenses engendrées par la présence d'éléments redondants. Ces dépenses proviennent non seulement des coûts initiaux des composants redondants mais aussi de l'entretien nécessaire pour les utiliser. Par ailleurs, la présence de composants redondants ne bénéficie pas toujours au système quand les pannes qu'ils doivent tolérer ne se produisent pas. Une bonne conception d'un système multi-processeurs doit permettre d'utiliser la redondance des composants soit pour la tolérance aux fautes soit pour augmenter la puissance de calcul quand aucune faute ne se produit.

La théorie de diagnostic de systèmes (*System-Level Diagnosis-SLD*) est une technique de tolérance aux fautes dont le but est d'essayer d'identifier les éléments fautifs dans un système donné. Cela se fait par déduction en utilisant les résultats de tests effectués avec ces éléments. Une fois que les éléments défaillants sont identifiés, le système est capable de les isoler, d'ignorer les résultats qu'ils produisent et d'effectuer une opération de réparation pour assurer la fiabilité du système à longue échéance.

Le SLD basé sur la comparaison est une variation de cette méthode où les tests utilisés pour faire le diagnostic sont en fait des comparaisons des résultats produits par les différents éléments pour une tâche donnée. Un des avantages de cette méthode est que la tâche testée peut effectuer un travail en rapport direct avec le système plutôt qu'être une tâche dédiée au diagnostic. De cette façon les opérations normales d'un système multi-processeurs peuvent être utilisées pour s'auto-diagnostiquer, ce qui rend le système plus efficace.

Les techniques de tolérance aux pannes basées sur la comparaison ne sont pas limitées au SLD. La redondance majoritaire (*Modular Redundancy with Voting-NMR*) repose finalement sur la comparaison des résultats des modules redondants pour assurer la capacité à masquer l'effet des défauts.

Nous présentons une approche nouvelle qui combine la technique classique NMR et la technique basée sur les comparaisons SLD. Notre but est d'exploiter la pluralité des processeurs dans les systèmes multi-processeurs pour assurer à la fois la tolérance aux fautes et une meilleure puissance de calcul.

Tout d'abord nous proposons un nouveau modèle pour le SLD qui est adapté aux larges systèmes à multi-processeurs. Ensuite en définissant une capacité de diagnostique dans ce nouveau modèle, nous classons les structures de systèmes en fonction de leur capacité à effectuer des diagnostics au moyen d'un faible nombre de comparaisons. Finalement nous proposons une méthode pour combiner NMR avec le diagnostic basé sur les comparaisons dans notre modèle de manière à obtenir des systèmes où un compromis entre la tolérance aux fautes et la puissance du système est facile à trouver.

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Glossary of Terms

Adaptive diagnosis. The identification of faulty PEs that is carried out so that the results of previous tests are used to determine which tests to perform in the future. In other words, the test assignment is not fixed before tests are carried out.

Candidate fault set. A set of vertices that for some purpose (e.g., as part of a diagnosis or demonstration of a proof) is considered to be the potential set of faulty nodes in the system. Therefore a candidate fault set contains at most f nodes.

Characterization problem: Given a certain value of f (the upper bound on the number of faulty PEs), what are the necessary and sufficient conditions for a multiprocessor system to be diagnosable?

Comparison test. A comparison of the task results of a task that has been executed on two distinct PEs, with identical values for input.

Comparison test outcome. Binary result of a comparison test. A value of 0 indicates that the two task results were identical; a value of 1 indicates that the two task results were different.

Consistent fault set. The set of at most f nodes that, if assumed faulty, are consistent with the actual syndrome obtained for a testing graph given the assumptions of the particular fault model of system-level diagnosis.

Diagnosability problem: Given a collection of PEs and a set of test assignments for each PE, what is the maximum number f of arbitrary PEs that can be faulty such that the set of faulty PEs can be *uniquely* identified on the basis of any given syndrome?

Diagnosis problem: Given a testing assignment for a system such that at most f PEs can always be uniquely identified on the basis of any given syndrome, does there exist an efficient procedure (algorithm) for identifying the faulty PEs?

***f*/*s*–diagnosis.** The identification of the at most *f* faulty PEs in *S* within a set of (at most) *s* PEs. This leaves the possibility of incorrectly diagnosing fault-free PEs as faulty, but a faulty PE is always identified [Fri75].

Fault. A disruption in the logical behavior of a PE. The faults considered in this thesis are those within one of the PEs, and which cause the digital output of the PE to be different from that which is specified for a given sequence of inputs [Dah88].

Intermittent fault. Temporary fault inside a PE whose conditions of activation cannot be reproduced or which occur rarely enough [Lap92]. The PE should be declared faulty until repaired.

One-step *f*-fault diagnosis. (Also called *diagnosis without repair*.) The identification of all of the at most *f* faulty PEs[1] in the system *S* from a given syndrome. The diagnosis is performed in one instance, without modifying anything in the system.

Permanent fault. A fault in a PE that causes the PE to *constantly* exhibit behavior that is different from its logical behavior [Dah88].

Processing Element (PE). An element of the system that is capable of processing information and making decisions. Typically this is the replicated element in a system where modular redundancy is applied. A PE is represented in terms of graph theory as a vertex *v*.

System. The collection of some number *n* of PEs, typically referred to as a multiprocessor system *S* and denoted in graph theory as a set $V = \{v_1, v_2, \ldots, v_n\}$.

Task. A set of instructions performed by a PE as part of real work done by the system.

Task result. The tangible result of the execution of a task on a single PE, which is used for comparison tests.

Test. A series of programs or other means of measuring the fault status of a PE [PMC67].

Test assignment. The complete collection of tests in *S* is represented by a graph $G = (V, E)$, where each node $v_i \in V$ is represented by a PE. If *G* is a directed graph, then each edge $(v_i, v_j) \in E$ if and only if v_i tests v_j. In the case where *G* is an undirected graph, $(v_i, v_j) \in E$ if and only if the task results obtained from v_i and v_j are used to make a comparison test.

1. In this thesis the number of faulty processing elements will be denoted *f* instead of the usual *t* because we will use *t* to denote tasks.

Test outcome. Binary result of a test. A value of 0 indicates that the test was passed; a value of 1 indicates that it was failed.

Transient fault. Temporary fault external to a PE but which affects it as a result of environmental interference or interaction [Lap92]. A transient fault does not necessarily imply that the PE should be declared faulty—a temporary shutdown may be warranted because of the unstable environment.

Sequential *f*-fault diagnosis. (Also called *diagnosis with repair.*) The identification of at least one of the at most *f* faulty PEs (if there exist any) in the system *S* from a given syndrome. The diagnosis is carried out in sequential rounds in which at least one faulty PE is guaranteed to be identified. Upon its identification the faulty PE is then replaced (or repaired). This process is repeated until no faulty PEs remain.

Syndrome. The complete set of test outcomes for the system *S*.

Glossary of Symbols

$A - B$	A less B (the elements of A that are not in B)
$\alpha_0(G)$	minimum vertex cover of G, 58
$H_1 \circ H_2$	corona of two graphs H_1 and H_2, 50
$f(v)$	set of subsets to which v belongs, 32
$\Gamma(v)$	set of nodes adjacent to v, 31
$\Gamma(X)$	set of nodes outside of X and adjacent to at least one node in X, 31
G_0	0-derived graph of G, 31
K_p	complete graph of p nodes, 29
$K_{m, n}$	complete bipartite graph formed of two subgraphs of m and n nodes respectively, 29
$\kappa(G)$	connectivity of G, 29
$\nu(G)$	cycle rank of G, 30
$\pi(G)$	cover of G, 32
$\sigma: E \rightarrow \{0, 1\}$	syndrome, 32

Chapter 1 Introduction

1.1 Overview

1.1.1 Inherent redundancy in multiprocessor systems

The computer systems of today are becoming more and more complex. The number of processors they contain is increasing along with the computational power of the processors themselves. To allow adequate interaction in these systems, the networks that connect the computers must furnish greater communication capacities. As the complexity of a system increases, so does the end-user's expectation about the system's reliability.

Ultimately the expected reliability of a system depends on what happens when the system behaves inappropriately. If lives of humans or large sums of money are at stake, then the system must be very reliable. Examples of ultra-reliable multiprocessor systems include the control systems for an airplane, a space ship or a nuclear reactor. Other systems, such as automated bank tellers or airline reservation systems, can be out of service for small periods of time without causing much more than some inconvenience. Nonetheless the distributed information in these systems is expected to remain consistent despite any down time that may occur.

The more complex a system is, the more difficult it is to achieve reliability. Fortunately multiprocessor systems have built-in redundancy. The multitude of processors and the equipment that allows connections between them can allow the system to perform more work than a single-processor system. But more importantly it is also this redundancy of equipment that enables the system to continue to function as needed, even though one ore more of the processors have stopped working correctly or there are problems in the network connecting the processors.

In any system the toleration of faults in the individual components is achieved through redundancy. Redundancy can have many forms: spatial (modular) redundancy, temporal redundancy, and redundancy of information. As described above, the plurality of the units in a multi-

processor system can be exploited in two ways: to tolerate faults or to increase the performance of the system. In some systems it is ideal to use the abundance of processors for both purposes.

1.1.2 Contents of this chapter

Section 1.1.3 discusses the problem of exploiting the plurality of resources in multiprocessor systems and gives the motivation for the work that we present in this dissertation. Section 1.1.4 provides a brief summary of the thesis defended in this dissertation, the contributions made as well as the results. Section 1.2 provides a brief introduction on the basic concepts of fault-tolerant, multiprocessor systems. Section 1.3 briefly describes the notion of comparison-based, system-level diagnosis. Section 1.4 describes each of the succeeding chapters of the thesis.

1.1.3 Problem and motivation

Any designer of a fault-tolerant system knows that redundancy is expensive. Expenses exist not only in the initial cost of the redundant components of the system, but also in the support needed to use these components or simply to have them in the system. Furthermore, the redundant components may not necessarily benefit the system when the faults that they were designed to tolerate have not occurred.

For example consider the spare tire in an automobile. It serves no purpose when the other four tires are in good shape. Yet the owners of the car paid money for the spare tire, and ideally they have to make sure that it always has air in it. Furthermore the tire takes up space and weighs down the car—think of all the wasted energy and excess air pollution due to spare tires! On the other hand, imagine a car that is designed to run very well with four tires, but its fault tolerance is such that if one of the four tires were to go flat, the car could manage to run adequately with only three tires—albeit at a slower speed because it is less stable. The “spare” tire in this case could be considered as any of the four normal tires. Essentially it enables the car to go faster because of the added stability. In this example the plurality of components in the system is put to use for both fault tolerance and higher performance.

In this thesis we consider the same form of problem, but in multiprocessor systems with their inherent abundance of processing elements. We consider ways in which the plurality of processors can be exploited for fault tolerance (in a technique known as *N modular redundancy* or NMR) and performance. One of the tools we use to aid us in our goal is a thirty-year-old concept known as *system-level diagnosis*. Specifically we are interested in comparison-based, system-level diagnosis.

Preparata, Metze and Chien [PMC67] introduced in 1967 the idea of system-level diagnosis as a method to determine which processing elements in a multiprocessor system are faulty. This clever technique is based on a strategy where some or all of the processing elements test other processing elements. A test has a result of either "0" (meaning the test was a success) or "1" (the test was a failure). In a variant of this strategy, which was proposed independently around 1980 by Chwa *et al.* [CH81b] and Malek [Mal80], instead of actively testing each other, the processing elements produce results for the same set of processing instructions (i.e., a *task*), and these results are compared. The comparison yields either a "0" if the results are identical or a "1" if the results are different. Thus the comparison replaces the test of the original strategy. Comparison-based, system-level diagnosis has the advantage that the processing instructions (tasks) used for comparison tests can be the real work of the system. This is obviously a good direction to move in toward our goal of efficient fault tolerance.

1.1.4 Thesis and contributions

In this dissertation we re-examine certain comparison-based strategies of system-level diagnosis. We extend and complement these strategies, and then re-apply them to the multiprocessor systems for which they were originally conceived. Our goal is to obtain comparison-based schemes for such systems where the inherent redundancy is used for both reliability and efficiency, but in a way that is better than in existing methods. The main contributions of this thesis are as follows.

- We describe and discuss the general concept of modular redundancy, and examine several specific techniques, putting special emphasis on the notion of comparison-based modular redundancy. We underscore the sometimes implicit assumptions made about the fault models in each of the methods, indicating when necessary the similarities to the fault models in system-level diagnosis.
- We examine the concept of system-level diagnosis, starting from its origins and proceeding to the specific contributions made in the areas of comparison-based, system-level diagnosis. Again we highlight the assumptions of the fault models in order to maintain proper perspective with modular redundancy techniques.
- We extend the characterization of $f/(n–1)$-diagnosable systems for comparison-based, system-level diagnosis, a problem that was first investigated by Xu *et al.* [XH90, Xu91]. We include a characterization for systems that require fewer comparisons than those previously introduced. The characterization of $f/(n–1)$-diagnosability is important in the

context of this thesis because this concept guarantees that at least one fault-free processor can be identified in a system, a goal that we show is crucial when considering the fault tolerance and throughput of a system.

- As part of the above extended characterization, we introduce a new class of $f/(n-1)$-diagnosable systems, which we call bi-corona systems. These systems are better than any previously proposed $f/(n-1)$-diagnosable systems that we are aware of, not only because they require fewer comparisons, but also because they allow the system to perform more than one task for which the comparison diagnosis can be performed, thereby increasing the potential throughput of the system.
- To complement the utility of these system structures, we provide and analyze an algorithm that adaptively diagnoses at least one fault-free node in a bi-corona system.
- The consideration of the two principal fault models (symmetric and asymmetric comparison models) common to modular redundancy and system-level diagnosis inspired us to develop a new model that is more general and that bridges the gap between the two models. In fact, these two accepted models become special cases of our proposed model, which we call the bounded symmetric comparison (BSC) model.
- Under the BSC model we characterize systems for which all faulty processors can be identified from a single set of comparison tests (one-step diagnosis). Returning to the theme of $f/(n-1)$-diagnosability, we also characterize such systems under the BSC model. Furthermore, we consider the issues of detecting and locating faulty processors under the BSC model.
- We distinguish between two types of diagnoses in fault-tolerant, multiprocessor systems: processor diagnosis and task diagnosis. We define the concept of task diagnosability, which corresponds to the notion of $f/(n-1)$-diagnosability in system-level diagnosis. This problem entails finding at least one node in the system such that its output can be judged as reliable.
- We characterize task-diagnosable systems for up to 9 processors, using an argument that is different and more intuitive than previous characterizations [Xu91]. Furthermore, we consider systems of disjoint chains as well as continuous chains of processors, something which has not been covered by Xu. Disjoint chains have an important advantage—each subchain of processors can execute a distinct set of calculations, and task diagnosis can still be performed. In this way the system's throughput is increased, as the processors do not all have to execute the same task.

- We consider adaptive diagnosis in a system using comparison-based, modular redundancy. We show that the number of comparisons can actually be reduced, but that this improvement may not be justified for a hardware implementation, as the switching hardware is too complex.
- We exploit the concept of task diagnosis in order to reduce the number of comparisons in a comparison-based, modular redundant system where the comparators themselves can be faulty and thus need to be redundant. We demonstrate once again how task diagnosis can be used with comparator fault tolerance and still reduce the number of comparisons necessary to tolerate comparator faults.
- In the context of modular redundant systems we introduce the notion of *multiple-task modular redundancy* (MTNMR). This method allows an existing NMR system to have a higher throughput by executing tasks optimistically. To further improve the system yield, we add the concept of comparison diagnosis to the basic MTNMR, which yields a method that we call *modular redundancy with comparison diagnosis* (NMRCD).
- We analyze in detail the MTNMR and NMRCD schemes, comparing and contrasting them to similar schemes with respect to their throughput and latency. The reliability of a system under this scheme is also analyzed.

1.2 Basic concepts of fault-tolerant, multiprocessor systems

1.2.1 Redundancy for fault tolerance

One technique for tolerating faults is known as *N modular redundancy* (NMR) *with voting*, and was inspired by ideas of von Neumann [Von56]. In this method a single processor (or module) is replicated to form a system of *N* processors. Each processor executes an identical set of calculations and produces a result. The system tolerates faulty processors by “masking” their output. A voting unit reads the *N* results produced by the set of redundant processors, and only produces the result which corresponds to that produced by at least the majority of the processors. The assumption with NMR is that some portion (greater than or equal to the majority) of the processors will be fault-free and that they always produce the same result for a set of calculations. If a processor is faulty its result is masked, for the result will be different from that produced by the majority.

Notwithstanding its effectiveness, modular redundancy is very costly when used exclusively for tolerating faults. The system produces the same amount of work as a single-module system. For this reason many designers have proposed actual fault-tolerant multiprocessor systems

[Skla76, WLGG78, HSLJ78, KWFT88] that utilize the plurality of elements in a system that is modular redundant for more than just fault tolerance. Furthermore, in its basic form, NMR merely prevents the system from producing faulty results; it does not identify the faulty processors. Over the long term, this can impede the system from tolerating faults. If faulty processors are not identified and then disabled or repaired, eventually the majority of processors will no longer be fault-free, and the NMR strategy will fail.

Another variant of spatial redundancy is known as *dynamic redundancy.* In this strategy, when faults have occurred, the system reconfigures itself to compensate for the faults. Typically, whenever a processor is identified as being faulty, a spare (and hopefully fault-free) processor is used in its place. Dynamic redundancy generally requires less resources than NMR and is therefore less costly, but it relies on the strong assumption that a faulty processor can be detected in a reliable manner. The identification of a faulty processor is done internally, either by a subunit of the processor itself or by errors in the processor's output.

In one dynamic redundancy technique, the system detects faults by comparing the output of two processors that have performed the same calculation. A fault has surely occurred if the two results are different, although it is not possible using this method alone to identify which of the two processors is faulty. An advantage of this method is that the useful work performed by the processors of the system is used to detect faults, as opposed to having to perform supplemental tests on the processors during which they are not available in the system. Therefore, comparisons are a convenient technique to check that everything is going well in a computer system.

1.3 Comparison-based, system-level diagnosis

1.3.1 Comparison-based, system-level diagnosis

An interesting technique to identify the faulty processors in a multiprocessor system was proposed by Preparata, Metze and Chien and is known as system-level diagnosis [PMC67]. In this strategy, the individual processors of a multiprocessor system perform diagnostic tests on each other. A centralized observer collects the results of all the diagnostic tests, and from these data, it identifies which processors are faulty. Preparata *et. al* showed how their method can always work for systems where the number of faulty processors is limited. As in the case of dynamic redundancy, the success of their method relies on the efficacy of the diagnostic test.

To improve this scheme, Chwa and Hakimi [CH81b] and also Malek [Mal80] proposed independently a comparison-based approach to system-level diagnosis. In their methods the diagnostic test is replaced by a comparison of two results performed by distinct processors for

the same calculation using identical values for input. As before, the advantage to this design is that the real work of the system can be used for the diagnostic information.

1.4 Organization of the thesis

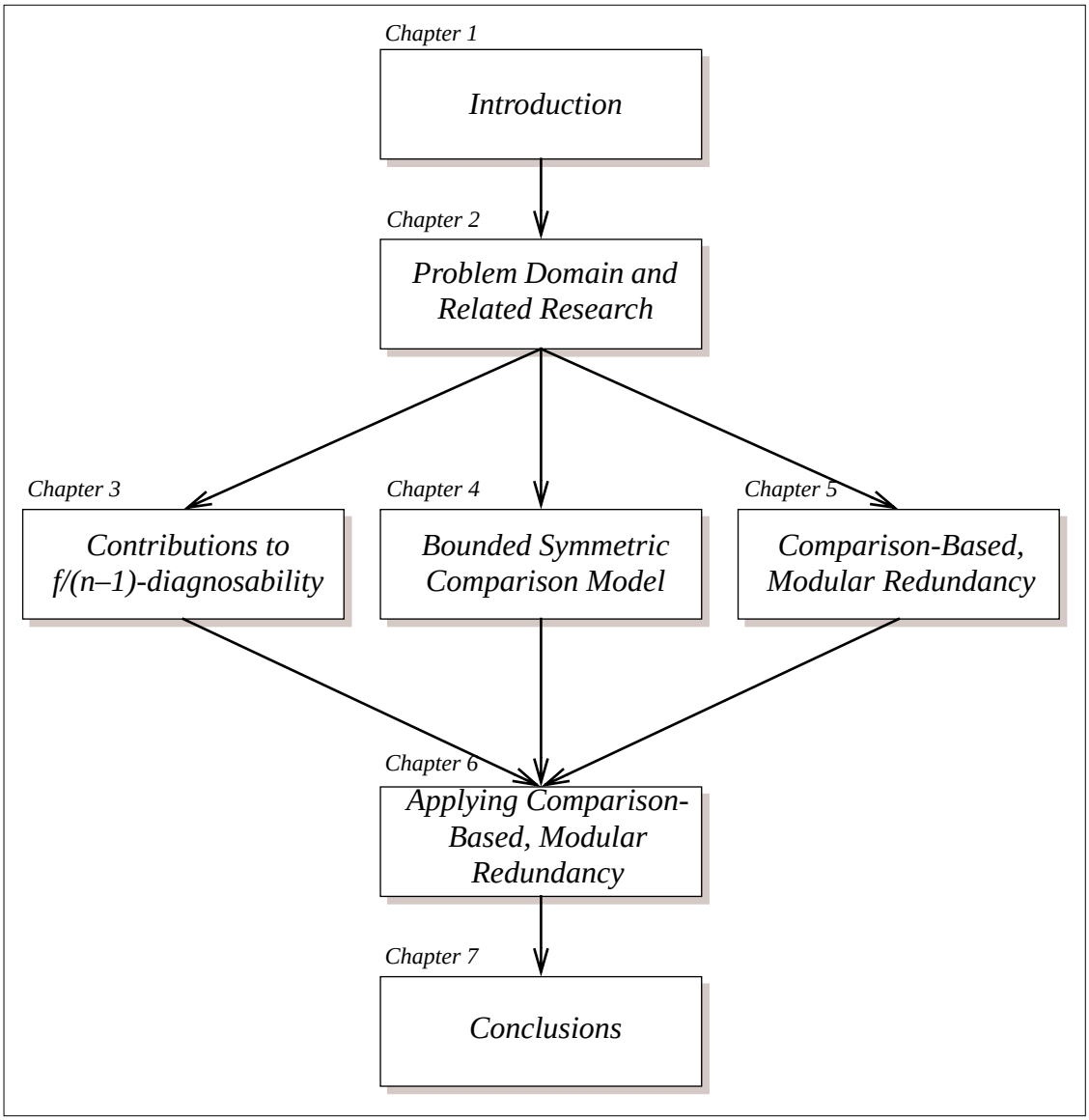


Figure 1.1 Layout of the thesis

1.4.1 Introduction

Figure 1.1 describes the layout of the thesis. Each of the following chapters is explained below.

1.4.2 Problem Domain and Related Research

Chapter 2 considers the domain of fault tolerance in multiprocessor systems. Specifically this

chapter describes the background of modular redundancy and system-level diagnosis.

1.4.3 Contributions to f/(n–1)-diagnosability

Chapter 3 presents the contributions of this dissertation in the area of $f/(n-1)$-diagnosability, a particular area of system-level diagnosis.

1.4.4 Bounded Symmetric Comparison Model

Chapter 4 introduces a new model for comparison-based, system-level diagnosis, which we call the *bounded symmetric comparison* (BSC) model. We provide several characterizations of systems under this model for system-level diagnosis.

1.4.5 Comparison-Based, Modular Redundancy

Chapter 5 contains contributions in the domain of comparison-based, modular redundancy. A detailed characterization of certain modular redundant systems under the comparison-based models of system-level diagnosis is given.

1.4.6 Applying Comparison-Based, Modular Redundancy

Chapter 6 proposes several applications of the work of the preceding chapters to demonstrate the usefulness of the contributions. We introduce two techniques for improving the throughput of a modular redundant system combining the ideas of comparison-based, modular redundancy and system-level diagnosis. We demonstrate the utility of the BSC model by characterizing these techniques under this model.

1.4.7 Conclusions

Chapter 7 summarizes the results and contributions of the thesis and indicates open problems and possibilities for further research.

Chapter 2 Problem Domain and Related Research

In Chapter 1 we motivate the solution of combining comparison-based, modular redundancy with system-level diagnosis. In this chapter, we present the important background of both of these areas of research.

This chapter is divided as follows. Section 2.1 gives a brief history of both areas of research. Section 2.2 lays down a foundation for the vocabulary and summarizes the approaches related to fault tolerance with modular redundancy. Section 2.3 performs a similar survey of the appropriate areas of system-level diagnosis. Section 2.4 concludes the chapter by discussing the contributions and underlining the important relations between modular redundancy and system-level diagnosis.

2.1 Introduction

Fault tolerance is generally achieved through the use of redundancy of hardware and/or software (spatial redundancy), of time (temporal redundancy), of information (value redundancy), or a combination thereof. Spatial redundancy means that individual elements of hardware and/or software that can be faulty are replicated. There exist three types of spatial redundancy: static, dynamic and hybrid.

Static redundancy is a technique that works by “masking” faults. This is performed by executing a distinct task on each of the replicated elements. Each element will yield a result for the task. Fault masking works by choosing a result to output for the system only if the result has been produced by the majority of the replicated elements. This method assumes that such a result is fault-free because the majority of the processors cannot produce an identical, faulty result. Any results not conforming to that of the majority are ignored and hence masked.

Dynamic redundancy involves two phases. In the first phase faulty elements of the system are identified in some manner. In the second phase the system recovers from the faulty units by switching over their operation to redundant, backup elements that are hopefully available and

fault-free. The success of this method relies on choosing a suitable number of spares, an effective technique for detecting faults, and a viable switching operation.

Hybrid redundancy, as its name implies, is a combination of static and dynamic redundancy. Essentially the system consists of a core of statically redundant elements, and several more spares are provided in case faults occur. Hybrid redundant systems can offer very high reliability provided that the number of spares used is appropriate and that the fault-detection and switching procedures are perfect. This technique is, however, susceptible to faults that occur during the detection of existing faults. As well, the procedure for switchover when faults have occurred becomes very complex as the number of spares increases.

Systems that use static redundancy, also known as *N-modular redundancy* (NMR), to tolerate faults are costly in redundant resources while only achieving the throughput of a system with just one element. This is because the technique in NMR relies upon masking faults. Theoretically, if the system were able to identify faulty elements in the system, it could simply ignore the output from these elements. This is the principle behind dynamic redundancy. In this way, the throughput of such a system with *N* processors could be augmented by the number of fault-free elements by scheduling various tasks on each fault-free element. As discussed above, the success of this improvement hinges on the system's ability to reliably detect and locate faulty modules.

This need for a correct identification of faulty modules is the prime mover of the theory of system-level diagnosis. It originated in a 1967 paper by Preparata, Metze and Chien [PMC67] in which they considered the automatic diagnosis of faulty modules in a multiprocessor system. In this fundamental scheme the modules of the system must be capable of testing, either independently or in groups, one or more other modules. The outcome of a *test* can be one of two possible values: 0 indicates a successful test result, and 1 indicates an unsuccessful test result. The complete set of the test results is called the *syndrome*. From a given syndrome, a *diagnosis* of the system is possible. A correct diagnosis involves identifying the faulty processing elements of the system.

2.2 Modular redundancy in fault-tolerant systems

2.2.1 Terminology and concepts

For clarity we present in this section some formal definitions and concepts related to modular redundancy that will be used throughout this thesis.

Processing Element (PE). An element of the system that is capable of processing information and making decisions. Typically this is the replicated element in a system where modular redundancy is applied.

Fault. A disruption in the logical behavior of a PE. The faults considered in this thesis are those within one of the PEs, and which cause the digital output of the PE to be different from that which is specified for a given sequence of inputs [Dah88].

Permanent fault. A fault (as above) in a PE that causes the PE to *constantly* exhibit behavior that is different from its logical behavior [Dah88].

Intermittent fault. Temporary fault inside a PE whose conditions of activation cannot be reproduced or which occur rarely enough [Lap92]. The PE should be declared faulty until repaired.

Transient fault. Temporary fault external to a PE but which affects it as a result of environmental interference or interaction [Lap92]. A transient fault does not necessarily imply that the PE should be declared faulty—a temporary shutdown may be warranted because of the unstable environment.

2.2.2 Modular redundancy

Many fault-tolerant systems are based on the principle of *N-Modular Redundancy* (NMR). In this scheme, which falls under the domain of static redundancy, a computation is performed on an odd number N of independent hardware units. The results of these redundant computations are combined in such a way as to mask the faults. Figure 2.1 shows a simple example of NMR.

Majority voting. NMR has a long history, going back as far as research done by Von Neumann [Von56]. Since that time, it has been applied to a large number of experimental and commercial systems [Joh89; SS92; HSLJ78; WLGG78; KWFT88; FCN95e]. In particular, these systems apply *majority voting* to the NMR scheme. Majority voting obtains a correct result if no more than $(N-1)/2$ of the redundant elements produce a faulty result. In other words, up to $(N-1)/2$ faulty elements can be tolerated by this technique. Furthermore, even if these $(N-1)/2$ faulty elements were to produce an identical, faulty result, it would still be masked by the majority voting scheme.

Majority voting has always been recognized as a significant part of the cost of an NMR system when the computational results are large sets of data, e.g., longwords, as opposed to a sin-

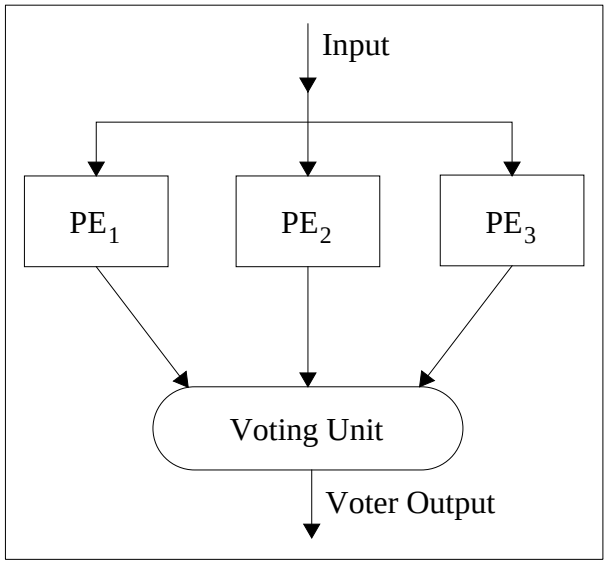


Figure 2.1 Majority voting in an NMR system with $n = 3$

gle bit of information. Furthermore, this cost is compounded if the replicated elements are distributed—there is significant overhead from the exchange of information during the voting process. In fact, if faults are arbitrary, distributed voting requires as many as $3f + 1$ processing elements to tolerate f failures, and requires $f + 1$ rounds of communication [LSP82].

Check words and signatures. To alleviate these problems, a technique was proposed in which the computation result itself is no longer used in the voting. Instead, the result is replaced by a *check word*, which is obtained from the computation result and is thus significantly smaller. Then these check words are used for the majority voting, resulting in a lower communication overhead as well as a less complicated voting procedure [Skla76, Wen82]. Although there is a chance of potentially selecting a faulty result when voting is done on check words rather than on the original results, if the method for computing the check words is carefully selected, then the chance of selecting a faulty output is low. Such a scheme is not suitable, however, if majority voting is implemented at the gate level, since there is a significant cost in computing the check words in hardware.

The goal of fault masking is neither to detect faults nor to identify the faulty elements—the results produced by faulty elements should simply be masked from the final output of the system. But if faulty elements are allowed to continue to function in an NMR system, they eventually will outnumber the fault-free elements, and the masking technique will have failed. Consequently other techniques have been proposed that permit the detection, isolation and removal of faulty elements in an NMR system.

Adaptive voting. In adaptive voting, each voted input n_i coming from the ith PE has a weight

a_i, which is updated according to the recorded history of disagreements that the PE has had with the majority of the other PEs. The voter decides upon an output value based on the sum $\sum a_i n_i$ and a threshold detector. In most practical digital systems, the a_i are either zero or one.

Self-purging redundancy. One form of adaptive voting is known as *self-purging redundancy* [Los76]. The system is organized as *N* redundant PEs that simultaneously perform identical operations. In the basic implementation, the boolean results of the redundant PEs are fed to a threshold voter, whose threshold equals 2. Any redundant unit is disconnected from the system if it disagrees with the vote-taker's output, that is, the majority of the PEs.

Figure 2.2 shows a self-purging redundant system with 5 processing elements. Each elementary switch (ES) compares the output of the system with the output of its corresponding processing element. If the two disagree, the ES isolates its PE from the rest of the system so that its input will no longer be a part of the voting process.

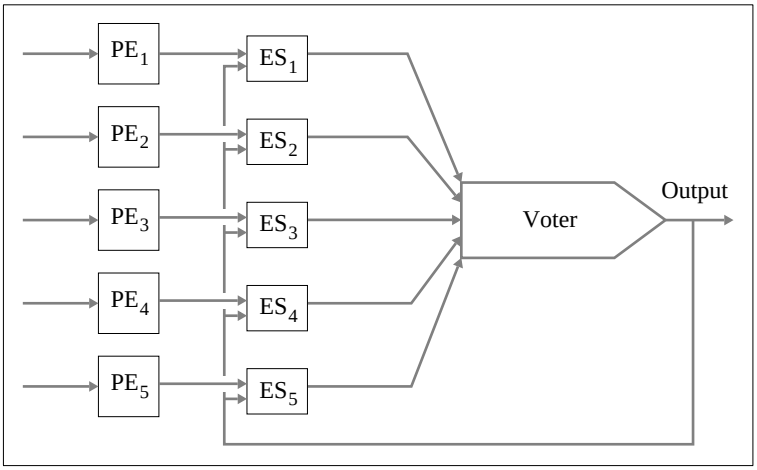


Figure 2.2 Self-purging redundancy with *N* = 5

It has been found that voting has many applications apart from NMR systems. These include Byzantine agreement [LSP82] and clock synchronization [KS84]. Many variations on how the voting is performed have also been proposed [Par91, Par94, BS94, SD89].

Sift-out redundancy. One alternative to conventional majority voting was proposed in which a comparison is done for every two results. This technique, called *sift-out modular redundancy,* was suggested by de Sousa *et al.* [dSM77, dSM78]. The system is arranged into *N* identical PEs, and the PEs perform simultaneous operations synchronously. Each PE remains active as long as it has not been identified as faulty. If a PE fails, it no longer contributes to the system, and the system then becomes a set of $N - 1$ PEs. This process is repeated each time a new failure occurs.

An example of sift-out modular redundancy is illustrated in Figure 2.3 for a system with five redundant elements. In this method, an assumption is made about faulty PEs that is somewhat different from that of conventional NMR: no more than one of the redundant elements may fail at the same instant of time. Under this assumption, the system can tolerate up to $N - 2$ failures. The complexity of the circuitry for comparison and selection in this method has been shown to be significantly less than that of conventional majority voting.

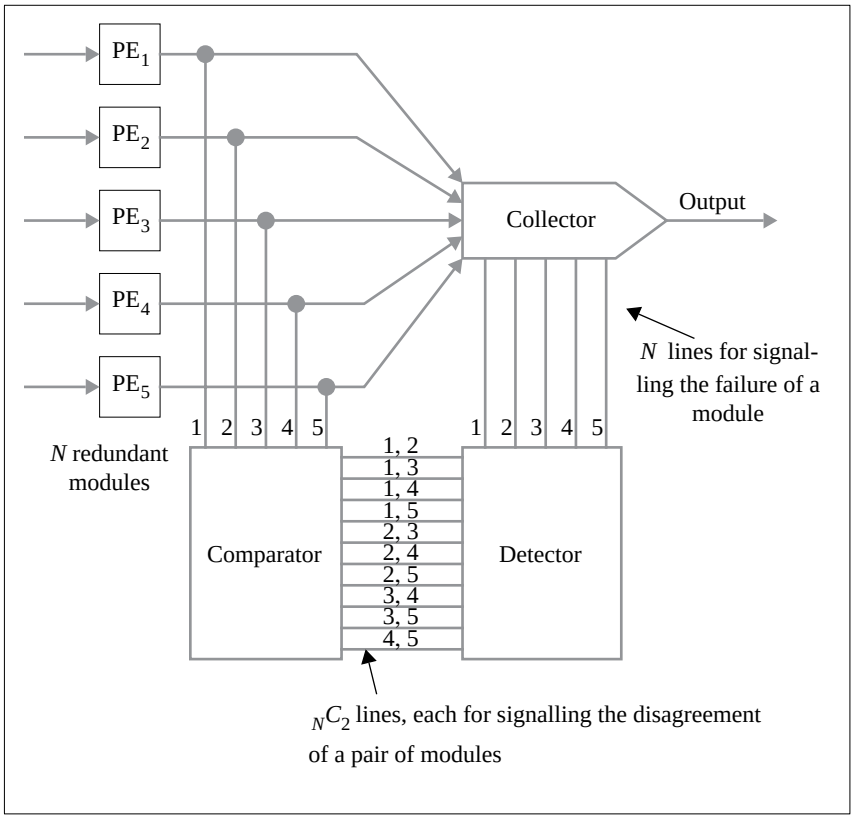


Figure 2.3 Sift-out redundancy with *N* = 5

Comparison-based modular redundancy. Drawing upon the more recent theory of system-level diagnosis, Y. Chen and T. Chen [CC90,Che92] explored a scheme called NMRC (*N-Modular Redundancy with Comparing*) in which the conventional majority voter is replaced by comparators plus auxiliary circuits. The authors characterized their approach for faults of both processing elements and comparators.

An NMRC system is composed of *N* identical PEs which execute the same task with identical inputs. The outputs of the PEs are connected directly to the system output through tri-state gates. Like the sift-out method describe above, NMRC has a set of comparators, where each of the *r* comparator's two inputs are connected to the output of a distinct PE. The output of the comparators form the fault syndrome, which is represented as a vector $S = (s_1, \ldots, s_r)$. The

syndrome can be decoded into a control vector $D = (d_1, \ldots, d_M)$, where $M \leq N$, of tri-state gates. One and only one component of *D* can be 1, and this selects a unique tri-state gate allowing the corresponding PE's output to pass through as the system output. An example of an NMRC system with three PEs is illustrated in Figure 2.4.

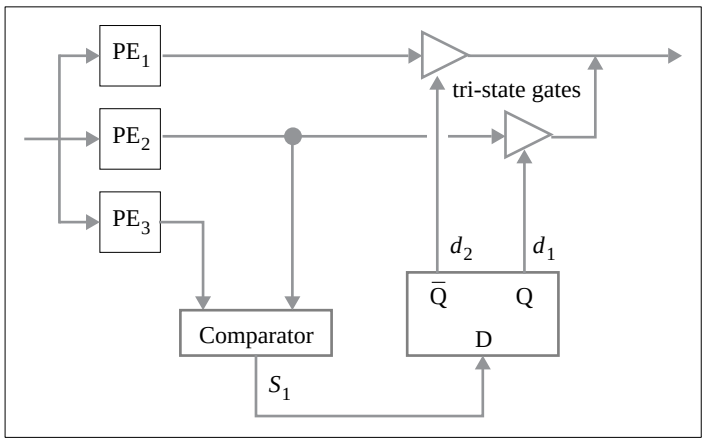


Figure 2.4 NMRC system with *N* = 3

Y. Chen and T. Chen found very low bounds on the number of comparisons, leading to a much more efficient NMR implementation than with majority voting or with the comparison-based, sift-out modular redundancy [dSM77, dSM78]. Furthermore, NMRC can tolerate faults in the comparators. NMRC systems can be tailored in various ways, for example, to tolerate a certain number of faults (from 0 to $N - 2$), where the cost of fault tolerance is paid in the number of comparators in the system. An example of an optimal, NMRC system can be seen in Figure 2.4 where only one comparator is used for three PEs. In this case, the system can tolerate only a single fault.

This kind of optimization is not without its drawbacks. For example, the system in Figure 2.4 can only detect a single fault, and this has implications when more than one PE is faulty (even though the system can tolerate a single fault). Assume first that the results produced by PE_2 and PE_3 differ because one of the two is faulty. The comparator detects this single fault, and the result produced by PE_1 is chosen by "inference" and thereby produced as system output without ever having been checked. Suppose next that PE_1 were *also* faulty, and hence the limit of the number of faults the system can tolerate is exceeded. The fault in PE_1 will never be detected, and its result will be output as though it were fault-free.

As mentioned, NMRC can also tolerate faulty comparators. This has important consequences, as the authors point out that two faulty PEs producing the same result can also be seen as a faulty comparator that produces an agreement result.

2.2.3 Assumptions about faulty processing elements in NMR systems

As is shown in the previous sections, the various techniques for redundancy do not always make the same assumptions about how PEs may fail in the system. From the above techniques, we can classify two distinct classes of models of how faulty PEs are expected to behave.

In the first class of models, as seen in NMR with voting, it is assumed that up to $(N-1)/2$ PEs may be simultaneously faulty, and that these faulty PEs may even produce an identical, faulty result without compromising the fault tolerance of the system. This class of models is analogous to the symmetric comparison model [CH81b] in system-level diagnosis that is discussed in Section 2.3.3 of this thesis.

For the other class of models, methods such as sift-out redundancy assume that simultaneous failures of PEs are not possible or that if simultaneous faults occur, the results produced by the faulty PEs are never identical. This assumption can be justified by considering the extremely small probability that such an event can occur. According to this assumption, two redundant units that have produced the same result can be considered as fault-free. This class of models is analogous to the asymmetric comparison model [Mal80] in system-level diagnosis, a discussion of which can be found in Section 2.3.3.

A list of various redundancy techniques and the assumptions made in their fault model is shown in Table 2.1.

Table 2.1 Number and type of faults tolerated by various redundancy techniques

Redundancy Technique	Max number of faulty results tolerated		
	sequential	simultaneous different	simultaneous identical
NMR with voting [Von56]	$(N-1)/2$	$(N-1)/2$	$(N-1)/2$
Self-purging [Los76]	$N-2$	—	—
Sift-out modular [dSM78]	$N-2$	—	—
NMRC [CC90, Che92]	$N-2$	up to $N-2$	up to $N-2$

2.3 System-level diagnosis

2.3.1 Terminology and concepts

In this section we present formal definitions and concepts related to system-level diagnosis that will be used throughout this thesis. Peculiarities between the definitions from Section 2.2.1 are noted as necessary.

Processing Element (PE). A PE is represented in terms of graph theory as a vertex *v.*

System. The collection of some number *n* of PEs, typically referred to as a multiprocessor system *S* and denoted in graph theory as a set $V = \{v_1, v_2, \ldots, v_n\}$.

Test. A series of programs or other means of measuring the fault status of a PE [PMC67].

Task. A set of instructions performed by a PE as part of real work done by the system.

Task result. The tangible result of the execution of a task on a single PE, which is used for comparison tests.

Comparison test. A comparison of the task results of a task that has been executed on two distinct PEs, with identical values for input.

Test outcome. Binary result of a test. A value of 0 indicates that the test was passed; a value of 1 indicates that it was failed.

Comparison test outcome. Binary result of a comparison test. A value of 0 indicates that the two task results were identical; a value of 1 indicates that the two task results were different.

Test assignment. The complete collection of tests in *S* is represented by a graph $G = (V, E)$, where each node $v_i \in V$ is represented by a PE. If *G* is a directed graph, then each edge $(v_i, v_j) \in E$ if and only if v_i tests v_j. In the case where *G* is an undirected graph, $(v_i, v_j) \in E$ if and only if the task results obtained from v_i and v_j are used to make a comparison test.

Syndrome. The complete set of test outcomes for the system *S*.

One-step *f*-fault diagnosis. (Also called *diagnosis without repair.*) The identification of all of the at most *f* faulty PEs[1] in the system *S* from a given syndrome. The diagnosis is performed in one instance, without modifying anything in the system.

Sequential *f*-fault diagnosis. (Also called *diagnosis with repair.*) The identification of at least one of the at most *f* faulty PEs (if there exist any) in the system *S* from a given syndrome. The diagnosis is carried out in sequential rounds in which at least one faulty PE is guaranteed to be identified. Upon its identification the faulty PE is then replaced (or repaired). This process is repeated until no faulty PEs remain.

1. In this thesis the number of faulty processing elements will be denoted *f* instead of the usual *t* because we will use *t* to denote tasks.

***f*/*s*–diagnosis.** The identification of the at most *f* faulty PEs in *S* within a set of (at most) *s* PEs. This leaves the possibility of incorrectly diagnosing fault-free PEs as faulty, but a faulty PE is always identified [Fri75].

Adaptive diagnosis. The identification of faulty PEs that is carried out so that the results of previous tests are used to determine which tests to perform in the future. In other words, the test assignment is not fixed before tests are carried out.

2.3.2 Validation (directed graph) models in system-level diagnosis

Any approach at system-level diagnosis is based on a model. Among the things that a model describes are 1) how the faulty and fault-free PEs of the system behave, 2) how tests' results are obtained or 3) how the diagnosis is carried out.

The primordial and best known model was formalized by Preparata *et al.* [PMC67] and is known as the PMC model or *symmetric invalidation* model. In graph theory, a system under this model is represented by a directed graph. After testing has been completed on a graph, a centralized arbiter interprets the syndrome and identifies each PE to be either faulty or fault-free.

Symmetric invalidation model. This model, also known as the PMC model, is based on the following assumptions:

1. all faults are permanent;
2. each test is operated by a single PE;
3. each PE has the capability of testing any other PE (provided a communication path exists between the two);
4. no PE tests itself;
5. a fault-free PE is always able to determine correctly the condition of a PE it is testing;
6. a faulty PE produces unreliable results;
7. there is an upper bound, *f*, on the number of faulty PEs in the system; and
8. all test data are gathered and analyzed by a centralized observer/distributor, and the diagnosis is distributed back to the system.

Some of the assumptions for the PMC model are not necessarily valid nor desired in a fault-tolerant, distributed system. As we shall see later, efforts have been made to remove these restrictions. The first assumption forbids intermittent and transient faults. Assumption 5 states that there is a test that allows 100% fault coverage, something that is impossible in reality.

Assumption 7 can be impractical for many fault situations. Finally, assumption 8 implies a high cost in terms of time, information exchange, and most importantly in system reliability, since it is directly dependent upon the reliability of the centralized observer/distributor. This last point has been considered by [Blou78] where a group of PEs was considered as the centralized observer/distributor in order to improve its reliability by using NMR.

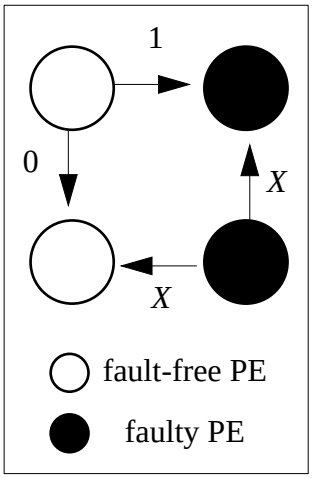


Figure 2.5 PMC test validity model

Preparata *et al*. showed that if as many as f PEs may be faulty (assumption 7), then the system must contain n PEs, where $n \geq 2f + 1$ to be diagnosable in all cases. Furthermore, they showed that each PE must be tested by at least f distinct other PEs. Figure 2.5 shows the possible test outcomes for the PMC model. This model is named the symmetric invalidation model because a faulty node produces unreliable test results for either faulty or fault-free tested PEs. The "symmetry" can be seen in the arrows leaving the faulty node in Figure 2.5—both yield "*X*" (either 0 or 1) test results.

An example system under the PMC model can be found in Figure 2.6. The PEs are denoted as *a* through *e*, and the edges (arcs) between two PEs indicate the testing direction. In the example, *a* tests *e*, which tests *d*, and so on. The result of the test is indicated by the a label on the edge corresponding to the test. Faulty PEs are denoted with a black color, e.g., *a* is faulty in Figure 2.6. A label of "*X*" indicates that the result of the test may be either a 0 or a 1, according to assumption 6 above, but without affecting the outcome of the diagnosis.

For the system in Figure 2.6, it is assumed $f = 1$ (assumption 7). Given this assumption, the testing assignment and the syndrome, it is possible to always identify *a* as a faulty PE. First, since *a* is faulty, then its test results do not necessarily reflect the true status of *e*, hence the "*X*" on the label from *a* to *e*. This means that the result could be either a "0" or "1" and the diagnosis would still indicate that *a* is the faulty node. To see how this works, observe that the

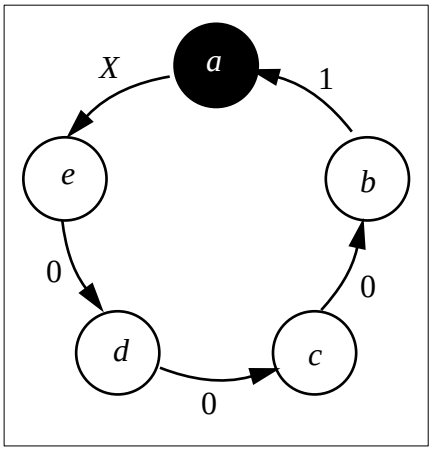


Figure 2.6 Example system under the PMC model

edge between *b* and *a* is labeled with a "1", which means in this model that *a* is faulty if *b* is fault-free. If *b* were hypothetically faulty then it would have to be the single faulty PE in the system, since $f = 1$. Essentially, then, diagnosing the system depends on deducing the status of *b.*

With the syndrome indicated in Figure 2.6, either *a* or *b* is faulty. If we assume *b* to be faulty, then *c* must be fault-free, since $f = 1$. Yet this contradicts the syndrome, as the edge from *c* to *b* is labeled with a "0" indicating that *b* is not faulty (assumption 5). Thus *b* is not faulty, and *a* must be faulty (assumption 6), regardless of the actual value of "*X*." In this example, if more than one PE is faulty, then diagnosis cannot be guaranteed under the PMC model, because each PE must be tested by at least *f* other PEs. The configurations for systems that allow unambiguous diagnosis in all cases under the assumptions of PMC model are said to be *t-diagnosable*, where *t* refers to *f*, the upper bound on the number of faulty PEs in the system.

Preparata *et al.* [PMC67] also studied cases where diagnosis was performed along with system repair. In a system where it is not feasible to carry out the diagnosis in multiple phases, then one must be able to diagnose all of the faulty PEs after a single round of testing. This is known as *one-step diagnosis* or *diagnosis without repair.* On the other hand, if the system is repairable, then only one faulty PE needs to be identified, if it exists. Once identified, the faulty PE can be repaired and the diagnosis continued until all faulty PEs have been identified and repaired. This is referred to as *k-step diagnosis* (where the diagnosis is carried out in *k* steps) or *sequential diagnosis* or *diagnosis with repair.*

Barsi, Grandoni and Maestrini [BGM76] proposed a model that is a modification of the PMC model, with the intention of having a more realistic representation of systems whose units have a rather complex structure. Their model, referred to as the BGM model or *asymmetric invalidation* model, is less general than the PMC model. A faulty PE can never cause tests

on another faulty PE to pass, yet it may cause tests on a fault-free PE to fail. The possible outcomes for the BGM model are shown in Figure 2.7. This model is named the asymmetric invalidation model because a faulty node produces unreliable test results *only* when it has tested a fault-free PE. The "asymmetry" can be seen in the arrows leaving the faulty node in Figure 2.7—one yields an "*X*" (either 0 or 1) test results, while the other yields always a 1.

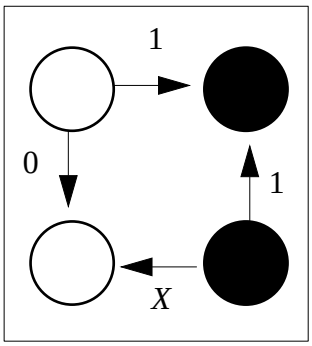


Figure 2.7 BGM test validity model

Asymmetric invalidation model. Known also as the BGM model, this model is based on the following assumptions:

1. all faults are permanent;
2. each test is operated by a single PE;
3. each PE has the capability of testing any other PE (provided a communication path exists between the two);
4. no PE tests itself;
5. a fault-free PE is always able to determine correctly the condition of a PE it is testing;
6. a faulty PE that tests a fault-free PE yields a result that is unreliable (the "*X*" in Figure 2.7 means that the result is either 0 or 1);
7. a faulty PE is always able to correctly determine the condition of a faulty PE it is testing;
8. there is an upper bound, *f*, on the number of faulty PEs in the system; and
9. all test data are gathered and analyzed by a centralized observer/distributor, and the diagnosis is distributed back to the system.

In summary, the asymmetric and symmetric invalidation models vary with respect to the way in which faulty PEs behave. The asymmetric invalidation model is more restricted in the sense that it allows for a fewer number of syndromes given a distinct set of faulty PEs. It is therefore easier to diagnose faulty PEs under this model.

2.3.3 Comparison (undirected graph) models in system-level diagnosis

The *test* in the invalidation models of system-level diagnosis typically consists of performing an action and then comparing the outcome of that action with a known value. If the test result disagrees with the known answer, then a fault has been uncovered. There is a problem with this method: the complexity of PEs makes it difficult to have a test that can determine indubitably and promptly whether the PEs are faulty or fault-free.

An alternate model that has gained much attention is called the comparison model, because tests are in fact the comparison of two task results produced on distinct PEs. Since the testing procedure is made more explicit, the comparison model is more specific than the PMC model. Also, as the task results used for comparison can be taken from tasks that perform real work, diagnosis can be carried out from the normal operation of the system.

When two results are compared, a fault can be detected but not diagnosed. When comparing more than two results, up to $\lceil n/2 \rceil - 1$ PEs can be diagnosed. When PEs have a low rate of failure, the comparison method is useful since it is less likely that two PEs will fail at the same time, and if they do, it is also less likely that they will fail in the same way. Hence, two PEs that execute identical, deterministic tasks should produce identical task results, except if one or possibly both of them has failed.

As in the other models for system-level diagnosis, intermittent faults can be difficult to brook in the comparison models. An intermittently faulty PE could produce correct results for certain tasks, and thus its faulty state is not detected. We discuss this issue more thoroughly in Section 2.3.5.

Unlike the PMC model where tests are carried out in rounds between the execution of system tasks, the comparison model allows comparison tests to be performed in conjunction with productive tasks. In this way, a fault is detected when it occurs, and the system's ability to contain faults is higher than in the PMC model. Furthermore, the comparison approach can be optimized if the results to be compared are complex. The results can be converted into *signatures*, e.g., check words or cyclic redundancy codes, and these can be compared instead of the results themselves.

One of the comparison models for system-level diagnosis was introduced by Malek [Mal80]. A method was presented for assigning edges in the graph of the system according to the comparisons to be carried out. More specifically, an edge exists between two nodes in an undirected graph if a comparison is to be performed between the results produced by the PEs corresponding to the two nodes. The syndrome is then generated simply by assigning a zero to

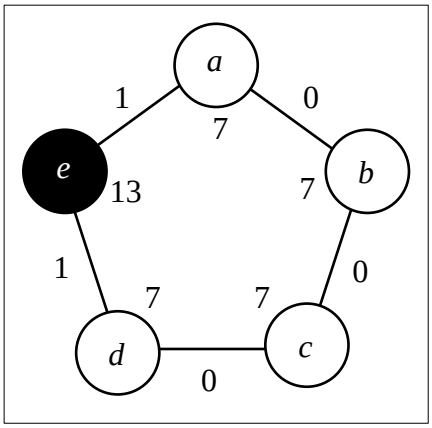


Figure 2.8 Example system for the comparison model

a comparison of two results that are identical, and a one to a comparison of two results that are different. Figure 2.8 illustrates this scheme, with the faulty node *e* that has produced a result (13) that is different from the result (7) produced by the remaining nodes *a*, *b*, *c*, *d*. The comparison outcomes are shown along the edges of the graph.

In this first model, Malek assumed that faulty PEs never produced identical results. This model is sometimes referred to as the asymmetric comparison model [BP92]. As in the PMC directed graph models, a centralized supervisor diagnoses the system by interpreting the syndrome.

Asymmetric comparison model. Proposed by Malek [Mal80], this model is based on the following assumptions:

1. all faults are permanent;
2. a faulty PE produces incorrect outputs for each of its given tasks;
3. two faulty PEs, when given the same inputs and task, *always* yield differing task results;
4. there is an upper bound, *f*, on the number of faulty PEs in the system; and
5. all task results are collected and compared by a centralized supervisor, and the diagnosis is distributed back to the system.

To overcome the problem of a centralized comparator, Maeng and Malek [MM81] considered using a third PE to actually do the comparison of results produced by two other PEs. This entailed a different fault model that is shown in Table 2.2, for it considers the behavior of a faulty comparator. Similar to the directed graph models, a faulty comparator produces unreliable results about the comparison. Although comparisons are specifically distributed in this model, the diagnosis may still be performed by a centralized unit.

Chwa and Hakimi [CH81b] propounded independently of Malek an approach using comparisons. Their model is sometimes called the symmetric comparison model [BP92].

Table 2.2 Fault model for comparisons in Maeng/Malek model

Comparator	PE 1	PE 2	Comparison test result
fault-free	fault-free	fault-free	0
fault-free	fault-free	faulty	1
fault-free	faulty	fault-free	1
fault-free	faulty	faulty	1
faulty	fault-free	fault-free	X
faulty	fault-free	faulty	X
faulty	faulty	fault-free	X
faulty	faulty	faulty	X

Symmetric comparison model. This model, which was proposed by Chwa and Hakimi [CH81b], is based on the following assumptions:

1. all faults are permanent;
2. a faulty PE produces incorrect task results for each of its given tasks;
3. two faulty PEs, when given the same inputs and task, *may* yield the same task result; and
4. there is an upper bound, *f*, on the number of faulty PEs in the system.

In summary, the two models vary with respect to the way in which faulty PEs behave. The asymmetric comparison model [Mal80] is more restricted in the sense that it allows for a fewer number of syndromes given a distinct set of faulty PEs. It is therefore easier to diagnose faulty PEs under this model. On the other hand, the symmetric comparison model [CH81b] is considered to be more realistic in systems where faulty PEs are capable of producing identical task results.

2.3.4 Alternative models

Kreutzer and Hakimi [KH83] presented two alternatives to the PMC and BGM models that are referred to respectively as the HK1 and HK2 models. In both models it is assumed that a test is composed of a large number of evaluations. We discuss these two models briefly as they are of some importance. Although they are based upon invalidation (directed graph) models, the assumptions about test results for the HK1 and HK2 models correspond respectively to those of the symmetric [CH81b] and asymmetric comparison [Mal80] models. Therefore the results for the HK models can be of use in our study of comparison models.

The HK1 model varies from the PMC model in that it is assumed that a faulty PE will

always evaluate a fault-free PE as faulty. This is due to the likely discrepancy between test results expected by the (faulty) testing PE and provided by the (fault-free) tested PE. It is still possible, however, for a faulty PE to evaluate another faulty PE as either fault-free or faulty since the test results expected could agree with those provided. The possible test outcomes for the HK1 model are shown in Figure 2.9.

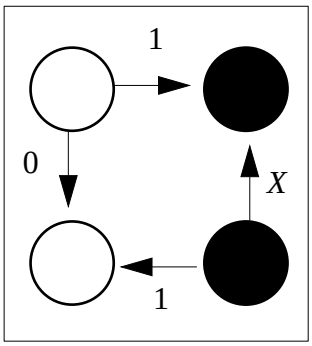


Figure 2.9 HK1 test validity model

The HK2 model is based on the BGM model, but Kreutzer and Hakimi improve on what they considered an unrealistic assumption of the BGM model—that a faulty PE and a fault-free PE may agree on test results. In the HK2 model, if one (or both) of the two PEs involved in a test is faulty, the tester will evaluate the tested unit as faulty, since it is assumed that the results will disagree. The possible test outcomes are as shown in Figure 2.10.

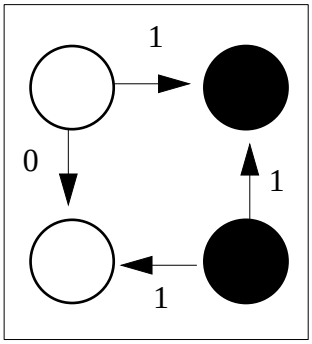


Figure 2.10 HK2 test validity model

2.3.5 Intermittent faults and probabilistic models

In the models discussed so far in this thesis, intermittent faults have not been discussed. Work has been done, however, to relax this constraint. In the PMC model, if intermittent faults are considered, then a fault-free unit can no longer be considered reliable when it judges the condition of another PE. This is because if the tested PE is intermittently faulty, it may actually pass the test and appear to be fault-free to the testing PE. This implies that the set of all test syn-

dromes produced by a system suffering from intermittent faults is a superset of all test syndromes possible in the same system where only permanent faults are allowed. Therefore, diagnostic algorithms designed for systems with only permanent faults may no longer be applicable.

To tackle this problem, Dahbura and Masson introduced the notion of *greedy diagnosis*, where the goal is to identify faulty PEs from a syndrome that is not one of the permanent-fault subset. This technique was originally applied to the PMC model [DM83a], but was later applied to the comparison model [DM83b]. Their algorithm requires that there be a bound to the number of intermittently faulty PEs as well as the number of PEs that can fail simultaneously. They showed that the problem was NP-complete in the most general of all cases.

Also for the comparison approach, Dahbura *et al.* [DSK87] proposed an efficient algorithm for identifying faulty PEs based on probabilities. The authors assume that for a given task, a faulty processor produces a result that does not necessarily have to be an incorrect result. As such this model can be seen to tolerate intermittently faulty PEs in the sense that faulty processors do not always produce incorrect results. It considers the probabilities for a faulty PE to produce each of the possible outcomes of a given task, and is similar in nature to the Malek model because it considers the likelihood of two faulty PEs producing an identical, incorrect result to be negligible. Although their strategy is not guaranteed to always identify the set of faulty PEs, they show that the probability of the cases where the algorithm fails is extremely small.

Kozlowski and Krawczyk [KK91] introduced a model that is based on comparisons and that tolerates hybrid faults, although it is not probabilistic. The authors assumed bounds on both the number of faulty PEs and the number of incorrect outcomes (i.e., fault-free PE evaluating a faulty PE to be fault-free), and they gave necessary and sufficient conditions for a comparison assignment to achieve complete and correct diagnosis. They presented an $\mathbf{O}[n|E|]$ algorithm for identifying faults.

2.3.6 Fundamental problems in system-level diagnosis

All work in system-level diagnosis can be considered as efforts to solve one or more of three following problems:

Characterization: Given a certain value of f (the upper bound on the number of faulty PEs), what are the necessary and sufficient conditions for a multiprocessor system to be diagnosable?

Hakimi and Amin [HA74] provided a general solution for one-step, f-diagnosable systems.

The number of PEs in these systems is $n \geq 2f + 1$, and there is the requirement that at each PE must be tested by at least *f* distinct other PEs. Furthermore, they showed that for each integer *p*, $0 \leq p < f$, every subset *X* of PEs, $|X| = n - 2f + p$ must be tested by more than *p* processing elements that are outside of *X*.

Huang *et al.* [HXC89] characterized systems that are sequentially *f*-diagnosable in the PMC model. Somani *et al.* [SAA87] contributed a generalized characterization theorem that provides necessary and sufficient conditions for a particular fault set of any size to be uniquely diagnosable under both the PMC (symmetric) and BGM (asymmetric) models, and where intermittent faults are considered or not. They showed that the *f*-diagnosability of a system only represents the performance of a diagnosis algorithm for a syndrome that represents the worst case. Furthermore, certain syndromes can be still uniquely *f*-diagnosable under the conditions they presented even though more than *f* faulty PEs may exist.

Diagnosability: Given a collection of PEs and a set of test assignments for each PE, what is the maximum number *f* of arbitrary PEs that can be faulty such that the set of faulty PEs can be *uniquely* identified on the basis of any given syndrome?

Sullivan [Sul84] gave a solution to the diagnosability problem under the assumptions of the PMC model. He developed an algorithm with a complexity $\mathbf{O}[|E|n^{1.5}]$, where $|E|$ is the number of tests, that can calculate the *f*-diagnosability of a given testing assignment. An improved version of this algorithm with a complexity $\mathbf{O}[nf^{2.5}]$ was proposed by Raghavan and Tripathi [RT91a]. These authors also showed [RT91b] that the problem of finding the diagnosability of systems that are repairable (sequentially diagnosable) is co-NP-complete.

Diagnosis: Given a testing assignment for a system such that at most *f* PEs can always be uniquely identified on the basis of any given syndrome, does there exist an efficient procedure (algorithm) for identifying the faulty PEs?

Fujiwara and Kinoshita showed that it is an NP-complete problem to determine a set of PEs of minimal cardinality that, if faulty, could produce a given syndrome on a graph with arbitrary testing assignments. Consequently, the arbitration of conflicting test results is also NP-complete.

For more restricted cases, efforts have been made to solve the diagnosis problem. Kameda, *et al.* [KTA75] proposed an $\mathbf{O}[f|E|]$ algorithm for *f*-diagnosable systems in which PEs are successively presumed to be faulty or fault-free. This supposition and the test syndrome impli-

cate the states of other PEs. If a contradiction is exposed, the algorithm falls back and tries again until it finds a consistent set of faulty PEs. Sullivan [Sul88] improved the Kameda *et al.* solution with an $\mathbf{O}[f^3 + |E|]$ algorithm. The best solution to this approach when considering worst-case efficiency was given by Dahbura and Masson [DM84a], with an $\mathbf{O}[n^{2.5}]$ algorithm in which an undirected graph G is created whose nodes are the PEs in the system and whose edges are the implied faulty sets of each PE. A more practical variation of their algorithm was also later given [DM84b].

A study was done by Dahbura *et al.* [DM84a] of the practical efficiency of the $\mathbf{O}[n^{2.5}]$ algorithm as opposed to the Kameda *et al.* [KTA75] procedure. They found that for $n \leq 30$, the Kameda procedure is almost always more efficient. Even when n is larger, the Kameda algorithm is more efficient on average than the Dahbura and Masson approach [DM84a,DM84b]. This is because for an obvious syndrome the Kameda algorithm does little back tracking whereas the $\mathbf{O}[n^{2.5}]$ algorithm executes a standard procedure for any syndrome.

More efficient methods for diagnosis exist when special classes of *f*-diagnosable systems are considered. Preparata *et al.* [PMC67] defined a class of structures known as $D_{\delta t}$ structures in which processing element u_i tests u_i if and only if $j - i = \delta m$ (modulo n), where $m = 1$, 2, …, t. It was shown that if δ and n are relatively prime, the system is one-step *f*-diagnosable. Several $\mathbf{O}[n^{2.5}]$ algorithms for systems where $\delta = 1$ were presented, namely by Meyer and Masson [MM78], Mallela [Mall80], and Chwa and Hakimi [CH81a]. Maheshwari and Hakimi [MH76] described Z_T systems. Chwa and Hakimi [CH81a] proposed the $D(n, t_0, X)$ class of systems, and Dahbura *et al.* [DMY85] defined a group of structures that are “self-implicating” each with diagnosis algorithms of complexity $\mathbf{O}[|E|]$. An $\mathbf{O}[|E|]$ algorithm for the most general class of testing graphs among these mentioned was proposed by Sullivan [Sul84]. These systems are known as *t*-vertex-connected digraphs and are a superset of the structures given by Dahbura *et al.* [DMY85].

The refinements to the original, PMC model proposed by Preparata *et al*. have led to many other solutions to the diagnosis problem. These solutions are addressed later in the chapter as needed.

2.3.7 Useful definitions and theorems

In this section we present several definitions and theorems concerning system-level diagnosis that are used throughout this thesis. First we recall several basic definitions and a theorem from graph theory, all of which apply in the context of this thesis to simple, undirected graphs (i.e., graphs with no self-loops or multiple edges) [Har72, Ber76].

Definition 2.1 A *walk* of a graph $G = (V, E)$ is an alternating sequence of nodes and edges $v_0, e_1, v_1, \ldots, v_{n-1}, e_n, v_n$, beginning and ending with nodes, in which each edge is incident with the two nodes immediately preceding and following it.

Definition 2.2 A walk is *closed* if $v_0 = v_n$ and is *open* otherwise.

Definition 2.3 A *cycle* is a closed walk whose $n \geq 3$ nodes are distinct.

Definition 2.4 A graph $G = (V, E)$ is a *bipartite graph* if its vertices can be partitioned into two disjoint sets $V = S_1 + S_2$ where every edge has one node in S_1 and the other in S_2.

Definition 2.5 A *path* is a walk whose *n* nodes (and thus necessarily all the edges) are distinct.

Definition 2.6 A *Hamiltonian path* is a path of a graph $G = (V, E)$ that includes all the nodes in *V.*

Definition 2.7 A graph $G = (V, E)$ is *connected* if there is a path between any two nodes $u \in V$ and $v \in V$.

Definition 2.8 The *connectivity* $\kappa = \kappa(G)$ of a graph *G* is the minimum number of points whose removal results in a graph that is not connected.

Definition 2.9 A graph $G = (V, E)$ is *complete* if every pair of nodes has an edge between them. A complete graph is denoted as K_p where $p = |V|$. Thus K_p has $(p(p-1))/2$ edges. A graph is a *complete bipartite graph* if for every $x \in S_1$ and $y \in S_2$ we have $(x, y) \in E$, and is denoted $K_{m,n}$, where $m = |S_1|$ and $n = |S_2|$.

Definition 2.10 A maximal connected subgraph of a graph $G = (V, E)$ is called a *connected component* of *G* or simply a c*omponent* of *G*.

An immediate consequence of Definition 2.10 is that a graph with a single component is connected.

Definition 2.11 [Har72] A *tree* is a connected graph with no cycles. A *spanning tree* of a graph *G* is a tree containing all vertices of *G*.

Definition 2.12 [Har72] In a connected graph *G* with a spanning tree *T*, a *chord* of *T* is a line of *G* which is not in *T*. Hence, the subgraph of *G* consisting of *T* and any chord of *T* has exactly one cycle.

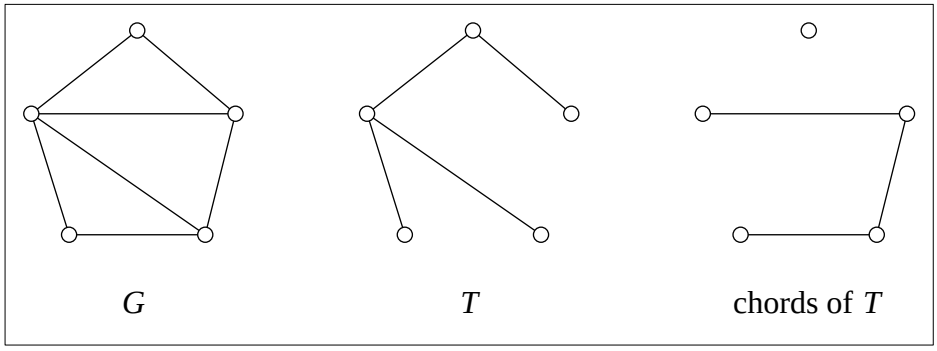


Figure 2.11 A graph *G*, a spanning tree *T* of *G*, and the three chords of *T*

Definition 2.13 [Har72] The *cycle rank* (also known as *cyclomatic number*) of a graph $G = (V, E)$ is equal to the number of chords in any spanning tree in *G*, and is represented as $\nu(G)$.

Figure 2.11 shows a connected graph *G*, a spanning tree *T* for *G*, and the chords for *T*. It can be seen how $\nu(G) = 3$ since it has three chords.

Theorem 2.1 For any graph $G = (V, E)$ with $n = |V|$, $m = |E|$ and k components, $\nu(G) = m - n + k \geq 0$.

Proof. Let $V = \{v_1, \ldots, v_n\}$ and $E = \{e_1, \ldots, e_m\}$. The theorem is proved recursively by constructing the sequence of graphs $G_i = (V, E_i)$, with $E_0 = \varnothing$ and $E_i = E_{i-1} \cup \{e_i\}$ for $i = 1, \ldots, m$. The theorem is proved for G_0 since $m = 0$, $k = n$ and $\nu(G) = -n + n = 0$.

Assume the theorem is proved for G_i and consider G_{i+1}. Then $\nu(G_i) = m_i - n + k_i \geq 0$, and there are two possible cases:

Case 1. $\{e_i\} = (u, v)$ *has its ends in two distinct components of* G_i. Then G_{i+1} has $m_{i+1} = m_i + 1$ edges, *n* nodes and $k_{i+1} = k_i - 1$ components. Hence

$\nu(G_{i+1}) = m_{i+1} - n + k_{i+1} = m_i + 1 - n + (k_i - 1) = m_i - n + k_i = \nu(G_{i+1}) \geq 0$.

Case 2. $\{e_i\} = (u, v)$ *has its ends in the same component of* G_i. Then G_{i+1} has $m_{i+1} = m_i + 1$ edges, *n* nodes and $k_{i+1} = k_i$ components. Hence

$\nu(G_{i+1}) = m_{i+1} - n + k_{i+1} = m_i + 1 - n + k_i = m_i - n + k_i + 1 \geq \nu(G_i) + 1 \geq 0$.

Hence in both cases $\nu(G_{i+1}) \geq \nu(G_i) \geq 0$. □

Corollary 2.1 For any graph *G*, $\nu(G) = 0$ if and only if there are no cycles in *G*.

Proof. It can be verified from the way the proof of Theorem 2.1 is constructed that there is a cycle when $\nu(G) > 0$. □

We now present several definitions that are useful in the context of comparison diagnosis. Formally we consider a system *S* that consists of a set $V = \{v_1, v_2, \ldots, v_n\}$ of *n* nodes. Each node $v_i \in V$ executes a replica of the same task and produces a result r_i. The result r_i produced by each node v_i is compared to the results produced by some other nodes. A comparison test link c_{ij} corresponds to the assignment of a comparison between nodes v_i and v_j. The complete collection of the c_{ij} is the *comparison testing assignment* that is represented by an undirected *comparison testing graph* $G = (V, E)$. Each vertex represents a node $v_i \in V$ and each edge $e_i = (v_i, v_j)$ is in *E* if and only if c_{ij} exists. With each edge is associated a test outcome $w_{ij} = 0(1)$ if the results r_i and r_j agree (disagree).

Definition 2.14 [GM95] With each node $v_i \in V$ we associate the set $\Gamma(v_i) = \{v_j : ((v_i, v_j) \in E)\}$, which is the set of all nodes adjacent to v_i.

Definition 2.15 [GM95] For a set of nodes $X \subseteq V$, we define $\Gamma(X) = \bigcup_{v_i \in X} \Gamma(v_i) - X$, the set of nodes that are outside of *X* and adjacent to at least one node in *X*.

Definition 2.16 [XH95] For an undirected graph $G = (V, E)$, a set of subsets of V, $\pi = \{V_1, \ldots, V_k\}$ where $V_i \subset V \ (1 \le i \le k)$ is called a *cover* of V if and only if $\bigcup_{i=1}^{k} V_i = V$. A cover π is called a *standard cover* if and only if $\bigcup_{i=1}^{k} V_i = V$ and for each j, $(1 \le j \le k)$, $\bigcup_{i=1, i \ne j}^{k} V_i \ne V$. In other words, if any subset V_i in a standard cover π is removed, then π is no longer a cover.

Definition 2.17 [XH95] Given a set of subsets of V, $\pi = \{V_1, \ldots, V_k\}$ where $V_i \subset V$ $(1 \le i \le k)$, with each node $v \in V$ we associate the set of subsets $f(v)$ to which v belongs, with $f(v) = \{V_i : (V_i \in \pi) \wedge (v \in V_i)\}$.

Definition 2.18 Let $G = (V, E)$ be an undirected graph. A function $\sigma: E \rightarrow \{0, 1\}$ is said to be a *syndrome* for G.

Definition 2.19 Let $G = (V, E)$ be an undirected graph for the comparison model and $G_0 = (V, E_0)$ be the graph derived from G by deleting all edges that compare to 1. G_0 is called the *0-derived graph* of G. Formally in $G_0 = (V, E_0)$, $E_0 \subseteq E$ and $E_0 = \{(v_i, v_j) : w_{ij} = 0\}$.

Definition 2.20 The graph $G_0 = (V, E_0)$ may be split into several components $G_{0,1} = (V_1, E_{0,1})$, …, $G_{0,k} = (V_k, E_{0,k})$ with $\bigcup_{i=1}^{k} V_i = V$, where a set of connected nodes V_i is called a *cartel* of G_0.

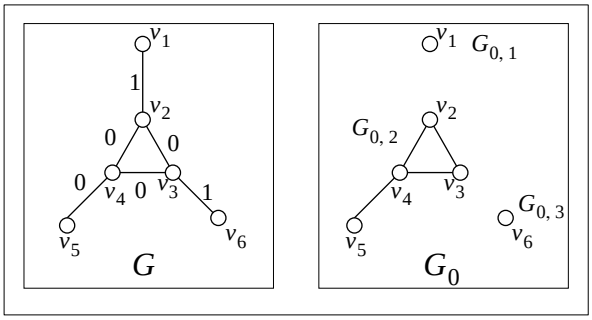


Figure 2.12 A graph *G* and its 0-derived graph G_0

Figure 2.12 shows an example of a graph G and its subsequent 0-derived graph G_0 with

components $G_{0,1}$, $G_{0,2}$ and $G_{0,3}$. In this example, the three sets $\{v_1\}$, $\{v_2, v_3, v_4, v_5\}$ and $\{v_6\}$ are all cartels of G_0.

2.4 Conclusion and contributions

This chapter has addressed the problem domain of this dissertation, namely that of modular redundancy and system-level diagnosis. As our thesis is oriented towards comparison-based diagnosis, we have concentrated on the techniques in modular redundancy that exploit comparisons. In addition, we have presented the background of system-level diagnosis, discussing the fundamental problems of characterization, diagnosability and diagnosis. We have intentionally focused on comparison-based diagnosis, as that is pertinent to this dissertation. We have underlined the utility of comparisons in modular redundant systems, namely that they permit the identification of faulty nodes, a point which is crucial for long-term fault tolerance.

As an important contribution of this dissertation, we expose the similarities between the fault models of modular redundancy and those of system-level diagnosis. We also point out that for system-level diagnosis, the alternative directed-graph models (HK models) are identical to the comparison-based models.

Chapter 3 Contributions to f/(n–1)-diagnosability

In this chapter we present several contributions to the area of $f/(n-1)$-diagnosability in system-level diagnosis. Section 3.1 introduces the concept of $f/(n-1)$-diagnosability and its importance in the context of this dissertation. Section 3.2 discusses how optimal systems can be characterized in terms of $f/(n-1)$-diagnosability. Section 3.3 presents a new class of optimal, $f/(n-1)$-diagnosable systems, including an algorithm for their diagnosis. Section 3.4 addresses the issue of sequential, $f/(n-1)$-diagnosability. Finally Section 3.5 concludes the chapter and summarizes the contributions.

3.1 Introduction

In the theory of system-level diagnosis much effort has been put forth in the study of various measures of diagnosability—the maximum number of faulty PEs a system can be subjected to and still remain diagnosable. A fundamental step was made by Friedman [Fri75] with the introduction of the notion of *f/s-diagnosability*: a system composed of n PEs is said to be f/s-diagnosable if, given a syndrome, the set of faulty PEs can be confined to a set of at most s PEs, provided that the number of faulty PEs does not exceed f. This work originally applied to the PMC testing model.

What interests us the most in f/s-diagnosability is the following: the $n - s$ PEs that are not in the set of s faulty PEs are identified as definitely fault-free. Obviously this is important since these fault-free PEs can be used either to diagnose adaptively the other possibly faulty PEs, or to determine that their output is fault-free.

For the case where $s = n - 1$, f/s-diagnosability consists of finding at least one fault-free PE in a system that has n PEs. The problem of f/s-diagnosability is notoriously difficult, but Xu and Huang [Xu91,XH90] achieved considerable success in characterizing the $f/(n-1)$-diagnosability conditions for several types of structures in the comparison testing models. They gave a few examples of systems that are $f/(n-1)$-diagnosable with two disjoint chains for application

in low-degree NMR systems, but these structures were not characterized for all system sizes. They presented optimal, $f/(n-1)$-diagnosable configurations for several classes of systems: chains, loops and $H_{2r,n}$ systems (a particular type of loop structure). In particular it was shown that for $n = 2f + 1$ chains are $f/(n-1)$-diagnosable for n up to $n = 9$ and loops are diagnosable for n up to $n = 13$.

In this section we extend the characterization of $f/(n-1)$-diagnosability under the symmetric comparison model specified in Section 2.3.3. We present a new class of systems and compare it to those presented by Xu *et al*.

3.1.1 Introductory definitions, theorems and lemmas

The following definition is pertinent to $f/(n-1)$-diagnosability.

Definition 3.1 Let $H = (F, L)$ be a subgraph of $G = (V, E)$ with $F \subseteq V$, $L \subseteq E$. The set F is a *consistent fault set* (CFS) for the syndrome σ in the symmetric comparison model if and only if all of the following hold:

1. $|F| \leq f$,
2. $\sigma(u, v) = 1$ for all $(u, v) \in E$, where $u \in V - F$ and $v \in F$,
3. $\sigma(u, v) = 0$ for all $(u, v) \in E$, where $u, v \in V - F$.

In other words, condition 1 implies that the maximum number of nodes in F cannot be more than the maximum number of allowed faulty nodes, f. Condition 2 states that any edge between a node outside of F and a node inside of F must have a value of 1, showing that the respective nodes' results disagree (see Figure 3.1). Condition 3 says that an edge between any two nodes which are outside of F must have a value of 0 indicating agreement between the results of the respective nodes (see Figure 3.1).

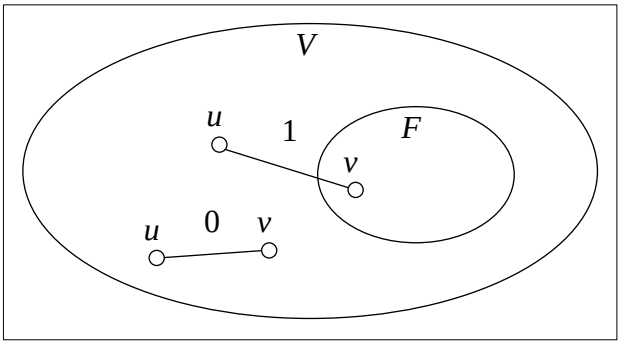


Figure 3.1 Conditions 2 and 3 for Definition 3.1

We present now some theorems and lemmas that are useful in our discussion of *f*/(*n*–1)-diagnosability. Xu *et al*. [XH90] gave a basic theorem that characterizes *f*/(*n*–1)-diagnosable systems that is repeated here without proof[1]:

Theorem 3.1 [XH90] A system S of $n \geq 2f + 1$ nodes represented by an undirected graph $G = (V, E)$ is *f*/(*n*–1)-diagnosable for the symmetric comparison model for up to f faults if and only if for any cover $\pi = \{V_1, \ldots, V_k\}$ of V where $|V_i| \leq f$, an edge (v_i, v_j) exists such that $f(v_i) \neq f(v_j)$ and $f(v_i) \cup f(v_j) \neq \pi$. □

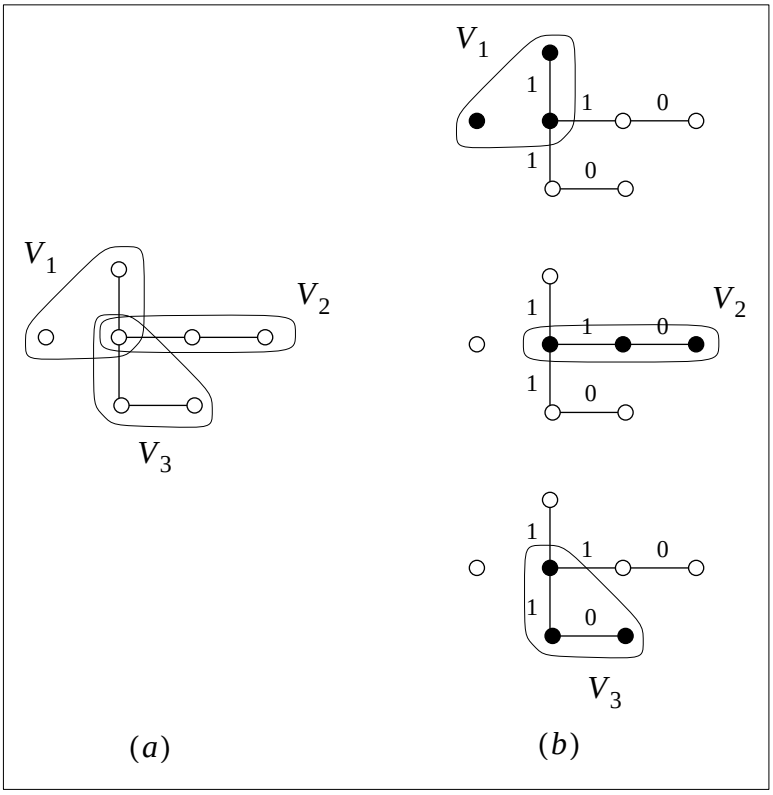


Figure 3.2 Example cover π where no edge (*u*, *v*) exists such that *f*(*u*)≠*f*(*v*) and *f*(*u*)∪*f*(*v*)≠π

In order to understand the necessity condition of this theorem, consider the undirected graph $G = (V, E)$ with $\pi = \{V_1, V_2, V_3\}$ as illustrated in Figure 3.2(a). In this example, no edge exists between any two nodes v_i and v_j such that $f(v_i) \neq f(v_j)$ and $f(v_i) \cup f(v_j) \neq \pi$. It is therefore possible to have a syndrome for the system under the symmetric comparison model such that *any* node can be faulty. More specifically, it is possible to have a syndrome such that given a cover where no edge exists between any two nodes v_i and v_j such that $f(v_i) \neq f(v_j)$

1. The proof can be found in Theorem 4.4 (page 73).

and $f(v_i) \cup f(v_j) \neq \pi$, any V_i can be a consistent fault set by Definition 3.1, as is shown in Figure 3.2b.

Now we present several important lemmas that further the characterization of *f*/(*n*–1)-diagnosable systems.

Lemma 3.1 [YM86] Consider an *n*-node, *f*/(*n*–1)-diagnosable system with an undirected comparison testing graph $G = (V, E)$ under the symmetric comparison model. For a consistent fault set F_1 where $|F_1| \leq f - 1$, the set $F_2 = F_1 \cup \{v_i\}$ is a consistent fault set if and only if $v_i \in V$ is such that $|\Gamma(v_i) \cap (V - F_1)| = 0$.

Proof. Necessity. Assume to the contrary that for a node v_i the set $F_2 = F_1 \cup \{v_i\}$ is a consistent fault set and that $|\Gamma(v_i) \cap (V - F_1)| \neq 0$. Then there is at least one comparison edge between v_i and $(V - F_1)$, as illustrated in Figure 3.3. If this edge compares to 1, then this outcome violates for F_1 the condition (3) of Definition 3.1, and F_1 cannot be a consistent fault set. If this edge compares to 0, then this outcome violates for F_2 condition (2) of Definition 3.1, and F_2 cannot be a consistent fault set.

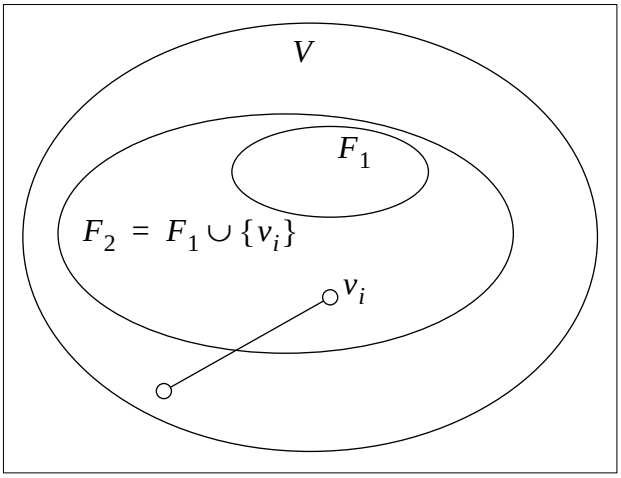


Figure 3.3 Proof of necessity for Lemma 3.1

Proof. Sufficiency. Assume to the contrary that for a node v_i the set $F_2 = F_1 \cup \{v_i\}$ is not a consistent fault set and that $|\Gamma(v_i) \cap (V - F_1)| = 0$, as shown in Figure 3.4. Then by Definition 3.1 we must have at least one comparison edge such that either (condition 2) $\sigma(u, v) = 0$ where $u \in V - F_2$ and $v \in F_2$ or (condition 3) $\sigma(u, v) = 1$ where $u, v \in V - F_2$. But this is

impossible because none of these two conditions are satisfied for F_1 which is a consistent fault set by definition. These two conditions cannot be satisfied for v_i since v_i has no comparison edges with $V - F_2$. □

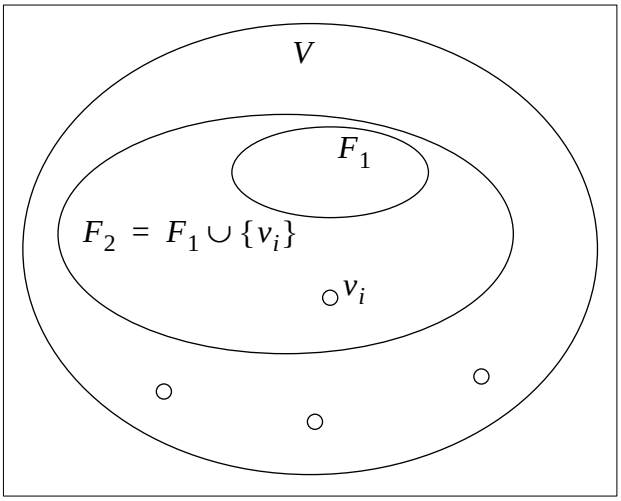


Figure 3.4 Proof of sufficiency for Lemma 3.1

Lemma 3.2 [XH90] If a system *S* represented by the comparison graph $G = (V, E)$ is *f*/(*n*–1)-diagnosable for the symmetric comparison model, then $|V| \geq 2f + 1$.

Proof. Assume that *G* is *f*/(*n*–1)-diagnosable and that $|V| < 2f + 1$. Then at least one fault-free node can be identified. Then add all edges to the graph such that the augmented graph G_0 is complete. in this new graph with $|V| < 2f + 1$ each node is connected to the identified fault-free node and can therefore be diagnosed unambiguously from this node. Hence the system should be one-step *f*-diagnosable with $|V| < 2f + 1$, which is impossible. □

Lemma 3.3 [XH90] Given any syndrome for a comparison graph $G = (V, E)$ under the symmetric comparison model, let F_1, F_2, …, F_k be consistent fault sets for the syndrome. Then a system *S* represented by $G = (V, E)$ is *f*/(*n*–1)-diagnosable if and only if $V - \bigcup_{i=1}^{k} F_i \neq \varnothing$

Proof. If $V - \bigcup_{i=1}^{k} F_i = \varnothing$ then the union of all the possible fault sets collects all the nodes in the system. Thus there is no way of deciding that one particular fault set is indeed

fault-free. If $V - \bigcup_{i=1}^{k} F_i \neq \varnothing$, then $V - \bigcup_{i=1}^{k} F_i$ is a set of fault-free nodes. Hence at least one fault-free node is identified thereby ensuring *f*/(*n*–1)-diagnosability. □

We introduce the following new lemma:

Lemma 3.4 Let $G = (V, E)$ be a comparison graph for the symmetric comparison model and $G_0 = (V, E_0)$ be the corresponding 0-derived graph with cartels V_i, $i = 1, \ldots, k$. Let $L_i = \{j \mid \exists (v, u)$ where $v \in V_i$ and $u \in V_j$ with $(v, u) \in E$ and $w(v, u) = 1\}$. Define $\Gamma_1(V_i) = \bigcup_{j \in L_i} V_j$. The set V_i pertains to a consistent fault set if $|\Gamma_1(V_i)| > f$.

Proof. All the nodes in any cartel have the same fault status since there is a path between any two nodes in a cartel, and each edge within a cartel yields a zero. If $|\Gamma_1(V_i)| > f$, then the nodes in V_i compare to 1 with more than *f* nodes. But no more than *f* nodes may be faulty. Hence the nodes in V_i are necessarily faulty because, otherwise, at least one node in $\Gamma_1(V_i)$ would be fault-free and would compare to 0 with a node in V_i. □

3.2 Towards a characterization of optimal *f*/(*n*–1)-diagnosable systems

In this section we seek to characterize *f*/(*n*–1)-diagnosable systems that have the minimum number of comparisons and the maximum number of distinct tasks. The advantage to such a system is obviously that fault-tolerance is attained with a minimal overhead for diagnosis and a maximal throughput of task results.

As a first approach we consider subdividing a system into two subgraphs, where a separate and distinct task is executed in each subgraph and the results of these tasks are used for comparison diagnosis. We note that comparisons can only be done among PEs of the same subgraph, as task results from different tasks cannot be used for comparisons. We start with a simple observation.

Theorem 3.2 Under the symmetric comparison model, a system with $n \geq 2f + 1$ nodes is *f*/(*n*–1)-diagnosable in a configuration with two subgraphs: one with a single isolated node and the other with $n - 1$ fully connected nodes (Figure 3.5).

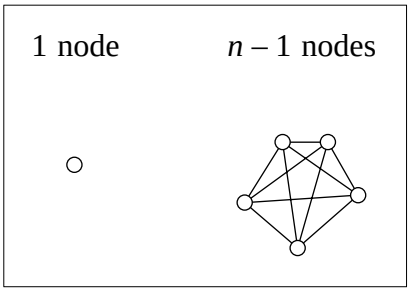


Figure 3.5 System configuration of Theorem 3.2

Proof. Assume first that $n = 2f + 1$ and let v_1 be the isolated node and $v_2, \ldots, v_n$ the $n-1 = 2f$ remaining nodes. Since the $n-1$ nodes $v_2, \ldots, v_n$ are fully connected, the existence of a path spanning any *k* nodes, $k = 2, \ldots, n-1$ is guaranteed. Thus if at most $f-1$ nodes are faulty, there is a path spanning the $f+1$ fault-free nodes. These fault-free nodes form a cartel of $f+1$ nodes. By definition, a cartel of more than f nodes must be composed of only fault-free nodes.

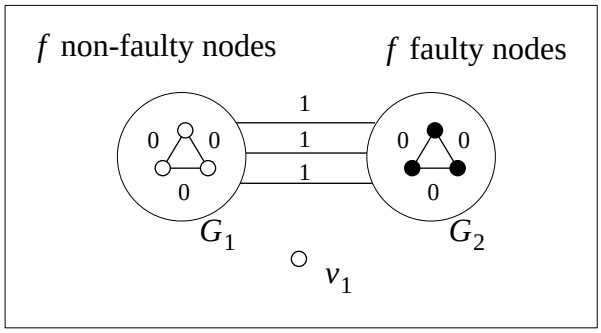


Figure 3.6 Limiting conditions on *f*/(*n*–1)-diagnosability with one isolated node for $n = 2f + 1$

If *f* nodes are faulty, the *f* non-faulty nodes will belong to a cartel of *f* nodes. First we consider when some of the *f* faulty nodes may test themselves to 1. If at least one pair of faulty nodes tests to 1, at least one of the *f* faults has been identified. Thus the cartel of *f* nodes is identified as fault-free, because one fault has been identified outside of the cartel.

In the case where the *f* faulty nodes test themselves to 0 we may have two *f*-node subgraphs G_1 and G_2, each a cartel of *f* nodes, with comparisons between the two subgraphs testing to 1 (Figure 3.6). Then there is no way of telling which cartel contains the non-faulty nodes. The diagnosis guarantees, however, that the maximum allowed number of faulty nodes has been reached, and that the *f* faulty nodes are all located either in G_1 or in G_2. By inference, therefore, the isolated node v_1 is fault-free[1].

For the case where $n > 2f + 1$, let $n = 2f + 1 + k$. Delete any *k* nodes from the graph.

This yields system with $n - k$ nodes where $n - k = 2f + 1$. Such a system is *f*/(*n*–1)-diagnosable as discussed above. □

We extend our characterization of system configurations for *f*/(*n*–1) diagnosability with the following theorem.

Theorem 3.3 Under the symmetric comparison model, an *n*-node system with $n = 2f + 1$ is not *f*/(*n*–1)-diagnosable in a configuration with three subgraphs: two with single isolated nodes and the third with full test connectivity among the remaining $n - 2$ nodes (Figure 3.7). If $n \geq 2f + 2$, however, the system is *f*/(*n*–1)-diagnosable.

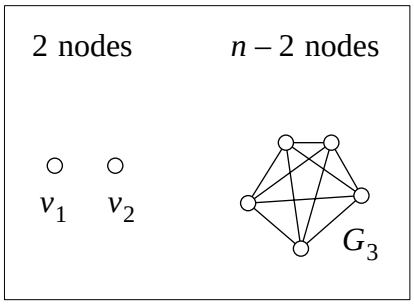


Figure 3.7 System configuration of Theorem 3.3

Proof. For the first part of the theorem, $n = 2f + 1$. Let v_1, v_2 be the isolated nodes and v_3, …, v_n the $n - 2 = 2f - 1$ remaining nodes in the fully connected subgraph G_3. First, assume that $f - 1$ nodes are faulty. Then if all the faults are in G_3 there is a cartel of *f* non-faulty nodes in G_3. Similarly there can be a cartel of no more than $f - 1$ faulty nodes in G_3. Thus there may be two cartels: one of *f* non-faulty nodes and one of $f - 1$ faulty nodes.

Assume now that *f* nodes are faulty. The *f* faults may be all located in G_3, and in the worst case, they form a cartel of *f* faulty nodes along with the cartel of $f - 1$ non-faulty nodes. The diagnoser may be faced with a syndrome in G_3 that may be interpreted either as meaning that there is in G_3 a group of *f* fault-free nodes and a group of $f - 1$ faulty nodes or the converse. There is no way to tell which one of these mutually exclusive hypotheses is correct, and the fault status of the isolated nodes v_1, v_2 cannot be deduced by inference, because one of the

1. The isolated node is judged fault-free under the assumptions of comparison-based, system-level diagnosis, without having actually been tested with another node. This theoretical approach may not necessarily be advisable in a practical system. The existence of *f* faults in the system does not guarantee the fault-free status of any other node—the assumed bound may actually have been exceeded.

hypotheses assume only $f - 1$ faults in G_3, leaving open the possibility that either v_1 or v_2 is faulty. Hence the system is not *f*/(*n*–1)-diagnosable.

For the second part of the theorem, $n \geq 2f + 2$. Let $n = 2f + 1 + k$, $k \geq 1$. Delete one of the isolated nodes and any $k - 1$ other nodes from the graph. This yields a system with $n - k$ nodes with $n - k = 2f + 1$, which is *f*/(*n*–1)-diagnosable by Theorem 3.2, since it has at most one isolated node. □

Corollary 3.1 For the symmetric comparison model, an *n*-node system with $n \geq 2f + 1 + k$ is *f*/(*n*–1)-diagnosable in a configuration with $k + 1$ isolated nodes and with full test connectivity among the remaining $n - k - 1$ nodes.

Proof. Delete *k* isolated nodes. Diagnose the rest of the system as per Theorem 3.2. □

Theorem 3.4 An *n*-node system with $n \geq 2f + 1$ is *f*/(*n*–1)-diagnosable for the symmetric comparison model in a configuration with two disjoint subgraphs $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$ where $G_1(V_1, E_1)$ is fully connected with *k* nodes, for $1 \leq k < n$, and $G_2(V_2, E_2)$ is fully connected with $n - k$ nodes (Figure 3.8).

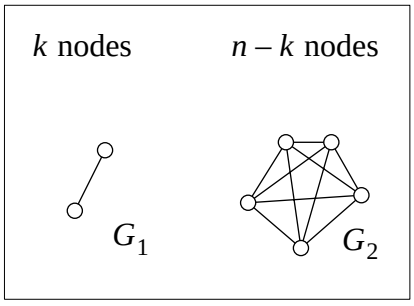


Figure 3.8 System configuration of Theorem 3.4

Proof. Let $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$ be the two subgraphs where V_1, V_2 are respectively the set of nodes v_i, v_j in $G_1(V_1, E_1)$, $G_2(V_2, E_2)$, and E_1, E_2 are respectively the set of edges. Note that $|V_1| \neq |V_2|$ when $n = 2f + 1$, since *n* is odd. Define C_1 and C_2, the largest cartel in G_1, G_2 respectively, with C_1: $\{C_1 \subseteq V_1$ and $\forall(v_i \in C_1, v_j \in C_1)\ w_{ij} = 0$, and $\forall(v_i \in C_1, v_j \notin C_1),\ w_{ij} = 1\}$ and C_2: $\{C_2 \subseteq V_2$ and $\forall(v_i \in C_2, v_j \in C_2)\ w_{ij} = 0$, and $\forall(v_i \in C_2, v_j \notin C_2),\ w_{ij} = 1\}$. Similarly define D_1 and D_2, the second largest cartel in

$G_1(V_1, E_1)$, $G_2(V_2, E_2)$ respectively, with D_1: $\{D_1 \subseteq V_1$, $|D_1| \leq |C_1|$ and $\forall(v_i \in D_1, v_j \in D_1)$ $w_{ij} = 0$, and $\forall(v_i \in D_1, v_j \notin D_1)$, $w_{ij} = 1\}$ and D_2: $\{D_2 \subseteq V_2$, $|D_2| \leq |C_2|$ and $\forall(v_i \in D_2, v_j \in D_2)$ $w_{ij} = 0$, and $\forall(v_i \in D_2, v_j \notin D_2)$, $w_{ij} = 1\}$. Define as F_1, F_2 the set of faulty nodes in G_1, G_2 respectively, and define $X_1 = V_1 - F_1$, $X_2 = V_2 - F_2$ the set of non-faulty nodes in G_1, G_2 respectively.

Assume first that $n = 2f + 1$. In the following we assume without loss of generality that $|V_1| < |V_2|$, which implies that $|V_1| \leq f$. Furthermore we assume that $|V_1| = k \geq 2$ and $|V_2| = n - k \leq n - 2$ since the other cases have already been dealt with. The diagnosis can be done as follows:

1. If all the links in the subgraph G_1 test to 1, discount G_1 and diagnose on G_2 only. Here the *f*/(*n*–1)-diagnosability is guaranteed because at most one node in G_1 can be fault-free. If more than one node were fault-free, at least two fault-free nodes would be connected since G_1 is fully connected. Thus we would have at least one link testing to 0. Hence the discounting of the *k* nodes in G_1 removes at least $k - 1$ faulty nodes, and the G_2 subgraph contains $n - k = 2f + 1 - k$ nodes out of which at most $f - k + 1$ are faulty. G_2 is then obviously *f*/(*n*–1)-diagnosable since $n - k = (2f + 1 - k) \geq 2f - 2k + 2 + 1$ for $k \geq 2$.
2. If the system has not been diagnosed in step 1, check whether $|C_2| \geq f + 1$. If this is the case then the nodes in C_2 are fault-free and the diagnosis is completed.
3. If the system has not been diagnosed in step 2, check whether $|C_1| = k$. Then the *k* nodes in G_1 are either all faulty or all fault-free. If the *k* nodes in G_1 are all faulty, then $|F_2| = f - k$ and since $|V_2| = n - k = 2f + 1 - k$, the set of non-faulty nodes $V_2 - F_2$ in G_2 is such that $|V_2 - F_2| = f + 1$. Therefore at least $f + 1$ nodes are non-faulty in G_2 and the system has already been diagnosed in step 2. If the *k* nodes in G_1 are all non-faulty, G_2 contains $n - k = 2f + 1 - k$ nodes out of which up to *f* may be faulty. Therefore less than $f + 1$ nodes are non-faulty in G_2. This means that the condition $(|C_2| < f + 1) \wedge (|C_1| = k)$ points to the *k* nodes in G_1 as being fault-free.

4. If the system has not been diagnosed in step 3, we first note that $|C_1| \geq |D_1| \geq 0$ and $|C_2| \geq |D_2| \geq 0$. Obviously we have $|C_1| + |C_2| \geq |X_1| + |X_2| \geq f + 1$ and we now show that this implies either $(C_1 = X_1) \wedge (C_2 \subseteq F_2)$, $(C_1 \subseteq F_1) \wedge (C_2 = X_2)$ or $(C_1 = X_1) \wedge (C_2 = X_2)$. To check this statement, assume to the contrary, with $(C_1 - X_1 = \varnothing) \wedge (C_2 - X_2 = \varnothing)$. Then we should have the following: $|C_1| + |C_2| + |X_1| + |X_2| \geq 2f + 2 = n + 1$. Yet this is impossible since the system contains only *n* nodes. Thus at least one of the sets C_1 or C_2 is fault-free. We now consider the two largest cartels C_1, D_1 and C_2, D_2 in the subgraphs G_1 and G_2 respectively. We prove that the pairs of sets C_1, D_2 and D_1, C_2 cannot be deemed simultaneously fault-free during the diagnosis. To check this statement, assume to the contrary. Then we would have simultaneously $|C_1| + |C_2| \geq f + 1$, $|C_1| + |D_2| \geq f + 1$ and $|D_1| + |C_2| \geq f + 1$, or equivalently $|C_1| + |C_2| = f + 1 + i$, $|C_1| + |D_2| = f + 1 + j$ and $|D_1| + |C_2| = f + 1 + k$, with $i, j, k \geq 0$. This would imply that $|C_1| + |C_2| + |C_1| + |D_2| + |D_1| + |C_2| = 3f + 3 + i + j + k$. But $|C_1| + |C_2| + |D_1| + |D_2| = 2f + 1 - l \leq n$, with $l \geq 0$. Thus we should have $2f + 1 - l + f + 1 + i = 3f + 3 + i + j + k$ or $1 + l + j + k = 0$ which is impossible.

Thus we can have only the 3 following diagnosable situations by retaining as fault-free only the set that is common to the pair C_1, C_2 and the pair C_1,D_2 or D_1,C_2 (if any):

1. $(|C_1| + |C_2| \geq f + 1) \wedge (|C_1| + |D_2| < f + 1) \wedge (|D_1| + |C_2| < f + 1)$, then $(|C_1| = X_1) \wedge (|C_2| = X_2)$,
2. $(|C_1| + |C_2| \geq f + 1) \wedge (|C_1| + |D_2| \geq f + 1) \wedge (|D_1| + |C_2| < f + 1)$, then $C_1 = X_1$
3. $(|C_1| + |C_2| \geq f + 1) \wedge (|C_1| + |D_2| < f + 1) \wedge (|D_1| + |C_2| \geq f + 1)$, then $C_2 = X_2$

This completes the proof of *f*/(*n*–1)-diagnosability for $n = 2f + 1$.

If $n > 2f + 1$, let $n = 2f + 1 + k$. Delete any *k* nodes from the system with the sole restriction that it remains structured as two disconnected subgraphs. This yields a system with $n - k$ nodes, where $n - k = 2f + 1$, and which is *f*/(*n*–1)-diagnosable as discussed above. □

We now extend Xu's characterization of *f*/(*n*–1)-diagnosable systems [Xu91] by considering

that a system can execute more than one task when $n > 2f + 1$. This is an improvement over systems proposed by Xu where only one task is used for comparison results. In the following theorem, we assume that a distinct and independent task is executed on each disjoint subgraph, thereby allowing the system to perform more work than if a single task were used for comparison results.

Theorem 3.5 For the symmetric comparison model, an *n*-node system with $n = 2f + 1 + k$ is *f*/(*n*–1)-diagnosable in any configuration where two disjoint, fully connected subgraphs contain a total of $2f + 1$ nodes, and where the remaining *k* nodes are located in up to *k* disjoint subgraphs.

Proof. Remove the *k* nodes in the corresponding disjoint subgraphs. The system is then diagnosed as per Theorem 3.4 on the two fully connected subgraphs that contain a total of $2f + 1$ nodes out of which no more than *f* may be faulty. □

The following theorem provides a limiting case for the characterization of *f*/(*n*–1)-diagnosable systems where more than one task is executed, one task per disjoint subgraph.

Theorem 3.6 For the symmetric comparison model, an *n*-node system with $n = 2f + 1$ is not *f*/(*n*–1)-diagnosable in any configuration with more than two disjoint, fully connected subgraphs.

Proof. Assume that $G(V, E)$ is split into 3 disjoint subgraphs $G_1(V_1, E_1)$, $G_2(V_2, E_2)$, $G_3(V_3, E_3)$. First, if $|V_1| \leq f$, $|V_2| \leq f$, $|V_3| \leq f$. Then $\pi = \{V_1, V_2, V_3\}$ is a cover that satisfies the cardinality conditions of Theorem 3.1. But there is no edge (v, u) such that $f(v) \neq f(u)$, since the three subgraphs are disconnected. Hence the system is not *f*/(*n*–1)-diagnosable.

Next, if one of the subsets such as V_1 has more than f nodes, then $|V_1| \leq f + r$, $1 \leq r \leq f - 1$ because there are two other subsets with at least one node in each of them. Thus we have $1 \leq |V_2| = s$, $1 \leq |V_3| = t$. This means that $s + r \leq f$ and $t + r \leq f$, and that we can define a cover $\pi = \{V_4, V_5, V_6\}$ as shown in Figure 3.9 where $|V_4| = f$, $|V_5| = r + s \leq f$, $|V_6| = r + t \leq f$. Then the only edges such that $f(v) \neq f(u)$ are in G_1, but for these edges $f(v) \cup f(u) = \pi$ so that the system is not *f*/(*n*–1)-diagnosable. □

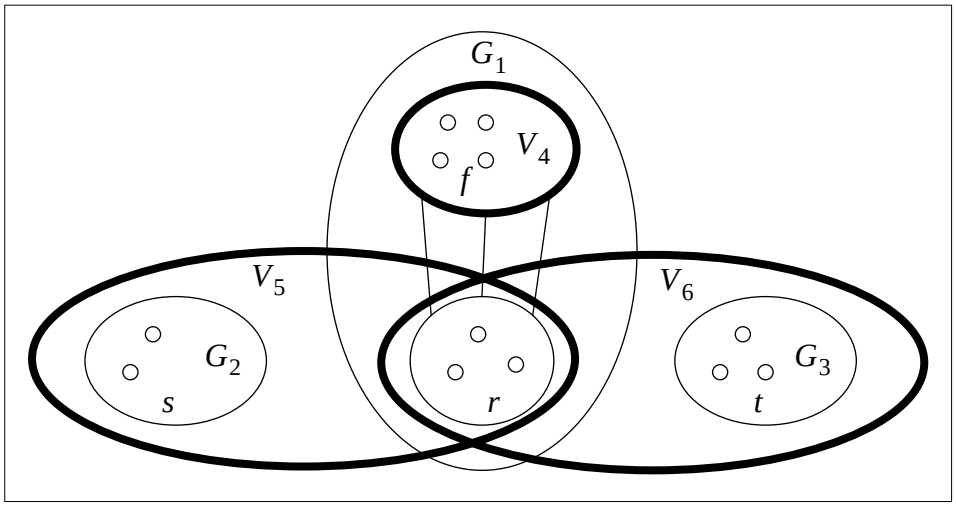


Figure 3.9 Impossibility proof of Theorem 3.6

The previous theorems indicate which structures are *f*/(*n*–1)-diagnosable. Theorem 3.4 is particularly important since it proves that a system with two subgraphs is *f*/(*n*–1)-diagnosable when the subgraphs are disjoint. On each disjoint subgraph a distinct and independent task can be executed, since such a topology implies that two results from differing tasks are never compared. In the following theorem we establish upper bounds on the number of comparisons required for *f*/(*n*–1)-diagnosability in such systems.

Theorem 3.7 For $n = 2f + 1$, an *n*-node system is *f*/(*n*–1)-diagnosable for the symmetric comparison model using no more than $M = (k(k-1) + (n-k)(n-k-1))/2$ comparisons when it is structured as two complete subgraphs G_1 and G_2 with *k* nodes in G_1 and $n-k$ nodes in G_2.

Proof. By Theorem 3.4 the system is *f*/(*n*–1)-diagnosable. Since each subgraph is complete, we have $(k(k-1))/2$ comparisons in G_1 and $((n-k)(n-k-1))/2$ comparisons in G_2. Hence

$$M = \frac{k(k-1) + (n-k)(n-k-1)}{2}. \tag{3.1}$$

The number of comparisons is obviously minimum, with value M_1, when the two subgraphs have approximately the same size, with $k = f$, $n-k = f+1$. Then

$$M_1 = \frac{f(f-1) + (f+1)f}{2} = f^2 = \frac{(n-1)^2}{4} \tag{3.2}$$

We now proceed with our characterization by showing that *f*/(*n*–1)-diagnosability cannot be guaranteed for more than two distinct tasks when $n = 2f + 1$ and that *f*/(*n*–1)-diagnosability can be assured under certain conditions for more than two tasks when $n > 2f + 1$.

Theorem 3.8 Let *S* be an *n*-module system with $n = 2f + 1$. This system is not *f*/(*n*–1)-diagnosable for the symmetric comparison model in any configuration with more than two disjoint subgraphs.

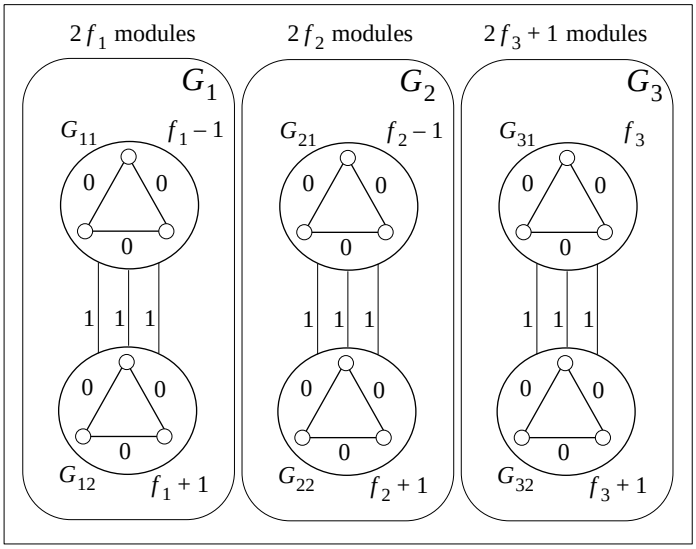


Figure 3.10 Impossibility proof for *f*/(*n*–1)-diagnosability with three tasks and $n = 2f + 1$

Proof. The theorem is proved by assuming the most favorable situation for diagnosis, with only three disjoint subgraphs $G_1(V_1, E_1)$, $G_2(V_2, E_2)$, $G_3(V_3, E_3)$ that are each complete as shown in Figure 3.10. Since *n* is odd, either each subgraph contains an odd number of nodes or just one subgraph contains an odd number of nodes. Assume first the latter case with $|V_1| = 2f_1$, $|V_2| = 2f_2$, $|V_3| = 2f_3 + 1$, and $f = f_1 + f_2 + f_3$. Then we might have the syndrome sketched in Figure 3.10 where the nodes in G_1, G_2, G_3 are clustered into two subgraphs such as G_{11}, G_{12} within which they test themselves to 0 while they test to 1 the nodes in the opposite subset. Then the results in at least one of the two subgraphs G_{i1}, G_{i2} pertaining to each G_i are all incorrect and the other subgraph is a candidate for containing the correct results. Since at least $f + 1$ results must be correct, the syndrome of Figure 3.10 can be interpreted as meaning that any of the combination of subgraphs G_{12}, G_{22}, G_{32}; G_{11}, G_{22}, G_{32}; G_{12}, G_{21}, G_{32}; G_{12}, G_{22}, G_{31}; is a candidate for containing the correct results since all these

combinations give at least $f + 1$ results. Yet no single subgraph is common to all these four combinations and can be selected unambiguously as being fault-free.

Hence the system is not *f*/(*n*–1)-diagnosable. A similar argument holds when G_1, G_2, G_3 contain an odd number of nodes. □

Theorems 3.4 and 3.5 show that up to $s + 2$ unrelated tasks can be executed concurrently in a system with $n = 2f + 1 + s$ nodes with the guarantee that a diagnosis allows the release of at least one correct result. The limiting case is reached for a system with k nodes in G_1, $2f + 1 - k$ nodes in G_2, and s isolated nodes. Then the system executes tasks T_1, T_2 on G_1, G_2 and tasks T_3, …, T_{s+2} on the s isolated nodes. Theorem 3.4 guarantees that one correct result of either T_1 or T_2 is released at the end of the execution.

This does not mean, however, that the results of T_3, …, T_{s+2} are always undecided; these results can be validated by inference, for instance, when the diagnosis has determined that all incorrect results lie within G_1 and G_2. This can be seen in the trivial example of a 5-node system, with $n = 5$ and $f = 1$ as shown in Figure 3.11. Then the system can be organized to process 4 tasks, with 3 tasks on isolated nodes and one task on a pair of nodes, with a single comparison between the results of this last task. Then if the two results in the pair are different, the single fault has been located in the pair. By inference the three results produced by the isolated nodes are correct.

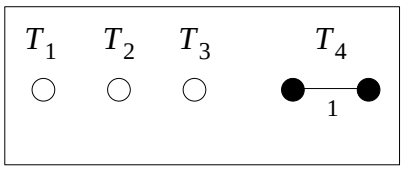


Figure 3.11 Example system with *n* = 5 PEs and where *f* = 1

Up to now we have assumed that the two disjoint subgraphs G_1 and G_2 are complete. But this is not really necessary. For $n = 2f + 1$, the concept of *f*/(*n*–1)-diagnosability means that up to f faults are located within at most $2f$ units. Hence, under the symmetric comparison model, any procedure that identifies at least one fault within at least two units is directly applicable to *f*/(*n*–1)-diagnosability. We now use this result to reduce the upper bound on the number of comparisons required for *f*/(*n*–1)-diagnosability. Theorem 3.4 was proved by identifying in each subgraph the two largest cartels that test to 1 with units outside the cartel. This condi-

tion can still be satisfied with a much lower number of comparisons than those found in this theorem. This leads us to define a more efficient class of structures for *f*/(*n*–1)-diagnosability as follows.

3.3 Bi-corona systems

We introduce a new class of *f*/(*n*–1)-diagnosable structures, which we call *bi-corona* systems, where the set of *n* PEs is split into two disjoint subsets, each with a special structure. As before, the PEs in a subset can only be compared with other PEs of the same subset; two PEs from different subsets cannot be compared. Furthermore, the special structure of each subset yields a particular pairing of PEs. The motivation for such a structure is twofold. Firstly it reduces the number of comparisons needed for *f*/(*n*–1)-diagnosability. Secondly, under a comparison model, a system structured in this way can concurrently execute several distinct tasks, provided that each distinct task is assigned to all the PEs of a distinct subset. In each subset, the results of the task's execution are compared. The set of comparisons in all subsets can be used to obtain a syndrome for *f*/(*n*–1)-diagnosability. The task results from fault-free nodes is also useful as output from the system.

We now recall a formal definition of the structure we have chosen for our subgraphs.

Definition 3.2 [Har72] The *corona* $H_1 \circ H_2$ of two graphs H_1 and H_2 is defined as the graph H obtained by taking one copy of H_1 (which has p_1 vertices) and p_1 copies of H_2, and then joining the ith point of H_1 to every point in the ith copy of H_2 (Figure 3.12).

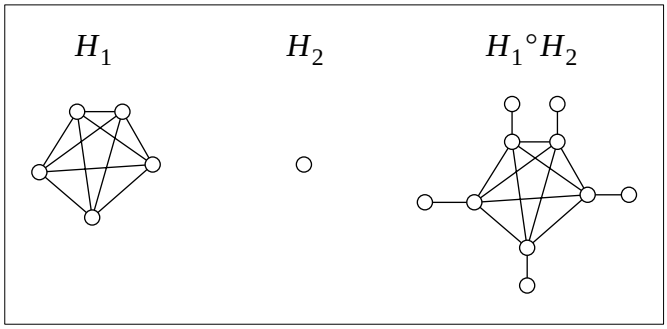


Figure 3.12 Two graphs and a corona

Definition 3.3 We define a *bi-corona system* as a system of *n* nodes (where *n* is odd) with a comparison testing graph $G = (V, E)$ that is structured as two disjoint subgraphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ with *k* nodes (where *k* is even) in G_1 executing a task T_1

and $n-k$ nodes in G_2 executing a possibly unrelated task T_2. G_1 is structured as the corona $K_{k/2}{}^{\circ}K_1$ which has $k/2$ *internal nodes* and $k/2$ *external nodes* (Figure 3.13). The set of internal nodes are fully connected ($K_{k/2}$), and each has a single connection to a distinct external node (K_1). Similarly G_2 is structured as the corona $K_{\lceil (n-k)/2 \rceil}{}^{\circ}K_1$, but since $n-k$ is odd, there are only $\lfloor (n-k)/2 \rfloor$ copies of the single-node graph K_1. This leaves a single, internal node (that we call the *lone* node) that is not connected to an external node (Figure 3.13).

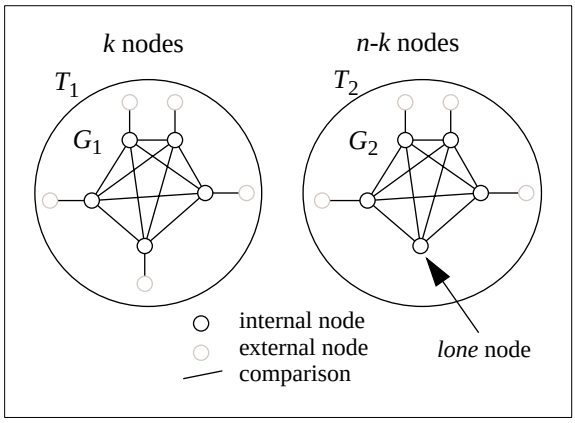


Figure 3.13 *f*/(*n*–1)-diagnosable, bi-corona structure

In order to demonstrate how bi-corona systems require fewer comparisons, we first present the following diagnosis algorithm for bi-corona systems that we have derived from the algorithm developed by Hakimi *et al*. [HS84] for adaptive diagnosis. This procedure guarantees that a fault-free node is identified if the total number, *n*, of nodes is

$$n = 2f + 1. \tag{3.3}$$

Algorithm 3.1 Identification of a fault-free node in a bi-corona system for the symmetric comparison model[1].

Step 0. A fault counter, FC, is initialized with the value 0.

Step 1. For each pair of internal-external nodes in G_1 and G_2 for which the comparison outcome is 1, discard the pair from the system for the rest of the diagnosis and increment FC by 1.

1. This algorithm was implemented in the C programming language, and the source code can be found in Appendix A.1.

Step 2. If FC = f then STOP. The remaining node (lone node) in G_2 is fault-free[1].

Else if FC = $f - 1$ then STOP. Any remaining pair of nodes in either G_1 or G_2 is fault-free.

Step 3. For each pair of remaining internal-internal nodes in G_1 and G_2 (excluding pairs formed with the lone node) for which the comparison outcome is 1, discard the pair of internal nodes as well as their corresponding external nodes and increment FC by 2.

Step 4. If FC = f then STOP. The lone node in G_2 is fault-free.

Else if $|V_1| > f - \text{FC}$ then STOP. The remaining nodes in G_1 are fault-free.

Else if $|V_2| > f - \text{FC} + 1$ then STOP. The remaining nodes in G_2 (excluding the lone node) are fault-free.

Step 5. Compare the lone node in G_2 to all remaining internal nodes of G_2. If any of the outcomes of these comparisons yields 1 then STOP. All the remaining nodes in G_1 are fault-free.

Else STOP. The remaining nodes in G_2 are fault-free.

This algorithm is not necessarily optimal, but merely shows the $f/(n-1)$-diagnosability of bi-corona systems.

Theorem 3.9 Consider a bi-corona system with $n = 2f + 1$ nodes, at most f of which are faulty. The system is $f/(n-1)$-diagnosable (i.e., a fault-free node is identified) for the symmetric comparison model with Algorithm 3.1. The diagnosis requires no more than $M = f + (k(k-2) + (2f - k + 2)(2f - k))/8$ comparison tests.

Proof. First we verify that the algorithm always terminates, and that it correctly identifies a fault-free node. We note that when a pair of nodes yields a comparison test of 1, at least one fault has been identified since at least one of the nodes is faulty. From the structure of the algorithm it is obvious that the algorithm always terminates as there are no loops. Let us consider the seven cases for which the algorithm terminates:

1. This is another case where a node is deemed fault-free before it has actually been tested.

Case 1. *The algorithm stops in Step 2 with* $\mathrm{FC} = f$*, and the lone node in* G_2 *is diagnosed as fault-free.* Since $\mathrm{FC} = f$, none of the internal-external pairs in either subgraph has survived Step 1 of the algorithm. Therefore, the lone node must be fault-free since the *f* faults have been accounted for.

Case 2. *The algorithm stops in Step 2 with* $\mathrm{FC} = f - 1$*, and the surviving pair of nodes in either* G_1 *or* G_2 *is diagnosed as fault-free.* If $\mathrm{FC} = f - 1$ then this implies that the surviving nodes are one internal-external pair (in either G_1 or G_2) and the lone node in G_2. As only one faulty node remains unidentified, it must be the lone node; a single faulty node in the remaining pair would have caused the pair to be discarded in Step 1.

Case 3. *The algorithm stops in Step 4 with* $\mathrm{FC} = f$*, and the lone node in* G_2 *is diagnosed as fault-free.* The reasoning here is the same as in Case 1. That is, no internal-external pairs remain and all *f* faults have been accounted for.

Case 4. *The algorithm stops in Step 4 with* $|V_1| > f - \mathrm{FC}$*, and the remaining nodes in* G_1 *are diagnosed as fault-free.* At this point, every pair (internal-internal and internal-external) of nodes remaining in G_1 yields a comparison outcome of 0. This means that the remaining pairs are either all faulty or all fault-free. Yet since at most $f - \mathrm{FC}$ of the remaining nodes can be faulty, if the number of remaining nodes in G_1 is more than $f - \mathrm{FC}$, then they must all be fault-free.

Case 5. *The algorithm stops in Step 4 with* $|V_2| > f - \mathrm{FC} + 1$*, and the remaining nodes in* G_2 *(excluding the lone node) are diagnosed as fault-free.* This case is similar to that of Case 4, but for the subgraph G_2. At this point, every pair of nodes (except those involving the lone node) remaining in G_2 yields a comparison outcome of 0. Therefore the remaining pairs must be either all faulty or all fault-free. Since there remain at most $f - \mathrm{FC}$ faulty nodes, if the number of remaining nodes in G_2 is more than $f - \mathrm{FC} + 1$ (accounting for the lone node which has not been considered), then they must all (excluding the lone node) be fault-free.

Case 6. *The algorithm stops in Step 5 with the lone node comparing to 0 with all internal nodes in* G_2*, and the remaining nodes in* G_2 *are diagnosed as fault-free.* If the algorithm has reached Step 5, then exactly $f - \mathrm{FC}$ nodes remain in G_1. Given that at any time, the total number of remaining nodes in the system is $n' = 2(f - \mathrm{FC}) + 1$, then there are

$n' - (f - \mathrm{FC}) = f - \mathrm{FC} + 1$ nodes remaining in G_2. If comparisons between the lone node and each remaining internal node in G_2 fail to yield any 1s, then the $f - \mathrm{FC} + 1$ remaining nodes in G_2 can only be all fault-free, as there can never be more than $f - \mathrm{FC}$ faulty nodes.

Case 7. *The algorithm stops in Step 5 with the lone node comparing to 1 with one or more of the internal nodes in G_2, and the remaining nodes in G_1 are diagnosed as fault-free.* If the outcomes of the comparisons done in Step 5 are not all 0, then at least one of the remaining nodes in G_2 is faulty. Consequently *at most* $f - \mathrm{FC}$ of the remaining nodes in G_2 are fault-free. This implies that the $f - \mathrm{FC}$ nodes remaining in G_1 (see Case 6) are all fault-free.

Now we prove the maximum number of comparisons. Assume without loss of generality that, from Definition 3.3, k is even and hence $n - k$ is odd. With the testing graph defined by Definition 3.3 the total number of comparisons is

$$\begin{aligned} M &= \frac{k}{2} + \frac{k(k-2)}{8} + \frac{n-k-1}{2} + \frac{(n-k+1)(n-k-1)}{8} \\ &= f + \frac{k(k-2) + (2f-k+2)(2f-k)}{8} \end{aligned} \tag{3.4}$$

as stated in the theorem. □

The number of comparisons is obviously minimized when the two subgraphs have approximately the same size. If f is even and $k = f$, then $n - k = f + 1$. Substituting $k = f$ in equation (3.4) the number, M_1, of comparisons.

$$M_1 = \frac{f(f+4)}{4} \tag{3.5}$$

For f odd, $k = f + 1$ and $n - k = f$, and the number, M_2, of comparisons M_2 is given by

$$M_2 = \frac{f(f+4) - 1}{4} \tag{3.6}$$

Xu and Huang [XH90,Xu91] characterized f/(n–1)-diagnosability with the symmetric comparison model for regular structures in the form of chains, loops and $H_{2r,n}$ systems. An $H_{2r,n}$

system has n nodes, and each node v_i is connected to all its neighbors v_j such that $i - r \le j \le i + r$ where indices are defined modulo n. Thus an $H_{2r,n}$ system yields nr comparisons. Xu *et al*. showed that an $H_{2r,n}$ system is f/(n–1)-diagnosable for $f \ge 6$ when $r \ge (f-1)/5$. This means that for $n = 2f + 1$ and $r = (f-1)/5$, then $r = (n-3)/10$, which gives a total number M_3 of comparisons:

$$M_3 = \frac{n(n-3)}{10} = \frac{(2f+1)(f-1)}{5} \tag{3.7}$$

For large n the number of tests in our approach is a little less than two-thirds of that given by Xu [Xu91].

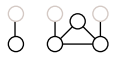
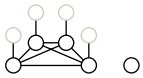
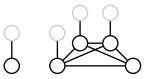
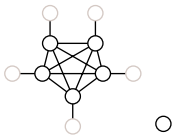
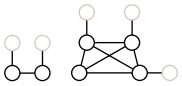
Another structure, called a *2-star structure*, requiring only $2f - 1$ comparisons is also mentioned by Xu *et al.* [XH90]. Although the authors imply no limiting assumptions with respect to the fault model, at least some of these 2-star structures can only be diagnosed with the asymmetric comparison model. Ironically some of the 2-star structures proposed by the authors are not f/(n–1)-diagnosable according to Theorem 3.1 (page 37) which originates from the same article [XH90]. The example in Figure 3.2 (page 37) is in fact a 2-star structure (given in [XH90], Table I, $f = 3$) that is not f/(n–1)-diagnosable, as it fails to meet the conditions of Theorem 3.1. Other 2-star structures can also be shown to be not f/(n–1)-diagnosable by similar means. Therefore, we do not consider the results reported by Xu [XH90] for the number of comparisons needed for 2-star structures to be in the context of the symmetric comparison model.

The new approach to diagnosability corresponding to Theorem 3.9 provides a rich set of new structures that are f/(n–1)-diagnosable with the symmetric comparison model. For small systems some of these structures reduce to the two disjoint chains indicated by Xu *et al.* [XH90,Xu91]. Table 3.1 shows examples of f/(n–1)-diagnosable structures for $n = 7, 9, 11$.

We now extend our approach to systems based on the bi-corona structure where $n > 2f + 1$.

Theorem 3.10 A bi-corona system with $n = 2f + 1 + s$ nodes, with $s > 0$, is f/(n–1)-diagnosable for the symmetric comparison model using no more than $M = f + (k(k-2) + (2f - k + 2)(2f - k))/8$ comparisons, $k \le 2f$ when it is structured as two subgraphs G_1 and G_2 with k nodes in G_1 and $2f + 1 - k$ nodes in G_2 where internal

Table 3.1 New structures that are *f*/(*n*–1)-diagnosable under the symmetric comparison model

n	*f*	*M*	Structure	*k*
7	3	6		6
				2
9	4	10		8
				2
		8		6
				4
11	5	15		10
		11		6
		12		4

nodes are fully connected, and the *s* additional nodes are connected arbitrarily among themselves and with G_1, G_2.

Proof. Remove *s* nodes in such a way that the $2f + 1$ remaining nodes form a bi-corona system with *k* nodes in G_1 and $2f + 1 - k$ nodes in G_2 where node pairs are fully connected. According to Theorem 3.9 this system is *f*/(*n*–1)-diagnosable with $M = f + (k(k-2) + (2f - k + 2)(2f - k))/8$ comparisons. □

Theorem 3.9 guarantees only that the correct result of one of the two tasks T_1 and T_2 is released at the end of the diagnosis. The second result may or may not be validated by the diagnosis. Nonetheless we show later in Chapter 5 of this thesis that such a diagnosis is sufficient when applied to fault-tolerant systems.

3.4 Sequential *f*-fault diagnosability

The notion of *f*/(*n*–1)-diagnosability refers to systems for which at least one *fault-free* node can

be identified. Closely related to this concept is the notion of sequential *f*-fault diagnosability where the goal is to find at least one *faulty* node. We will use the following theorem as a tool to characterize sequential *f*-fault diagnosability as a function of *f*/(*n*–1)-diagnosability:

Theorem 3.11 [XH95] A system with *n* nodes, where the test assignment graph *G* is connected, is sequentially *f*-fault diagnosable if it is *f*/(*n*–1)-diagnosable.

Proof. Perform the diagnosis in order to find at least one fault-free node v_i. Since the graph is connected, v_i is connected to at least another node v_j. If the two nodes compare to 1, v_j is faulty. Then repair it and re-diagnose the repaired system. If the two nodes compare to 0, v_j is non-faulty. In this case v_j is used to test another node and the algorithm continues until either a faulty node is identified or all the nodes have been tested to be fault-free. □

With Theorem 3.11, *f*/(*n*–1)-diagnosability can easily be used to prove sequential *f*-fault diagnosability in bi-corona systems:

Theorem 3.12 A system with $n \geq 2f + 1$ nodes is sequentially *f*-fault diagnosable using no more than $M = f + 1 + (k(k-2) + (2f-k+2)(2f-k))/8$ comparisons when it is structured as a bi-corona system with two subgraphs G_1 and G_2 with *k* nodes in G_1 and $n-k$ nodes in G_2 where internal nodes are fully connected, and where there is a single comparison link between G_1 and G_2.

Proof. By Theorems 3.9, 3.10 and 3.11 the system is *f*/(*n*–1)-diagnosable with $f + (k(k-2) + (2f-k+2)(2f-k))/8$ comparisons. Moreover, the system is connected since there is one comparison link between G_1 and G_2. Hence the system is sequentially *f*-fault diagnosable with $M = f + 1 + (k(k-2) + (2f-k+2)(2f-k))/8$ comparisons since a link between G_1 and G_2 ensures that the graph is connected.

When the sequential *f*-fault diagnosis is performed on *f* (or $f + 1$) systems, the respective numbers of comparisons become with equations (3.5) and (3.6):

$$M_1 = \frac{f(f+4)}{4} + 1 \text{ for } f \text{ even} \tag{3.8}$$

$$M_2 = \frac{f(f+4)-1}{4} + 1 \text{ for } f \text{ odd} \tag{3.9}$$

□

The complexity of the diagnosis performed in this manner is $\mathbf{O}[n^2]$, while the lower bound on the number of comparisons for non-adaptive, sequential, *f*-fault diagnosis has been reported in [Pel92] to be $n - \lfloor n/(2f+1) \rfloor$. The complexity of the latter is $\mathbf{O}[n]$, and the method yields $n-1$ comparisons for $n = 2f+1$. Thus the implementation of sequential *f*-fault diagnosis through *f*/(*n*–1)-diagnosis is inefficient for large *n*.

When *n* is small, however, this is not the case, as only 4, 6, 9, 12 comparisons are required for bi-corona systems with 5, 7, 9, 11 nodes, while the lower bound is 4, 6, 9, 10 for the same number of nodes with the optimum approach. Furthermore, as shown in Section 5.4.2, it is attractive to add the property of sequential *f*-fault diagnosability to a comparison-based NMR system since for the incremental cost of just one additional comparison, the capability of identifying at least one faulty node is gained.

3.5 *f*/(*n*–1)-diagnosability for the asymmetric comparison model

We now give a theorem for *f*/(*n*–1)-diagnosability for the asymmetric comparison model [Mal80] that is patterned after a similar theorem proposed in [HN84] for the asymmetric invalidation model.

Definition 3.4 Let $G = (V, E)$ be an undirected graph. A subset $V_1 \subset V$ is called a *vertex cover* of *G* if for every edge $(u, v) \in E$, $u \in V_1$, $v \in V_1$, or $u, v \in V_1$.

Definition 3.5 We define $\alpha_0(G)$ as a *minimum vertex cover* of *G*. That is, $\alpha_0(G)$ is a vertex cover with the smallest number of vertices.

Definition 3.6 Let $G - v$ denote the graph obtained by deleting vertex *v* and all edges incident at *v*.

We now establish the following general theorem on *f*/(*n*–1)-diagnosability for the asymmetric comparison model.

Theorem 3.13 Let S be a system with n nodes where the tests are represented by the undirected graph $G = (V, E)$. The diagnosis identifies a fault-free node under the asymmetric comparison model if and only if either

1. There exists an edge $(u, v) \in E$ with $\sigma(u, v) = 0$. In this case u and v are fault-free.
2. There exists a node $v \in V$ such that $\alpha_0(G - v) \geq f$. In this case v is fault-free.

Proof. If $\sigma(u, v) = 0$, u and v are fault-free in the asymmetric comparison model. Assume now that there exists no edge with weight 0 but that $\alpha_0(G - v) \geq f$. As such, all edges have weight 1. If F is a consistent fault set, any edge is either in F or linking a vertex in F and a vertex in $V - F$. Thus any edge has either one or both its vertices in F. Since the system contains at least one fault-free node v, any consistent fault set F must contain a vertex cover of $G - v$. This vertex cover has cardinality at least equal to $\alpha_0(G - v)$ and therefore is at least equal to f. Thus there is no consistent fault set containing v that is of cardinality equal or less than f, which implies that v is fault-free.

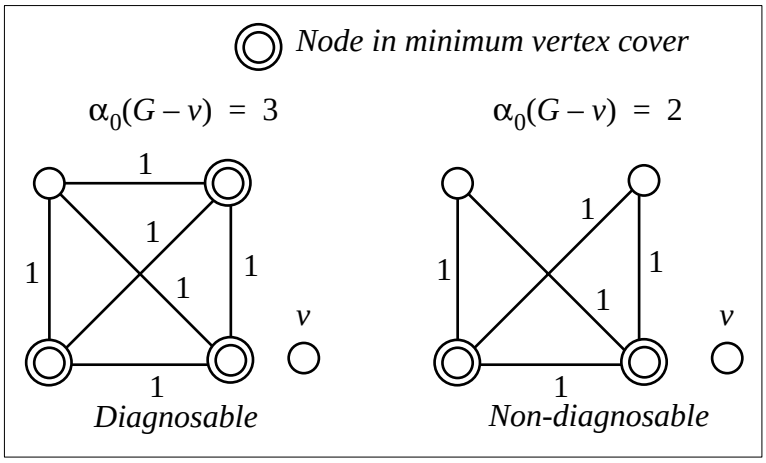


Figure 3.14 Example of the application of Theorem 3.13, for a system where *f* = 3

Now let us suppose that, given the graph $G = (V, E)$, there exists a node v which can be identified as a fault-free node. This means that there exists no consistent fault set $F \subset V$ which contains v with $|F| \leq f$. A consistent fault set must contain at least one vertex u_1, v_1 for each edge (u_1, v_1) where $\sigma(u_1, v_1) = 1$ and neither vertex u_0, v_0 of an edge (u_0, v_0) where $\sigma(u_0, v_0) = 0$ (according to conditions 2 and 3 of Definition 3.1). Thus if there is an edge $(u, v) \in E$ with $\sigma(u, v) = 0$, v can be identified as fault-free. But suppose there is no edge

whose weight is 0. Then each consistent fault set must contain at least one vertex u, v of each edge (u, v); that is, the consistent fault set must be a vertex cover V_1 of G. This implies, therefore, that for every cover V_1 of G which contains v, we must have $|V_1| > f$. Thus if we eliminate the vertex v, this implies that $\alpha_0(G - v) \geq f$. □

An application of Theorem 3.13 is illustrated in Figure 3.14 for $n = 5$ and $f = 3$.

3.6 Conclusion and contributions

This chapter has addressed the problem domain of system-level diagnosis and the specific areas of comparison-based diagnosis as the context of this thesis. We have presented several contributions to the theory of system-level diagnosis: (1) an extension to the characterization of f/(n–1)-diagnosable systems under the symmetric comparison model, (2) a new class of f/(n–1)-diagnosable structures that we call bi-corona systems, (3) a diagnosis algorithm for bi-corona systems.

Our extension to the characterization of f/(n–1)-diagnosable structures is novel as our approach is general enough to apply to systems of any size, which as far as we know was not done previously. The bi-corona systems derived from this extended characterization require fewer comparisons and are therefore more efficient than other existing structures. The application of the extended characterization to the technique of adaptive diagnosis is complemented with an algorithm that finds a fault-free node in a bi-corona system.

We address the issues of sequential f-fault diagnosability as a complementary contribution to our f/(n–1)-diagnosable structures. If long term fault tolerance is to be acquired, it is important to identify the faulty nodes and remove them. Finally we contribute a general theorem for f/(n–1)-diagnosability under the asymmetric comparison model.

3.7 Related publications

[FCN96] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, “Fault Tolerance with Comparison-Based Modular Redundancy,” in *Proceedings of the 14th IASTED International Conference Applied Informatics*, Innsbruck, Austria, 1996.

Chapter 4 Bounded Symmetric Comparison Model

This chapter presents a new model for comparison-based, system-level diagnosis, which we call the bounded symmetric comparison (BSC) model. Section 4.1 motivates the need for this new model and discusses the advantages of it. Section 4.2 introduces some definitions that will be used in the discussions of diagnosability of the BSC model that follow in Section 4.3 and Section 4.4. The latter of these two sections is most important to the thesis, for it is here were we provide a theorem with necessary and sufficient conditions for f/(n–1)-diagnosablility under the BSC model. Section 4.5 then investigates which structures are f/(n–1)-diagnosable under the BSC model. In Section 4.6 we present several algorithms for f/(n–1)-diagnosis. Sections 4.7 and 4.8 deal with the issues of fault detectability and fault locatability respectively in the context of the BSC model. Section 4.9 concludes the chapter with a summary of the contributions.

4.1 Introduction

As described in Section 2.3.3, there are two well established comparison models that consider how an outcome of a comparison may be interpreted. The first model, called the symmetric comparison model proposed by Chwa and Hakimi [CH81b], assumes that two faulty processors can produce either identical or different results for a given task. This means that an outcome of 1 implies that at least one of the two processors whose results are compared was definitely faulty. An outcome of 0, however, implies either that both processors were fault-free or both were faulty.

In this model it has been shown that for a system to be diagnosable, $n \geq 2f + 1$. In systems where n is small, e.g., $n = 5$, and where the same task is executed on all PEs as per *N-modular redundancy* (NMR), the symmetric comparison model is suitable since the $f \leq (n-1)/2$, e.g., $f \leq 2$, faulty PEs are potentially capable of producing identical, faulty results. Yet, as n

increases, it becomes less likely that f faulty PEs can produce identical results. Furthermore, when n is large it is unnecessary and inefficient to execute the same task on all n PEs. For these reasons the symmetric comparison model is inappropriate in large multiprocessor systems.

The asymmetric comparison model proposed by Malek [Mal80] assumes that two faulty PEs *always* produce differing results. Because PEs are complex units, it is improbable that the faulty results from two faulty PEs agree. Systems under this model are more easily diagnosable—one condition for diagnosability is $n \geq f + 1$. But this model is not necessarily realistic for systems where the probability of two faulty PEs producing an identical task result is non-negligible.

In systems with a large number of PEs, it can be assumed that $n >> 2f + 1$. Therefore, a more realistic and more efficient strategy (under the symmetric comparison model) is to partition the system as k groups of $2f + 1$ PEs, such that $n = k(2f + 1)$, and to execute k tasks in parallel, with one task per group. Thus the redundancy of the system is used to tolerate faults and to improve throughput. Under the symmetric comparison model the majority quorum of $f + 1$ identical results for the same task is retained but only applies to any single group. But a much lower number, $f = (n - k)/2k$, of the n PEs are allowed to be faulty, compared to systems where $n = 2f + 1$.

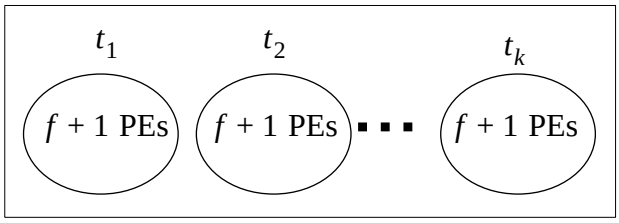


Figure 4.1 Optimistic execution policy with *k* parallel tasks

This arrangement can be improved, as was proposed by Lombardi [Lom84,Lom85], by organizing the system into k groups of just $f + 1$ PEs and executing one task in each group (Figure 4.1), again with a quorum of $f + 1$ identical results for a valid result. Under the symmetric comparison model of Chwa and Hakimi, a result of the task executed by a group is valid only if all PEs in the group have produced it; otherwise the result is undecided. Any such ambiguous results are later resolved with a centralized diagnosis using the global syndrome, with the *a priori* assumption that at most f PEs can be faulty in the entire system. The total number of PEs is $n = k(f + 1)$, and the number of PEs that are allowed to be faulty is then $f = (n - k)/k$.

Following the strategy of Lombardi, we propose a variation of the comparison model that represents actual systems more realistically than the conventional, symmetric comparison model. Our model accommodates faulty PEs from each of the two classic models, but an upper bound is imposed on the number PEs that can produce identical, faulty results. We appropriately call this model the *bounded symmetric comparison* (BSC) model.

Table 4.1 Elementary task diagnosis under the BSC model

$w_{i,j}$	Task pair diagnosis
0	$r_i = r_j$, either both are correct or both are incorrect (no more than f_2 can be incorrect)
1	$r_i \neq r_j$, either both are incorrect, or one is correct and the other incorrect

The BSC model describes a system composed of *n* PEs of which a maximum of f_1 may be faulty, and where the upper bound on the number of faulty PEs producing identical, faulty results is f_2[1], where $f_2 \leq f_1$. Ideally such a system is divided into *k* groups of $f_2 + 1$ PEs (Figure 4.1 with $f = f_2$). Each group is assigned a distinct task, and the PEs execute their task concurrently. Two distinct PEs of the same group yield the task results r_i and r_j for the task assigned to their group. A comparison of these two task results yields a test outcome, $w_{i,j}$, according to the rules shown in Table 4.1.

The BSC model is similar to the symmetric comparison model in that two task results produced by faulty PEs may be identical and therefore yield an outcome of 0 (Table 4.1). In the BSC model, however, the number of such symmetrically faulty results is bounded to f_2. When $f_1 = f_2$ the model is *equivalent* to the symmetric comparison model, which we have shown to be inappropriate for large systems. Therefore, we are mostly interested in the cases where $f_2 < f_1$. The BSC model reduces to the asymmetric comparison model when $f_2 = 1$. This means that no two faulty nodes can produce identical results and therefore yield an outcome of 0. The relationship between the BSC, symmetric comparison and asymmetric comparison models is illustrated in Figure 4.2.

One of the advantages of this approach is to consider structures of a system under the BSC model that minimize the number of comparisons while keeping as many subgroups (on which

1. The restriction on the number of faulty PEs producing identical, faulty results applies only to those PEs whose results are actually compared.

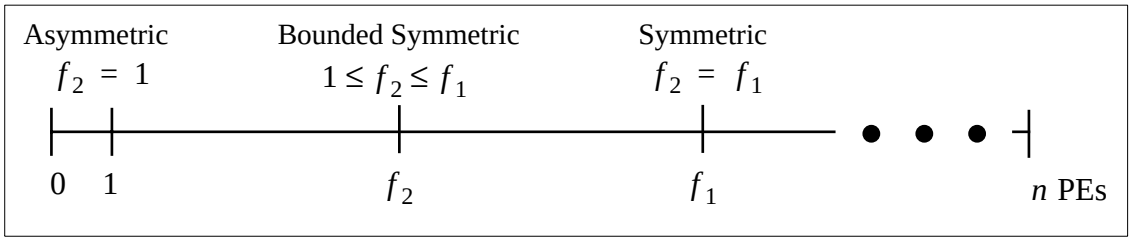


Figure 4.2 Relationship between BSC and asymmetric/symmetric comparison models

distinct, simultaneous tasks are executed) as possible. In this way, the system can perform more work but without forfeiting fault tolerance.

4.2 Diagnosability in the BSC model

In this chapter we consider several aspects concerning the diagnosability of the BSC model and how it relates to the asymmetric and symmetric comparison models. As much as possible we situate the results we obtain for the BSC model for the special cases of $f_2 = 1$ (asymmetric comparison model) or $f_2 = f_1$ (symmetric comparison model). First we introduce some definitions and theorems for the problem of diagnosability under the BSC model.

4.2.1 Useful definitions

We now introduce some definitions that are necessary in the context of the BSC model.

Definition 4.1 A system is *one-step diagnosable* for the BSC model if and only if, given any comparison syndrome, all faulty nodes can be uniquely identified provided that the number of faulty nodes in the system does not exceed f_1 and no more than $f_2 \le f_1$ faulty nodes are allowed to produce faulty results that are identical and hence compare mutually to 0.

Definition 4.2 Let $H = (F, L)$ be a subgraph of $G = (V, E)$ with $F \subseteq V$, $L \subseteq E$. Let $H_0 = (F, L_0)$ be the 0-derived graph of H with the cartels C_1, …, C_k for the syndrome σ. The set F is a *consistent fault set* (CFS) for the syndrome σ in the BSC model if and only if all of the following hold:

1. $|F| \le f_1$,
2. $\sigma(u, v) = 1$ for all $(u, v) \in E$, where $u \in V - F$ and $v \in F$,
3. $\sigma(u, v) = 0$ for all $(u, v) \in E$, where $u, v \in V - F$

4. $|C_i| \leq f_2$ for all cartels C_i.

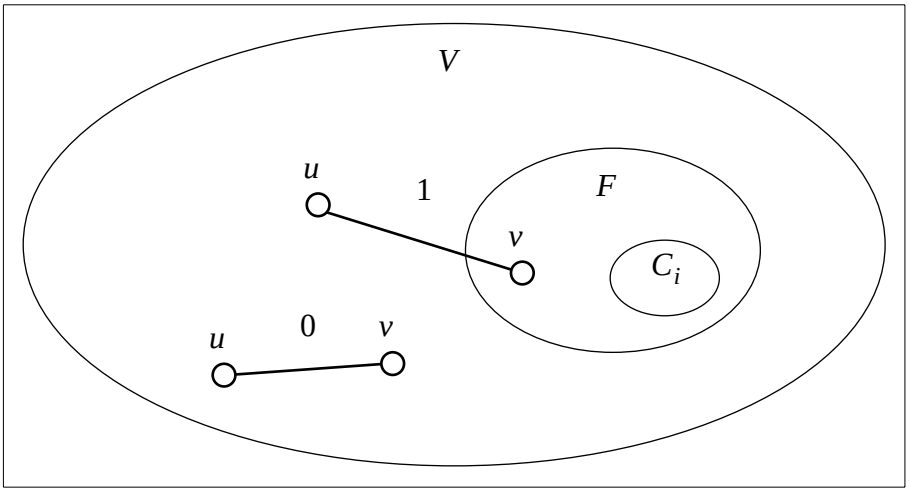


Figure 4.3 Conditions 2 and 3 for Definition 4.2

In other words, condition 1 implies that the maximum number of nodes in *F* cannot be more than the maximum number of faulty nodes allowed, f_1. Condition 2 states that any edge between a node outside of *F* and a node inside of *F* must have a value of 1, showing that the respective nodes' results disagree (see Figure 4.3). Condition 3 says that an edge between any two nodes which are outside of *F* must have a value of 0 indicating agreement between the results of the respective nodes (see Figure 4.3). Condition 4 states that for any cartel, C_i, there are never more than f_2 nodes. This last condition stems from the fact that for a cartel of more than f_2 nodes, the nodes can only be fault-free, since by definition of the BSC model, f_2 is the maximum number of faulty nodes that can produce identical, incorrect task results.

Note that Definition 4.2 applies immediately to the symmetric comparison model by setting $f_2 = f_1$ and to the asymmetric comparison model by setting $f_2 = 1$. For the symmetric model, we see that Definition 4.2 is identical to Definition 3.1 on page 36, since condition 4 of Definition 4.2 is always true when $f_2 = f_1$.

Throughout this chapter we often refer to *candidate fault sets*. A candidate fault set is defined as a set of vertices that for some purpose (e.g., as part of a diagnosis or demonstration of a proof) is considered to be the potential set of faulty nodes in the system. Therefore a candidate fault set contains at most f_1 nodes.

In the following section, we introduce some theorems that provide a partial characterization of one-step diagnosable systems for the BSC model.

4.3 One-step diagnosability for the BSC model

The following theorems provide sufficient conditions for one-step diagnosability (Definition 4.1) for the BSC model. Although these results are not part of the main thrust of the thesis, we include them for their contribution to the characterization of the BSC model.

We start out with a theorem defining two necessary conditions for one-step diagnosability. Immediately we see that if $f_2 < f_1$, the number of nodes necessary for diagnosability is reduced from that required in the symmetric comparison model, where $n = 2f_1 + 1$ and $f_2 = f_1$.

Theorem 4.1 If a system *S* with *n* nodes is one-step diagnosable for the BSC model (Definition 4.1), then

1. $n \geq f_1 + f_2 + 1$ and
2. every node is linked to at least f_1 other units.

Proof. For condition 1 of the theorem, assume the comparison testing graph is complete and $n = f_1 + f_2 + 1$. Since a total of at most f_1 nodes are faulty, there is a cartel of at least $n - f_1 = f_2 + 1$ non-faulty nodes. These nodes are diagnosed as fault-free since no more than f_2 faulty nodes are allowed to form a cartel. The faulty nodes can be diagnosed from one of the identified non-faulty nodes since the graph is complete.

If $n < f_1 + f_2 + 1$, take $n = f_1 + f_2$. Assume the comparison testing graph is complete and that f_1 nodes are faulty. Then $n - f_1 = f_2$ nodes are non-faulty and form a cartel. But there are at least f_2 faulty nodes, since $f_1 \geq f_2$ so that there might also be a cartel of f_2 faulty nodes. In this case, there is no way to sort the faulty nodes from the non-faulty nodes since we have two cartels of f_2 nodes, with one fault-free and one non-faulty. Thus the system is not diagnosable if $n < f_1 + f_2 + 1$.

Now consider condition 2 of the theorem. If this condition is not satisfied, there is at least one subgraph of the system where one node v_i is linked with less than f_1 other units. Assume the limiting condition where v_i is linked with $f_1 - 1$ other units. Then there may be a syndrome in this subgraph where all 1s are produced on the comparison edges of v_i. As such the

fault status of v_i cannot be decided if all the $f_1 - 1$ units linked to v_i are faulty.

Yet if v_i is linked with f_1 other units, then there are two possible worst cases where the diagnosis can be performed, since no more than f_1 units are faulty. In the first case, v_i is faulty, and thus at least one of the units, v_j, linked to it is non-faulty. Once v_j is determined to be fault-free by other tests, the fault status of v_i is decided unambiguously. In the second case, v_i is fault-free, and all the units linked to it are faulty. But once all of these f_1 units are determined to be faulty by other tests, the fault status of v_i is decided unambiguously as faulty by inference. □

The following theorem provides necessary and sufficient conditions for one-step diagnosability under the BSC model.

Theorem 4.2 A system *S* represented by the undirected graph $G = (V, E)$ is one-step diagnosable with the BSC model if and only if for every two distinct sets F_1, F_2 with $F_1 \subset V$, $F_2 \subset V$ and $|F_1| \leq f_1$, $|F_2| \leq f_1$, one of the following conditions holds:

1. There is an edge between a node in $V - (F_1 \cup F_2)$ and a node in $(F_1 \cup F_2) - (F_1 \cap F_2)$.
2. A connected component in either $Z_1 = F_1 - (F_1 \cap F_2)$ or $Z_2 = F_2 - (F_1 \cap F_2)$ contains more than f_2 nodes.

Proof. Necessity. Assume that neither condition 1 nor 2 is satisfied for a particular pair of sets F_1, F_2. We will show that for some test results, F_1 and F_2 are two consistent fault sets. Let $Y = F_1 \cap F_2$ and $X = V - (F_1 \cup F_2)$ as shown in Figure 4.4.

There is no edge $(u, v) \in E$, $u \in V - (F_1 \cup F_2)$, $v \in (F_1 \cup F_2) - (F_1 \cap F_2)$ and no component in Z_1 or Z_2 with more than f_2 nodes. Note that $|Y| \leq f_1$, since $|F_1| \leq f_1$, $|F_2| \leq f_1$. Thus we can assume that *Y* contains only faulty units since it pertains to the two sets F_1 and F_2 that are candidates for being consistent fault sets. Let *X* contain only fault-free units. Then we might have the syndrome sketched in Figure 4.4 where there is no edge between *X* and

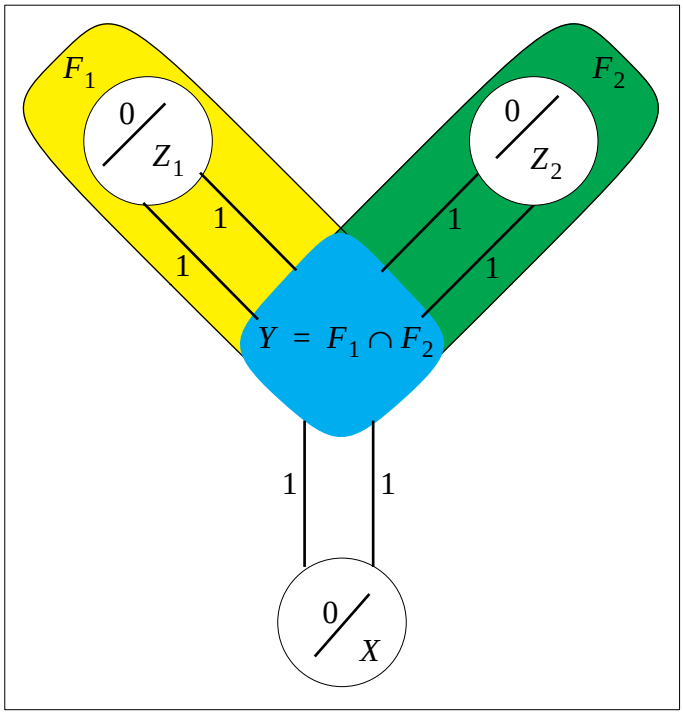


Figure 4.4 Proof of diagnosability for Theorem 4.2

either Z_1 or Z_2 and where no cartel in either Z_1 or Z_2 has cardinality greater than f_2. It can be seen easily that the two sets F_1 and F_2 satisfy all the conditions set forth by Definition 4.2. Thus F_1 and F_2 are two different consistent fault sets and the system is not one-step diagnosable.

Proof. Sufficiency. Assume that the system is not one-step diagnosable and the conditions 1 or 2 are satisfied for any pair of sets F_1, F_2, with $F_1 \subset V$, $F_2 \subset V$ and $|F_1| \leq f_1$, $|F_2| \leq f_1$. Since the system is not one-step diagnosable there should be at least one pair of sets F_1, F_2 that are both consistent fault sets.

Suppose first that for two sets F_1, F_2, condition 1 is satisfied, with $(u, v) \in E$, $u \in X = V - (F_1 \cup F_2)$, $v \in Z_1 \in (F_1 \cup F_2) - (F_1 \cap F_2)$. Then if $\sigma(u, v) = 0$, F_1 is not a consistent fault set by condition (2) of Definition 4.2. If $\sigma(u, v) = 1$, then F_2 is not a consistent fault set by condition 3 of Definition 4.2. Hence for the system to be non-diagnosable, there can be no edge between X and either Z_1 or Z_2, which is in contradiction with condition 1 of the theorem.

Suppose now that the system is not one-step diagnosable for a particular pair of sets F_1, F_2 and that condition 2 is satisfied. Then the sets F_1, F_2 must be both consistent fault sets. Note

first that for any two nodes *u*, *v* such that $u \in Z_1$, $v \in Z_1$, $Z_1 \in (F_1 \cup F_2) - (F_1 \cap F_2)$, the syndrome must be consistent with F_2. Similarly, the same conclusions can be drawn for the nodes in Z_2 with respect to the consistency of F_1. Thus $\sigma(u, v) = 0$, by condition 3 of Definition 4.2. However, condition 2 of the theorem states that a connected component of more than f_2 nodes exists in either Z_1 or Z_2. If this component is in Z_1, then F_1 cannot be consistent since it contains a cartel of more than f_2 nodes (condition 4 of Definition 4.2). Hence condition 2 of the theorem is in contradiction with the system not being one-step diagnosable. The same argument applies when *u* and *v* are in Z_2, which completes the proof of sufficiency. □

Note that when $f_1 = f_2$, Theorem 4.2 reduces to the corresponding theorem reported in [Pel91] for one-step diagnosability with the symmetric comparison model. In this case, condition 2 disappears since $|Z_1| \leq f_2$ and $|Z_2| \leq f_2$. When $f_2 = 1$, Theorem 4.2 reduces to the corresponding theorem reported in [Pel91] for one-step diagnosability with the asymmetric comparison model. In this case, condition 2 simply means that there is no edge within either Z_1 or Z_2.

As a final remark on Theorem 4.2, we note that condition 2 is satisfied either when $|Z_1| > f_2$ and Z_1 is connected, or when $|Z_2| > f_2$ and Z_2 is connected. By Theorem 2.1 (page 30), Z_1 is connected when it has a single component, with $p = 1$. Then $m \geq |Z_1| - 1$, where *m* is the number of edges in Z_1. Since $|Z_1| > f_2$, this implies that $m \geq f_2$. For the other alternative of condition 2 we must have $m \geq |Z_2| - 1$, where *m* is the number of edges in Z_2.

4.4 *f*/(*n*–1)-diagnosability under the BSC model

As we have pointed out in Chapter 3, one of the benefits of *f*/(*n*–1)-diagnosability is to maximize the throughput of a fault-tolerant system where several distinct tasks are concurrently executed. The primary interest is in guaranteeing the identification of at least one fault-free task result per set of executions, provided the stated limits on the number and type of faulty nodes for the BSC model (f_1, f_2) are not exceeded. This leads to the following definition:

Definition 4.3 A system composed of *n* nodes under the BSC model (where no more than f_1 nodes are faulty of which at most f_2 can produce identical, faulty results to form a cartel) is said to be *f/(n–1)-diagnosable* if, for any syndrome σ, the identification of at least one fault-free node is guaranteed.

From Definition 4.3 the following necessary condition for *f*/(*n*–1)-diagnosability is obtained.

Theorem 4.3 If a system with *n* nodes is *f*/(*n*–1)-diagnosable under the BSC model, then

$$n \geq f_1 + f_2 + 1, \text{ with } f_1 \geq f_2 \tag{4.1}$$

Proof. Assume the comparison testing graph is complete and $n = f_1 + f_2 + 1$. Since, by Definition 4.3, a total of at most f_1 nodes are faulty, there are at least $n - f_1 = f_2 + 1$ fault-free nodes that produce identical test results. These nodes are diagnosed as fault-free since no more than f_2 faulty nodes are allowed to produce identical results in a single subset.

If $n < f_1 + f_2 + 1$, take $n = f_1 + f_2$. Assume f_1 faulty nodes. Then $n - f_1 = f_2$ nodes are fault-free and test to 0. But at least f_2 nodes are faulty, since $f_1 \geq f_2$. Thus there may be f_2 faulty nodes that produce identical results. In this case there is no way to distinguish the faulty nodes from the fault-free nodes since there are two sets of f_2 nodes that yield comparison tests of all zeros, with one fault-free and one faulty. Thus the system is not *f*/(*n*–1)-diagnosable if $n < f_1 + f_2 + 1$. □

As previously stated, if $f_1 = f_2 = f$, the BSC model is equivalent to the symmetric comparison model. As should be expected equation (4.1) reduces to $n \geq 2f + 1$, for $f_1 = f_2 = f$. On the contrary, if $f_2 = 1$, no two faulty nodes are allowed to produce identical, faulty results that yield a comparison outcome of 0. This describes the asymmetric comparison model originally proposed by Malek [Mal80]. In this case, equation (4.1) becomes $n \geq f_1 + 2$, with $f = f_1$, as reported [Mal80]. With these results one can see that the BSC model provides a continuum between the symmetric and the asymmetric comparison models. Therefore, the

characterization of the BSC model can be expected to yield results that stand between those corresponding to the symmetric and asymmetric comparison models.

In recalling Theorem 3.1 (page 37) one can see clearly that a system that satisfies the conditions for this theorem when $f = f_1$ can be also *f*/(*n*–1)-diagnosable for the BSC model, since the BSC model is less demanding than the symmetric comparison model to which Theorem 3.1 applies. Hence we seek to relax the conditions of Theorem 3.1 in the case of the BSC model.

Before characterizing *f*/(*n*–1)-diagnosability under the BSC model, the following remarks are merited. In one-step diagnosability as shown above, the connectivity must be such that every allowable syndrome must be consistent with a *single* fault set. The complete fault set is identified unambiguously, and, by inference, all the other nodes are considered fault-free.

In contrast to this technique, a system that is *f*/(*n*–1)-diagnosable within the BSC model may have limited connectivity because of the disjoint groups of tasks (as explained in Chapter 3), which results in several candidate fault sets $F_1, \ldots, F_k$ that can by mutually consistent with a given syndrome. By Lemma 3.3 (page 39), if $V - \bigcup_{i=1}^{k} F_i \neq \varnothing$ (the union of all the candidate fault sets does not cover all of the nodes in *V*), the nodes in $V - \bigcup_{i=1}^{k} F_i$ (the nodes not covered by the candidate fault sets) are fault-free, which ensures *f*/(*n*–1)-diagnosability.

The key to characterizing *f*/(*n*–1)-diagnosability, then, lies in characterizing the consistency of multiple candidate fault sets for each possible syndrome. In the case of *f*/(*n*–1)-diagnosability under the BSC model, *any* set of sets $F_1, \ldots, F_k$ with $|F_i| \leq f_1$ is *a priori* a set of candidate fault sets. Thus we need some mechanisms to invalidate some of these candidate fault sets if we are to identify fault-free nodes. In the following we investigate the issues for invalidating the consistency of candidate fault sets in the BSC model.

The first mechanism we investigate is that of determining the presence of an edge such that, no matter what the edge's value in a given syndrome, other candidate fault sets cannot be consistent. This mechanism is not new, and was introduced by Xu *et al.* [XH90] for *f*/(*n*–1)-diagnosability in the symmetric comparison model. We present it here, nonetheless, in the context of the BSC model.

A second mechanism for invalidating candidate fault sets is unique to the BSC model. In this mechanism, the goal is to show that at least one candidate fault set is not consistent by the existence of a connected component of more than f_2 nodes.

4.4.1 Elimination of candidate fault sets by the existence of certain edges

Xu and Huang [XH90] characterized $f/(n–1)$-diagnosability under the symmetric comparison model in a fashion similar to the following. Let us first consider two candidate fault sets F_1 and F_2, with $|F_1| \le f_1$, $|F_2| \le f_1$, $F_1 \ne F_2$ (Figure 4.5). Let $X = F_1 \cap F_2$. Assume first that

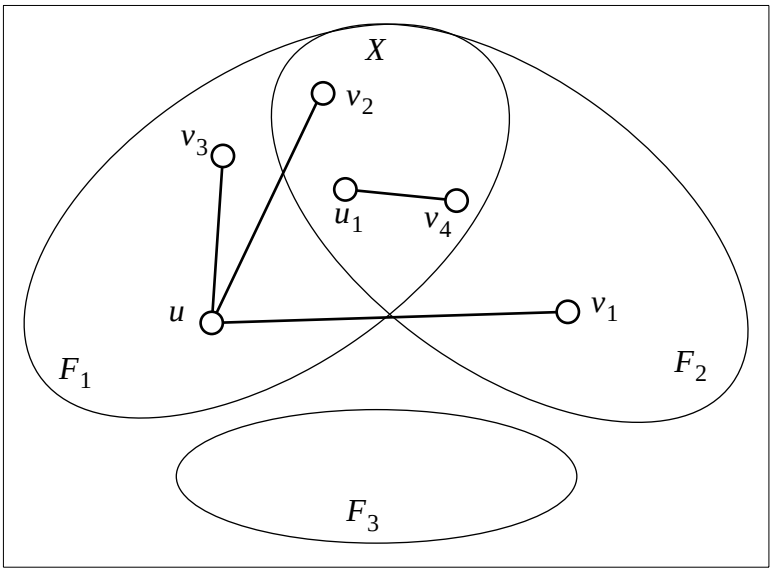


Figure 4.5 Breaking the mutual consistency of candidate fault sets by Theorem 4.4

there is a comparison link $(u, v_1) \in E$ between a node $u \in F_1 - X$ and a node $v_1 \in F_2 - X$. Then if $\sigma(u, v_1) = 0$, by Definition 4.2, this is inconsistent with either F_1 being the actual fault set or F_2 being the actual fault set. For the same edge, however, if there is a third candidate fault set F_3, $|F_3| \le f_1$ which does not contain u and v_1, a test outcome $\sigma(u, v_1) = 1$ is inconsistent with F_3 being the actual fault set. Thus any edge such as (u, v_1) prohibits the three candidate fault sets F_1, F_2, F_3 from being mutually consistent, regardless of the syndrome. Note that this edge (u, v_1) is such that $f(u) \ne f(v_1)$ and $f(u) \cup f(v_1) \ne \pi = \{F_1, F_2, F_3\}$.

Suppose now that there is an edge (u, v_2) such that $u \in F_1 - X$, $v_2 \in X$. In this case, an outcome $\sigma(u, v_2) = 0$ is consistent with F_3 being the actual fault set. But the outcome $\sigma(u, v_2) = 0$ invalidates the consistency of F_1 if F_2 is fault-free, and vice versa. On the other hand, the outcome $\sigma(u, v_2) = 1$ prohibits F_3 from being consistent. Thus, regardless of the comparison outcome, any edge such as (u, v_2) prohibits the mutual consistency of F_1, F_2,

F_3. Note that, as before, this edge is such that $f(u) \neq f(v_2)$ and $f(u) \cup f(v_2) \neq \pi = \{F_1, F_2, F_3\}$.

Suppose this time that there is an edge (u, v_3) such that $u \in F_1 - X$, $v_3 \in F_1 - X$. A comparison outcome $\sigma(u, v_3) = 1$ prevents F_3 from being consistent. Yet a comparison outcome $\sigma(u, v_3) = 0$ does not deny the consistency of F_1, F_2, F_3[1]. Thus, with such an edge in the graph, the consistency of the three fault sets cannot be unconditionally denied; it depends upon the syndrome that is produced. Note that the edge (u, v_3) is such that $f(u) = f(v_3)$ and $f(u) \cup f(v_3) \neq \pi = \{F_1, F_2, F_3\}$.

Finally, suppose that there is an edge (u_1, v_4) such that $u_1 \in X$, $v_4 \in X$. Again, the consistency of the three candidate fault sets is not broken when $\sigma(u_1, v_4) = 0$ according to Definition 4.2. Note that the edge (u_1, v_4) is such that $f(u_1) = f(v_4)$ and $f(u) \cup f(v_2) \neq \pi$, where $\pi = \{F_1, F_2, F_3\}$. Hence we recall the following theorem from Xu *et al.* [XH90], which is identical to Theorem 3.1 (page 37) for the symmetric comparison model. It provides a necessary and sufficient condition for eliminating candidate fault sets under the assumptions of the symmetric comparison model, i.e., when $f_1 = f_2$. Yet, as we shall show in Section 4.4.2, additional conditions for invalidating candidate fault sets are obtained by the BSC model, that is, where $f_1 \geq f_2 \geq 1$.

Theorem 4.4 Assume $f_1 = f_2$ (symmetric comparison model), and consider a cover $\pi = \{F_1, \ldots, F_k\}$ of V with candidate fault sets F_i such that $|F_i| \leq f_1$. Some of the candidate fault sets F_i are not consistent with a common syndrome (and can thus be disallowed once identified) if there exists an edge $(u, v) \in E$ which satisfies the following condition:

$$(f(u) \neq f(v)) \wedge ((f(u) \cup f(v)) \neq \pi) \tag{4.2}$$

Proof. Necessity. Assume to the contrary that a cover π of V exists such that for any edge $(u, v) \in E$, $f(u) = f(v)$ or $f(u) \cup f(v) = \pi$ (Figure 4.6). We shall show that the k candidate

1. This is under the assumption that u and v_3 do not pertain to a cartel of more than $f_2 + 1$ nodes, as we shall see in the following section.

fault sets $F_1, \ldots, F_k$ in π can be consistent fault sets for a common syndrome, if the syndrome is constructed as follows. For all $(v_i, v_j) \in E$, if $f(v_i) = f(v_j)$, then let $\sigma(v_i, v_j) = 0$, else if $f(v_i) \neq f(v_j)$ and $f(v_i) \cup f(v_j) = \pi = V$, then let $\sigma(v_i, v_j) = 1$.

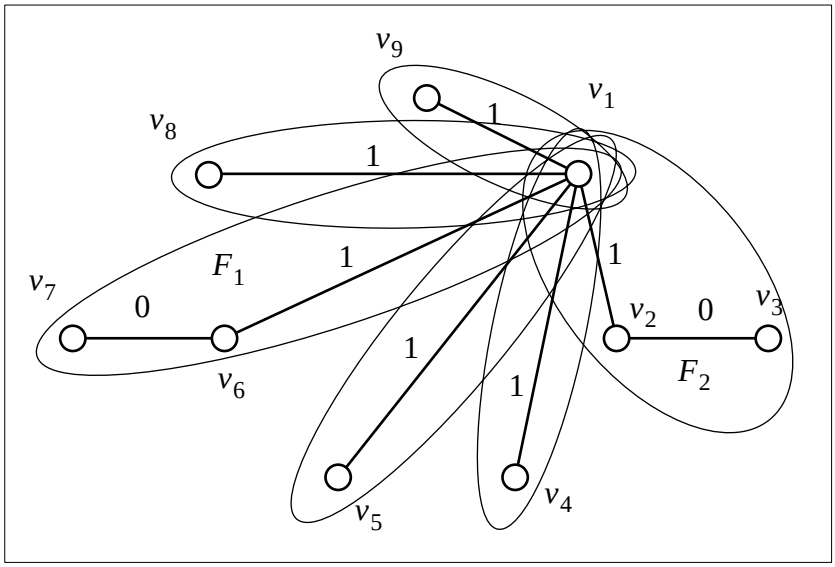


Figure 4.6 Cover π where for any edge (v_i, v_j), $f(v_i) = f(v_j)$ or $f(v_i) \cup f(v_j) = \pi$, with a syndrome where all candidate fault sets are consistent

In the following, refer to Figure 4.6. For the first condition of the syndrome assignment, $f(v_i) = f(v_j)$. Consider any two distinct, candidate fault sets, e.g., F_1 and F_2, and any edge, e.g., (v_2, v_3), in one of these subsets, e.g., F_2. Since $f(v_2) = f(v_3)$, if $\sigma(v_2, v_3) = 1$, F_1 cannot be a consistent fault set according to condition 3 of Definition 4.2. As the proof states, however, by setting $\sigma(v_2, v_3) = 0$, either all the nodes in F_2 are fault-free and all the nodes in F_3 are faulty (or vice versa). Thus setting, setting $\sigma(v_i, v_j) = 0$ in the syndrome whenever $f(v_i) = f(v_j)$ allows all candidate fault sets to be consistent fault sets, and hence disallows *f*/(*n*–1)-diagnosability. More precisely, for any syndrome σ, if $v_i \in V - F_i$ and $\sigma(v_i, v_j) = 0$, then $v_j \in V - F_i$, since $f(v_i) = f(v_j)$.

For the second condition of the syndrome assignment, consider an edge such that $f(v_i) \neq f(v_j)$ but $f(v_i) \cup f(v_j) = \pi = V$, e.g., (v_1, v_2). If $\sigma(v_1, v_2) = 1$, then this is compatible with the two candidate fault sets F_1 and F_2 being consistent fault sets by condition 2 of Definition 4.2. Moreover, for any syndrome σ, any $(v_i, v_j) \in E$ and $F_i \in V$, if $v_i \in V - F_i$ and $\sigma(v_i, v_j) = 1$, then $v_j \in F_i$, since $f(v_i) \cup f(v_j) = V$, which is compatible with all subsets

being consistent fault sets. Again, if $F_1, \ldots, F_k$ are consistent fault sets for the syndrome constructed, the system cannot be *f*/(*n*–1)-diagnosable. □

Proof. Sufficiency. Assume that for any cover $\pi = \{F_1, \ldots, F_k\}$ of V an edge (v_i, v_j) exists such that $f(v_i) \neq f(v_j)$ and $(f(v_i) \cup f(v_j)) \neq \pi$, and, contrary to the theorem, that $F_1, \ldots, F_k$ are consistent. Then there exists a syndrome such that for all fault sets $F_1, \ldots, F_k$ that are consistent for this syndrome, $\bigcup_{i=1}^{k} F_i = \pi = V$, since not a single candidate fault set F_i can be disallowed because of being inconsistent.

According to the above assumption, an edge (v_i, v_j) exists such that $f(v_i) \neq f(v_j)$ and $f(v_i) \cup f(v_j) \neq \pi$. Since $f(v_i) \neq f(v_j)$ and π is a cover of *V,* however, there is a candidate fault set F_i such that $F_i \in f(v_i)$ and $F_i \notin f(v_j)$. Thus v_i belongs to F_i but v_j does not. Then for the edge (v_i, v_j), if $\sigma(v_i, v_j) = 0$, F_i is not a consistent fault set by condition 2 of Definition 4.2, since $v_i \in F_i$ and $v_j \in V - F_i$. This is a contradiction.

Moreover, since $f(v_i) \cup f(v_j) \neq \pi$, and π is a cover that contains only candidate fault sets, there is a candidate fault set F_i such that $v_i \notin F_i$ and $v_j \notin F_i$. Then for the edge (v_i, v_j), if $\sigma(v_i, v_j) = 1$, by condition 3 of Definition 4.2, F_i cannot be a consistent fault set, which is also a contradiction. □

4.4.2 Elimination of candidate fault sets by the existence of connected components

So far we have not taken into account the f_2 parameter of the BSC model in our characterization of *f*/(*n*–1)-diagnosability. The advantage of considering f_2 is that it allows invalidating any candidate fault set where a cartel *C* of more than f_2 nodes exists. As such, there are fewer syndromes that are not diagnosable. For the symmetric comparison model $f_1 = f_2$, and since $|F_i| \leq f_1$, no candidate fault set can be invalidated, because no cartel *C* of more than f_2 nodes can exist inside a single candidate fault set F_i. Thus Theorem 4.4 is the only way to invalidate candidate fault sets when $f_1 = f_2$, and its condition is necessary and sufficient.

But the situation is obviously different in the BSC model when $f_2 < f_1$. Consider again the

syndrome shown in the system of Figure 4.6, assuming $f_1 = 3$. If we assume also that $f_2 = 1$, then suddenly the consistency of some of the fault sets (e.g., F_1 and F_2) can be invalidated using condition 4 of Definition 4.2. In other words, the syndrome that we generated to show that the system was not *f*/(*n*–1)-diagnosable is actually diagnosable, since we see that there are cartels of more than f_2 nodes in two of the candidate fault sets. This is in spite of the fact that the conditions of Theorem 4.4 are not satisfied.

We now explore the invalidation mechanism that is specific to the BSC model. In the preceding discussion we have shown that in a situation with three candidate fault sets F_1, F_2, F_3, the mutual consistency of these candidate fault sets was not denied by a comparison link between two nodes u, v such that either $u, v \in F_1 - (F_1 \cap F_2)$ or $u, v \in (F_1 \cap F_2)$. Thus we concentrate on these two cases which are illustrated in Figure 4.8.

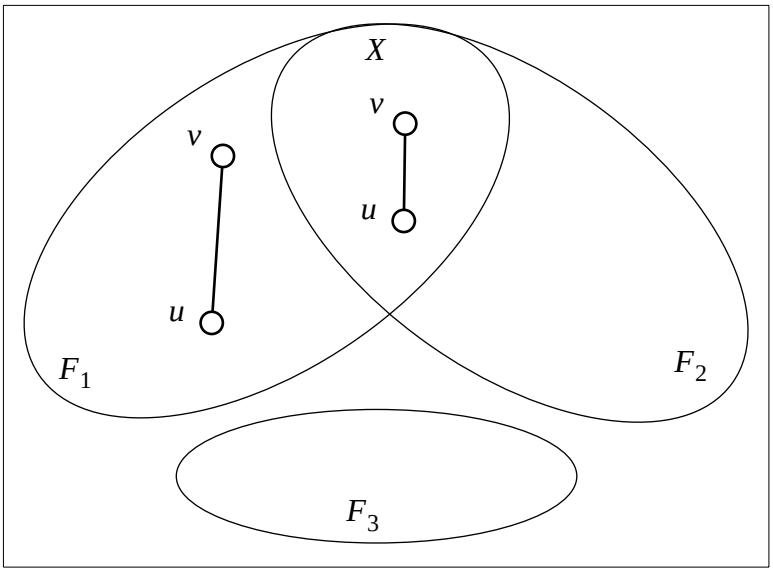


Figure 4.7 Remaining cases where mutual consistency of candidate fault sets is not denied for any syndrome σ

We point out first that if any component forms a cartel *C* of more than f_2 nodes (that is, all comparisons in the component yield 0), then the nodes of that component are asserted as fault-free by definition. Thus the only interesting case is when there are connected components where some of the comparisons yield 1 without leaving any cartel(s) of more than f_2 nodes.

Consider the first of the two remaining situations where there is a connected component C_1 in $F_1 - (F_1 \cap F_2)$ such that $|C_1| > f_2$ (Figure 4.9). Then if C_1 is a cartel, the nodes in C_1 are fault-free, and the diagnosis is completed. If C_1 is not a cartel, however, then some edges in

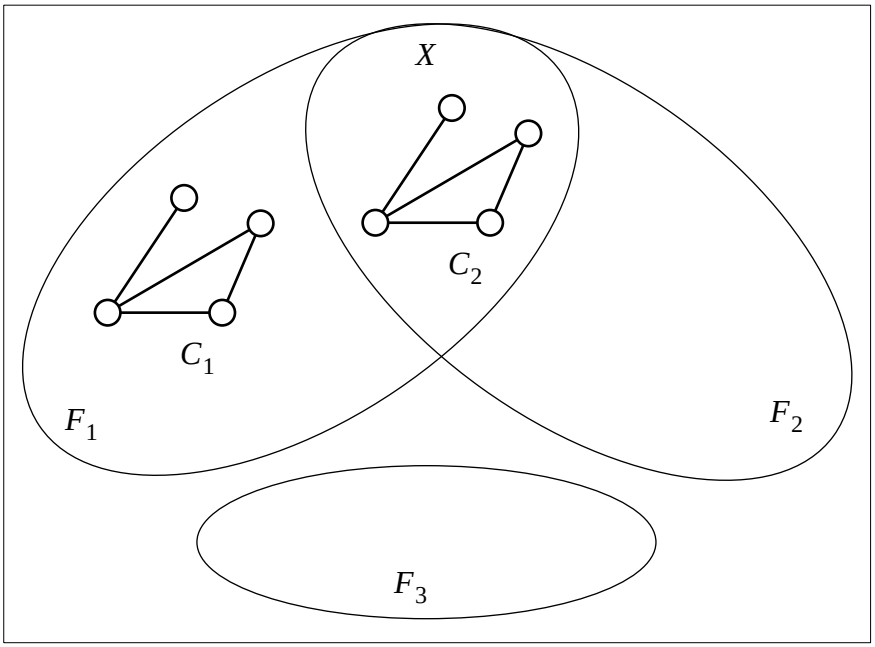


Figure 4.8 Connected components that deny the mutual consistency of candidate fault sets

C_1 yield 1, and by condition 3 of Definition 4.2, the candidate fault sets F_2 and F_3 are inconsistent with F_1 for this syndrome. Thus, no matter what the actual syndrome, the candidate fault sets cannot be mutually consistent when there is a connected subgraph $H = (C, A)$, $A \subseteq E$, $C = \{u_1, \ldots, u_p\}$, $u_1, \ldots, u_p \in V$, $|C| = p > f_2$ such that for any $(u_i, u_j) \in A$, $f(u_i) = f(u_j)$.

Similarly, for the second remaining case described above, the candidate fault sets cannot be mutually consistent if a connected component C_2 such that $|C_2| > f_2$ is included in $F_1 \cap F_2$ (Figure 4.9). There is an exception to this case, however, when all the nodes in C_2 are included in all the candidate fault sets of a cover. The reason is obvious—fault sets can only be discounted if the nodes in C_2 form a cartel. That is, if any edge between two nodes of C_2 yields 1, then all of the conditions of Definition 4.2 are respected (since the edge is not across the boundary of any candidate fault set), and no candidate fault set can be discounted.

For any component C for which some of the nodes are in different candidate fault sets (Figure 4.9), the conditions of Theorem 4.4 apply, as there are edges now that exist such that they disallow the consistency of at least one of the fault sets. This leads us to the following lemma.

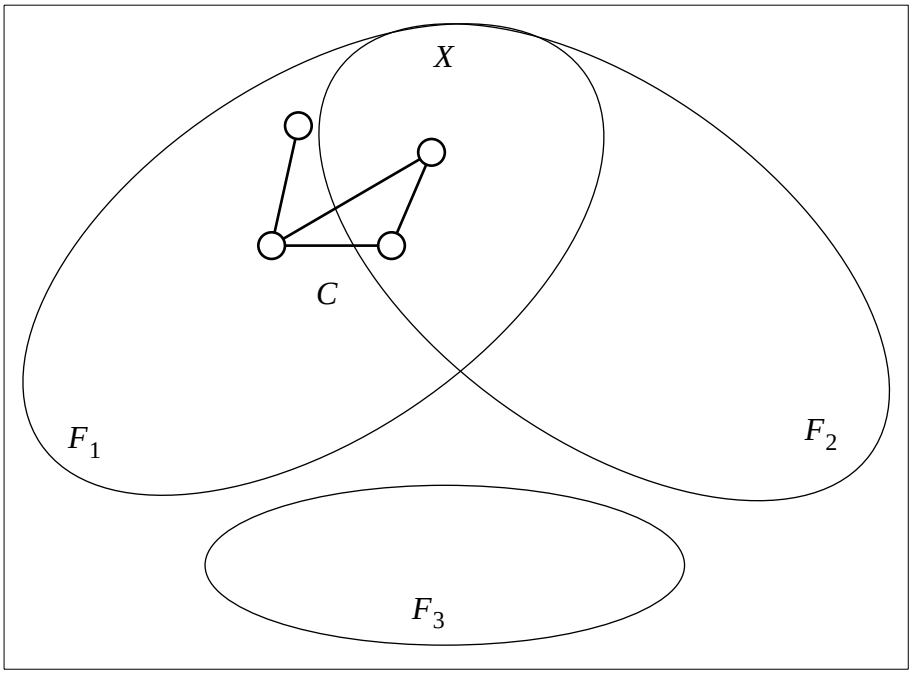


Figure 4.9 Breaking the mutual consistency of candidate fault sets by Lemma 4.1 when some of the nodes of a connected component are included into several candidate fault sets

Lemma 4.1 The candidate fault sets F_i in a cover of candidate fault sets $\pi = \{F_1, \ldots, F_k\}$ of V are not consistent with a common syndrome for the BSC model if there is a connected component $C = \{u_1, \ldots, u_p\}$, $u_1, \ldots, u_p \in V$, $|C| > f_2$ such that $\bigcup_{i=1}^{p} f(u_i) \neq \pi$.

Proof. If there is at least one connected component $C = \{u_1, \ldots, u_p\}$, $u_1, \ldots, u_p \in V$, $|C| = p > f_2$ that is not included in every candidate fault set pertaining to the cover under consideration, then either the nodes in C form a cartel of more than f_2 nodes, or some of the nodes in C test to 1 and by inference the sets which do not contain nodes from C are disallowed from being consistent fault sets. □

The basic theorem of $f/(n–1)$-diagnosability for the BSC model can now be derived directly from Theorem 4.4 and Lemma 4.1. In essence this theorem states that a system is $f/(n–1)$-diagnosable if for any cover π of V the mutual consistency is broken either by the conditions corresponding to the symmetric comparison model given by Theorem 4.4 or by the condition specific to the BSC model that corresponds to Lemma 4.1.

Theorem 4.5 A system S represented by an undirected graph $G = (V, E)$ is $f/(n–1)$-diagnosable for the BSC model if and only if for any cover $\pi = \{V_1, \ldots, V_k\}$ of V where $|V_i| \leq f_1$, at least one of the following conditions is satisfied:

1. an edge (u, v) exists such that $(f(u) \neq f(v)) \wedge ((f(u) \cup f(v)) \neq \pi)$
2. a connected component C exists such that $C = \{u_1, \ldots, u_p\}$, $u_1, \ldots, u_p \in V$, $|C| = p > f_2$ and for any $u_i \in C$, $\bigcup_{i=1}^{p} f(u_i) \neq \pi$.

Proof. Necessity. Assume to the contrary that a cover exists such that no edge exists satisfying condition 1 and no connected component exists satisfying condition 2. This situation is illustrated in Figure 4.10 for $f_1 = 3$ and $f_2 = 2$. We can see that the syndrome given in the figure is a case where all candidate fault sets in the cover are consistent fault sets according to Definition 4.2. Hence the system is not *f*/(*n*–1)-diagnosable.

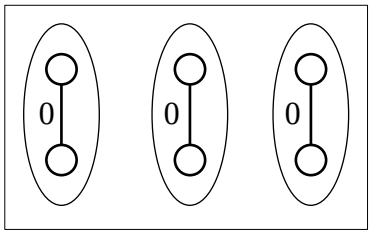


Figure 4.10 Cover π with a syndrome where all candidate fault sets are consistent for $f_1 = 3$, $f_2 = 2$

Proof. Sufficiency. Assume that for a given cover π there exists an edge satisfying condition 1 or a connected component satisfying condition 2, but, contrary to the theorem, the system is not *f*/(*n*–1)-diagnosable. This implies that all of the candidate fault sets of the cover $\pi = \{V_1, \ldots, V_k\}$ are consistent for any given syndrome. We have shown in Theorem 4.4 that if condition 1 exists, this is a contradiction. To complete this proof, therefore, we need to show that if condition 2 is satisfied, this assumption is also a contradiction. Note that since condition 1 also applies to the edges in *C*, we only consider possibilities for *C* that do not contain edges satisfying condition 1. That is, we do not consider connected components whose nodes are contained in different candidate fault sets, as illustrated in Figure 4.9. Given such a *C*, either all of its edges yield 0, or at least one of its edges yields 1. We show that in either case, a contradiction occurs.

If all the comparisons in *C* yield 0, then *C* forms a cartel of more than f_2 nodes, and the system is *f*/(*n*–1)-diagnosable, and this is a contradiction. But some of the comparisons in *C* may yield 1. Looking again at Figure 4.8 (which gives two possible connected components satisfying condition 2 and whose edges do not satisfy condition 1), we can see that if any edge in

C yields a 1, then at least one of the candidate fault is not consistent, which is again a contradiction of the assumption that the system is not *f*/(*n*–1)-diagnosable. □

For precision we specify that when $\bigcup_{i=1}^{p} f(u_i) \neq \pi$ for all $u_i \in C$, and an edge (u_i, v_j) exists such that $u_i \in C$, $u_j \in C$, $f(u_i) \neq f(u_j)$, then condition 1 of Theorem 4.5 is satisfied. Thus condition 2 is needed only when $f(u_i) = f(u_j)$ for every u_i, u_j in *C*.

Along the lines of Theorem 4.5, we propose the following conjecture, though no proof is provided:

Conjecture 4.1 A system *S* represented by an undirected graph $G = (V, E)$ is *f*/(*n*–1)-diagnosable for the BSC model if and only if for any cover $\pi = \{V_1, \ldots, V_k\}$ of *V* where all $|V_i| = f_1$, at least one of the following conditions is satisfied:

1. an edge (u, v) exists such that $(f(u) \neq f(v)) \wedge ((f(u) \cup f(v)) \neq \pi)$
2. a connected component *C* exists such that $C = \{u_1, \ldots, u_p\}$, $u_1, \ldots, u_p \in V$, $|C| = p > f_2$ and for any $u_i \in C$, $\bigcup_{i=1}^{p} f(u_i) \neq \pi$.

With Theorem 4.5, a simple way of investigating the diagnosability of a system consists in checking condition 1 only on the covers $\pi = \{F_1, \ldots, F_k\}$ of *V* such that no candidate fault set $F_i \in \pi$ contains any component *C*, with $|C| > f_2$.

4.4.3 Examples

To illustrate the diagnosability conditions defined by Theorem 4.5 we present several examples. Figure 4.11 illustrates a simple case of a system with $n = 4$, $f_1 = 2$, $f_2 = 1$. In Figure 4.11a there is a cover with the two candidate fault sets F_1 and F_2, and four edges between F_1 and F_2 such that $f(u) \neq f(v)$. For example, $f(v_1) = F_1$ and $f(v_2) = F_2$. Thus the first part of condition 1 of Theorem 4.5 is satisfied with, for example, the existence of the edge (v_1, v_2) such that $f(v_1) \neq f(v_2)$. For this edge, however, $f(v_1) \cup f(v_2) = \pi = V$. Thus condition 1 of Theorem 4.5 is not satisfied. Condition 2 of Theorem 4.5 is also not satisfied since there is no component *C* such that $|C| > f_2 = 2$ and $\bigcup_{i=1}^{p} f(u_i) \neq \pi$. Hence the system

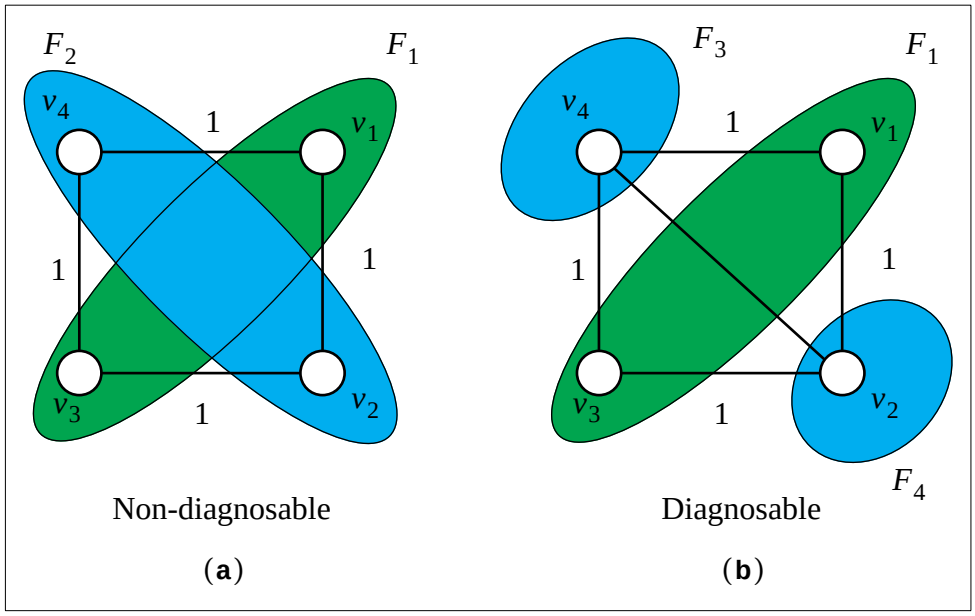


Figure 4.11 Diagnosability conditions of Theorem 4.5 for $n = 4$, $f_1 = 2$, $f_2 = 1$

is not $f/(n-1)$-diagnosable as can be verified easily by noting that there is no way of determining which of F_1 or F_2 is faulty for a syndrome of all 1s.

In Figure 4.11b the system has an additional comparison edge between v_2 and v_4. For a cover such that one of the candidate fault sets is F_2 (from Figure 4.11a), the system is $f/(n-1)$-diagnosable by condition 2 of Theorem 4.5. Thus F_2 can be broken into the two sets F_3, F_4, and the system becomes $f/(n-1)$-diagnosable under condition 1 of Theorem 4.5.

Figure 4.12 illustrates another application of Theorem 4.5 for $n = 7$, $f_1 = 3$, $f_2 = 1$.

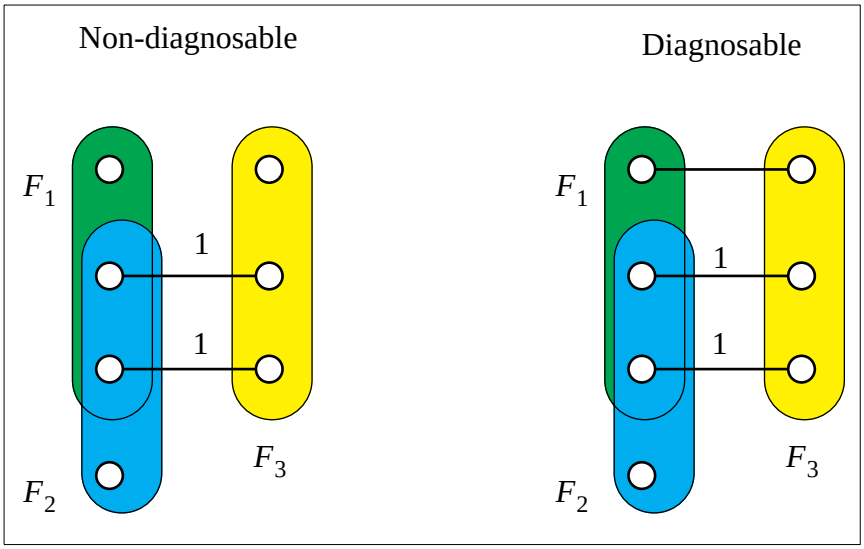


Figure 4.12 Diagnosability conditions of Theorem 4.5 for $n = 7$, $f_1 = 3$, $f_2 = 1$

We consider now a 7-node system with the symmetric comparison model and $f_1 = 3$

(Figure 4.14). This system is not *f*/(*n*–1)-diagnosable for the cover π shown in Figure 4.13, because for the only two edges such that $f(u) \neq f(v)$, we have $f(u) \cup f(v) = \pi$.

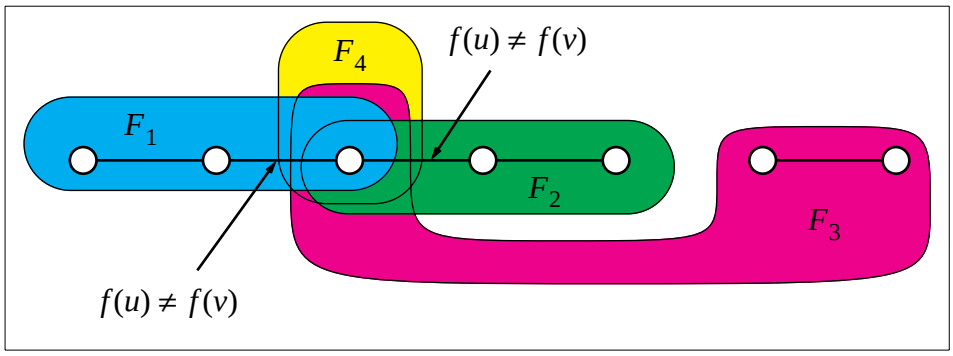


Figure 4.13 Proof of non-diagnosability for the system of Figure 4.14 with $n = 7$, $f_1 = 3$, $f_2 = f_1$

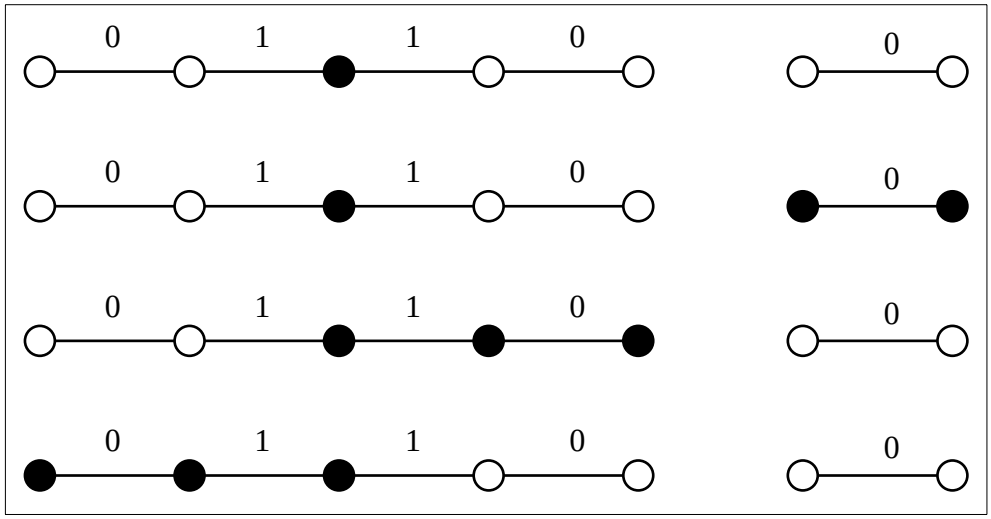


Figure 4.14 Example of non-diagnosable system for $n = 7$, $f_1 = 3$, $f_2 = f_1$

For the BSC model with $f_2 = 2$, it would seem that the system should be *f*/(*n*–1)-diagnosable since we have components with 3 nodes in them. Yet the system is still not *f*/(*n*–1)-diagnosable because the clause $\cup_{i=1}^{p} f(u_i) \neq \pi$ of condition 2 of Theorem 4.5 is not satisfied. This is verified in Figure 4.14 where there is no cartel of three nodes that could resolve the diagnosis.

Figure 4.15 shows a second example of a 7-node system that is not *f*/(*n*–1)-diagnosable for $f_1 = 3$, $f_2 = 3$. Again the only two edges such that $f(u) \neq f(v)$, are also such that $f(u) \cup f(v) = \pi$, which precludes *f*/(*n*–1)-diagnosability by condition 1 of Theorem 4.5. Note that when $f_2 = 2$ the system remains non-diagnosable because none of the candidate fault sets contains a component of more than 2 nodes.

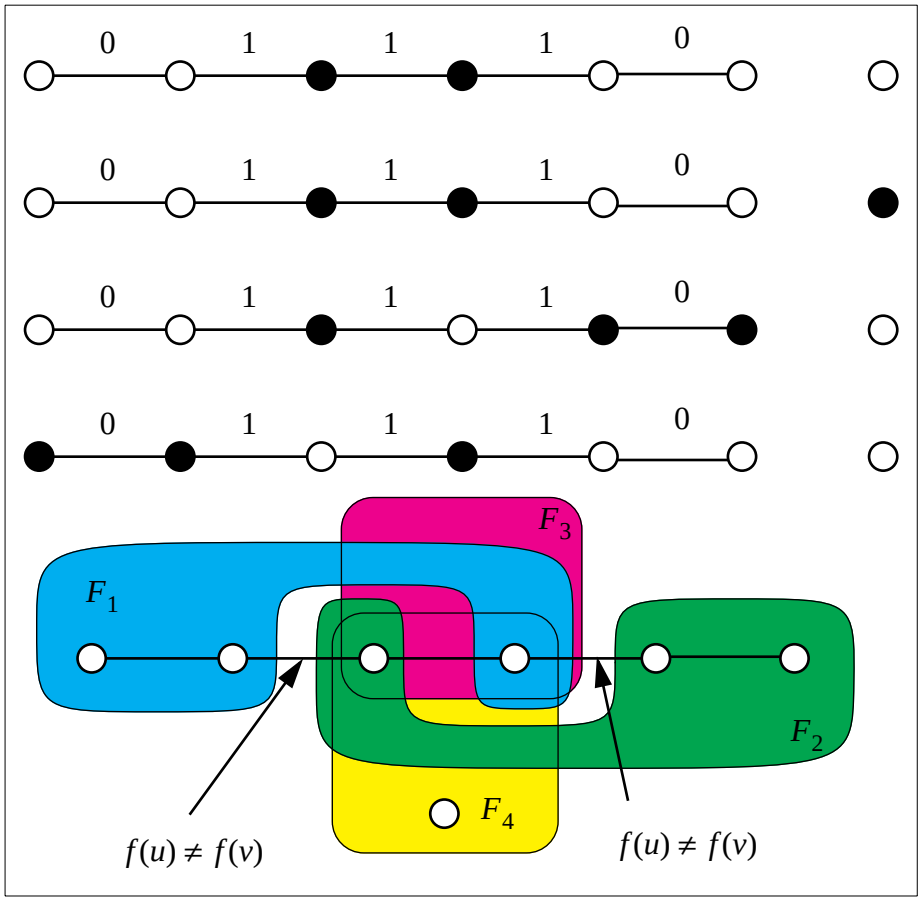


Figure 4.15 Example of non-diagnosable system for $n = 7$, $f_1 = 3$, $f_2 = f_1$

4.5 Structures that are *f*/(*n*–1)-diagnosable under the BSC model

We now investigate which general structures are *f*/(*n*–1)-diagnosable under the BSC model.

Theorem 4.6 A system with $n = f_1 + f_2 + 1$ nodes is *f*/(*n*–1)-diagnosable under the BSC model in a configuration with one isolated node in subgraph G_1 and $n-1$ fully connected nodes in a second disjoint testing subgraph $G_2 = K_{n-1}$.

Proof. By Theorem 4.3 at least $n - f_1 = f_2 + 1$ nodes are fault-free, and any subset of $f_2 + 1$ nodes that produce identical results must be fault-free. There are two possible cases of how these $f_2 + 1$ nodes can be situated in this configuration. Assume first that the (at least) $f_2 + 1$ fault-free nodes are located in G_2. Then in G_2 there is a subset of (at least) $f_2 + 1$ nodes that produce identical results, and these nodes are necessarily fault-free. Assume now that only f_2 of the $f_2 + 1$ fault-free nodes are located in G_2, and the remaining fault-free

node is the isolated node in subgraph G_1. In this case, there can be no subsets in G_2 of more than f_2 nodes that produce identical results. By inference, the isolated node in G_1 is fault-free. □

Theorem 4.7 Under the BSC model, a system S represented by an undirected graph $G = (V, E)$, with $n = f_1 + f_2 + 1$ nodes, where $f_2 < f_1$, is not f/(n–1)-diagnosable in any configuration with two disjoint, fully connected testing subgraphs G_1 and G_2 where each subgraph contains more than one node.

Proof. By Theorem 4.3 at least $n - f_1 = f_2 + 1$ nodes are fault-free, and any subset of $f_2 + 1$ nodes that produce identical results can only be fault-free. Assume that exactly f_1 nodes are faulty, which implies that exactly $f_2 + 1$ nodes are faulty. If we distribute the $f_2 + 1$ fault-free nodes such that one is in G_1 and f_2 are in G_2, then there is no subset in G_1 and/or in G_2 of more than f_2 nodes that produce identical results.

Thus in the worst case there is no way of deciding where some fault-free nodes could be located; there could be several subsets of f_2 nodes yielding comparison tests of all zeros, but only one of which that corresponds to fault-free nodes. Hence the system is not f/(n–1)-diagnosable. □

The cases that correspond to $n = f_1 + f_2 + 1$, $f_2 < f_1$ concern systems in which a majority of the nodes may be faulty; this is not realistic in most practical situations. Therefore we henceforth restrict our discussion to the cases that correspond to the condition $n \geq 2f_1 + 1$ with $f_2 < f_1$. In addition we shall seek to minimize the number of comparisons required for f/(n–1)-diagnosis by considering structures composed of a number of disjoint subgraphs. This leads to the following theorem:

Theorem 4.8 Under the BSC model, a system S represented by an undirected graph $G = (V, E)$, with $n = 2f_1 + 1$ nodes, where $f_2 < f_1$, is f/(n–1)-diagnosable in a configuration with k disjoint, fully connected subgraphs $G_1 = (V_1, E_1)$, …, $G_k = (V_k, E_k)$, each of

$2f_2$ nodes, plus another disjoint, fully connected subgraph $G_{k+1} = (V_{k+1}, E_{k+1})$ of $q = 2b + 1$ nodes, with $q < 2f_2$ and $k = \lfloor f_1 / f_2 \rfloor = (f_1 - b)/f_2$ where $b < f_2$.

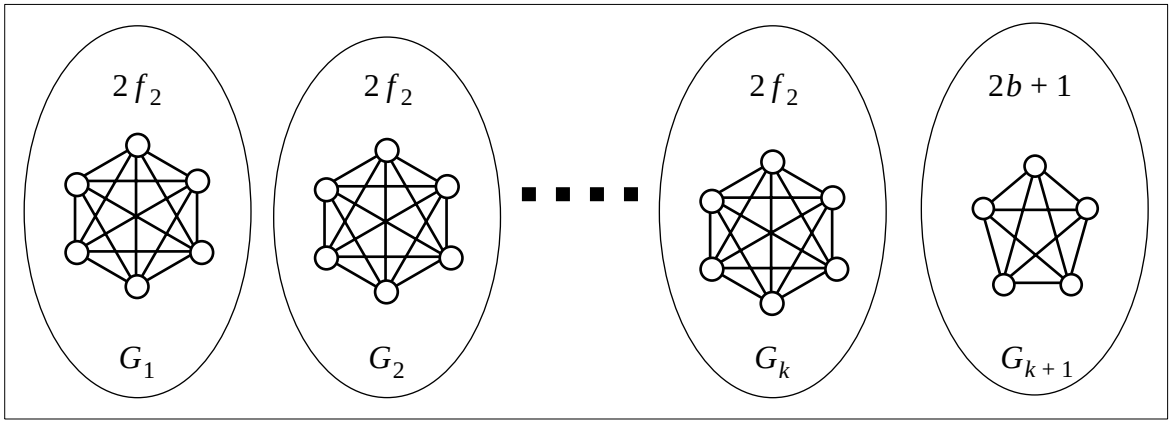


Figure 4.16 System described by Theorem 4.8 where $f_2 = 3$

Proof. The basic idea for the diagnosis is that if there is in any subgraph a cartel *C* of more than $f_2 + 1$ nodes, these nodes are asserted as fault-free and the diagnosis is completed. Since we strive to minimize the number of comparisons by structuring the system in as many disjoint, identical subgraphs as possible, we organize the system into *k* subgraphs G_1, …, G_k of $f_2 + r$ nodes plus one isolated subgraph G_{k+1} that collects the *q* residual nodes, with $q = n - k(f_2 + r) < f_2 + r$. As we have at most f_1 faulty nodes, if we set $kr \geq f_1$, there is always at least one cartel *C* with at least $f_2 + 1$ fully connected, fault-free nodes in one of the subgraphs G_1, …, G_k of $f_2 + r$ nodes. The only exception is when $kr = f_1$, and exactly *r* nodes are faulty in each of the subgraphs G_1, …, G_k of $f_2 + r$ nodes. But since we have *k* such subgraphs and $kr = f_1$, the supply of faulty nodes is exhausted and by inference the residual subgraph G_{k+1} is fault-free.

This last situation is illustrated by the case where there is no cartel of $f_2 + 1$ nodes in any of the subgraphs G_1, …, G_k. Let

$$kr \geq f_1 \tag{4.3}$$

Since the system contains exactly $2f_1 + 1$ nodes we must have

$$n = k(f_2 + r) + q = 2f_1 + 1, \quad 0 \le q < f_2 + r \tag{4.4}$$

Clearly equation (4.3) can be satisfied only if $r \ge 1$. Both equation (4.3) and equation (4.4) yield

$$kf_2 \le f_1 + 1 - q, \quad 0 \le q < f_2 + r \tag{4.5}$$

We now explore the possible values of *k* and *r* for the range of *q*, with $0 \le q < f_2 + r$. It can be seen immediately that for $r < f_2$, equation (4.3) and equation (4.4) have a solution only for the trivial case where $k = 1$. For $r = f_2$, we must have with equation (4.3) $kf_2 \ge f_1$. Hence equation (4.3) and equation (4.4) become

$$kf_2 \ge f_1 \tag{4.6}$$

Since the system contains $2f_1 + 1$ nodes there must be

$$n = 2kf_2 + q = 2f_1 + 1, \quad 0 \le q < 2f_2 \tag{4.7}$$

where equation (4.7) can be satisfied only if $kf_2 = f_1$ and $q = 1$. Hence

$$\text{If } f_2 \mid f_1, \text{ then } k = \frac{f_1}{f_2},\ r = f_2,\ q = 1 \tag{4.8}$$

Otherwise, if f_2 does not divide f_1, then we can attempt to set *k* as

$$k = \left\lfloor \frac{f_1}{f_2} \right\rfloor \tag{4.9}$$

Since $k = \lfloor f_1 / f_2 \rfloor \le f_1 / f_2$, if we set $r = f_2$, then $kr = kf_2 \le f_1$ and equation (4.6) is not satisfied. Yet the system can still be diagnosed for some values of *k* and *q*. In order to establish the corresponding conditions, let

$$f_1 = kf_2 + b, \quad 0 < b < f_2 \tag{4.10}$$

Then equation (4.4) becomes

$$2f_1 - 2b + q = 2f_1 + 1, \quad 0 \le q < 2f_2 \tag{4.11}$$

$$q - 2b = 1, \quad 0 \le q < 2f_2 \tag{4.12}$$

which can be satisfied for any allowable value of *b.*

If no cartel of more than f_2 nodes appears in any of the *k* subgraphs G_1, …, G_k, this means that at least f_2 nodes are faulty in each of these subgraphs. Therefore the number n_f of faulty nodes that are localized within G_1, …, G_k is given by

$$n_f \ge k f_2 = f_1 - b. \tag{4.13}$$

Thus we have at most *b* faulty nodes in G_{k+1}. Recall that

$$|G_{k+1}| = q = 2b + 1. \tag{4.14}$$

Since at most *b* nodes are faulty in G_{k+1}, at least $b + 1$ nodes are fault-free in G_{k+1}. Thus if no cartel of more than f_2 nodes appears in any of the *k* subgraphs G_1, …, G_k, and $q = 2b + 1$, then any component of more than *b* nodes testing to 0 in G_{k+1} is fault-free. With equation (4.12), setting $q = 2b + 1$ satisfies equation (4.4). Hence the system is always *f*/(*n*–1)-diagnosable under the conditions set forth by the theorem. □

Another example of a system that adheres to Theorem 4.8 is shown in Figure 4.17.

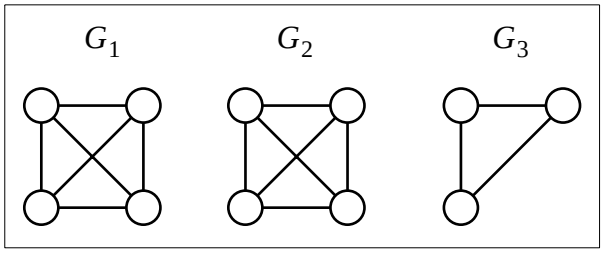


Figure 4.17 Example system for Theorem 4.8 with $f_1 = 5$, $f_2 = 2$, $n = 11$, $k = 2$

For the asymmetric comparison model $f_2 = 1$ and $k = f_1$. Then Theorem 4.8 reduces to the well known result that the system is *f*/(*n*–1)-diagnosable with f_1 pairs of nodes plus one

isolated node. For the symmetric comparison model, $f_2 = f_1$ and $k = 1$. Here Theorem 4.8 states that the system is *f*/(*n*–1)-diagnosable with a subgraph of $2f_1$ nodes plus one isolated node, a result that is a direct consequence of Theorem 3.10 (page 55). When $n > 2f_1 + 1$, we have the following corollary:

Corollary 4.1 Under the BSC model a system with $n \geq 2f_1 + 1$ nodes, where $f_2 < f_1$, is *f*/(*n*–1)-diagnosable in a configuration with k disjoint, fully connected subgraphs $G_1 = (V_1, E_1), \ldots, G_k = (V_k, E_k)$, each of $2f_2$ nodes, plus another disjoint, fully connected subgraph $G_{k+1} = (V_{k+1}, E_{k+1})$ of $q = 2b + 1$ nodes and $n - 2f_1 - 1$ isolated nodes, where $q < 2f_2$, $k = \lfloor f_1 / f_2 \rfloor = (f_1 - b)/f_2$ and $b < f_2$.

Proof. The proof follows trivially from Theorem 4.8: if $n > 2f_1 + 1$, the $2f_1 + 1$ nodes corresponding to $G_1, \ldots, G_{k+1}$ are diagnosed according to Theorem 4.8, which guarantees that at least one fault-free node is discovered. The remaining nodes are not diagnosed, except when f_1 faults have been found in $G_1, \ldots, G_{k+1}$, in which case these nodes are fault-free. □

An example of a system adhering to Corollary 4.1 is shown in Figure 4.18.

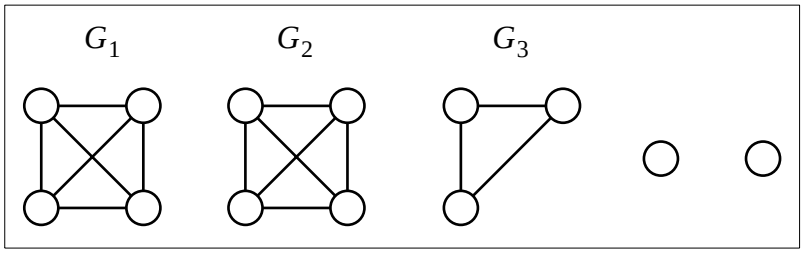


Figure 4.18 Example system for Corollary 4.1 with $f_1 = 5$, $f_2 = 2$, $n = 13$, $k = 2$

Corollary 4.2 Under the BSC model a system with $n \geq 2f_1 + 1$ nodes, where $f_2 < f_1$, is *f*/(*n*–1)-diagnosable according to Corollary 4.1 with no more than $M = (f_1 - b)(2f_2 - 1) + b(2b + 1)$ comparisons, with $f_1 = kf_2 + b$, $b < f_2$.

Proof. The proof follows immediately from Corollary 4.1 by noting that the subgraphs $G_1, \ldots, G_{k+1}$ are complete. □

It should be noted that for values of f_2 that are relatively small compared to f_1, the num-

ber of comparisons is drastically reduced with respect to that required for the symmetric comparison model.

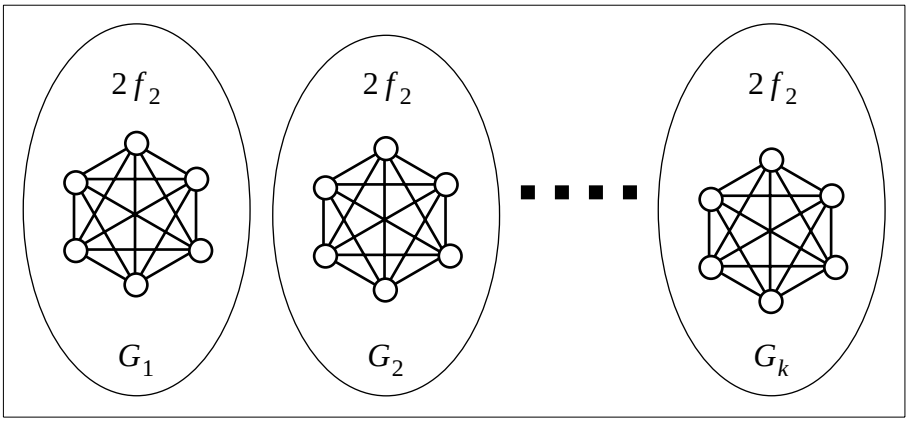


Figure 4.19 Heart of the system described by Corollary 4.1 for $f_2 = 3$

With Corollary 4.1 the heart of the system is a set of *k* subgraphs G_1, …, G_k, each with $2f_2$ fully connected nodes as shown in Figure 4.19. We now seek to relax the requirement for full connectivity in these subgraphs.

The *f*/(*n*–1)-diagnosis proceeds by considering in turn each of these subgraphs G_i. Then there are only two possible outcomes. If there is a cartel of more than f_2 nodes in G_i, then these nodes are necessarily fault-free and the *f*/(*n*–1)-diagnosis is completed. If there is no cartel of more than f_2 nodes in G_i, then the count of the minimum number of faulty nodes identified so far is increased by f_2, and G_i is no longer considered in the rest of the diagnosis.

We can slightly reduce the number of comparisons by using the following theorem originally established in the context of network diagnosis.

Theorem 4.9 If in a complete graph *G* of *m* vertices, no more than $m-2$ edges are removed, then there still exists a Hamiltonian path in G', the subgraph obtained by the removal of edges from *G*.

Proof. The proof to this theorem is in [CN95]. □

In our case, if there are $f_2 + 1$ fault-free nodes in the subgraph G_i, these fault-free nodes are still connected after removing some edges. If G_i is initially fully connected, any component C of $f_2 + 1$ nodes is also fully connected. Thus we can use Theorem 4.9 with

$m = f_2 + 1$. Then if we remove any $m - 2 = f_2 - 1$ comparison links from G_i, we are sure that *C* is still connected even if the removal of edges is done in the most harmful way. Hence we can remove any of up to $f_2 - 1$ comparison links from each fully connected subgraph G_1, …, G_k without jeopardizing the diagnosability. This yields the following corollary:

Corollary 4.3 Under the BSC model, a system with $n \geq 2f_1 + 1$ nodes, where $f_2 < f_1$, is *f*/(*n*–1)-diagnosable according to Corollary 4.1 and Theorem 4.9 with no more than $M = (f_1 - b)(2f_2^2 - 2f_2 + 1)/f_2 + b(2b + 1)$ comparisons, with $f_1 = kf_2 + b$, $b < f_2$.

For the asymmetric comparison model, $f_2 = 1$ and the number of comparisons is the same as with Corollary 4.2. This is not surprising since each subgraph reduces to a simple comparison link in this case. For $f_2 = 2$, we have with Corollary 4.3 $2f_2^2 - 2f_2 + 1 = 5$, as opposed to $2f_2^2 - f_2 = 6$ for Corollary 4.2. Thus the approach proposed with Corollary 4.3 may reduce the total number of comparisons by 10% to 20% with respect to a fully connected structure.

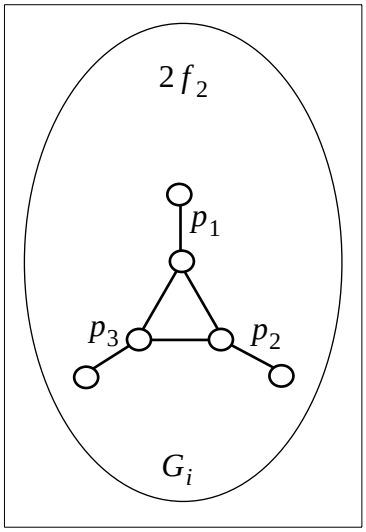


Figure 4.20 Corona subgraph with fully connected pairs for $f_2 = 3$

Another approach to reduce the number of comparisons would be to organize the subgraphs as f_2 pairs of nodes p_1, …, p_{f_2} and where only one node of each pair forms a fully connected sub-subgraph, as illustrated in Figure 4.20. This structure was proposed for the symmetric comparison model in Section 3.3, “Bi-corona systems,” on page 50. In this strategy, the system is structured identically to that of Corollary 4.1, except that the *k* subgraphs are now coronas as

defined below. As such, the system requires even fewer comparisons as shown in the following corollary.

Corollary 4.4 Under the BSC model a system with $n \geq 2f_1 + 1$ nodes, where $f_2 < f_1$, is *f*/(*n*–1)-diagnosable in a configuration with k disjoint subgraphs $G_1 = (V_1, E_1)$, …, $G_k = (V_k, E_k)$, each containing $2f_2$ nodes that are structured as a corona $K_{f_2} \circ K_1$, plus another disjoint, fully connected subgraph $G_{k+1} = (V_{k+1}, E_{k+1})$ of $q = 2b + 1$ nodes and $n - 2f_1 - 1$ isolated nodes, where $q < 2f_2$, $k = \lfloor f_1 / f_2 \rfloor = (f_1 - b)/f_2$ and $b < f_2$. The diagnosis requires no more than $M = (f_1 - b)(f_2 + 1)/2 + b(2b + 1)$ comparisons, with $f_1 = kf_2 + b$, $b < f_2$.

Proof. The proof follows from the number of connections in a corona $K_{f_2} \circ K_1$, which can be expressed as $(f_2(f_2 - 1))/2 + f_2$, or $(f_2(f_2 + 1))/2$. With k such subgraphs, where $k = (f_1 - b)/f_2$, plus the remaining subgraph G_{k+1} with $b(2b + 1)$ comparisons, we obtain the result of $M = (f_1 - b)(f_2 + 1)/2 + b(2b + 1)$ comparisons. □

As before, for the asymmetric comparison model, $f_2 = 1$ and the number of comparisons is the same as with Corollary 4.2. For $f_2 \geq 2$ the number of comparisons is practically halved with Corollary 4.4, with respect to Corollary 4.2 or Corollary 4.3.

Returning to the idea of the maximum number of edges that can be removed from a complete graph with it still being *f*/(*n*–1)-diagnosable, we present a fundamental theorem for *f*/(*n*–1)-diagnosability in the BSC model. This theorem is based upon a theorem introduced by Chutani *et al.* [CN95].

Theorem 4.10 If $f(n, k)$ denotes the maximum number of edges that can be arbitrarily removed from a complete graph $G = (V, E)$ with n nodes such that there remains at least one chain of k nodes in the resulting subgraph, irrespective of the edges removed, then

$$f(n, k) = \frac{n(n-1)}{2} - g(n, k) - 1 \tag{4.15}$$

with

$$g(n,k) = \frac{\left\lfloor \frac{n}{k-1} \right\rfloor (k-1)(k-2)}{2} + \frac{\left(n - \left\lfloor \frac{n}{k-1} \right\rfloor (k-1)\right)\left(n - \left\lfloor \frac{n}{k-1} \right\rfloor (k-1) - 1\right)}{2}. \quad (4.16)$$

Proof. With equation (4.15) and equation (4.16)

$$f(n,k) = \frac{(2n-k+1)\left\lfloor \frac{n}{k-1} \right\rfloor (k-1) - \left(\left\lfloor \frac{n}{k-1} \right\rfloor\right)^2 (k-1)^2}{2} - 1 \quad (4.17)$$

□

We now consider an *n*-node system *S* with comparison testing graph $G = (V, E)$. This system will be *f*/(*n*–1)-diagnosable under the BSC model if we can guarantee that the connectivity is such that there is always a chain spanning any $f_2 + 1$ nodes. Thus we can start with a fully connected graph of *n* nodes and use Theorem 4.10 to determine the maximum number of edges that can be removed from this graph without destroying the *f*/(*n*–1)-diagnosability.

Since at least $n - f_1$ nodes are fault-free, any set of $n - f_1$ nodes must still contain a path of at least $f_2 + 1$ nodes after removal of $f(n,k)$ edges. Thus we replace n by $n - f_1$ and k by $f_2 + 1$ in equation (4.17):

$$f(n,k) = \frac{(2n - 2f_1 - f_2)\left\lfloor \frac{n-f_1}{f_2} \right\rfloor f_2 - \left\lfloor \frac{n-f_1}{f_2} \right\rfloor^2 f_2^2}{2} - 1. \quad (4.18)$$

Hence a fully connected graph of *n* units remains *f*/(*n*–1)-diagnosable after any $f(n,k)$ edges have been removed, with $f(n,k)$ defined by equation (4.18). Assume first that $n = f_1 + f_2 + 1$. Then

$$f(n,k) = \frac{(f_2+2)\left\lfloor \frac{f_2+1}{f_2} \right\rfloor f_2 - \left\lfloor \frac{f_2+1}{f_2} \right\rfloor^2 f_2^2}{2} - 1. \quad (4.19)$$

If $f_2 = 1$, equation (4.19) becomes $f(n,k) = 0$, and the system may no longer be *f*/(*n*–1)-

diagnosable if even a single edge is removed. If $f_2 > 1$ and $n = f_1 + f_2 + 1$, equation (4.18) becomes

$$f(n, k) = f_2 - 1 \tag{4.20}$$

and only $f_2 - 1$ edges can be removed from the complete graph without jeopardizing the diagnosability.

Assume now that $n = 2f_1 + 1$. Then equation (4.18) becomes

$$f(n, k) = \frac{(2f_1 + 2 - f_2)\left\lfloor \frac{f_1 + 1}{f_2} \right\rfloor f_2 - \left\lfloor \frac{f_1 + 1}{f_2} \right\rfloor^2 f_2^2}{2} - 1 \ . \tag{4.21}$$

Assume first that f_2 divides $(f_1 + 1)$. Then equation (4.21) becomes

$$f(n, k) = \frac{(2f_1 + 2 - f_2)(f_1 + 1) - (f_1 + 1)^2}{2} - 1 = \frac{(f_1 + 1 - f_2)(f_1 + 1)}{2} - 1 \tag{4.22}$$

Thus in this case the number of edges that can be removed from a complete graph without jeopardizing the diagnosability becomes significant.

Theorem 4.10 has important implications. First, it can be used in the context of network fault management in order to determine the bound on the number of links that can be allowed to be faulty in a diagnosable system. In this case, one must use bidirectional testing instead of unidirectional testing as discussed by Chutani *et al*. [CN95]. This is not a major issue since most communication links are actually bidirectional. In the context of fault tolerance, Theorem 4.10 can be used to determine the maximum number of comparators that are allowed to be faulty.

4.5.1 Parallel systems executing multiple tasks

The approach defined by Theorem 4.8 is important because it allows the design of parallel systems that process several independent tasks concurrently, one per disjoint subgraph, where all nodes of each subgraph execute their task in a period of time known as a cycle. This parallel execution of independent tasks is possible because no comparisons are performed between disjoint subgraphs. In this manner, the results of different jobs are never used for comparison.

Because they are *f*/(*n*–1)-diagnosable, such systems produce at least one fault-free result. This result is guaranteed to be released after the diagnosis that takes place at the end of each cycle, under the proviso that no more than f_1 nodes are faulty. This approach can be generalized by designing systems that process *m* tasks concurrently, out of which at least *p* are diagnosed as fault-free (i.e., produced by nodes diagnosed as fault-free) at the end of any cycle, provided no more than f_1 nodes are faulty. In the context of *f*/(*n*–1)-diagnosis under the BSC model, we characterize as follows this generalized approach that has improved guarantees of performance over a single-task system.

Definition 4.4 A system is said to be *f/(n–1)-diagnosable for p tasks* under the BSC model if at least *p* nodes pertaining to different task groups of nodes are diagnosed as fault-free.

With Definition 4.4 a system that conforms to Theorem 4.8 is *f*/(*n*–1)-diagnosable for 1 task. Starting from this base, a system can be rendered *f*/(*n*–1)-diagnosable for *p* nodes, each in a different task group by using one of the following approaches.

1. Augment a system that is *f*/(*n*–1)-diagnosable for 1 task where $n = 2f_1 + 1$ by adding $p-1$ components of $2f_2$ nodes without changing f_1. Then $n = 2f_1 + 1 + 2(p-1)f_2 > 2f_1 + 1$.
2. Augment the size of each subgraph and keep $n = 2f_1 + 1$.

The first solution is trivial and will not be discussed further. The last solution is characterized by the following theorem.

Theorem 4.11 Consider a system *S* under the BSC model with $n = 2f_1 + 1$ nodes, where $f_2 < f_1$, If the system *S* is configured with *k* disjoint, fully connected subgraphs $G_1 = (V_1, E_1)$, …, $G_k = (V_k, E_k)$, each of $2f_2 + p - 1$ nodes, plus another disjoint, fully connected subgraph $G_{k+1} = (V_{k+1}, E_{k+1})$ of $q = 2b + 1$ nodes, with $q < 2f_2$ and $k = \lfloor f_1 / f_2 \rfloor = (f_1 - b)/f_2$ where $b < f_2$, then *S* is *f*/(*n*–1)-diagnosable for *p* tasks.

Proof. As in the proof of Theorem 4.8, if there is a cartel *C* of more than f_2 nodes in any subgraph, the corresponding nodes are asserted as fault-free and the diagnosis is completed. Again, we organize the system into *k* subgraphs G_1, …, G_k, each of $f_2 + r$ nodes plus one

isolated subgraph G_{k+1} that collects the q residual nodes, with $q = n - k(f_2 + r) < f_2 + r$. Since we have at most f_1 faulty nodes, if we set $kr \geq f_1 + p - 1$, there are at least p subgraphs in $G_1, \ldots, G_k$ each with a cartel C of at least $f_2 + 1$ fully connected fault-free nodes. The only exception is when $kr = f_1 + p - 1$, and exactly r nodes are faulty in $k - (p-1)$ subgraphs G_i, while exactly $r - 1$ nodes are faulty in the remaining $p - 1$ subgraphs $G_1, \ldots, G_k$ of $f_2 + r$ nodes. In this case only $p - 1$ subgraphs contain a cartel that is identified as fault-free. We note in this case, however, that $G_1, \ldots, G_k$ contain all the possible f_1 faulty nodes, since $(k - (p-1))r + (p-1)(r-1) = kr - (p-1) = f_1$. Hence by inference, the residual subgraph G_{k+1} is fault-free. This last situation is characterized by the fact that there are only $p - 1$ subgraphs pertaining to $G_1, \ldots, G_k$ which contain a cartel of $f_2 + 1$ nodes in any of the subgraphs. Let

$$kr \geq f_1 + p - 1 \tag{4.23}$$

Since the system contains $2f_1 + 1$ nodes we must have

$$n = k(f_2 + r) + q = 2f_1 + 1, \quad 0 \leq q < f_2 + r \tag{4.24}$$

Both equation (4.23) and equation (4.24) yield

$$kf_2 \leq f_1 + 2 - p - q, \quad 0 \leq q < f_2 + r \tag{4.25}$$

We now explore the possible values of k and r for the range of q, with $0 \leq q < f_2 + r$. It can be seen immediately that for $r < f_2 + p - 1$, equation (4.23) and equation (4.24) have a solution only for the trivial case where $k = 1$. For $r = f_2 + p - 1$, there must be with equation (4.23) $kf_2 + k(p-1) \geq f_1 + p - 1$. Hence equation (4.23) becomes:

$$kf_2 \geq f_1 + (p-1)(1-k) \tag{4.26}$$

Since the system contains $2f_1 + 1$ nodes we must have

$$n = 2kf_2 + k(p-1) + q = 2f_1 + 1, \quad 0 \le q < 2f_2 \tag{4.27}$$

If $kf_2 = f_1$ equation (4.27) can be satisfied only if $k(p-1) + q = 1$. Hence

$$\text{If } f_2 | f_1, \text{ then } k = \frac{f_1}{f_2}, \quad r = f_2 + p - 1, \quad q = 1 \tag{4.28}$$

If f_2 does not divide f_1 evenly, then we can attempt to set *k* as

$$k = \left\lfloor \frac{f_1}{f_2} \right\rfloor \tag{4.29}$$

Since $k = \lfloor f_1 / f_2 \rfloor \le f_1 / f_2$, if we set $r = f_2 + p - 1$, then $kr = kf_2 \le f_1$ and equation (4.26) is not satisfied. Yet the system can still be diagnosed for some values of *k* and *q*. In order to establish the corresponding conditions, let

$$f_1 = kf_2 + b, \quad 0 < b < f_2 \tag{4.30}$$

Then equation (4.24) becomes

$$2f_1 - 2b + q = 2f_1 + 1, \quad 0 \le q < 2f_2 \tag{4.31}$$

$$q - 2b = 1, \quad 0 \le q < 2f_2 \tag{4.32}$$

which can be satisfied for any allowable value of *b.*

If no cartel of more than f_2 nodes appears in any of the *k* subgraphs G_1, …, G_k, this means that at least f_2 nodes are faulty in each of these subgraphs. Therefore the number n_f of faulty nodes that are localized within G_1, …, G_k is given by

$$n_f \ge kf_2 = f_1 - b \tag{4.33}$$

Thus we have at most *b* faulty nodes in G_{k+1}. Assume that

$$|G_{k+1}| = q \ge 2b + 1 \tag{4.34}$$

Since at most *b* nodes are faulty in G_{k+1}, at least $b + 1$ nodes are fault-free in G_{k+1}. Thus if no cartel of more than f_2 nodes appears in any of the *k* subgraphs G_1, …, G_k, and $q \geq 2b + 1$, then any cartel of more than *b* nodes in G_{k+1} is fault-free. With equation (4.32), we see immediately that setting $q = 2b + 1$ allows satisfying equation (4.24). Hence *p* fault-free nodes (each in a distinct subgroup) can always be identified under the conditions set forth by the theorem. □

We introduce the following theorem that describes another configuration of a system that is *f*/(*n*–1)-diagnosable under the BSC model with multiple, independent tasks that are distinct.

Theorem 4.12 A system with $n = k(f_2 + r)$ nodes, where $r \geq 1$ and $f_2 < f_1$ is *f*/(*n*–1)-diagnosable in a configuration with *k* disjoint, fully connected subgraphs G_1, …, G_k, each of $f_2 + r$ nodes, with $rk \geq f_1 + 1$. (Figure 4.21)

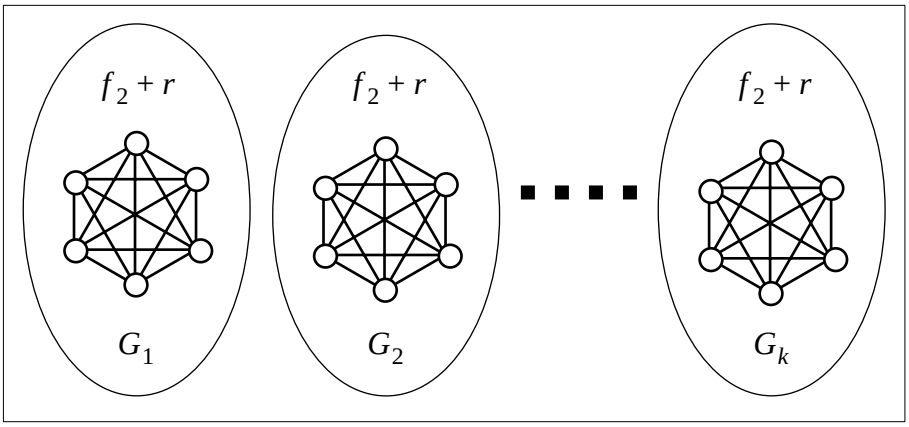


Figure 4.21 System described in Theorem 4.12

Proof. We note first that the necessary condition $n \geq f_1 + f_2 + 1$ of Theorem 4.3 is satisfied, since $n = k(f_2 + r)$, or $n = k(f_2 + r) = kf_2 + rk \geq kf_2 + (f_1 + 1) \geq f_1 + f_2 + 1$. There are up to f_1 faults in any configuration in the *k* disjoint subgraphs. If no more than $r - 1$ nodes are faulty in a single subgraph, then there is in this subgraph a set of at least $f_2 + 1$ fault-free nodes that produce identical results; this syndrome can only be produced by fault-free nodes. Hence the system is *f*/(*n*–1)-diagnosable if we can guarantee that no more than $r - 1$ nodes are faulty in a single subgraph. But since no more than $f_1 = rk - 1$ nodes can be

faulty, in the worst case there are $k-1$ subgraphs with r faults and the remaining subgraph with $r-1$ faults. The system is therefore $f/(n-1)$-diagnosable. □

For instance, with $f_2 = 2$ and $r = 2$, the system is split into disjoint subgraphs of four nodes. If the system is made up of only two subgraphs with $k = 2$, then $f_1 = 3$ and the system is $f/(n-1)$-diagnosable with up to 3 faults. With $k = 3$, the system becomes $f/(n-1)$-diagnosable with up to 5 faults.

The following theorem describes a simple configuration of a system that is $f/(n-1)$-diagnosable under the BSC model with multiple, distinct tasks executed, one per disjoint subgraph.

Theorem 4.13 Consider a system of n nodes of which no more than f_1 are faulty and $n = f_1 + m f_2 + 1$, $m \geq 1$. This system is $f/(n-1)$-diagnosable using the BSC model in a configuration with m fully connected subgraphs V_1, …, V_m, and one isolated node. Let $q = f_2 + \lfloor f_1/m \rfloor$. There are $(n-1-qm)$ subgraphs V_i that are such that $|V_i| = q+1$ and the remaining $m-n+qm+1$ subgraphs are such that $|V_i| = q$. The system is diagnosed with a maximum of $N = ((2n-2-mq-m)q)/2$ comparisons. (Figure 4.22)

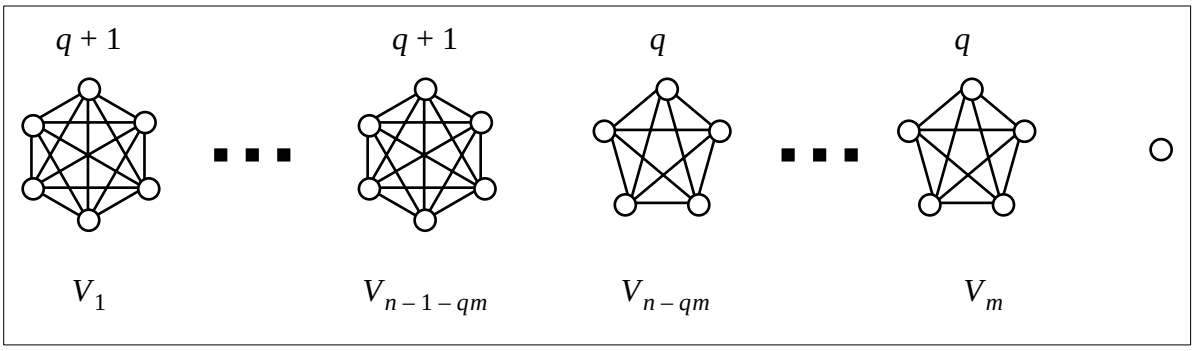


Figure 4.22 System described in Theorem 4.13

Proof. Diagnosability. Assume first that in one of the subgraphs there is a cartel with more than f_2 nodes. Then these nodes are fault-free and the diagnosis terminates early. Suppose now that no cartel with more than f_2 nodes is found in any subgraph. Then at least $q+1-f_2$ faulty nodes are identified within each subgraph of cardinality $q+1$ and at least $q-f_2$ faulty nodes are identified within each subgraph of cardinality q. Hence the number M of identified faults is

$$M = (n-1-qm)(q+1-f_2) + (m-n+qm+1)(q-f_2) \tag{4.35}$$

and with $n = f_1 + mf_2 + 1$

$$M = n-1-mf_2 = f_1 \tag{4.36}$$

Since f_1 faulty nodes are identified in the m subgraphs and no more than f_1 nodes can be faulty, the isolated node v_n is fault-free by inference.

Proof. Complexity. Since all the subgraphs are fully connected, the number N of comparisons is given by

$$N = \frac{(n-1-qm)(q+1)q}{2} + \frac{(m-n+qm+1)(q-1)q}{2} = \frac{(2n-2-mq-m)q}{2} \tag{4.37}$$

□

For the asymmetric comparison model, with $f_2 = 1$, we note that if $n = f_1 + f_2 + 1 = f_1 + 2$, then $m = 1$ and $q = f_1 + f_2 = f_1 + 1$. In this case we have $N = f_1(f_1+1)/2$, a result that we shall later confirm in the proof of Theorem 4.15. For the same model, with $f_2 = 1$, $n = 2f_1 + 1$, we have $m = f_1$, $q = 2$ and $N = f_1$, which is the result reported in [HN84]. Note also that for $mf_2 = f_1$, Theorem 4.13 reduces to Theorem 4.8.

In the general case f_2 does not divide $n-1-f_1$, and one must take $m = \lfloor (n-1-f_1)/f_2 \rfloor$. A development similar to that given in Theorem 4.13 gives the conditions for *f*/(*n*–1)-diagnosability.

4.6 Algorithms for *f*/(*n*–1) diagnosis in the BSC model

In this section we present three algorithms for adaptive *f*/(*n*–1)-diagnosis of systems under the BSC model. In typical one-step diagnosis, a set of comparison tests is chosen and then carried out, after which an algorithm attempts to identify the faulty units assuming that the number of faulty units does not exceed some limit. In adaptive diagnosis, however, the goal the test assignment is not necessarily pre-defined, but can be adapted as the test results are obtained.

We remind the reader that the goal in $f/(n-1)$-diagnosis is simply to find at least one fault-free node in the system.

4.6.1 Algorithm 1: adaptive $f/(n-1)$ diagnosis

The first algorithm is for adaptive $f/(n-1)$-diagnosis under the BSC model, and is patterned after the algorithm of Hakimi *et al.* [HS84] for the PMC model. The following assumptions are made with respect to the algorithm.

Let S be a system with n nodes $v_1, \ldots, v_n \in V$ and $n \geq 2f_1 + 1$. Take any subset U of V with $U \subseteq V$ and $|U| = 2f_1 + 1$. For each node $v \in U$, define $\Delta_0(v)$ as the number of faulty nodes that would be identified as a consequence of v being faulty. Let F be the minimum number of nodes that have been identified as faulty as a result of the preceding tests. Let L_1 and L_2 be two lists where each element is a pair $(v, \Delta_0(v))$ and where the elements are arranged from the top by decreasing weights of $\Delta_0(v)$.

The following algorithm is related to Algorithm 3.1 (page 51) in the sense that it identifies a fault-free node in the system. Algorithm 3.1, however, works only for bi-corona structures under the symmetric comparison model and is not an adaptive algorithm. The following algorithm works for the general case of systems under the BSC model and is adaptive.

Algorithm 4.1 Adaptive $f/(n-1)$-diagnosis algorithm[1] for the BSC model with $n \geq 2f_1 + 1$.

Step 0. Initialization.

$\forall v_i \in U, \Delta_0(v_i) = 1$

$L_1 \leftarrow (v_1, \Delta_0(v_1)), (v_2, \Delta_0(v_2)), \ldots, (v_{2f_1+1}, \Delta_0(v_{2f_1+1}))$; $|L_1| = 2f_1 + 1$

$L_2 \leftarrow \varnothing$

$F = 0$

Step 1. Do

Take the first two members $(u, \Delta_0(u))$ and $(v, \Delta_0(v))$ of L_1 and delete them.

(a) If $w(u, v) = 0$, then

$\Delta_0(v) = 2\Delta_0(v)$,

1. This algorithm was implemented in the C programming language, and the source code can be found in Appendix A.2.

create the pair $(v, \Delta_0(v))$ and place it at the top of L_2.

If $\Delta_0(v) > f_2$, or $\Delta_0(v) > f_1 - F$, then v is fault-free. STOP.

(b)else if $w(u, v) = 1$, then

$F = F + \Delta_0(v)$.

If $L_2 \neq \varnothing$, then

let $(x, \Delta_0(x))$ be the first element of L_2

else

let $(x, \Delta_0(x))$ be the first element of L_1.

If $\Delta_0(x) > f_1 - F$, x is fault-free. STOP.

Repeat until either $|L_1| \leq 1$ or $\Delta_0(u) > \Delta_0(v)$.

Step 2. $L_1 \leftarrow (L_2$ followed by $L_1)$

$L_2 \leftarrow \varnothing$.

GO TO STEP 1.

It should be noted that the algorithm goes through a number of cycles before its termination. Each cycle is defined as being the computations between the beginning of Step 1 and either a STOP or a GO TO STEP 1 instruction. Note that the test $w(u, v)$ is performed at the jth cycle if and only if $\Delta_0(u) = \Delta_0(v) = 2^{j-1}$.

Theorem 4.14 Let S be a system of n nodes of which no more than f_1 are faulty, of which no more than f_2 may produce identical, faulty results. Suppose the tests applied to S satisfy the assumptions of the BSC model for the values of f_1 and f_2. If $n \geq 2f_1 + 1$, then S is $f/(n-1)$-diagnosable with Algorithm 4.1, and at most $2f_1 - f_1 2^{-\lfloor \log_2 f_2 \rfloor}$ comparisons are required.

Proof. First we verify that any time Algorithm 4.1 terminates, it correctly identifies a fault-free node. Let us consider the cases for which the algorithm stops.

Case 1. *The algorithm stops in Step 1.* (a) If $\Delta_0(v) > f_2$, then there are more than f_2 nodes that compare to 0. These nodes are fault-free by definition of the BSC model. If $\Delta_0(v) > f_1 - F$ and v were faulty then we would have more than f_1 faulty nodes, which contradicts the hypothesis

for the model. (b) If $\Delta_0(x) > f_1 - F$ and x were faulty then we would have more than f_1 faulty nodes, which again contradicts the assumptions for the model. Hence the algorithm correctly identifies v in part (a) or x in part (b) as fault-free.

Now we prove that Algorithm 4.1 requires at most $2f_1 - f_1 2^{-\lfloor \log_2 f_2 \rfloor}$ tests. We first note that if the f_1 comparisons in cycle 1 produce the outcome 1, the algorithm terminates at the end of cycle 1. On the other hand, if the comparisons produce only 0s, f_1 tests have been performed during cycle 1 and the weight of the nodes is $\Delta_0 = 2$. Then $\lfloor f_1/2 \rfloor$ tests are performed in cycle 2, and the weight of the nodes is $\Delta_0 = 4$. Thus if all tests give result 0 the weight of the nodes is $\Delta_0 = 2^i$ at any cycle i. Therefore, the algorithm stops at cycle i on the condition $\Delta_0 = 2^i \geq f_2 + 1$. Let $f_2 = 2^a + b$, with $b \leq 2^a - 1$. Then $f_2 + 1 \leq 2^{a+1}$ and the maximum number of cycles N_C when all test results are 0 is given by $N_C = i = \lfloor \log_2 f_2 \rfloor + 1$. Note that if in a given cycle there is a mix of 1 and 0 test results, this can only reduce the number of cycles by depriving the next cycle of pairs on which to execute the testing, with a consequence of an earlier termination. Thus the maximum number of cycles s is given by:

$$s = \lfloor \log_2 f_2 \rfloor + 1 \tag{4.38}$$

Let T_j be the number of tests performed during the jth cycle. Obviously $T_1 = f_1$ and $T_j \leq \lfloor T_{j-1}/2 \rfloor$. Assuming first that $f_1 = 2^r$, the maximum number of tests N_f is given by

$$N_f = \sum_{j=0}^{s} 2^{r-j} = 2^{r+1}(1 - 2^{-s}) \tag{4.39}$$

For the symmetric comparison model with $f_2 = f_1$ this gives $s = r + 1$ and

$$N_f = 2f_1 - 1 \tag{4.40}$$

For the asymmetric comparison model, with $f_2 = 1$, we have $s = 1$ and

$$N_f = f_1 \tag{4.41}$$

For any f_2 we have with equation (4.38)

$$N_f = 2f_1 - 2f_1 2^{-\lfloor \log_2 f_2 \rfloor - 1} = 2f_1 - f_1 2^{-\lfloor \log_2 f_2 \rfloor} \tag{4.42}$$

Let ν be the number of 1s in the binary representation of f_1. Then $f_1 = \sum_{i=1}^{\nu} 2^{a_i}$ and the maximum number of tests N_f is given by

$$N_f = \sum_{i=1}^{\nu} \sum_{j=0}^{\min(s, a_i)} 2^{a_i - j} \tag{4.43}$$

which is difficult to evaluate in closed form. We note, however, that since at each cycle $T_j \leq \lfloor T_{j-1}/2 \rfloor$, the number of comparisons given in equation (4.42) is an upper bound. Hence $N_f \leq 2f_1 - f_1 2^{-\lfloor \log_2 f_2 \rfloor}$. □

4.6.2 Algorithm 2: adaptive *fi*(*n*–1) diagnosis when $f_2 < f_1$

When $f_2 < f_1$ and $n \geq f_1 + f_2 + 1$, the adaptive testing can be performed on just $f_1 + f_2 + 1$ nodes instead of $2f_1 + 1$ nodes as in Algorithm 4.1. We cannot use, however, the same algorithm, because more than half the nodes in the system may be faulty, which no longer respects the conditions of Theorem 4.14. Hence we can no longer discard the pairs that test to 1, else all the pairs could be discarded before a fault-free node is isolated. Intuitively, when $n = f_1 + f_2 + 1$, we need a near complete testing graph[1] to diagnose the system in the worst case, for all nodes may need to be compared in pairs before a cartel of $f_2 + 1$ nodes is isolated. This fits with the adaptive algorithm given by Hakimi *et al.* [HN84] for the symmetric invalidation model.

This worst-case situation is slightly alleviated by performing the testing on a complete subgraph of $n - 1$ nodes. As such, if at least $f_2 + 1$ fault-free nodes are present in the subgraph, a

1. As shown with equation (4.20), one needs a complete testing graph out of which at most f_2–1 edges have been removed arbitrarily.

fault-free node is found. If no fault-free node has been identified in the subgraph, then the isolated node is deemed by inference to be fault-free. This yields a simple algorithm for diagnosis that is given below.

In the following algorithm consider a system with n nodes $v_1, \ldots, v_n \in V$ and $n \geq f_1 + f_2 + 1$. The algorithm starts with any subset $S \subset V$, where $S = V - \{v_{f_1 + f_2 + 1}, \ldots, v_n\}$, $|S| = f_1 + f_2$, and $S = \{v_1, \ldots, v_{f_1 + f_2}\}$.

Algorithm 4.2 Adaptive $f/(n–1)$-diagnosis algorithm for the BSC model with $n \geq f_1 + f_2 + 1$.

Step 0. Initialization.

$S = U$ with $U \leftarrow V - \{v_n\}$, $|U| = f_1 + f_2$

$L \leftarrow \varnothing$; $|L| = 0$; $F \leftarrow \varnothing$; $|F| = 0$; $f = 0$; $A \leftarrow \varnothing$; $t_2 = f_2$

Step 1. Select an arbitrary node $v_i \in U$, place v_i at the top of list L, set $|L| \leftarrow |L| + 1$ and $U \leftarrow U - \{v_i\}$

Step 2. Take any $v_j \in U$, and seek the result of the comparison $\sigma(v_i, v_j)$ of the top member v_i of L with v_j.

(a) If $\sigma(v_i, v_j) = 1$, place v_j on top of list A and set $U \leftarrow U - \{v_j\}$, $f \leftarrow f + 1$.

If $f > f_1 - |F|$, set $F \leftarrow F + L$, $|F| = |F| + |L|$. If $|F| < f_2$, set $t_2 = |F|$.

If $|F| \geq f_1$, STOP; $V - F$ is fault-free, F is faulty.

Else if $|F| < f_1$

If $|U| + |A| \leq 1$, STOP; $v_{f_1 + f_2 + 1}, \ldots, v_n$ are fault-free.

Else $|U| + |A| > 1$, set $f = 0$, $L \leftarrow \varnothing$, $U \leftarrow U + A$, $A \leftarrow \varnothing$. Return to Step 1.

Else if $f \leq f_1 - |F|$,

If $|U| \neq 0$ return to Step 2.

Else if $|U| = 0$,

If $|A| \leq 1$, STOP; $v_{f_1 + f_2 + 1}, \ldots, v_n$ are fault-free.

Else $|A| > 1$, set $f = 0$, $L \leftarrow \varnothing$, $U \leftarrow A$, $A \leftarrow \varnothing$. Return to Step 1.

(b) Else if $\sigma(v_i, v_j) = 0$, set $U \leftarrow U - \{v_j\}$. Place v_j on top of list L.

If $|L| \geq t_2 + 1$, STOP; the members of L are fault-free.

Else $|L| < t_2 + 1$,

If $|U| = 0$,

If $|A| \leq 1$, STOP; $v_{f_1 + f_2 + 1}, \ldots, v_n$ are fault-free.

Else $|A| > 1$, set $f = 0$, $L \leftarrow \varnothing$, $U \leftarrow A$, $A \leftarrow \varnothing$. Return to Step 1.

Else $|U| \neq 0$, return to Step 2.

Informally the algorithm takes a node v_1 and compares it successively to the $n-2$ other nodes in *S*. If all comparisons outcomes are 1, v_1 is discarded and the process continues from v_1 to the remaining $|S| - 3$ other nodes in *S*. The algorithm continues recursively until either all the comparisons in a complete graph of *S* are exhausted or f_1 faulty nodes are diagnosed. Here a node is deemed unambiguously to be faulty when it tests to 1 with $f_1 + 1$ other nodes.

When the tests produce an outcome of 0, the algorithm attempts to build in L a chain of nodes testing mutually to 0. Then the algorithm stops when $|L| = t_2 + 1$, that is, when the nodes in L form a cartel of more than f_2 nodes and are thus diagnosed as fault-free. When the attempt to build such a chain is unsuccessful, the nodes in L are thrown away and the algorithm is continued until the remaining nodes are exhausted. Finally if no fault-free node has been identified in *S*, the isolated nodes $v_{f_1 + f_2 + 1}, \ldots, v_n$ are deemed to be fault-free by inference.

Theorem 4.15 Let *S* be a system of *n* nodes of which no more than f_1 are faulty, of which no more than f_2 may produce identical, faulty results. Suppose the tests applied to *S* satisfy the assumptions of the BSC model for the values of f_1 and f_2. Then Algorithm 4.2 correctly identifies a fault-free node if and only if $n \geq f_1 + f_2 + 1$. When $n = f_1 + f_2 + 1$, the algorithm requires at most $N = (f_1(f_1 + 3))/2$ comparisons.

Proof. Correctness. Here we must prove that the algorithm terminates and that it always identifies correctly a fault-free node if and only if $n \geq f_1 + f_2 + 1$. By Theorem 4.3, the sys-

tem cannot be *f*/(*n*–1)-diagnosable if $n < f_1 + f_2 + 1$. Thus we need only to prove the correctness for $n \geq f_1 + f_2 + 1$.

Note that the algorithm initiates a new cycle each time it goes through step 2. Initially $|U| = |S| = f_1 + f_2$. During each cycle, the algorithm goes recursively through step 3 by performing each time a comparison between the node that is currently at the top of *L* and one of the other surviving nodes in *L*. Thus the number of passes though step 3 is bounded at each cycle by $|U|$, which is implemented in the algorithm by the stopping conditions and the return to step 2 conditions that are based on the predicate $|U| = 0$. Moreover, each time the algorithm goes through step 2, one node is removed from *U* and it can be verified that it is never put back into *U* since *L* is set to $\varnothing$ before any return to step 2. Hence the algorithm executes no more than $f_1 + f_2 - 1$ cycles that are themselves bounded by *U*, which guarantees the termination.

In order to prove that a fault-free node is always correctly identified we consider first the mechanism for identifying a faulty node. Here a node v_i at the top of *L* is deemed to be faulty if it tests to 1 with $f_1 + 1$ other nodes. This is obviously correct since out of these $f_1 + 1$ other nodes at least one must be fault-free.

But this fault-free node should compare to 0 with v_i if it were fault-free. Hence v_i is faulty. Thus the stopping condition on $|F| \geq f_1$ where the nodes in *F* are identified as indicated is correct. Similarly the stopping condition on $|L| \geq t_2 + 1$ with $t_2 = f_2$ for $|F| \geq f_2$ and $t_2 = |F|$ for $|F| < f_2$ is correct since it deems as fault-free a chain of fault-free nodes that test to 0.

The algorithm normally explores a complete graph and depart from this behavior in only two occasions. The first one is when the node v_i currently at the top of *L* is deemed to be faulty. In this case the current cycle is aborted since it would make no sense to continue the testing of other nodes from the faulty v_i. Thus this does not reduce the diagnosis capability of the algorithm. The second occasion where the algorithm departs from exhaustive testing is when v_i test to 0 with another node v_j. Then the current cycle is suspended, v_j is placed at the top of *L* and the testing is resumed, with v_j replacing v_i as tester. Here it would make no sense to test all nodes in *U* since v_j is producing the same results as v_i. Hence limiting the tests by

v_j to only the nodes not already tested by v_i does not reduce the diagnosis capability of the algorithm. Thus the algorithm actually performs an exhaustive search for both faulty and faulty nodes and it fails only when the f_1 faulty nodes are located in *S*. But then the isolated nodes are fault-free by inference.

Now let us consider a proof of the complexity of the algorithm. Assume first that $n = f_1 + f_2 + 1$, with $f_2 = 1$. In this case the BSC model reduces to the asymmetric comparison model. Then if f_1 nodes are faulty all comparisons in *S* give a result 1 if one of the two fault-free nodes is v_n. Since $|S| = f_1 + f_2$, the node v_i at the top of list *L* is compared with no more than f_1 other nodes. Hence the stopping conditions and branching conditional on $f > f_1 - |F|$ are never invoked. Thus the algorithm stops either because a single comparison yields a 0 or because all possible comparisons in *S* have been performed. The last case provides the bound on the number of comparisons, which is here $(f_1(f_1 + 1))/2$.

Consider now the general case with $|S| = f_1 + f_2$. Let n_1 be the number of fault-free nodes in *S* and n_2 be the number of faulty nodes in *S*, with $n_1 + n_2 = |S| = f_1 + f_2$. The algorithm can at most generate a complete graph in a total of $f_1 + f_2 - 1$ cycles, with $((f_1 + f_2)(f_1 + f_2 - 1))/2$ comparisons. We now show, however, that this is not a lower bound on the maximum number of comparisons. Note that there are at least $f_2 + 1$ fault-free nodes in the system. Hence $n_1 \geq f_2$.

Suppose first that $n_1 \geq f_2 + 1$. Then $n_2 \leq f_1 - 1$. Denote by v_i the node that is on top of list *L* at the current cycle. We remark that when the algorithm happens to select its first fault-free node at the beginning of a cycle, the algorithm terminates in the current cycle since it constructs in this cycle a chain of at least $t_2 + 1$ nodes testing to 0. In the worst case this happens immediately after the algorithm has exhausted the supply of n_2 faulty nodes, at cycle $n_2 + 1 \leq f_1$. But if v_i tests to 1 with all faulty and fault-free nodes, then the algorithm performs $f_1 + 1$ comparisons at cycle 1, f_1 comparisons at cycle 2, …, 3 comparisons at cycle each cycle $f_1 - 1$. At the end of cycle $f_1 - 1$, we have $|F| = 1$ and the next cycle requires a single comparison to deem as fault-free the current node v_i. Thus the algorithm terminates

after N comparisons, with

$$N = 1 + \sum_{i=1} f_1^{-1}(f_1 + 2 - i) = f_1\frac{(f_1 + 3)}{2} - 1 \tag{4.44}$$

If v_i tests to 0 with some faulty nodes, then the algorithm is shortened and the number of comparisons is reduced.

Suppose now that $n_1 = f_2$. Then $n_2 = f_1$ and the algorithm requires just f_1 cycles to identify all faulty nodes. Hence the algorithm terminates after N comparisons, with

$$N = \sum_{i=1} f_1(f_1 + 2 - i) = f_1\frac{(f_1 + 3)}{2} - 1 = f_1\frac{(f_1 + 3)}{2} \tag{4.45}$$

□

4.6.3 Algorithm 3: alternate adaptive *fl*(n–1) diagnosis when $f_2 < f_1$

The last algorithm presented in this section is a non-standard, adaptive algorithm. Generally in adaptive diagnosis, the graph is assumed to be fully connected so that potentially any test between two nodes can be performed. In this algorithm, however, this assumption is not made. Instead the goal is to conclude the diagnosis of at least one fault-free node by identifying a cartel of more than f_2 nodes.

Hakimi *et al.* [HN84] showed that in the worst case a fault-free node cannot be found with less than $f_1((f_1 + 1)/2)$ comparisons when $f_1 = 1$ and $n = f_1 + f_2 + 1$. When $n \geq f_1 + f_2 + 1$, however, the bound on the maximum number of comparisons can be reduced by resorting to a divide-and-conquer strategy.

Let $V^- = V - \{v_n\}$. The basic idea is to split V^- into m fully connected subsets whose cardinalities are approximately $f_2 + f_1/m$ as discussed in Theorem 4.13 and illustrated in Figure 4.22 on page 98. Then if the faults are spread evenly among the m subsets, none of them is diagnosable, but the isolated node v_n is deemed by inference to be fault-free. If the faults are not spread evenly among the m subsets, then at least one of them has at least $f_2 + 1$ fault-free nodes that are diagnosed as fault-free since they compare to 0. In order to formalize this idea,

suppose first that

$$n = f_1 + f_2 + 1 + (m-1)f_2 = f_1 + mf_2 + 1, \qquad m \geq 1 \tag{4.46}$$

which defines *m* as

$$m = \frac{n-1-f_1}{f_2} \tag{4.47}$$

Define a partition of V^- into *m* subsets $V_1, \ldots, V_m$ whose cardinalities differ by at most one. More precisely, let

$$q = \left\lfloor \frac{n-1}{m} \right\rfloor = f_2 + \left\lfloor \frac{f_1}{m} \right\rfloor \tag{4.48}$$

$$|V_i| = q + 1 \text{ for } i = 1, 2, \ldots, (n-1-qm) \tag{4.49}$$

$$|V_i| = q \text{ for } i = (n-qm), \ldots, m \tag{4.50}$$

It can be verified easily that $\sum_{i=1}^{m} |V_i| = n-1$. Let $V_i = \{v_{i1}, \ldots, v_{ir}\}$ with $r = q$ or $q+1$ and $i = 1, \ldots, m$.

We now present the algorithm which is similar to one presented by Hakimi *et al.* [HN84] for the asymmetric comparison model. We have adapted it with the above reasoning for the BSC model.

Algorithm 4.3 Adaptive *f*/(*n*–1)-diagnosis algorithm for the BSC model with $n \geq f_1 + f_2 + 1$

Step 0. Initialization.

V_i defined as above. $E \leftarrow \varnothing$.

Step 1. Seek the results $\sigma(v_{ik}, v_{il})$ for $i = 1, \ldots, m$ and $k < l$ with $1 \leq k \leq r$. If any cartel is found with cardinality greater than f_2, STOP. The nodes in this cartel are fault-free. Otherwise continue until all tests in a complete subgraph for each subset are completed

Step 2. STOP. v_n is fault-free.

The behavior of the algorithm is somewhat similar to that of Algorithm 4.2. Informally the algorithm takes a node $v_{1,1}$ of the subset V_1 and compares it successively to the $r-1$ other nodes in V_1. If all comparisons outcomes are 1, $v_{1,1}$ is discarded, and the process continues from $v_{1,1}$ to the remaining $|V_1|-2$ other nodes in V_1. The algorithm continues recursively for all V_i until either a cartel of more than f_2 nodes is found in V_i, or no cartels are identified and the remaining, isolated node V_1 is deemed fault-free by inference that f_1 faults have been located in the *m* subsets V_i.

Theorem 4.16 Let *S* be a system of *n* nodes of which no more than f_1 are faulty, of which no more than f_2 may produce identical, faulty results. Suppose the tests applied to *S* satisfy the assumptions of the BSC model for the values of f_1 and f_2. Then Algorithm 4.3 correctly identifies a fault-free node if and only if $n \geq f_1 + f_2 + 1$, using at most $N = ((2n-2-mq-m)q)/2$ comparisons.

Proof. Correctness. We must prove that the algorithm terminates and that it always identifies correctly a fault-free node if and only if $n \geq f_1 + f_2 + 1$. This part of the proof is virtually identical to that of Theorem 4.15.

Proof. Complexity. This is a direct consequence of the number of comparisons proved in Theorem 4.13.

Algorithm 4.3 has the advantage over the previous adaptive algorithms in that full connectivity of the graph is not required. A fully connected graph implies that a comparison between any two nodes is possible. Therefore, all nodes in the system must execute the same task, because the task result is to be used for the comparison. Because the constraint of full connectivity is not present with Algorithm 4.3, the nodes of each subgraph can execute a distinct task that does not have to be the same for each subgraph, a technique discussed in Section 4.5.1 and also later in Chapter 6. Comparisons between two nodes of different subgraphs can never occur in Algorithm 4.3.

4.7 Fault detectability in the BSC model

In this section we address the subject of fault detectability in the BSC model, which is given in the following definition:

Definition 4.5 A system is fault detectable under the BSC model if a fault can always be detected, possibly without being localized, provided no more than f_1 nodes are faulty.

Definition 4.6 We define an *n*-node *star* as a tree whose one vertex, called the *center*, has degree $n-1$. Each of the other $n-1$ vertices is called a *leaf* and has degree 1. Similarly, a star is defined as a graph $K_{1,\,n-1}$. Figure 4.23 illustrates an 8-node star, $K_{1,\,7}$.

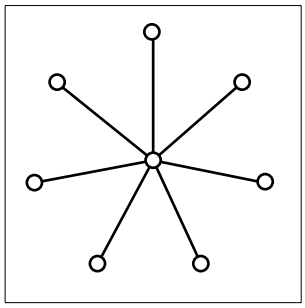


Figure 4.23 A star with 8 nodes

We are interested in determining a lower bound on the number of comparison tests needed to detect faulty nodes. Therefore, the following theorem is proposed:

Theorem 4.17 The minimum number of tests sufficient to detect faulty nodes in the BSC model with at most f_1 faulty nodes among *n*, $f_1 \le n-1$ for $f_2 > 1$ and $f_1 \le n$ for $f_2 = 1$ is $n - \lfloor n/(f_2+1) \rfloor$, both for the adaptive and the non-adaptive diagnosis.

Proof. We show a non-adaptive fault-detection algorithm using $n - \lfloor n/(f_2+1) \rfloor$ tests and prove that fewer tests are not sufficient even in an adaptive diagnosis. Let $k = \lfloor n/(f_2+1) \rfloor$. By assumption $k \ge 1$. Partition the set V of all nodes into subsets V_1, …, V_k such that each V_i, $i < k$ has exactly f_2+1 nodes and V_k contains the rest of them. Thus every subset has at least f_2+1 nodes. This implies that $n = k(f_2+1) + a$, $0 \le a < f_2$.

Perform the comparisons in such a way that the induced subgraph on nodes from each V_i is

a star that is not connected to the stars corresponding to the other V_j, as shown in Figure 4.24. Clearly this requires $n - \lfloor n/(f_2 + 1) \rfloor$ tests. If at least one of the results is 1, a faulty node is detected. If all results are 0, then all nodes are fault-free since no more than f_2 faulty nodes are allowed to test to 0 in a connected graph.

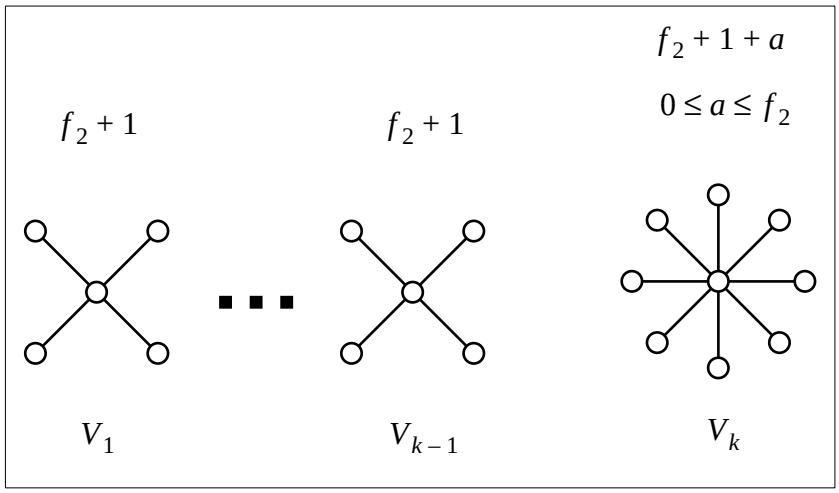


Figure 4.24 System described in proof of Theorem 4.17, $f_2 = 4$, $a = 4$

Suppose that less than $n - \lfloor n/(f_2 + 1) \rfloor$ tests are performed in an adaptive way, giving all results 0. Then one of the cartels *C* of the resulting graph may have less than $f_2 + 1$ elements. It is thus impossible to detect faulty nodes since the nodes in *C* could be either all faulty or all fault-free. This shows that $n - \lfloor n/(f_2 + 1) \rfloor$ is the minimum number of tests for both adaptive and non-adaptive testing. □

Note that Theorem 4.17 reduces to the corresponding theorem reported by Pelc [Pel92] for the symmetric comparison model by setting $f_2 = f_1 = f$. Similarly, Theorem 4.17 reduces to the corresponding theorem reported by Pelc [Pel92] for the asymmetric comparison model by setting $f_2 = 1$. In that case, the minimum number of tests becomes $n - \lfloor n/2 \rfloor = \lceil n/2 \rceil$.

4.8 Fault locatability in the BSC model

In this section we consider the notion of fault locatability in the BSC model, which is defined as follows:

Definition 4.7 A system is fault locatable under the BSC model if at least one faulty node can always be localized, provided no more than f_1 nodes are faulty.

From this definition it can be seen that fault locatability is the first step in sequential fault diagnosis, where the repair of the faulty node is not considered.

Lemma 4.2 If an *n*-node system is fault-locatable, then $n \geq f_1 + f_2 + 1$.

Proof. Assume $n < f_1 + f_2 + 1$. Then there could be only f_2 fault-free nodes and possibly several induced subgraphs with cardinality less than $f_2 + 1$ and testing to 0 making it impossible to locate even a single node. □

Theorem 4.18 The minimum number of comparisons performed in an non-adaptive way, sufficient to locate a faulty node in the BSC model with at most f_1 faulty nodes among *n*, where $n \geq f_1 + f_2 + 1$ is $n - \lfloor n/(f_1 + f_2 + 1) \rfloor$.

Proof. First we consider the sufficiency of the condition of Theorem 4.18. Let $k = \lfloor n/(f_1 + f_2 + 1) \rfloor$. By assumption $k \geq 1$. Partition the set V of all nodes into subsets $V_1, \ldots, V_k$ such that each V_i, $i < k$ has exactly $f_1 + f_2 + 1$ elements and V_k contains the rest of them. Thus every subset has at least $f_1 + f_2 + 1$ elements. This implies that $n = k(f_1 + f_2 + 1) + a$, $0 \leq a < (f_1 + f_2)$.

Perform comparisons in such a way that the induced subgraph on elements from each V_i is a star that is not connected to the stars corresponding to the other V_j. Clearly this requires $n - \lfloor n/(f_1 + f_2 + 1) \rfloor$ tests. If, in at least one component, more than f_1 comparisons yield 1, then the center node is faulty. If in each component at least f_1 comparisons yield 0, then the center must be fault-free in each component. If at least one comparison yields 1, the leaf involved in this test is faulty. If all comparisons yield 0, then all nodes are fault-free.

Now we consider the necessity of the condition in Theorem 4.18. Suppose than less than $n - \lfloor n/(f_1 + f_2 + 1) \rfloor$ tests are performed in an non-adaptive way. Then at least one of the components *C* of the resulting graph has less than $f_1 + f_2 + 1$ elements. Assume that all tests

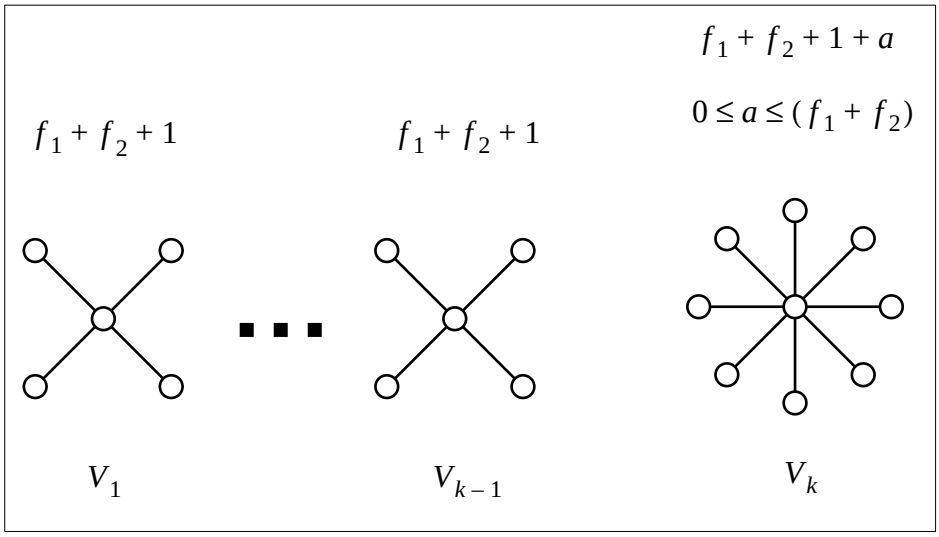


Figure 4.25 System described in proof of Theorem 4.18, $f_1 = 2$, $f_2 = 2$, $a = 4$

in all components other than *C* give result 0, indicating that all corresponding elements are fault-free. Because $|C| \le f_1 + f_2$, there is an isolated subgraph *C* with less than $f_1 + f_2 + 1$ nodes out of which up to f_1 are faulty, that is, not fault locatable by Lemma 4.2. This shows that $n - \lfloor n/(f_1 + f_2 + 1) \rfloor$ is the minimum number of tests for non-adaptive one-fault localization. □

Note that Theorem 4.18 reduces to the corresponding theorem reported in [Pel92] for the symmetric comparison model by setting $f_2 = f_1 = f$. Similarly, Theorem 4.18 reduces to the corresponding theorem reported by Pelc [Pel92] for the asymmetric comparison model by setting $f_2 = 1$.

4.9 Conclusion and contributions

This chapter has presented a new model for comparison-based, system-level diagnosis, which we call the bounded symmetric comparison (BSC) model. This new model encompasses the two of the existing models for comparison-based systems. But the true advantage of the BSC model over the symmetric and asymmetric comparison models is that it applies to systems with a large number of processors where neither of these limited models is very appropriate.

We contribute some necessary and some sufficient conditions of one-step diagnosable systems under the BSC model. More importantly, we characterize *f*/(*n*–1)-diagnosable systems (Chapter 3) under our new model, which is useful for the application of this model to fault-tolerant systems with a large number of processors. Perhaps the most important aspect of our

model is that it permits the diagnosis of subdivided systems where multiple tasks are concurrently executed. This is more useful in large systems where it is impractical to expect that all processors will execute the same task whose result is used for comparison diagnosis. Furthermore, these structures have a secondary advantage that the number of comparisons necessary for diagnosis is lower.

We have proposed three adaptive algorithms for $f/(n-1)$-diagnosis under the BSC model. The last of these algorithms is applicable to systems where the testing assignment is such that there are several distinct, disjoint subgraphs, on which distinct tasks can be executed. That is, it can be used on the parallel, multiple-task systems for which the BSC model is ideal.

Finally, we have addressed the issues of fault detectability and locatability for BSC model. Namely we give theorems for the minimum number of comparisons necessary to detect and localize faults under the assumptions of the BSC model.

4.10 Related publications

[FN96a] C.P. Fuhrman and H.J. Nussbaumer, “A New Comparison Model in System-Level Diagnosis,” to appear in *Proceedings of the International Conference on Parallel and Distributed Processing Techniques and Applications (PDPTA'96)*, Sunnyvale, California, USA, 1996.

[FN96b] C.P. Fuhrman and H.J. Nussbaumer, “Comparison Diagnosis in Large Multiprocessor Systems,” to appear in *Proceedings of the Fifth Asian Test Symposium (ATS'96)*, Hsinchu, Taiwan, 1996.

Chapter 5 *Comparison-Based, Modular Redundancy*

In this chapter we study the problem of modular redundancy where comparisons are used to mask, detect and locate faults. In Section 5.1 we describe the strategy of masking faults in terms of a diagnosis—fault masking attempts to diagnose the correct results of those produced by multiple processing elements that have executed the same task. We give this perspective on fault masking the name of *task diagnosis*. We compare and contrast task diagnosis with the more classical concept of *processor diagnosis*, and we show how the two are interrelated. Processor diagnosis, which we show to be a derivative of task diagnosis, is important for long-term toleration of faults, since faulty elements must be identified and removed or repaired.

In section Section 5.2 we characterize systems that are task-diagnosable under the symmetric comparison model. Section 5.2 establishes the minimum number of comparisons required for NMR systems with an odd number n of processing elements and $n \leq 11$. In section 5.3, we consider comparison-based NMR with adaptive diagnosis. In section 5.4, we discuss the conditions that must be met for fault detection, an important practical topic not discussed by Xu [Xu91]. We consider in section 5.5 the case where comparators can be faulty.

In Section 5.2 we characterize task diagnosis with comparisons for the fault model known as the symmetric comparison model of system-level diagnosis. This model is analogous to the fault model of modular redundancy systems where majority voting or similar fault-masking techniques are used. The characterization presented in Section 5.2 are extensions of the work done by Xu [Xu91].

In Section 5.3 we consider the notion of adaptive diagnosis in comparison-based, modular redundant systems. In Section 5.4 the notion of fault detection is addressed. Section 5.5 considers how to tolerate faulty comparators in a comparison-based NMR system. Finally, Section 5.6 concludes this chapter and summarizes the contributions.

5.1 Introduction

Consider an NMR system with 5 processing elements (PEs) as illustrated in Figure 5.1. Each PE v_i executes the same task *t*, and produces a result r_i. A voting unit collects the five results, and in turn produces the result that has been produced by the majority of the PEs, provided the fault tolerance of the system has not been exceeded. This 5-PE system can tolerate up to two faulty PEs.

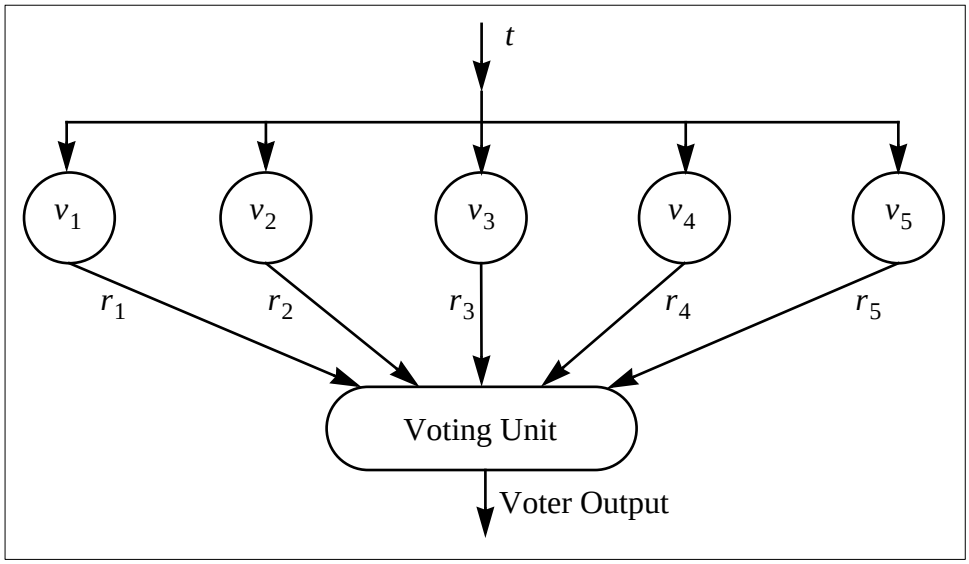


Figure 5.1 Majority voting in an NMR system with *n* = 5

NMR can be applied at the circuit level, or on larger PEs such as computers with their associated local memory. In the rest of this chapter the computation performed by a PE between two cycles of voting will be called a *task*, even though a task in the usual sense has no meaning at the circuit level. With this notation, each PE v_i processes in the time interval between two cycles of voting the replica t_i of task *t* and produces a result r_i that is sent to the voting unit.

The first issue that needs to be addressed when discussing NMR systems concerns the fault model (see Section 2.2.3). The voter in a conventional NMR system deems a result to be correct if it has been produced by the majority of the PEs. This policy is predicated upon the assumption that the probability of a majority of the PEs being faulty and producing the same incorrect result is very small. The occurrence of common-mode faults such as a strong electromagnetic interference that affects all PEs may defeat this policy. But experience has shown that with proper care, NMR is in practice a very effective means of masking most faults whether they be permanent, transient or intermittent.

First let us consider the inferences that can be drawn from the outcome w_{ij} from the com-

parison between two results r_i and r_j as illustrated in Figure 5.2. We adopt the notation used in system-level diagnosis [PMC67], with $w_{i,j} = 0$ if the two results are identical, and $w_{i,j} = 1$ if the two results are different. We assume that the comparison is performed by a unit that is distinct from the two PEs that produce the results r_i and r_j. In a conventional NMR system there must be a total of $n = 2f + 1$ PEs, if up to *f* faulty PEs are to be tolerated. Therefore, the majority of the PEs are supposed to be fault-free, and these $f + 1$ PEs are assumed to produce identical results. Given these assumptions, the comparison outcomes can be interpreted as follows. A comparison outcome $w_{i,j} = 0$ can mean that r_i and r_j are both correct. On the other hand, the conventional NMR system tolerates up to *f* incorrect results, and these may have an identical, incorrect value. Thus the comparison outcome $w_{i,j} = 0$ can also mean that r_i and r_j are both incorrect. A comparison outcome $w_{i,j} = 1$ means that at least one of

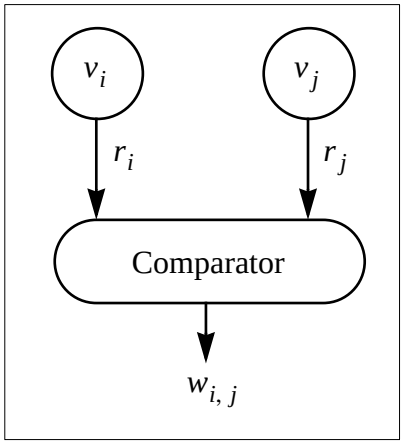


Figure 5.2 Comparison outcome between two results r_i and r_j

the results (possibly both) is (are) incorrect.

As summarized in Chapter 2, the above inferences drawn from comparison results in the NMR fault model are identical to those of the model introduced by Chwa *et al*. [CH81b] for comparison-based, processor diagnosis in system-level diagnosis. System-level diagnosis was originally conceived as a tool for fault tolerance by diagnosing the faulty processors. The comparison model described above is also called the *symmetric comparison model* [Pel91] in system-level diagnosis and is discussed in detail in Section 2.3.3.

In the rest of this chapter and unless otherwise mentioned, the work is assumed to apply to the fault model of conventional NMR. That is, a task result will be dubbed as *correct* if it has been produced identically by a majority of the PEs. This does not mean that this result is neces-

sarily error-free, because the majority of the PEs may not be fault-free if the fault tolerance of the system has been exceeded. Nevertheless the result produced by the majority of the PEs is considered correct within the limitations of the NMR fault model.

5.1.1 Task diagnosis

In the initial scheme for NMR with voting, the voter was implemented as a majority circuit. Since then many other approaches have been proposed for selecting a result that corresponds to the majority. These techniques include sifting and sorting, and are summarized in Section 2.2 of this dissertation. Be that as it may, in all of these approaches, in one way or another, either explicitly or implicitly, the results r_i are always *compared*, and an adjudicator *selects* the correct result as a function of the outcomes of the comparisons. In the conventional approach with the majority circuit the comparison and selection are done simultaneously. In this chapter, however, we consider the different approaches where the comparison and selection processes are separate, as proposed by others [CC90, Xu91, Che92].

In such techniques where comparing tasks and selecting the correct task are separate operations, fault masking can be seen as a form of diagnosis: the outcome of the comparisons can be considered as a syndrome that is then used to diagnose the correctness of the task results. In this dissertation, we shall refer to this concept as *task diagnosis*. Task diagnosis rivals the more conventional concept of *processor diagnosis* (which is the basis of comparison-based, system-level diagnosis), in which the comparison syndrome is used to diagnose the PEs that perform the work of the system. Figure 5.3 illustrates the two concepts of processor and task diagnosis.

As the figure indicates, the two types of diagnosis are interrelated. First, task diagnosis can be seen as derivative of processor diagnosis, since the correct task results can be obtained from those PEs which are deemed fault-free. Conversely processor diagnosis can be obtained from task diagnosis. It is important to note, however, that in an approach based on task diagnosis, if a PE has produced a correct result, this PE is not necessarily fault-free. A particular task's execution might not always expose the fault(s) in a PE if they exist. This concept can be thought of as imperfect fault coverage.

As we have discussed in Chapter 2, the primary goal of an NMR system is to find one correct task result by masking any incorrect results. A secondary (and desirable) goal is to detect and localize the faulty PEs. Thus the NMR operation can be cast into a problem of task diagnosis, as we have defined it above. As in some implementations of NMR, processor diagnosis for detection and localization of faulty PEs may or may not be performed. Therefore the success of

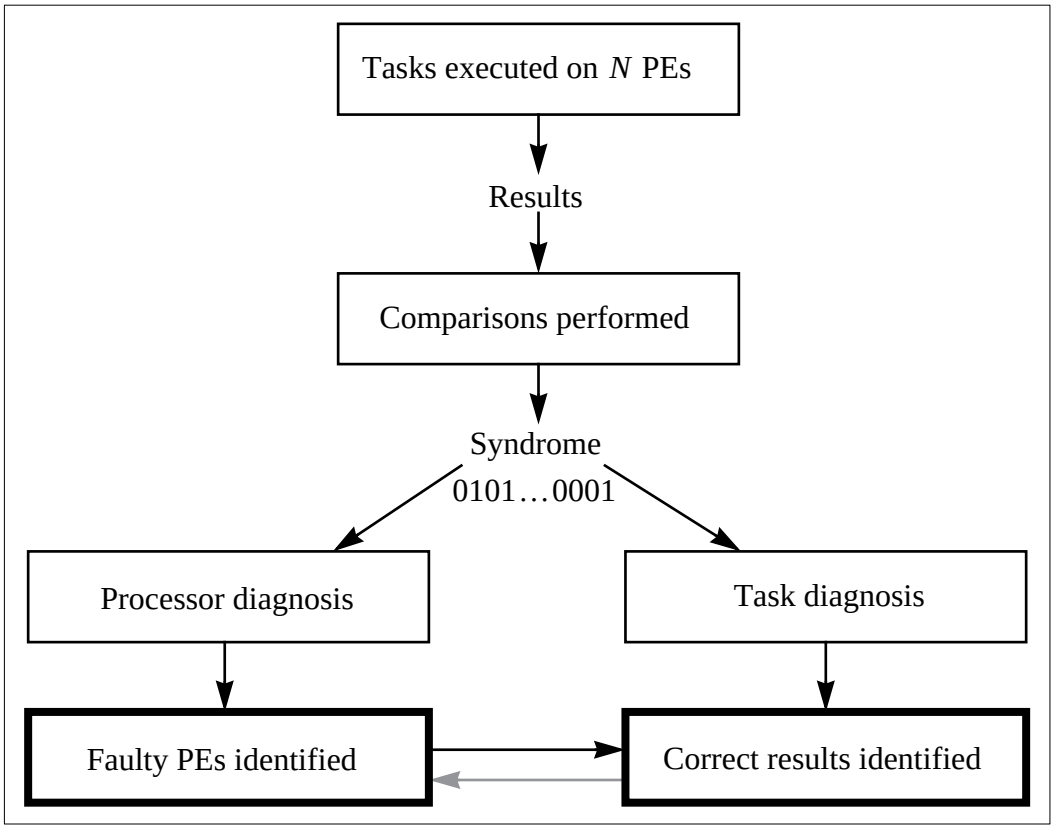


Figure 5.3 Processor diagnosis versus task diagnosis

the basic operation of an NMR system is not hindered by the imperfect fault coverage that is inherent in a given task's execution.

5.1.2 Processor diagnosis in comparison-based, system-level diagnosis

As described above, comparison-based, system-level diagnosis can be used to achieve fault tolerance by processor diagnosis alone. But it has also been applied to modular redundant systems [CH81b, Lom84, Lom85]. In either case, a task is executed on the *N* processing elements in the system, and the task results are compared in order to produce the syndrome, as shown in Figure 5.3. The tasks in question may be tasks that comprise actual work done by the system and whose results may ultimately be used by a practical application. The tasks may also be specialized, diagnostic tasks that have been designed to maximize the chance of detecting a fault in a PE (thus improving the fault coverage) and whose results are solely used for the diagnosis.

In processor diagnosis via system-level diagnosis, at the point after which the comparisons have been performed and the syndrome is available, no attempt is made to release a correct task result. Instead, it is only after the phase of processor diagnosis, when the fault-free PEs have presumably been identified, that the correctness of task results produced by these PEs is

asserted[1]. Therefore, in the approach where system-level diagnosis is used for processor diagnosis, the fault tolerance—or the task diagnosis—is a by-product of the processor diagnosis. This is in contrast with the NMR approach, where processor diagnosis can be thought of as a by-product of task diagnosis.

5.1.3 Combining task and processor diagnosis

NMR techniques based purely on task diagnosis can be combined with system-level diagnosis (processor diagnosis). This was, after all, the original idea that motivated the theory of system-level diagnosis. Consequently the system can be improved if the phase in which real-work tasks are executed is followed by a phase of *partial* task diagnosis, where task results that can be asserted as unambiguously fault-free are immediately released. If all task results cannot be released at this stage, the phase of partial diagnosis is followed by a phase of processor diagnosis, and a final phase of task diagnosis. This technique is the same method proposed for many systems [CH81b, Lom84, Lom85, FCN95d, FCN94] and is illustrated in Figure 5.4.

Let us assume that the syndrome (Figure 5.3) produced by the comparisons is sufficient to assert unambiguously that one of the task results is correct, with diagnosability conditions that will be established in Section 5.2. Such an NMR system can be organized for $n = 5$ as shown in Figure 5.5, where the dotted lines indicate the comparisons between task results. The voter is replaced by a set of comparators that produce the set of comparison test results w_{ij} constituting the syndrome. A task diagnosis is then performed on the syndrome, and if the fault tolerance of the system has not been exceeded, this diagnosis indicates a correct task result, such as r_2, which is then obtained by a selector.

It may seem at first glance that such an approach is much more complicated than NMR with majority voting. In fact, this approach is not as complicated as it may seem for several reasons. Firstly, the diagnosis is performed on the test outcomes, w_{ij}, which are boolean values, and the diagnosis algorithm is simple for practical systems where n is small. Secondly, the costliest part of the system in Figure 5.5 lies with the comparators when the task results are large set of data. Moreover we will show in Section 5.2 that surprisingly few comparisons are required for task diagnosis, and that this leads to a better solution, for there is a significant reduction of communication and processing costs compared with the conventional scheme of majority vot-

1. Note that once a PE has been deemed to be fault-free, it is trivial to deem that the execution of a task on this PE was correct, provided the system fault status has not changed during the whole process.

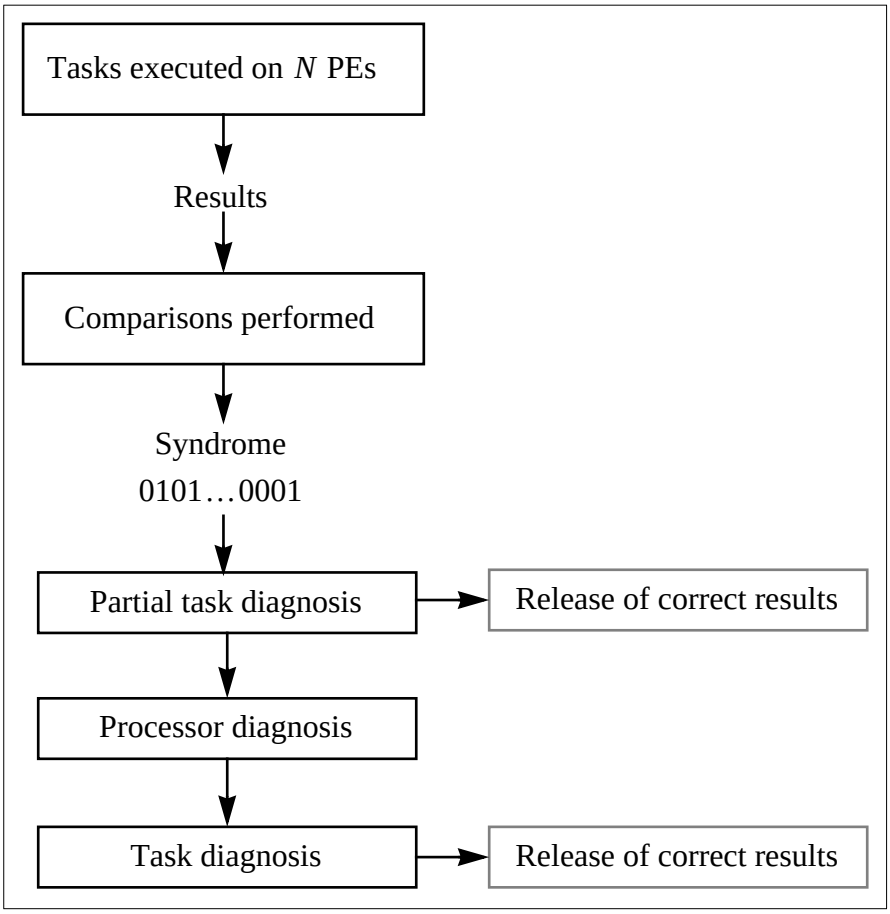


Figure 5.4 Combining task and processor diagnosis yields early release of correct task results

ing.

To summarize, by combining the ideas of task diagnosis and processor diagnosis in comparison-based, modular redundant systems, we hope to obtain a strategy in which task results can be released earlier, but without over complicating the resulting system or jeopardizing the fault tolerance. In order to prove that this is possible, we consider first the characterization of comparison-based, NMR systems from the perspective of the goal of task diagnosis. Xu [Xu91] succeeded in characterizing comparison-based, NMR systems in the case of regular structures such as chains or loops. This important work applies to the NMR analog of the symmetric comparison model. In Chapter 3 of this thesis, we made several contributions to comparison-based, system-level diagnosis. In this chapter, we apply those contributions to the concept of comparison-based, modular redundancy. In the following section, we draw upon Xu's work [Xu91] to explore several important issues related to the implementation of NMR systems in the case of the conventional NMR fault model.

5.2 Modular redundancy with the symmetric comparison model

In this section we characterize comparison-based, modular redundancy in terms of the fault

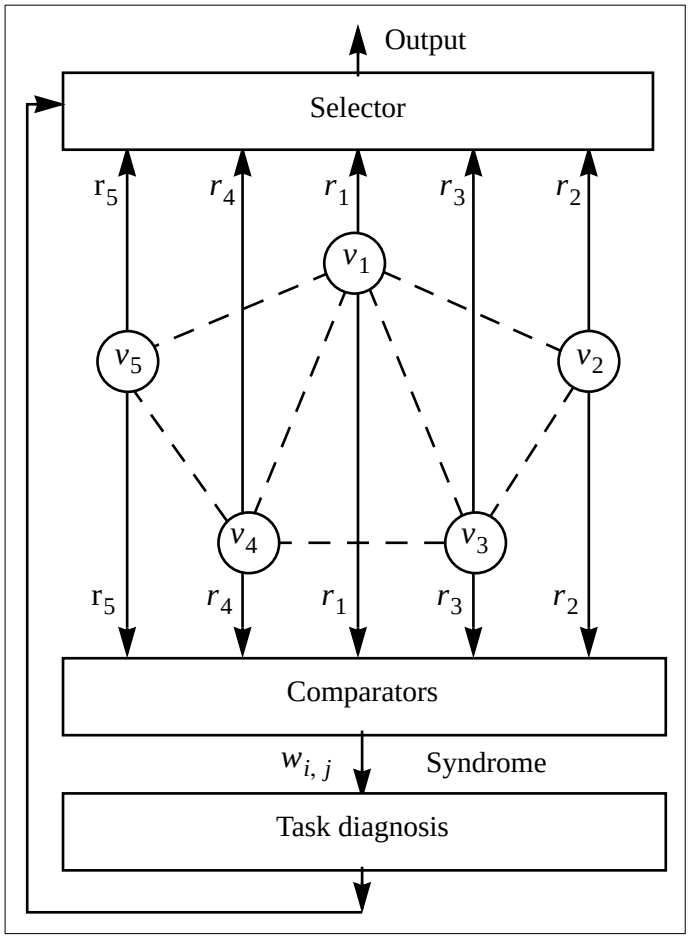


Figure 5.5 N-Modular Redundancy with comparison and task diagnosis by a central unit

model that is known as the NMR fault model, or similarly, the symmetric comparison model of system-level diagnosis.

5.2.1 Useful definitions

This section defines concepts used throughout this section. Much of the work in this section concerns particular structures of systems, which are defined here for clarity.

Definition 5.1 A comparator c compares the task results r_i and r_j produced by PEs v_i and v_j respectively, and yields a value $w_{i,j} = 0$ if the two results are identical, and $w_{i,j} = 1$ if the two results are different. For simplicity, a comparator is represented by an edge (dashed line) in the figures.

Definition 5.2 [Xu91] A *chain* in a comparison-based NMR system of PEs is an alternating sequence of distinct PEs and comparators $v_0, c_1, v_1, \ldots, v_{k-1}, c_k, v_k$, beginning and ending

with PEs, in which each comparator is connected with the two PEs immediately preceding and following it.

Xu and Huang [XH90] used the notion of $H_{2r,n}$ structures, where $r > f/2$ and $f > 1$.

Definition 5.3 [Xu91] A structure in a comparison-based NMR system of PEs is said to be an $H_{2r,n}$ structure where there exists a comparator between each pair of PEs v_i, v_j, where $i = 0, \ldots, n-1$ and where if and only if $i - r < j < (i + r) \bmod n$.

An example of an $H_{2r,n}$ structure is illustrated in for $n = 7$, $f = 3$ and $r = 2$.

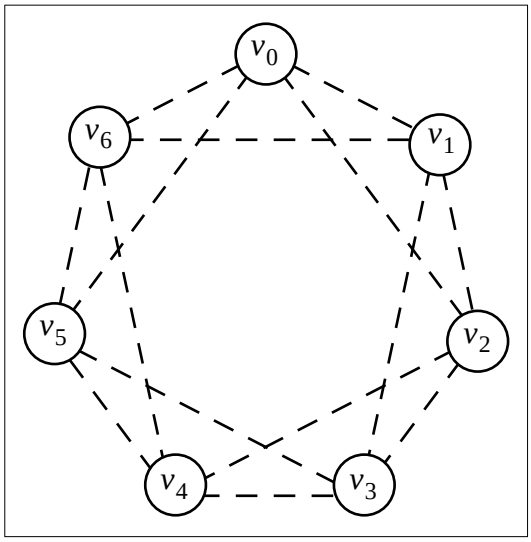


Figure 5.6 Example of an $H_{2r,n}$ system for $r = 2$, $n = 7$

Definition 5.4 [Xu91] A *loop* in a comparison-based NMR system of PEs is an $H_{2r,n}$ structure where $r = 1$.

We define formally the concept of task diagnosability, which is equivalent to the concept of *f*/(*n*–1)-diagnosability from Chapter 3.

Definition 5.5 Consider a system composed of *n* PEs where each PE executes a copy of the same task *t*. The system is said to be *task-diagnosable* if, given a syndrome of comparison outcomes for the *n* task results for *t*, at least one fault-free task result can be identified, provided that the number of faulty PEs does not exceed *f*.

5.2.2 Characterizations

Modular redundancy with the symmetric comparison model has been addressed by Xu [Xu91]. The author casts this problem in the framework of system-level diagnosis as one of *f*/*s*-*diagnosability* [Fri75], where up to *f* faulty PEs in a *n*-PE system must be located within a set of *s* PEs, with $s \geq f$. Xu considered the case where the problem consists in finding just one fault-free PE, i.e., $s = n - 1$. Hence, if the system is *f*/(*n*–1)-*diagnosable*, all the faulty results are guaranteed to be within an identified set of $n - 1$ results, leaving one result that can be deemed unambiguously to be correct. As shown in Definition 5.5, this concept is equivalent to task diagnosability, since the task result produced by the fault-free PE is guaranteed to be correct.

Despite the notorious difficulty of the problem of *f*/*s*-diagnosability, Xu achieved considerable success in characterizing the diagnosability conditions for regular structures, e.g., chains, loops and $H_{2r,n}$ structures. In particular it was shown that for $n = 2f + 1$, chains are task-diagnosable for up to $n = 9$, and loops are task-diagnosable for up to $n = 13$.

In this section, we restrict our discussion to small systems with $n \leq 11$ and $n = 2f + 1$ since many modular redundant systems fall into this class. Our results are not significantly new with respect to Xu [Xu91], but our theorems use a more intuitive argument. We consider disjoint chains as well as continuous chains, while Xu [Xu91] only presented results concerning continuous chains and for a single case of $n = 5$. Disjoint chains are important, for as we later show, a distinct task can be executed on the nodes of each disjoint part of the graph. We establish the lower bounds on the number of comparisons required to diagnose at least one correct task for chains. We also establish the characterization theorems by a simple recursive argument that differs from [Xu91]. This characterization is also similar to that of Chen *et al*. [CC90,Che92], except that it applies in our work to the symmetric comparison model as opposed to the asymmetric comparison model.

We begin by the following lemma drawn from Chen *et al*. [CC90,Che92] since it applies to both the symmetric and the asymmetric comparison models.

Lemma 5.1 A system is task diagnosable with one comparison when $n = 3$ and $f = 1$.

Proof. Assume for instance that the comparison is performed between v_2 and v_3 (Figure 5.7). There are two possible outcomes of this comparison. If $w_{2,3} = 0$, then results r_2 and r_3

are correct. With $w_{2,3} = 0$, they are either both correct or both incorrect. Since no more than one result is allowed to be incorrect, they must both be correct. If $w_{2,3} = 1$, then at least one of the results r_2 and r_3 is incorrect. This exhausts the maximum supply of incorrect task results. Hence by inference the r_1 result is correct. □

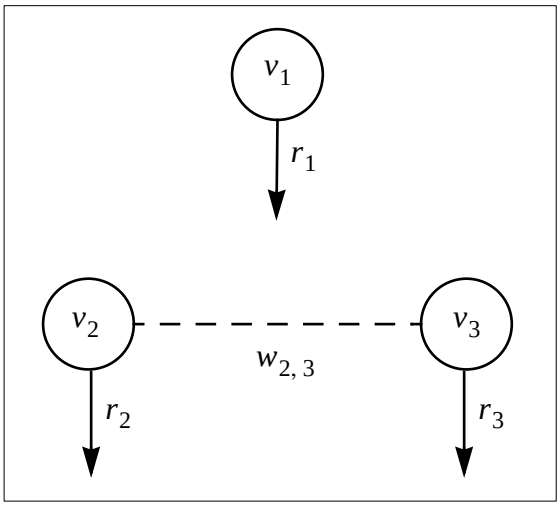


Figure 5.7 System described in proof of Lemma 5.1

We now characterize systems that are task-diagnosable for up to 9 PEs under the assumptions of the symmetric comparison model of system-level diagnosis.

Theorem 5.1 Consider a system *S* that has *n* PEs and where $n = 2f + 1$. The system is task-diagnosable with $n - 2$ comparisons for the symmetric comparison model for $n \leq 7$.

Proof. This theorem is proved by a recursive argument starting with $n = 3$. First, augment the 3-PE system to a 5-PE system by bringing in two additional PEs v_4 and v_5, with a comparison between them as shown in Figure 5.8. Consequently $n = 5$ and $f = 2$.

Consider first the comparison between v_4 and v_5. If $w_{4,5} = 1$, there is at least one incorrect result in the pair leaving at most one incorrect result in the set of results r_1, r_2, r_3. Thus, the group of PEs comprised of r_1, r_2, r_3 is task diagnosable by Lemma 5.1.

Assume now that $w_{4,5} = 0$. If $w_{2,3} = 1$, by the preceding argument, the set r_1, r_4, r_5 is task diagnosable. The only difficult case is when $w_{4,5} = 0$ and $w_{2,3} = 0$. An additional comparison is then needed, for instance between v_3 and v_4, to be able to diagnose the system. If

$w_{3,4} = 0$, then the chain v_2, v_3, v_4, v_5 produces correct results since no more than two results can be incorrect. If $w_{3,4} = 1$, there is at least one incorrect result in the pair v_3, v_4, and there is also an incorrect result that agrees with it in either v_2 or v_5. Therefore, by inference r_1 is correct as all faults have been accounted for.

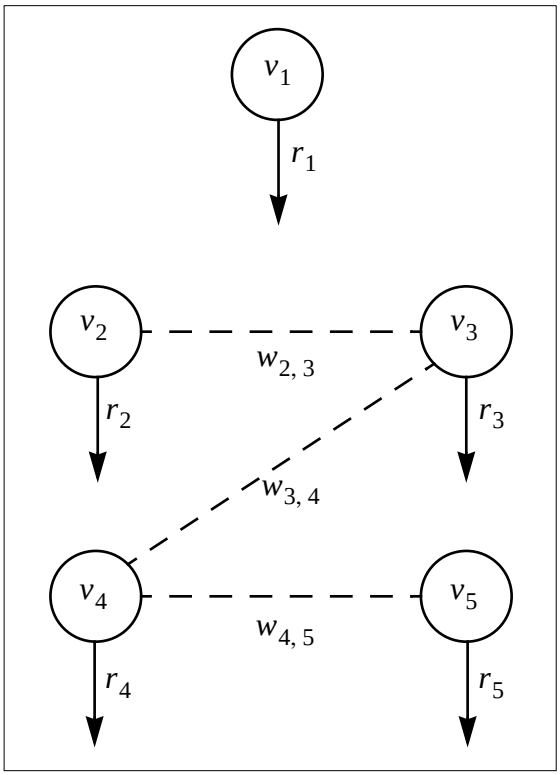


Figure 5.8 System described in proof of Theorem 5.1

The additional comparison could have been performed between v_1 and v_2 instead of between v_3 and v_4, yielding a second diagnosable configuration for 5-PE system. This is illustrated in Figure 5.9.

To complete the proof, we now consider a system where $n = 7$. Since a 7-PE system is to be diagnosed with only $n-2 = 5$ comparisons, the testing graph must have two disjoint chains. This yields three candidate structures with groups of 1,6; 2,5; 3,4 PEs as shown in Figure 5.10c-e. It can be easily verified that the two structures illustrated in Figure 5.10c and Figure 5.10d are not diagnosable with the syndromes shown in the figure. This leaves the structure of Figure 5.10e with a chain v_1, v_2, v_3 of 3 PEs and a chain v_4, v_5, v_6, v_7 of 4 PEs.

This structure is diagnosable if either v_4, v_5 or v_6, v_7 or both test to 1, since discounting one of these pairs yields a 5-PE system with at most two incorrect results that is diagnosable as

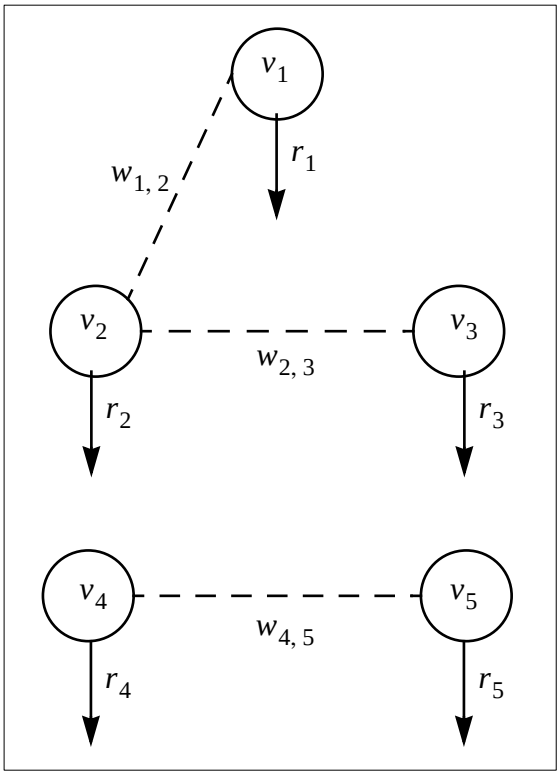


Figure 5.9 Alternative configuration for system described in proof of Theorem 5.1

a structure of the type shown in Figure 5.10b. This leaves the case where $w_{4,5} = 0$ and $w_{6,7} = 0$. Then if $w_{5,6} = 0$, the all-zero chain v_4, v_5, v_6, v_7 produces four correct results, since no more than three incorrect results testing to 0 can be produced by the system. If $w_{5,6} = 1$, then either r_4, r_5 or r_6, r_7 are incorrect and by inference r_1, r_2, r_3 are correct. □

It can be shown easily that a 9-PE system is not diagnosable with two disjoint chains. Adding one more comparison yields the following theorem:

Theorem 5.2 Consider a system *S* that has *n* PEs and where $n = 2f + 1 = 9$, $f = 4$. The system is task diagnosable for the symmetric comparison model with $n - 1 = 8$ comparisons when it is arranged as a continuous chain of 9 PEs.

Proof. The system structure is a chain as shown in Figure 5.10f. If one or more of the comparisons $w_{1,2}, w_{4,5}, w_{5,6}, w_{8,9}$ yield 1, then removing a single pair of PEs that test to 1 yields a 7-PE system with structure like Figure 5.10e. Such a structure is diagnosable by Theorem 5.1 for $f = 3$ since no more than 3 results are incorrect, and there is always a sufficient number of comparisons.

Suppose now that all the pairs $w_{1,2}, w_{4,5}, w_{5,6}, w_{8,9}$ yield 0. Then results of the center

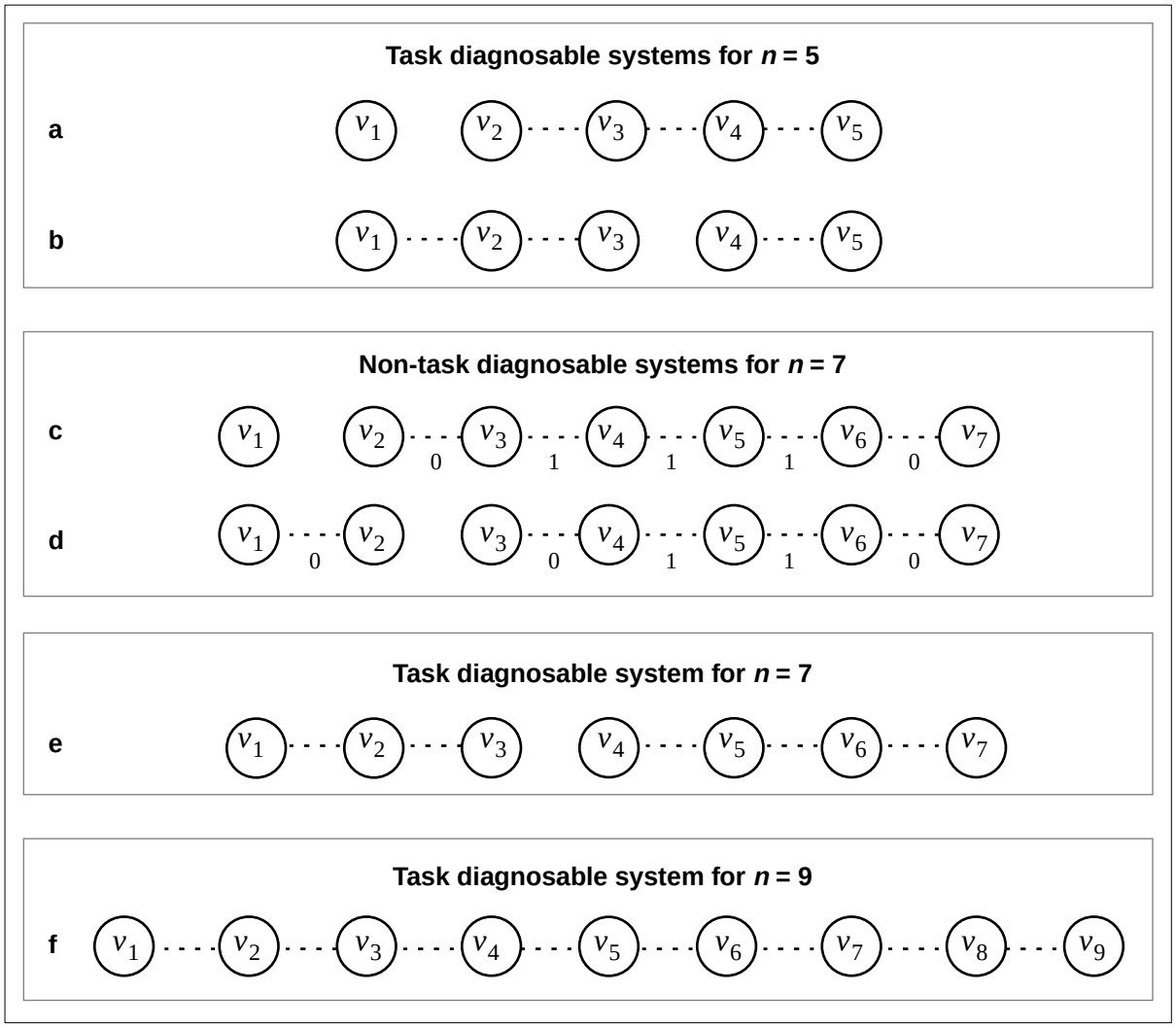


Figure 5.10 Diagnosability of configurations for $n = 5, 7, 9$

triplet r_4, r_5, r_6 are either all correct or all incorrect, because their values are identical. Consider the left-most and right-most result triplets r_1, r_2, r_3 and r_7, r_8, r_9. If all comparisons in at least one of these triplets, say r_1, r_2, r_3, yield 0, then the triplets r_4, r_5, r_6 and r_1, r_2, r_3 cannot be both incorrect, since no more than 4 results can be incorrect. So, consider now the comparisons between groups of triplets. If $w_{3,4} = 0$, the results $r_1, r_2, r_3, r_4, r_5, r_6$ are all correct. If $w_{3,4} = 1$, then either the triplets r_1, r_2, r_3 are all incorrect, or the triplets r_4, r_5, r_6 are incorrect (both sets could not be incorrect as that would imply $f = 6$). Consequently no more than one additional result can be incorrect, and r_8, r_9 are correct since $w_{8,9} = 0$.

This leaves the case where both $w_{2,3} = 0$ and $w_{7,8} = 0$, which is resolved by considering the pairs r_3, r_4 and r_6, r_7. If both $w_{3,4} = 0$ and $w_{6,7} = 0$, then obviously r_3, r_4, r_5, r_6, r_7 are all correct. If only one pair, say r_3, r_4, tests to 0, then either the pair r_1, r_2 or the triplet

r_4, r_5, r_6 is incorrect. Yet the triplet r_4, r_5, r_6 cannot be incorrect, because, as $w_{6,7} = 1$, this would imply that r_7 would be thus correct and test to 0 with r_8. This leaves finally the case with a syndrome of 01100110 for which r_4, r_5, r_6 are necessarily correct since otherwise either r_3, r_4 and/or r_6, r_7 would test to 0. Hence the 9-PE system is task diagnosable as a chain with 8 comparisons as proved in [Xu91] by other means. □

This in turn means that an 11-PE system is task diagnosable as a loop with 11 comparisons, since any pair in the loop that tests to 1 can be disallowed leaving a 9-processor chain with no more than 4 incorrect results. This yields the following corollary:

Corollary 5.1 Consider a system *S* that has *n* PEs and where $n = 2f + 1 = 11$, $f = 5$. The system is task diagnosable for the symmetric comparison model with 11 comparisons when it is arranged as a loop.

Note that the bounds established above for the symmetric comparison model differ from those reported by Chen *et al*. [CC90, Che92] for the asymmetric comparison model. With the latter model, two results that are identical are necessarily correct. Therefore, an NMR system can be implemented with $(n-1)/2$ disjoint comparisons between pairs of PEs plus an isolated PE. This contrast in the number of comparisons need for both approaches is shown in Table 5.1 for NMR systems of up to 11 PEs.

Table 5.1 Comparisons required for asymmetric vs. symmetric comparison models

n	Asymmetric comparison model	Symmetric comparison model
3	1	1
5	2	3
7	3	5
9	4	8
11	5	11

5.2.3 Task diagnosability in efficient comparison-based systems

In Section 3.3, “Bi-corona systems,” on page 50, we have seen examples of comparison-based systems with $2f + 1$ PEs. These systems are efficient in the sense that the number of comparisons have been reduced and more than one task can be executed on the system. In a bi-corona

system, however, only $f + 1$ PEs (internal nodes) are needed for providing the output result of the system, while the remaining f PEs (external nodes) are in effect used solely as checkers. We now characterize this property by starting with the following two definitions.

Definition 5.6 In a comparison-based NMR system a PE is said to be a *producer* when it is one of the candidates for being selected to produce the system output by the comparison-based algorithm.

In a hardware implementation, a producer is connected to the NMR output bus through a tri-state gate.

Definition 5.7 In a comparison-based NMR system a PE is said to be a *checker* when its output is used solely for internal testing within the NMR system and therefore is never selected as the external output of the NMR system.

In a hardware implementation, a checker is not connected to the NMR output through a tri-state gate.

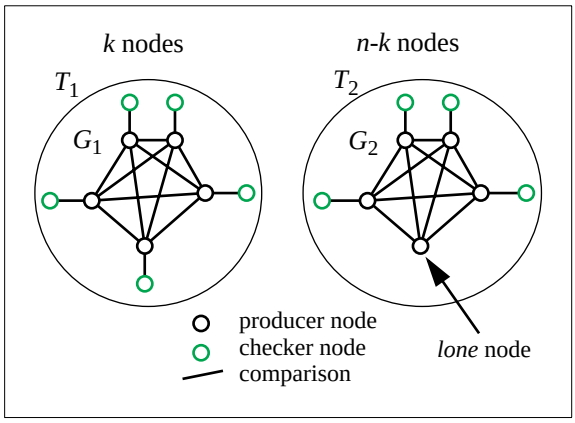


Figure 5.11 Producer and checker nodes in a bi-corona structure

For a given NMR system, there are obvious advantages in minimizing the number of producers with respect to the number of checkers. In particular, reducing the number of producers in turn reduces the number of tri-state gates that can become very significant when the results are a large set of data. In a VLSI (Very Large Scale Implementation), the checkers may be located in the same chip as the other PEs with which they compare. In this case, the chip requires no output pins for the checker. Finally the checkers might be simpler than the producers if the comparison testing can be made on a check word/signature or on a subset of the result data. Then if there is a way for the producer to produce directly the signature or the subset of

result data without first producing the result, the checker can be simplified.

We now characterize the class of bi-corona systems defined by Theorem 3.10 on page 55 for producer and checker PEs.

Theorem 5.3 Consider a task-diagnosable, bi-corona system (Theorem 3.10) with $n = 2f + 1 + s$ nodes, where $s \geq 0$. The minimum number of producers is equal to $f + 1$, and these producers are the internal nodes (see Figure 3.13 on page 51) in each of the subgraphs of the bi-corona system.

Proof. First, it is obvious that at least $f + 1$ producers are required. Since up to *f* arbitrary PEs may be faulty, a fixed connection topology that assigns only *f* PEs as producers would not work if these *f* PEs were faulty. This is because the diagnosis Algorithm 3.1 would designate a checker node as fault-free, and this checker node would be unable to provide the system output since it is not connected.

We now prove that $f + 1$ producer nodes are sufficient and that these nodes can be identified *a priori*. The proof is a direct consequence of the way Theorems 3.9 and 3.10 were established. Assume first that $n = 2f + 1$. In step 1 of the diagnosis Algorithm 3.1, all internal-external pairs testing to 1 are counted and discarded from the rest of the diagnosis. At the end of step 1, all surviving internal-external pairs test to 0, with a maximum of *f* such pairs. If there is no surviving internal-external pair, the lone node is deemed to be fault-free. Otherwise, the node finally selected as fault-free will be either the lone node or one node in one of the internal-external pairs testing to 0. The two nodes in each internal-external pair testing to 0 have the same fault status, so that only one node per pair need be deemed a producer node. Since there are *f* pairs plus a single node, this yields a total of $f + 1$ producer nodes that are identified *a priori*.

If $n = 2f + 1 + s$, $s > 0$, then *s* nodes can be removed from the system in a way that reduces the system to the $(2f + 1)$-node system discussed above. Hence only $f + 1$ producer nodes are required, and these are identified *a priori*. □

5.3 Comparison-based NMR with adaptive diagnosis

In the preceding sections we have assumed implicitly that the comparisons required for task diagnosis were predefined for a given system. For hardware-implemented fault tolerance the topology for comparison testing should indeed be fixed, as this makes it unnecessary to con-

nect the comparators to the PEs through a switching network. On the other hand, for software-implemented fault tolerance as in SIFT [WLGG78], adaptive task diagnosis could be an interesting proposition if it significantly decreases the number of comparisons.

Adaptive system-level diagnosis was introduced by Hakimi *et al*. [HN84] as an alternative to using a predetermined test pattern. It was then refined [HS84] and extended [KH83] to cover the case of comparison testing. With the comparison method, diagnosis is performed concurrently with testing, and the next comparison is applied between two processors that are chosen as a function of the inferences drawn from the test results already collected. The basic goal is to identify first one fault-free PE and then to compare this PE to the rest of the PEs.

The first part of adaptive comparison diagnosis as reported by Kreutzer and Hakimi [KH83] applies immediately to task diagnosis in the comparison-based NMR setting since it reports results on a model that is identical to the symmetric comparison model. Kreutzer and Hakimi [KH83] present an algorithm to find one fault-free PE with no more than $2f - v(f)$ comparisons, where $v(f)$ is the number of 1s in the binary representation of *f*. Moreover they showed that $2f - v(f)$ is a lower bound on the number of comparisons. This algorithm is discussed in the setting of a directed testing graph (PMC model), but with some minor modifications, it applies to the comparison model as well.

Generally speaking, the Kreutzer-Hakimi [KH83] algorithm consists first in performing comparison tests on *f* distinct pairs of PEs, leaving one isolated PE. The pairs testing to 1 contain at least one faulty PE, and the pairs testing to 0 contain either two faulty or two non-faulty PEs. At the end of this first cycle a counter *C* is set to a value equal to the number of pairs testing to 1. This counter indicates the minimum number of PEs that have been identified as faulty within a pair.

If the *f* pairs test to 1 during the first cycle, all the faulty PEs have been located within the pairs, which indicates that the isolated PE is fault-free. Otherwise some pairs have tested to 0, and the pairs testing to 1 are discounted and a second cycle of testing is initiated. In this second cycle one PE of the remaining pairs is compared to a PE of another remaining pair and the algorithm continues as in cycle 1, but by counting 2 faults if the test yields a 1 outcome. The algorithm stops either when counter *C* reaches the value *f* or when a chain of $f + 1$ PEs tests to 0.

An interesting property of the algorithms given in [HS84, KH83] is that the number of cycles is at most $\lfloor \log_2 f \rfloor + 1$, where the *f* comparisons in the first cycle can be predetermined.

Thus using an adaptive algorithm for comparison-based NMR is not as bad as it would seem, for practical NMR systems are usually limited to a small number of redundant processors.

Table 5.2 Comparisons required for adaptive vs. non adaptive comparison diagnosis

n	Adaptive [KH83]		Non-adaptive
	Comparisons	Cycles	Comparisons
3	1	1	1
5	3	2	3
7	4	2	5
9	7	3	8
11	8	3	11

First we note that the bound $2f - v(f)$ on the number of comparisons and the bound $\lfloor \log_2 f \rfloor + 1$ on the number of cycles translate into the number of comparisons and cycles shown in Table 5.2 for systems with up to 11 PEs. Table 5.2 also shows the number of comparisons needed for the symmetric fault model in a non-adaptive setting. This result shows the number of comparisons is not reduced by resorting to adaptive diagnosis for 3-PE and 5-PE systems and that the number of comparisons is only slightly reduced for higher order systems.

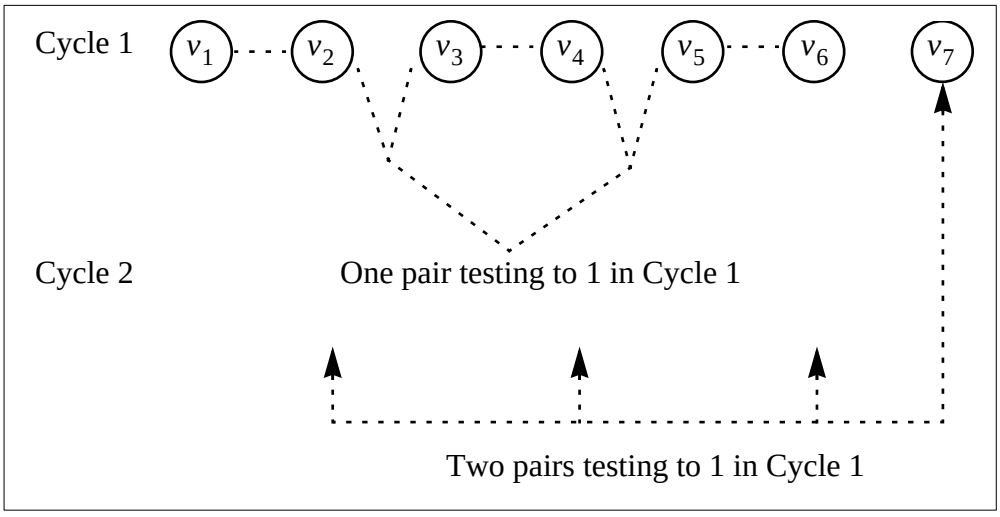


Figure 5.12 Adaptive task diagnosis for a comparison-based NMR system with $n = 7$

In order to contrast the additional complexity of adaptive diagnosis, consider, for example, a 7-PE system $v_1, \ldots, v_7$ that produces task results $r_1, \ldots, r_7$. Connections are statically assigned for the $f = 3$ comparisons in the first cycle, with comparisons done on the result pairs r_1, r_2; r_3, r_4; r_5, r_6 as shown in Figure 5.7. If the three pairs test to 1, then r_7 is neces-

sarily fault-free.

If only one of the pairs tests to 1, at least one fault is identified within this pair, and at most 2 faults remain in the rest of the PEs. Then the two remaining pairs that test to 0 must be compared, which requires one comparison out of three possible options. If the outcome of this comparison is 0, the four corresponding results are correct. Otherwise the two faults are located in one of the pairs and r_7 is necessarily fault-free.

Finally, if two pairs test to 1 in the first cycle, the isolated PE v_7 must be compared with the pair testing to1, which requires one comparison out of three possibilities. If this comparison yields a 0 then the three corresponding results are correct. Otherwise, the result in the pair that is not directly connected to r_7 is necessarily fault-free.

Thus the 7-PE system is task diagnosable with 4 comparisons done in an adaptive manner instead of with 5 done in a non-adaptive way. But this slight improvement is achieved at the expense of an added switching complexity that does not seem to be justified for a hardware implementation.

5.4 Fault detection

The fundamental, comparison-based, modular redundant system guarantees that a single correct result is found if the fault tolerance capacity of the system is not exceeded, but it does not provide a complete diagnosis of all the tasks. For practical systems, fault masking must be complemented by fault detection and localization, and repair to ensure that the number of faulty PEs does not exceed the limit for the system. Fault detection can be achieved by using circuitry for this purpose or by running diagnostic programs in the “background” on the PEs. Yet for several reasons, it remains desirable to expose the faults as a by-product of executing the tasks that comprise the principal function of the system.

Firstly, there may be a significant latency between the occurrence of the fault and its detection by a diagnostic program. This increases the chance that a system can fail when, in rare instances, nearly concurrent faults arise. Secondly, if the fault is transient or intermittent, it may actually affect a task result and then go undetected when the specialized fault detection is invoked. Thirdly, some permanent faults might produce erroneous task results without being detected by the auxiliary fault detection process, because of the difference in fault coverage of the two.

In the setting of Section 5.1, the detection and localization of faults is a problem of proces-

sor diagnosis. The factor of fault coverage becomes important in the sense that an incorrect task result implies that the PE that produced it is faulty. However, no conclusions can be drawn on the fault status of the PE if it executed a task that produced a correct result, since, if the PE were faulty, the task might not have exercised the faulty part of the PE.

5.4.1 One-step fault diagnosis

Ideally a system should be able to identify all faulty PEs as soon as the faults occur, provided no more than f PEs are faulty. This principle corresponds to the notion of *one-step f-fault diagnosis* in system-level diagnosis [PMC67]. In a conventional NMR system, an approximation of one-step f-fault diagnosis is achieved by comparing the output of the voter to the output of each PE with disagreement detectors [MA70]. The PE that produced an incorrect result is identified and can be replaced automatically by a spare. Thus faults are handled appropriately, provided they are properly exposed by the execution of the task. The additional cost incurred with this scheme is significant: n comparators are needed as well as some switching hardware if spares are to replace faulty units.

In the case of comparison-based NMR, the lower bounds on the number of comparisons have not yet been established. However, an adaptive algorithm for locating all faulty PEs with the symmetric comparison model was shown [KH83] to use at most k_n comparisons, with

$$k_n = n + f - 3 \tag{5.1}$$

Kreutzer and Hakimi [KH83] indicate that $n + f - 3$ is not a best upper bound on the number of comparisons for adaptive comparison diagnosis. They conjecture that

$$k_n \leq \left\lfloor \frac{n}{2} \right\rfloor + \left\lceil \frac{\left\lceil \frac{n}{2} \right\rceil - (f-2)}{2} \right\rceil \tag{5.2}$$

Note that the upper bound, equation (5.2), is equal to $n + f - 3$ for $n = 2f + 1$. We first restate the result in equation (5.1) for $n = 2f + 1$ with a simple recursive argument. We show that equation (5.1) applies only if $n > 5$, which was not stated in [KH83].

Theorem 5.4 Consider a system with n PEs, where $n \geq 2f + 1$. For $n > 5$ the system is one-step f-fault diagnosable for adaptive comparison diagnosis with no more than $n + f - 3$ com-

parisons. For $n = 3$ and $n = 5$, the adaptive diagnosis requires no more than 2 and 5 comparisons respectively.

Proof. The theorem is first proved for $n = 2f + 1$ and later extended to $n \geq 2f + 1$. We begin by proving that the system is not diagnosable with $n + f - 3$ comparisons for $n \leq 5$.

For $n = 3$, $f = 1$, equation (5.1) gives $k_3 = 1$. Yet two comparisons are obviously needed, since a single comparison is insufficient to determine the fault status of the PE that has not been tested when the two other PEs give identical results.

For $n = 5$, $f = 2$ equation (5.1) yields $k_5 = 4$. However, in the worst case, five comparisons are needed to adaptively diagnose a 5-PE system $v_1, \ldots, v_5$. This is shown by taking any two PEs, e.g., v_1, and v_2. If the comparison outcome is a one, either one or both of the PEs are faulty. Thus there is at most one faulty PE in the three remaining PEs v_3, v_4, v_5. In the worst case, v_3, v_4, v_5 cannot be diagnosed with less than two comparisons. If the diagnosis on v_3, v_4, v_5 indicates that they are all fault-free, v_1, v_2 can still be both faulty, and the determination of the fault status of v_1, v_2 requires comparing each v_1 and v_2 to one of the identified fault-free PEs. This gives a total of 5 comparisons.

We now prove that a system is diagnosed with the stated bound of $k_n = n + f - 3$ comparisons for $n = 7$, $f = 3$. First the PEs in the pairs v_1, v_2; v_3, v_4; v_5, v_6 are compared. If the three comparison outcomes are 1, only one PE in each pair is faulty, and since all possibly faulty PEs are located within the subset $v_1, \ldots, v_6$, PE v_7 is fault-free. The fault status of $v_1, \ldots, v_6$ is then asserted with three additional comparisons between v_7 and v_1, v_3, v_5 respectively for a total of 6 comparisons.

If the three comparison outcomes on v_1, v_2; v_3, v_4; v_5, v_6 are 0, then the PEs in each pair are either both faulty or both fault-free, and the number of faults in the subset $v_1, \ldots, v_6$ can only be even, that is, 0 or 2. Then the 5-PE subset $v_2, \ldots, v_6$ has at most two faults and can be diagnosed as indicated earlier with 5 comparisons, out of which 2 have already been performed in the first step. The fault status of v_1 is the same as that of v_2, and the fault status of v_7 is determined by an additional comparison, for a total of 7 comparisons.

If only one of the pairs v_1, v_2; v_3, v_4; v_5, v_6 tests to 1, say v_1, v_2, then one PE of each of the two pairs v_3, v_4; v_5, v_6 which test to 0 are compared, e.g., v_4, v_5. If this comparison yields a 0, v_3, …, v_6 are fault-free, and the fault status of the three remaining PEs v_1, v_2, v_7 is found by comparing them with one of the fault-free PEs which yields a total of 7 comparisons.

If v_4, v_5 compare to 1, one of the pairs v_3, v_4; v_5, v_6 is faulty, and the other one is fault-free. Then v_7 is fault-free, and comparing its result with those produced by v_1, v_6 completes the diagnosis since only one PE can be faulty in the pair v_1, v_2. Here the diagnosis is performed with 6 comparisons.

Finally if two pairs such as v_1, v_2; v_3, v_4 test to 1 in the first phase of the diagnosis then there is at least one faulty PE in each of these pairs, with possibly two faulty PEs in one of the pairs. Then the remaining pair that tests to 0, here v_5, v_6, has its two PEs deemed to be fault-free. Thus v_4 and v_2 can be tested with v_5. If one of the test outcomes is 0, the opposite PE in the pair is by inference faulty, and the diagnosis can be completed by testing with v_5 the single PE v_7 and the remaining PE in the other pair. If both tests between v_5, v_4 and between v_5, v_2 yield a 1, at most one PE out of the subset v_1, v_7, v_3 can be faulty, and it is diagnosed by the two comparisons v_1, v_7 and v_7, v_3. This yields again 7 comparisons, which completes the part of the proof of the diagnosability of the 7-PE system with 7 comparisons.

We now show by a recursive argument that a system with n PEs is diagnosed with $k_n = k_{n-2} + 3$ comparisons where k_{n-2} is the number of comparisons required to diagnose a system with $n-2$ PEs. We begin by performing the $f = (n-1)/2$ comparisons v_1, v_2; v_3, v_4; …; v_{2f-1}, v_{2f}. If at least one of these pairs tests to 1, one faulty PE has been identified in the pair, and at most $f-1$ PEs are faulty in the remaining $n-2$ PEs. Hence these $n-2$ PEs are diagnosed with k_{n-2} comparisons, out of which $f-1 = (n-1)/2-1$ have already been done during the first step. The two PEs that test to 1 can then be diagnosed with 2 comparisons performed from one of the PEs already identified as fault-free. Hence the total number of comparisons is given by $k_n = k_{n-2} + 2 + f - f + 1 = k_{n-2} + 3$.

If the f pairs $v_1, v_2; v_3, v_4; \ldots; v_{2f-1}, v_{2f}$ test to 0, then there is a set of $n-1$ PEs with an even number of up to $f-1$ faults if f is odd. In this case the subset of the $n-2$ PEs $v_2; v_3, v_4; \ldots; v_{2f-1}, v_{2f}$ is diagnosed with $k_{n-2}-(f-1)$ comparisons since the $f-1$ comparisons $v_3, v_4; \ldots; v_{2f-1}, v_{2f}$ have already been performed in the first step.

Thus the fault status of v_1 is equal to that of v_2 since it tests to 0 with v_2, and the fault status of the single PE v_{2f+1} is assessed with one additional comparison from a PE already identified as fault-free, yielding a total of $k_{n-2}+2$ comparisons.

Finally if the f pairs $v_1, v_2; v_3, v_4; \ldots; v_{2f-1}, v_{2f}$ test to 0 after the first step, and f is even, then the second step consists in testing the pairs with the $f-1$ comparisons $v_2, v_3; v_4, v_5; \ldots; v_{2f-2}, v_{2f-1}$. If these $f-1$ comparison outcomes are 0, the PEs $v_1, v_2; v_3, v_4; \ldots; v_{2f-1}, v_{2f}$ are all fault-free, and the diagnosis is completed by comparing v_{2f+1} with one of these PEs. If at least one of the comparisons $v_2, v_3; v_4, v_5; \ldots; v_{2f-2}, v_{2f-1}$, for example, v_4, v_5, yields an outcome of 1, two faults have been located within one of the corresponding pairs, for example, $v_3, v_4;.v_5, v_6$. Then discounting these two pairs yields a system with $n-4$ PEs, on which the same algorithm can be repeated. This reduction is done at the cost of $2f-1$ comparisons, out of which $2f-6$ are used for diagnosing the system with $n-4$ PEs. Hence, in this case, $k_n = k_{n-4}+5$.

In the worst case

$$k_n = k_{n-2}+3 \tag{5.3}$$

with $k_7 = 7$ for $f = 3$, odd and $k_5 = 5$ for $f = 2$, even. The relation $k_n = n+f-3$ is established by noting that it becomes $k_n = 3(n-1)/2-2$ for $n = 2f+1$, which means that $k_n = (3((n-2)-1))/2-2+3 = k_{n-2}+3$ as in equation (5.3). Since a 7-PE system is diagnosed with the stated bound of $k_n = n+f-3 = 7$ comparisons, systems of any size can be diagnosed within the $n+f-3$ bound by successive reductions down to a 7-PE system. The single exception is when all pairs test to 0 in the first step and f is even. Then the reduction takes only 5 comparisons instead of 6 to go repetitively from an n-PE system to an $(n-4)$-PE

system down to a 5-PE system, which is diagnosed with 5 comparisons. This completes the proof of the theorem for $n = 2f + 1$.

When $n = 2f + 1 + 1 \geq 2f + 1$, the diagnosis can be first performed with $n - 1 + f - 3$ on $n - 1 = 2f + 1$ PEs chosen arbitrarily. The diagnosis is then completed with l additional comparison between one identified fault-free PE and the l PE left over from the first step. Thus the system is diagnosed with $k_n = n + f - 3$ comparisons.

For non-adaptive, one-step *f*-fault diagnosis the minimum number *k* of comparisons sufficient to locate all faulty units when $n = 2f + 1$ is bounded by

$$\frac{fn}{2} \leq k \leq fn \tag{5.4}$$

This is a consequence of a system being diagnosable if [Pel92]:

1. every PE is linked to all other PEs and
2. every PE is linked to at least *f* other PEs.

5.4.2 Sequential diagnosis

The next best thing to one-step, *f*-fault diagnosis, where the goal is to detect all faulty units, is being able to detect and localize at least one faulty unit. With this approach, called *sequential f-fault diagnosis*, at least one faulty unit out of possibly several is detected and identified [PMC67, Pel92]. This allows the triggering of a repair action or switching a spare to replace the identified faulty unit. Pelc [Pel92] showed that the minimum number of comparisons required for one-step, sequential *f*-fault diagnosis is equal to $n - \lceil n/(f + 1) \rceil + 1$ in the adaptive case, and $n - \lfloor n/(2f + 1) \rfloor + 1$ in the non-adaptive case. For $n = 2f + 1$ this gives $n - 1$ comparisons in both cases, with the number of comparisons given in Table 5.3 for $n = 3, 5, \ldots, 11$. One-step, sequential *f*-fault diagnosis is a reasonable goal in many practical situations where nearly coincident faults are very unlikely to occur. This is because it still allows bringing in a spare automatically.

As a final expediency, a system could merely detect that at least one fault has occurred without possibly being able to locate any faulty unit. In such a system, automatic replacement of a faulty unit is no longer directly possible. With this approach fault detection could prompt the execution of diagnostic programs that would, upon eventual identification of a faulty PE, alert a repair team or automatically replace the failed PE.

Table 5.3 Number of comparisons for various types of diagnosis with $n = 2f + 1$

Diagnosis	n				
	3	5	7	9	11
Adaptive task diagnosis	1	3	4	7	8
Non-adaptive task diagnosis	1	3	5	8	11
Adaptive one-step *f*-fault diagnosis	2	5	7	10	13
Non-adaptive one-step *f*-fault diagnosis	2	5	11	18	
Adaptive sequential *f*-fault diagnosis	2	4	6	8	10
Non-adaptive sequential *f*-fault diagnosis	2	4	6	8	10
Adaptive (*n*–1)-fault detection	2	4	6	8	10
Non-adaptive (*n*–1)-fault detection	2	4	6	8	10
Non-adaptive task & sequential *f*-fault diagnosis + (*n*–1)-fault detection	2	4	6	8	11

It should be first noted that a conventional NMR system equipped with disagreement detectors is capable of detecting up to $n-1$ faults. This is a very desirable feature since an alarm is raised even if the fault tolerance capability is exceeded and comparison based NMR should provide this *(n–1)-fault detection* feature. Pelc [Pel92] has shown that the minimum number of comparisons required for (*n*–1)-fault detection is equal to $n - \lfloor n/(f+1) \rfloor$ for both adaptive and non-adaptive detection, which translates to $n-1$ comparisons when $n = 2f + 1$ (Table 5.3).

Table 5.3 shows that with only a slight increase in the number of comparisons over that required for simple task diagnosis, it should be possible to achieve both (*n*–1)-fault detection and one-step, sequential *f*-fault diagnosis, with 2, 4, 6, 8, 11 comparisons for systems made up of 3, 5, 7, 9, 11 PEs respectively. The following theorem shows that the three goals of task diagnosis, (*n*–1)-fault detection and one-step, sequential *f*-fault diagnosis are indeed achieved for $n = 2f + 1$ when the PEs are arranged in a chain of up to $n = 9$ and in a loop for $n = 11$:

Theorem 5.5 A chain of *n* PEs, with $n = 2f + 1$ and $n \leq 9$ is task diagnosable, (*n*–1)-fault detectable and one-step, sequentially *f*-fault-diagnosable with $n-1$ comparisons. All three diagnoses can be performed simultaneously.

Proof. Theorem 5.2 proves that chains are task diagnosable for $n \leq 9$. Fault detection is ensured if at least one comparison outcome is 1.

The conditions of Theorem 5.5 state that fault detection is guaranteed if there is at least one fault-free PE. In a chain, this fault-free PE is connected to at least one other PE by a comparator. Thus ultimately if one to $n - 1$ PEs are faulty, there is at least one comparison between one faulty PE and one non-faulty PE that yields a 1 comparison outcome. This proves that the system is (n-1)-fault detectable.

The chain is obviously sequentially f-fault-diagnosable because the task diagnosis property guarantees that at least one fault-free PE is identified. Starting from this PE there is always a succession of comparisons that ends with a comparison between a fault-free PE and a faulty PE, if any. This allows the identification of at least one faulty PE and ensures sequential f-fault diagnosability. □

For $n = 11$, Corollary 5.1 guarantees the task-diagnosability of a loop. The same arguments as in the above proof show that such a loop is both (n–1)-fault-detectable and sequentially f-fault-diagnosable. Hence the following corollary:

Corollary 5.2 A loop of n PEs with $n = 2f + 1 = 11$ is simultaneously task diagnosable, (n-1)-fault-detectable and one-step, sequentially f-fault-diagnosable with n comparisons.

5.5 Comparator fault tolerance

In a modular redundancy system that is implemented with the minimum number of comparisons, each comparator is a single point of failure. For most systems where fault tolerance is important, this design is unacceptable. We discuss here the requirements on the number of comparisons and on the testing topology to ensure that the same fault tolerance degree can be guaranteed on the comparators and on the PEs. Again we show that the comparison-based approach is clearly superior to the conventional NMR scheme when the fault tolerance of the comparators is considered.

NMR systems with majority voting tolerate faults in the voter by replicating it and using majority voting on the replicated voter outputs. With this approach a triplicated voter tolerates one voter fault, and a 5-voter scheme tolerates up to two voter faults. The same technique is directly applicable to comparison-based modular redundancy as illustrated in Figure 5.13 for a 3-PE system. For a system with n PEs where $n = 2f + 1$, each comparator is replicated n

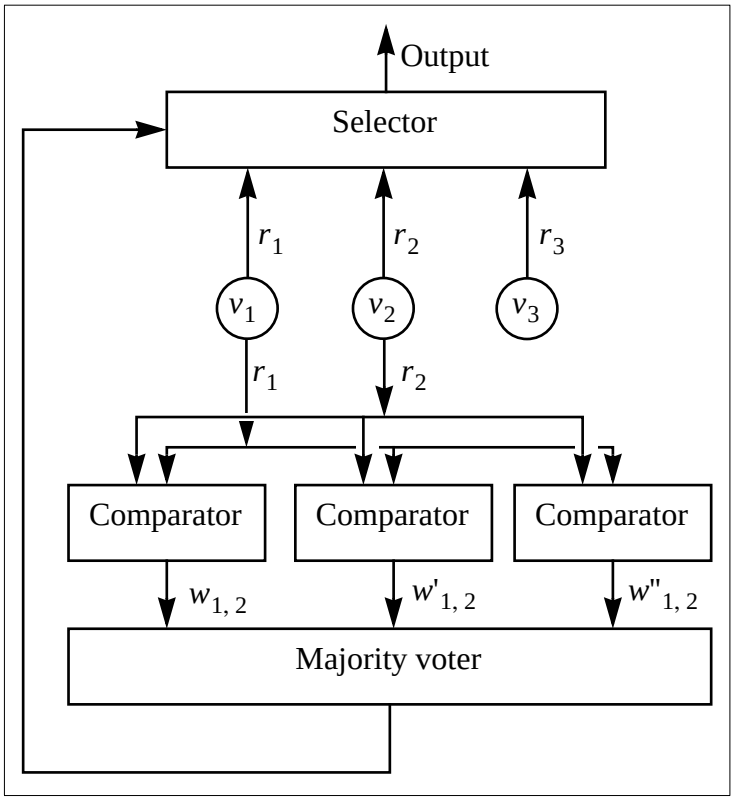


Figure 5.13 Comparison-based TMR with triplicated comparators and majority voting on comparison results

times, and the comparison outcomes $w_{i,j}$ are majority voted to produce final comparison results that are fault-free for up to $f = (n-1)/2$ comparator faults in the same group. Note that we use majority voting to mask faulty comparison results. The cost of the majority voting is negligible here since it is performed only on the binary comparison outcomes w_{ij} and not on the possibly bulky computation results r_i. If k_n is the number of comparisons required for a system with non-replicated comparators and up to *f* comparator faults must be tolerated within a group, the number of comparisons k'_n for a system that tolerates both PE and comparator faults becomes

$$k'_n = nk_n \qquad (5.5)$$

Hence the cost of making the system more robust by replicating the comparisons is very significant for large systems. However, this naive replication approach may be thought of as an overkill since it protects against *f* faults occurring within the replicas of a single comparator.

It might therefore be expected that this scheme can be improved by combining task diagnosis with comparator diagnosis. In the rest of this section we make the following assumptions concerning the behavior of a faulty comparator:

1. A faulty comparator produces either a 0 (stuck-at-0 fault) or a 1 (stuck-at-1 fault) outcome regardless of the relative values of the results that are compared.
2. A faulty comparator does not modify the results that it compares. Thus we assume here that a comparator fault is contained within the comparator and does not propagate to the output lines of the PEs connected to it.

We now consider the task diagnosability of low-order, comparison-based NMR systems where comparators may be faulty. We define a system with *n* PEs to be *f_1-f_2-task diagnosable* if at least one correct result can always be identified when up to f_1 results are incorrect and up to f_2 comparators are faulty. We now prove the following lemma for a simple 3-PE system.

Lemma 5.2 A 3-PE system is 1-1-task diagnosable with a number *C* of comparisons such that $C \geq 3$.

Proof. In order to prove that the system is not diagnosable for $C < 3$, we assume $C = 2$. Then if the two comparisons are not replicated, the system is organized as a chain, with comparator c_1 for the pair r_1, r_2 and comparator c_2 for the pair r_2, r_3. Assume r_1 is incorrect and c_1 is a stuck-at-0 faulty comparator. The resulting 00 syndrome is not distinguishable from the 00 syndrome produced by a fault-free system. Hence there is no way to guarantee that a correct result is selected, and the system is not task diagnosable.

Assume now that the two comparisons are performed on the same result pair, say r_1, r_2. Then, if one comparator is faulty, its output may be different from that produced by the non-faulty comparator. However, there is no way to tell which one is faulty and which one is fault-free. Therefore the comparison outcome is undecided, which leaves the possibility that any one of the PEs may be faulty. In this case the system is also not task-diagnosable, and this concludes the proof that task diagnosability cannot be achieved with 2 comparisons when one PE and one comparator may simultaneously be faulty.

We show now that the system can be task diagnosable with 3 comparisons, and we give the configurations for which this is achieved. It has already been shown that the system is task diagnosable if it is organized with one triplicated comparison. Another possibility is to have 3 distinct comparisons c_1, c_2, c_3 on the pairs r_1, r_2; r_2, r_3; r_3, r_1 respectively as illustrated in Figure 5.14c. But then the syndrome is 111 for any of the 3 fault patterns in which one pair such as r_1, r_2 produces mismatching results because the comparator is faulty, while the oppo-

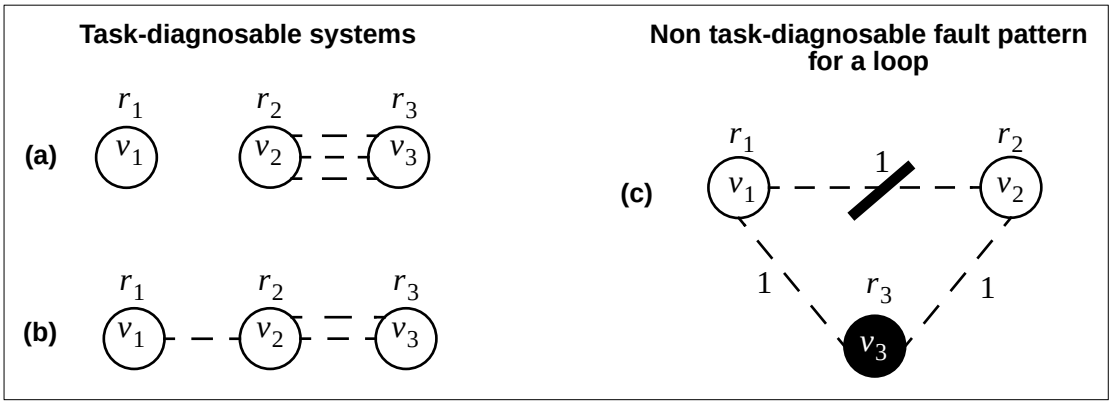


Figure 5.14 Task diagnosability with PE and comparator faults for a 3-PE system

site result such as r_3 is incorrect (Figure 5.14c). Hence the loop is not task diagnosable.

The last possibility consists in comparing on one pair such as r_1, r_2 with a single comparator, and comparing with duplicated comparators on another pair, such as r_2, r_3 (Figure 5.14b). Here the duplicated comparators are either fault-free or one is faulty and behaving as non-faulty if they produce the same result. Therefore the system is task diagnosed with the common comparison result, by ignoring the single comparison on r_1, r_2. If the duplicated comparators produce different outcomes, then one of them is faulty, and by inference, the single comparator on r_1, r_2 is fault-free, which allows the task diagnosis of the system. This proves that task diagnosis is possible with the configurations of Figure 5.14a and Figure 5.14b in the presence of up to one PE fault plus on comparison fault. □

It should be noted that the structure of Figure 5.14b is clearly superior to that of Figure 5.14a since it ensures fault detectability in addition to task diagnosis for the same number of comparisons.

5.6 Conclusions and contributions

This chapter has addressed the concept of comparison diagnosis in the framework of modular redundant systems. We started with a minor contribution by characterizing modular redundant systems including loops and chains (disjoint and continuous) for the symmetric comparison model. By considering this problem from the point of view of task diagnosis as opposed to processor diagnosis, we were able to propose methods for improving NMR systems through the exploitation of comparison diagnosis.

In order to justify the introduction of comparisons into an NMR system, several issues must be considered. We have addressed in this chapter the issues of imperfect comparators as well as

the potential complexity added to the system by this method. We have shown that the resulting comparison-based systems are not necessarily more complex than basic NMR systems.

We have introduced several theorems for task diagnosability, which are in effect extensions and applications of the theorems presented in Chapter 3 for the concept of $f/(n-1)$-diagnosability.

5.7 Related publications

[FCN94] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, "Modular Redundancy with Comparison Diagnosis," Technical Report 94/72, Swiss Federal Institute of Technology, Lausanne, October, 1994.

[FCN95e] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, "A Multiple-Task Approach to Efficient Fault Tolerance," in *Proceedings of the 1995 Pacific Rim International Symposium on Fault-Tolerant Systems*, pp. 106-111, Newport Beach, California, 1995.

[FCN96] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, "Fault Tolerance with Comparison-Based Modular Redundancy," in *Proceedings of the 14th IASTED International Conference Applied Informatics*, Innsbruck, Austria, 1996.

Chapter 6 Applying Comparison-Based, Modular Redundancy

In this chapter, we present the application of the theory proposed in the previous chapters of this dissertation. In Section 6.1 we present the concept of *N-Modular Redundancy with Comparison Diagnosis* (NMRCD), a technique which exploits an optimistic task execution policy as well as the concepts of task and processor diagnosis to better utilize the redundancy of a modular redundant system. In Section 6.2 we characterize NMRCD systems under the symmetric comparison model of system-level diagnosis. Then, in Section 6.3 we characterize NMRCD systems in the bounded symmetric comparison model, which we proposed in Chapter 4. In Section 6.4 we present a comparative analysis of NMRCD systems with respect to the NMR techniques. Finally, we conclude the chapter by summarizing the contributions in Section 6.5.

6.1 N-Modular Redundancy with Comparison Diagnosis

In this chapter we introduce the NMRCD (N-Modular Redundancy with Comparison Diagnosis) approach [FCN95c, FCN95d, FCN95a, FCN94] that we will show to be significantly more efficient than conventional NMR in terms of average throughput and execution delay. In essence this technique is based on the optimistic execution of several independent tasks as a group within a core NMR system. Typically, in an n-node system with $n = 2f + 1$, each task is executed on only $f + 1$ processors when the system is fault-free, that is, most of the time. When faults occur, additional redundant executions allow sorting the correct task results from the incorrect ones. With this scheme a comparison-based approach is much better suited than a majority voting circuit because it provides the flexibility to adapt the voting and diagnosis algorithm to a varying fault situation. The characterizations of the $f/(n-1)$-diagnosability and task-diagnosability that were given in the previous chapters apply directly to the NMRCD scheme. These characterizations define which structures are applicable to NMRCD and under which conditions they are applicable. In this section we extend the original NMRCD concept

to concurrent tasks with different degrees of fault tolerance, and we characterize NMRCD systems.

The degree of fault tolerance of a task is defined as follows:

Definition 6.1 The degree of fault tolerance *d* of a task is the minimum number of results computed by replicas of this task that must be identical in order for this result to be deemed as correct.

With this definition, a task that must be executed on an NMR system with $n = 2f + 1$ has a degree of fault tolerance $f + 1$.

To present this principle we consider two independent tasks t_1 and t_2 that must be executed in a fault-tolerant manner. We assume that the fault tolerance degrees of t_1 and t_2 are 2 and 3 respectively. The basic idea behind the NMRCD scheme is to group the execution of two or more independent tasks on a core NMR system with a number of nodes that is equal to that required for the task with the largest fault tolerance degree.

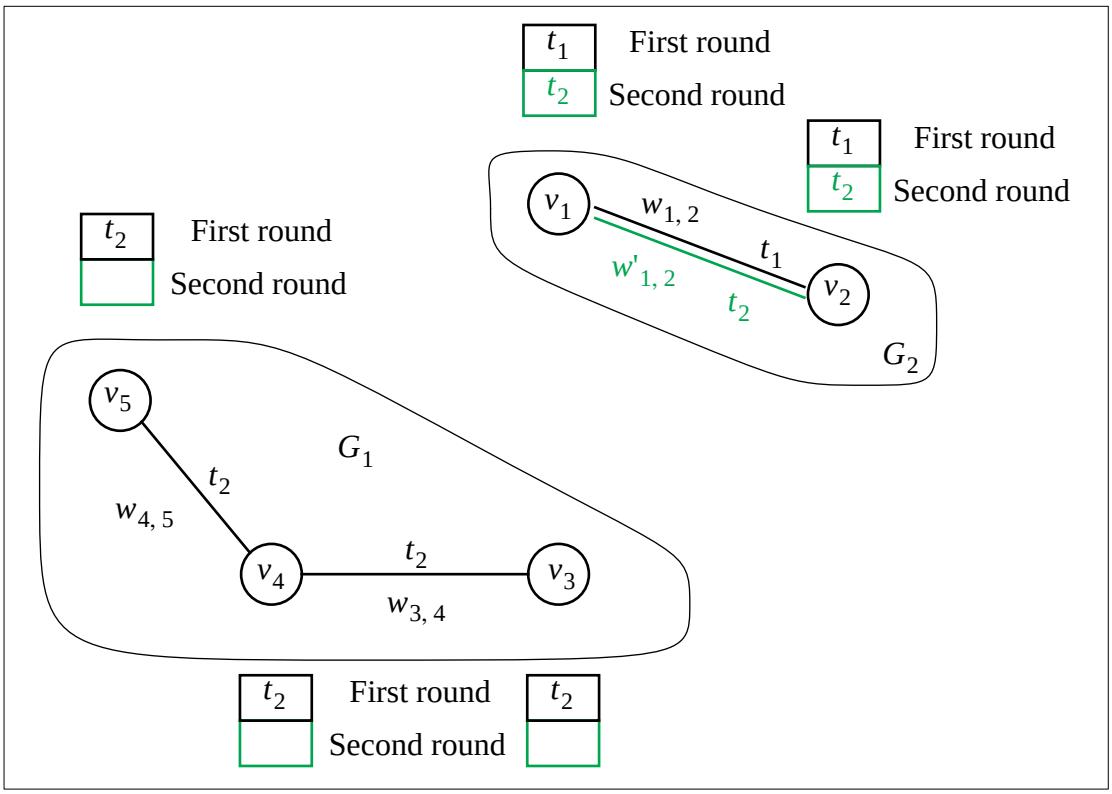


Figure 6.1 Example of testing configuration for NMRCD system

There are two tasks, and the system must have 5 nodes since t_2 with a degree 3 of fault tol-

erance should normally be executed on a 5-PE NMR system. Thus if t_1 and t_2 have approximately the same execution time we can adopt an optimistic execution policy where the tasks t_1 and t_2 are run concurrently on the 5-PE system, as shown in Figure 6.1. Task t_1 executes on v_1, v_2, while t_2 executes on v_3, v_4, v_5. Note that at the end of this first round of execution only the results produced by copies of the same task can be compared. Thus, in a comparison-based approach the comparison testing configurations is split in two disjoint testing configurations, one with v_1, v_2, and the other one with v_3, v_4, v_5. We assume here that comparisons are performed between v_1, v_2; v_3, v_4; v_4, v_5 and produce the binary comparison outcomes $w_{1,2}$, $w_{3,4}$, $w_{4,5}$ respectively. If the system is fault-free, v_1, v_2 produce an identical result, and $w_{1,2} = 0$ indicates this result as being correct, since only two identical results are sufficient for validation in a TMR system.

Similarly v_3, v_4, v_5 produce the same common result that is deemed to be correct in the context of a 5-PE NMR system. Thus the results from the execution of t_1 and t_2 can both be released at the end of the first round. Note that in order to produce the two results in a single round, 8 PEs would have been required with the conventional NMR approach, or equivalently two rounds of execution would be needed with a 5-PE system.

Consider now what happens when faults occur in the system of Figure 6.1. Up to two faults must be tolerated for t_2 and up to one fault must be tolerated for t_1. These faults are signaled by one or more 1s in the comparison outcomes $w_{1,2}$, $w_{3,4}$, $w_{4,5}$. Obviously a second round of execution is needed in order to ensure that the quorum of identical results is reached for both t_1 and t_2.

6.1.1 Multiple Task NMR

A first simple solution consists in simply executing during the second round each task on the processors on which it did not run during the first round (Figure 6.2). With this approach that we have introduced in [FCN95c, FCN95e, FCN95d] under the name of Multiple Task N-Modular Redundancy (MTNMR), the results of the two tasks are available in each node at the end of the second round, if those that were produced during the first round are stored. Thus task results can be compared between any two nodes and the testing subgraph is potentially fully connected. This means that the system is *f*-diagnosable, and that all correct and incorrect

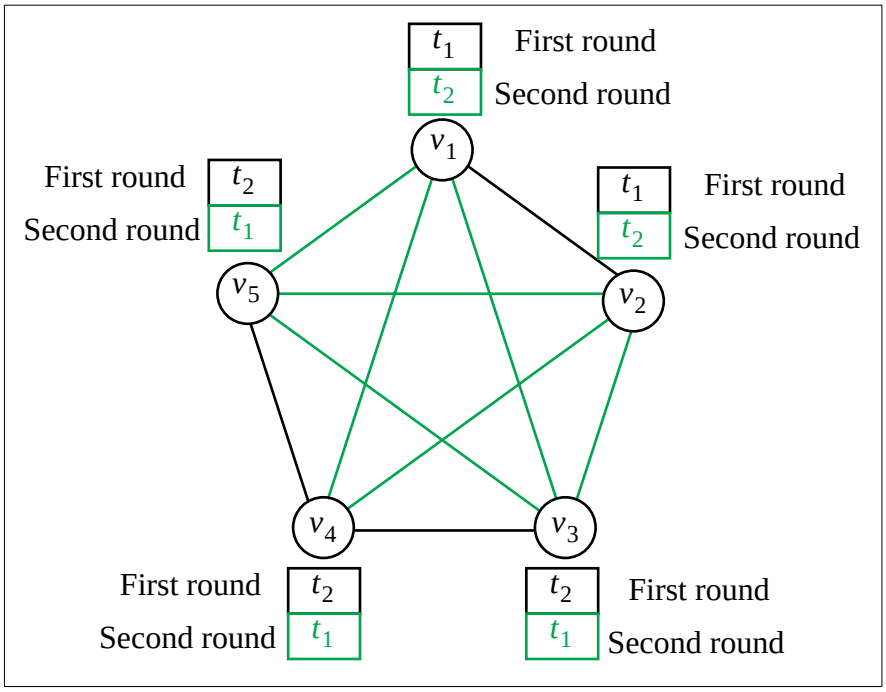


Figure 6.2 Example of testing configuration for MTNMR system

results can be identified provided no more than two results are incorrect.

The system releases in the worst case the two task results at the end of the second round, that is, in no more time than would be required with the conventional NMR approach. However, well designed systems rarely encounter faults, so that the MTNMR system operates with a single round of execution for two tasks most of the time. This yields a gain in throughput that is close to double that of a conventional system.

6.1.2 Adding task diagnosis

The MTNMR scheme can be significantly improved by resorting to task diagnosis (as described in Section 5.1.1 on page 120) in order to release at least one task result at the end of the first round, and in order to minimize the number of task replicas that must be executed in the second round, if any. We introduced this technique under the name of N-Modular Redundancy with Comparison Diagnosis (NMRCD) [FCN94]. We will first introduce informally this approach on the simple example of Figure 6.1. We will then use the basic theorems on $f/(n–1)$-and task-diagnosability to characterize the NMRCD technique and to present an extension of the NMRCD scheme to the case of tasks with different degrees of fault tolerance.

6.1.3 Example of NMRCD

Let us reconsider the 5-PE system of Figure 6.1 that produces the 3-bit syndrome $w_{1,2}$, $w_{3,4}$,

Table 6.1 Task diagnosis in the system of Figure 6.1 (t_1: $d = 2$, t_2: $d = 3$)

Syndrome			Diagnosis at the end of round 1	Executions at round 2
$w_{1,2}$	$w_{3,4}$	$w_{4,5}$		
1	0	0	t_1 undecided, t_2 released.	t_1 on v_5
0	1	0	t_1 released, t_2 undecided.	t_2 on v_1, v_2
0	0	1	t_1 released, t_2 undecided.	t_2 on v_1, v_2
1	1	0	t_1 undecided, t_2 released.	t_1 on v_5
0	1	1	t_1 released, t_2 undecided.	t_2 on v_1, v_2
1	0	1	t_1 undecided, t_2 released.	t_1 on v_3
1	1	1	t_1 undecided, t_2 released.	t_1 on v_5
0	0	0	t_1 released, t_2 released.	*none*

$w_{4,5}$ at the end of round 1. The seven possible syndromes corresponding to the occurrence of faults are shown in Table 6.1 with the corresponding diagnosis. Obviously the result of task t_1 can be released at the end of round 1 if $w_{1,2} = 0$, since the fault tolerance degree of t_1 is 2.

If $w_{1,2} = 1$ at least one of the nodes v_1, v_2 produced an incorrect result and the result of t_1 cannot be released. Then if $w_{3,4} = 0$ and $w_{4,5} = 0$, v_3, v_4, v_5 produced a common result for t_2 that is necessarily correct since the quorum of 3 identical executions has been reached. Finally if either or both of the two outcomes $w_{3,4}$ and $w_{4,5}$ are equal to 1, this implies that only one of the PEs v_3, v_4, v_5 has produced an incorrect result, since one has already been identified within the pair v_1, v_2, and at most two incorrect results are allowed in the system. Thus even in this worst case the result of task t_2 can be released. This implies that with any allowable fault situations at least one task result is released at the end of round 1.

At this point it is immediately apparent that the NMRCD approach is superior to the corresponding MTNMR scheme, since the release of at least one correct task result is guaranteed to be available at the end of round 1 with NMRCD, whatever the allowable fault situation. On the other hand, this is possible with MTNMR only when at least one of the groups of tasks is executed correctly on all its replicas. The reason for this difference lies in the fact that MTNMR

relies entirely on a single mechanism, quorum voting, for validating task results. NMRCD, however, supplements quorum voting with task diagnosis for validating one task result at the end of round 1 when quorum voting is defeated.

This result is intriguing since we use here the results of the execution of one task to validate the execution of another task. In order to check the validity of this approach, assume that there is the following syndrome at the end of round 1: $w_{1,2} = 0$, $w_{3,4} = 1$, $w_{4,5} = 1$. Then the result of task t_1 is validated under the quorum majority rule of 2 that applies to task t_1. The partial syndrome $w_{3,4} = 1$, $w_{4,5} = 1$ means that one or two results of task t_2 are incorrect and the validation is deferred until additional executions at round allow the validation of a correct result for t_2.

Since tasks t_2 and t_1 do not expose the PE faults identically, the syndrome $w_{1,2} = 0$, $w_{3,4} = 1$, $w_{4,5} = 1$ could very well have been produced when only one of the PEs v_3, v_4, v_5 is faulty and has produced one incorrect result on task t_2 while one of the PEs v_1, v_2 was faulty but its fault was not exposed by the execution of task t_1. But the validation of the result of task t_1 is then entirely justified in this case, since it is based on the quorum majority rule that applies to t_1.

Assume now that there is the following syndrome at the end of round 1: $w_{1,2} = 1$, $w_{3,4} = 1$, $w_{4,5} = 1$. Then one incorrect result of task t_1 has been exposed at the end of round 1, and if only hardware faults are allowed, one PE in the pair v_1, v_2 is faulty. Then at most another PE can be faulty in the group v_3, v_4, v_5 and with $w_{3,4} = 1$, $w_{4,5} = 1$ this faulty PE can only be v_4. Thus the diagnosis which points to the results of task produced by v_3, v_5 as being correct is justified. This behavior is the same whether the hardware faults be permanent, transient or intermittent since these faults do not differ for task diagnosis but differ in the way they are sorted and recovered from. The same approach remains valid if software faults are allowed provided no more than two hardware or software faults occur in the system.

We now proceed with our simple example by discussing the requirements for re-executions at round 2 (Table 6.1). We first note that if the result of task t_1 is undecided at the end of round 1, this corresponds to the case $w_{1,2} = 1$ where the two nodes v_1, v_2 have produced at least

one incorrect result. But then the execution of task t_1 on another PE is necessarily correct since t_1 is supposed to be executed in TMR mode where at most one fault is allowed. Hence when $w_{1,2} = 1$ the only requirement for t_1 in round 2 is a single additional execution of t_1, for instance on v_5. When $w_{1,2} = 0$, the result of task t_2 is undecided at the end of round 1 if at least one of the results produced by v_3, v_4, v_5 is incorrect. Then if t_2 is executed on v_1, v_2 in the second round, t_2 has been executed on the five processors in the two rounds, which guarantees the task diagnosis of the result of t_2 since the testing graph is a chain at the end of round 2.

Note that if we insist that the result of task t_1 be validated as a task of fault degree 2 even when two PEs are faulty, then t_1 must be executed on both v_3 and v_5 when the syndrome is $w_{1,2} = 1$, $w_{3,4} = 0$, $w_{4,5} = 0$ at the end of round 1.

It is evident that when two tasks that have a degree of fault tolerance of 2 and 3 respectively are executed as group on a 5-PE system in an NMRCD fashion, only a single round of execution is required when the system is non-faulty. Moreover, the two results are validated at the end of round 2, with no more than two executions at round 2. Then even in the worst case where a second round of execution is required, at least three PEs are available in this round for the execution of other tasks. Compare this now with what happens when the same two tasks are executed on the same 5-PE system. In this case task t_2 could be executed on the 5 PEs during round 1 and task t_1 on 3 PEs during round 2. Thus the system would use a total of 8 PE slots to process the two tasks and would release only one task result per round. On the other hand, the corresponding NMRCD system uses only 5 PE slots and releases the two task results in the first round when the system is non-faulty, that is, most of the time. When faults occur the NMRCD is still superior to the conventional NMR system since it uses at most 7 PE slots while still releasing one task result per round.

6.2 Characterization of NMRCD systems with the symmetric comparison model

The preceding example shows that task diagnosis on disjoint testing graphs is a key issue in designing efficient NMRCD systems. We now use the characterization theorems for $f/(n-1)$-diagnosis of Section 3.2 and the task-diagnosis characterization of Section 5.2.3 to establish

the existence conditions for efficient NMRCD systems. We begin with the following theorem that is a direct generalization of the preceding NMRCD example.

Theorem 6.1 Let t_1 and t_2 be two independent tasks that have approximately the same execution time on identical PEs and have fault tolerance degrees $f + 1$ and f respectively. Then the two tasks can be processed in an NMRCD system with n PEs, where $n = 2f + 1$ in at most two rounds of execution with the following behavior:

1. If the system is non-faulty, then correct results of the two tasks are released at the end of the first round.
2. if up to f faults occur in the system, then the correct result of one of the tasks is released at the end of the first round and the correct result of the second task is released at the end of the second round.
3. The system requires no more than $M = (3f^2 + 12f - 7)/8$ comparisons for f odd and $M = (3f^2 + 10f)/8$ comparisons for f even.
4. The comparison testing assignment is fixed.
5. In the worst case at most $3f + 1$ PE slots are used to process the two tasks over the two rounds.

Proof. In the first round task t_2 is executed on $f + 1$ PEs and task t_1 is executed on the remaining f PEs. At the end of the first round, the results of task t_2 can be compared two by two, and the results of task t_1 can also be compared two by two, thus forming two disjoint comparison testing subgraphs G_1 and G_2. Assume the two subgraphs are organized with comparisons between pairs of results, and a complete testing subgraph among pairs and among one isolated result in one of the subgraph, as stated for bi-corona systems in Theorem 3.9 (page 52).

Then if the global syndrome is all 0s, the quorum for majority voting has been reached for the two tasks, and the two correct task results corresponding to tasks t_2 and t_1 are released at the end of round 1. If some 1s appear in the syndrome, then some of the results are incorrect. But Theorem 3.9 guarantees task diagnosability provided no more than f faults have occurred. Thus one correct task result pertaining either to t_2 or t_1 is released at the end of round 1, and a second round of execution must be scheduled for the task whose result was not validated in

the first round. In the second round this task is executed on the PEs on which it was not executed in the first round, in such a way that this task has been executed at the end of the second round on $2f + 1$ PEs for t_2 and $2f - 1$ PEs for t_1. Assume for instance that task t_2 is executed in round 1 on the $f + 1$ PEs of subgraph G_1 and that its result cannot be released at the end of round 1. Then t_2 is executed at round 2 on the f PEs pertaining to subgraph G_2 (Figure 6.1). The task t_2 is used for the same comparison assignment that was used for t_1 in G_2 during the first round. Now there are two disjoint testing subgraphs for task t_2 that satisfy the conditions of Theorem 3.9. Hence t_2 is task diagnosable at the end of round 2 and a correct result is released for t_2. This proves points 1 and 2 of the theorem.

In order to address point 3 of the theorem, assume first that f is even. Then G_1 is structured as $f/2$ fully interconnected PE pairs that are themselves fully connected to one isolated PE, which yields $f(f + 6)/8$ comparisons. Similarly, G_2 is organized as $f/2$ fully interconnected pairs, giving $f(f + 2)/8$ comparisons. In the worst case t_1 is executed on G_2 in the first round and released at the end of this round. Then t_2 is executed on G_1 in the first round and executed on the f PEs in G_2 during the second round. For this case there are a total of $f(f + 6)/8 + 2f(f + 2)/8 = (3f^2 + 10f)/8$ comparisons. If f is odd, there are $(f^2 + 4f + 3)/8$ comparisons for G_1 and $(f^2 + 4f - 5)/8$ comparisons for G_2, which gives a total of $(f^2 + 4f + 3)/8 + (f^2 + 4f - 5)/8 = (3f^2 + 12f - 7)/8$ comparisons. Hence point 3 of the theorem is proven. Point 4 follows from the way the proof of point 3 was established.

Since the system executes in the second round on at most $f + 1$ PEs the task that was undecided in the first round, at least f PEs are available during the second round for executing other tasks. Hence at most $3f + 2$ PE slots are used over the two rounds to execute the group of two tasks. □

Note that with the example of the preceding section, $f = 2$. Thus we should have $M = 4$ comparisons and 7 PE slots in the worst case, which is exactly what we used in our example.

Up to now we have assumed that the degrees of fault tolerance of the two tasks were f and $f + 1$. We have already discussed [FCN94, FCN95a, FCN95c] the NMRCD approach for the

case where a group of *n* tasks with the same fault tolerance degree $f + 1$ is processed on a *n*-PE system with $n = 2f + 1$. We return briefly to this case in order to characterize more precisely the NMRCD scheme for tasks with identical degrees of fault tolerance. We first consider again the simple 5-PE NMRCD system of Section 6.1 (Figure 6.1). Assume that the two tasks t_1 or t_2 have the same fault tolerance degree $f + 1$. The only difference with the example is that the fault tolerance degree of t_1 is now $f + 1$ instead of *f*. This means that three identical results of t_1 are now needed to certify a result as correct. Therefore the result of task t_1 cannot be released at the end of round 1 if the system is fault free since t_1 has then been executed on only two PEs and the fault-diagnosis of t_2 does not provide any cues on the fault status of the two t_1 results. Thus an all 0 syndrome at the end of round 1 allows the release of task t_2. Task t_1 must then be executed in the second round on the $f + 1$ processors which did not process it at the first round. Apart from this difference the diagnosis at the end of round 1 is exactly the same as that of the preceding system (Table 6.1). Here, however, one task is always re-executed at the second round on either *f* or $f + 1$ processors. For the general case this gives the following theorem.

Theorem 6.2 Let t_1 and t_2 be two independent tasks that have approximately the same execution time on identical PEs and have the same fault tolerance degrees $f + 1$. Then the two tasks can be processed in an *n*-PE NMRCD system with $n = 2f + 1$ in at most two rounds of execution with the following behavior:

1. if up to *f* faults occur in the system, then the correct result of one of the tasks is released at the end of the first round and the correct result of the second task is released at the end of the second round.
2. The system requires no more than $M = (3f^2 + 12f + 1)/8$ comparisons for *f* odd and $M = (3f^2 + 14f)/8$ comparisons for *f* even.
3. The comparison testing assignment is fixed.
4. In the worst case at most $3f + 2$ PE slots are used to process the two tasks over the two rounds.

Proof. The proof is essentially the same as that of Theorem 6.1, the only differences being that only a single task result is released at the end of round 1 and that re-executions at round 2 always take place on at least f processors. □

If the two tasks had been processed sequentially in a conventional NMR mode, with one task executed on the n PEs at each round, one task result would have been released at the end of each of the two rounds, and $4f + 2$ computer slots would have been used instead of $3f + 2$ with the NMRCD scheme. In this case NMRCD do not allow for early release of tasks when the system is error-free but uses the PEs more efficiently.

We have shown in [FCN94, FCN95a, FCN95c] that the NMRCD approach can be implemented efficiently when a group of n tasks of fault tolerance degree $f + 1$ is processed on an n-PE system, with $n = 2f + 1$. In this case, the system releases an average of nearly two task results per round and in the worst case does not require more execution rounds than a conventional NMR system. The reader can refer to [FCN94, FCN95a, FCN95c] for details on this approach.

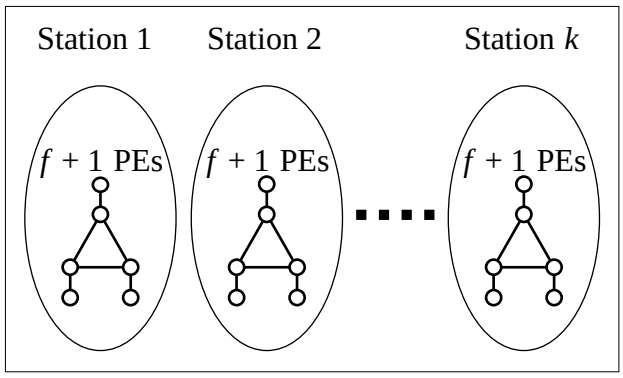


Figure 6.3 Comparison testing graph for a system defined by Theorem 6.2 for $f = 5$

Theorems 3.5 and 3.10 characterize $f/(n–1)$- or task-diagnosability for systems where $n > 2f + 1$. They show that when $n > 2f + 1$, task diagnosability can be achieved on structures with more than two disjoint comparison testing graphs. This gives a great flexibility in the allocation of tasks that in turn provides a rich set of design choices for NMRCD systems. Rather than exploring in detail the numerous possible alternatives for $n > 2f + 1$, we just consider here the special case of a system of k stations where each station contains $f + 1$ PEs as illustrated in Figure 6.3. Thus $n = k(f + 1)$, with $k > 1$. We will assume that the system executes a single task in each station and that a group of k distinct, independent tasks $t_1, \ldots, t_k$ of fault tolerance degree $f + 1$ is allocated for execution in the system. We will also assume that

the tasks have approximately the same execution time on the PEs. The NMRCD operation can be characterized as follows.

Theorem 6.3 Let a system with n PEs where up to f faults must be tolerated be organized as k stations of $f + 1$ PEs each, with $n = k(f + 1)$, $k > 1$. For f even, each station is structured as $f/2$ pairs of PEs, plus one isolated PE, with one comparison on each PE pair and a fully connected comparison testing graph among the pairs and the isolated PE. For f odd, each station is structured as $(f + 1)/2$ pairs of PEs, with one comparison on each pair of PEs and a fully connected testing graph among the pairs. There are k independent tasks (with approximately the same execution time) allocated for execution in the system. The k tasks are executed in the NMRCD mode with at most two rounds of execution under the following conditions:

1. If the system is non-faulty, then the correct results of the k tasks are released at the end of the first round.
2. If up to f faults occur in the system, then the correct results of $k - 1$ tasks are released at the end of the first round and the correct result of the undecided task is released at the end of the second round.
3. The system requires no more than $M = (k + 1)(f^2 + 6f)/8$ comparisons for f even and $M = (k + 1)(f^2 + 4f + 3)/8$ comparisons for f odd.
4. The comparison testing assignment is fixed.
5. In the worst case at most $(k + 1)(f + 1)$ PE slots are used to process the k tasks over the two rounds.

Proof. Each station is structured as a bi-corona system defined by Theorem 3.10, and therefore any pair of stations is task diagnosable, since the pair has only two disjoint subgraphs with the proper structure and no more than f results are incorrect out of a total of $2f + 2$ results. The diagnosis at the end of round 1 can be done as follows. Task diagnose any two stations. Then by Theorem 3.10, at least one correct result in one of the stations is validated. The diagnosis is continued in a similar manner, and it is easily seen that at most one task result remains unresolved at the end of round 1. Then the corresponding task is re-executed on another station at round 2 which guarantees the task diagnosis since the execution of the same task on two distinct stations is task diagnosable by Theorem 3.10. Note that the k task results are released at the end of the first round for a fault-free system since the quorum for majority voting is reached for every task. Thus conditions 1 and 2 are proved.

To establish condition 3, note that each station is implemented with $(f^2 + 6f)/8$ comparators. In the worst case the k stations execute the tasks in the first round and a single station must execute a task in the second round. Thus at most $M = (k + 1)(f^2 + 6f)/8$ comparisons are needed when f is even. If f is odd, each station is implemented with $(f^2 + 4f + 3)/8$ comparators, and this yields a total of $M = (k + 1)(f^2 + 4f + 3)/8$ comparisons in the worst case. This establishes condition 3.

Condition 4 follows immediately from the way the system is diagnosed. Since each station contains $f + 1$ PEs and at most one station re-executes a task at the second round, at most $(k + 1)(f + 1)$ PE slots are needed to execute the k tasks. □

We remark immediately that the NMRCD systems defined by Theorem 6.3 are optimum in the sense that the stations cannot contain less than $f + 1$ PEs. If the stations contain less than $f + 1$ PEs, then task diagnosis cannot be performed on a pair of two stations, since there may be up to f faults in a subsystem with less than $2f + 1$ PEs. Task diagnosis cannot be performed either on subsets of more than two stations, as proved by Theorem 3.8 (page 48). Thus if the stations contain no more than f PEs, the system is not task diagnosable in the first round and no task result can be released at the end of the first round. Hence, even if the system is non-faulty, with all-zero syndromes in each station, every task must be re-executed on some PEs at the second round. This is a very undesirable behavior because the time response of the system is then the longest in the most probable case, which is that of a non-faulty system.

One could also think of using more than $f + 1$ PEs in each station in the hope of ensuring that all task results are released at the end of the first round. This certainly increases the chance that all task results are released in a single round, but in the worst case one task must be re-executed at the second round if the number of PEs is less than $2f + 1$. In particular this is always the case when f faults occur in a single station, since the number of PEs is less than $2f + 1$. Thus the NMRCD systems are also optimum in the sense that no similar system with more than $f + 1$ PEs in each station guarantees that one additional round will never be re-executed in a second round. The only way to guarantee with this fault model that all results are released in one round is to fall back to a conventional NMR system where all stations contain at least $2f + 1$ PEs.

It can be seen easily that the system defined by Theorem 6.3 has a throughput that is about twice that of a comparable system operating in the conventional NMR mode, since each task is

executed on only $f + 1$ PEs in most cases, as opposed to $2f + 1$ PEs with the NMR scheme.

6.3 Characterization of NMRCD systems with the BSC model

In Chapter 4 we introduced the bounded symmetric comparison (BSC) model for comparison-based, system-level diagnosis. The BSC model represents large, fault-tolerant systems in a way that is more realistic than either the symmetric or asymmetric comparison model of system-level diagnosis. In this section we apply the BSC model to NMRCD systems.

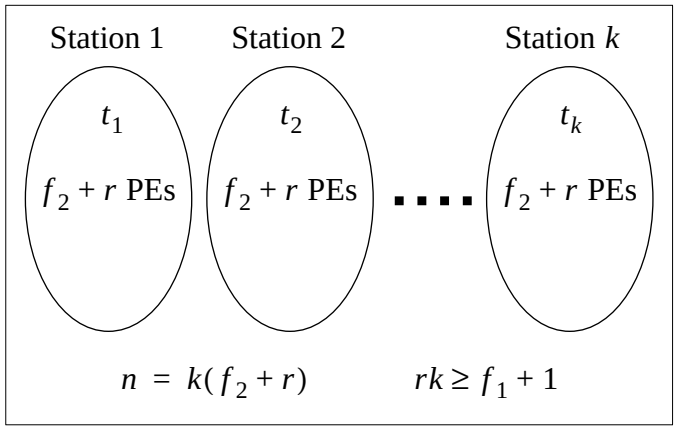


Figure 6.4 NMRCD system under the BSC model

Consider a system of *k* stations, where each station contains $f_2 + r$ PEs, with $r \geq 1$. Each station *i* is assigned a distinct, independent task t_i as illustrated in Figure 6.4. Under the BSC model, at most f_1 PEs can be faulty, but no more than f_2 PEs are allowed to produce identical, incorrect results that can yield comparison outcomes of 0. Note that for NMRCD systems, the limit of f_2 identical, faulty results obviously applies only to PEs in the same station, since comparisons are not performed between PEs that have not executed the same task. Thus $n = k(f_2 + r)$, with $k > 1$. The group of *k* distinct, independent tasks t_1, …, t_k are of fault tolerance degree $f_2 + 1$. We also assume that the tasks have approximately the same execution time on the PEs. The NMRCD operation described above can thus be characterized as follows.

Theorem 6.4 Consider a system with *n* PEs where up to f_1 faults must be tolerated. The system is organized as $k \geq 1$ stations of $f_2 + r$ PEs, with $n = k(f_2 + r)$, $rk \geq f_1 + 1$, $r \geq 1$. Futhermore, *k* independent tasks t_1, …, t_k of fault tolerance degree $f_2 + 1$ with approximately the same execution time are allocated for execution in the system. Initially one task is

allocated to each station. The *k* tasks are executed in the NMRCD mode with at most two rounds of executions under the following conditions:

1. If the system is non-faulty, then the correct results of the *k* tasks are released at the end of the first round.
2. If up to f_1 faults occur in the system, the correct results of $k - \lfloor f_1/r \rfloor$ tasks are released at the end of the first round and the correct result of the undecided tasks are released at the end of the second round.
3. The comparison testing assignment is fixed.
4. In the worst case at most $(k + \lfloor f_1/r \rfloor)(f_2 + r)$ PE slots are used to process the *k* tasks over the two rounds.

Proof. Point 1 is an obvious consequence of the fact that when no faults occur, the system can produce one task per station per round, for a total of *k* tasks at the end of the first round.

To address point 2, consider that the PEs in each station form a fully connected comparison testing graph. Note that under the BSC model, a station produces a result in the first round if and only if there is a cartel of more than $f_2 + 1$ nodes in the station. In other words, because of the degree of fault tolerance of the task, at least $f_2 + 1$ results must be identical in the station for it to produce a correct task result in one round. In the same vein, for a station of $f_2 + r$ PEs *not* to produce a correct result in a single round, at least *r* faulty PEs must be in the station. Thus for the worst case, where f_1 faults occur, assume that *r* of the f_1 faults occur in at most $\lfloor f_1/r \rfloor$ stations. Hence the correct results for $k - \lfloor f_1/r \rfloor$ tasks are released at the end of the first round. It is obvious that the correct results of the $\lfloor f_1/r \rfloor$ undecided tasks are released at the end of the second round in a way similar to that described in the proof of Theorem 6.1.

Point 3 is proven as follows. First, each task requires at most two rounds of execution. The first round occurs only on a single station, for which the testing assignment is obviously fixed. For the second round, the task must be executed on PEs which have not already executed. Again this testing assignment is decided *a priori* and is therefore fixed.

Finally point 4 of the characterization is proven by considering the worst case where the most tasks need to be re-executed in the second round. As a consequence of point 2 this occurs when there are the most number of stations that do not produce correct task results in the first round. These $\lfloor f_1/r \rfloor$ undecided tasks must be re-executed in the second round in order to obtain a task diagnosis. According to Theorem 4.3 (page 70) and Theorem 4.6 (page 83) which

correspond to task diagnosability under the BSC model, at least $f_1 + f_2 + 1$ task results are needed for each task to be surely diagnosed in the pessimistic case. Thus, in the worst case, at most $\lfloor f_1 / r \rfloor (f_1 + f_2 + 1)$ PE slots are needed. □

We demonstrate a system defined by Theorem 6.4 with the simple example where $f_2 = 2$ and $r = 2$ as illustrated in Figure 6.5. There are *k* tasks whose degree of fault tolerance is 3. Each of these tasks is executed on a station of only $f_2 + r = 4$ PEs, for a total of *k* stations. We compare a 4-PE system to that of a conventional 5MR operation which would be required for tasks of an equivalent degree of fault tolerance. Only $f_2 = 2$ faults are allowed to produce the same incorrect result, but assume that up to $f_1 = 3$ faults may occur in the system. The system can then be structured with $k = 2$ stations of $f_2 + r = 4$ PEs that execute respectively the tasks t_1 and t_2 in the first round.

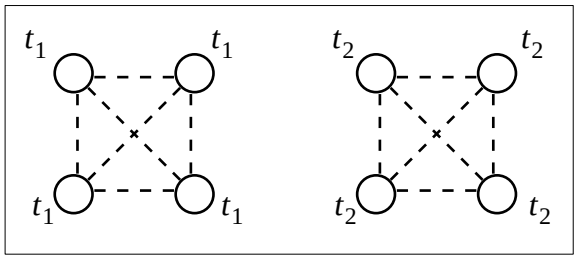


Figure 6.5 Example NMRCD system under BSC model with $f_1 = 3$, $f_2 = 2$, $n = 8$

In the worst case, since $f_1 = 3$, only one station can contain more than one faulty node and hence not produce a correct result. Thus at least one task is always released at the end of the first round. If necessary, the task that was not released in the first round can be re-executed on just two PEs of the station on which it was not executed in the first round. For this task, over the span of two rounds of execution, there is a fully connected testing graph with $f_1 + f_2 + 1 = 6$ results, and therefore the task diagnosability is guaranteed by Theorem 4.3 (page 70) and Theorem 4.6 (page 83).

In the best case only 8 executions are needed, instead of 10 for a conventional NMR system with two tasks and 5 PEs. Furthermore, the worst case situation is no worse than with a conventional NMR system.

Assume now that up to $f_1 = 5$ faults may occur in the system. The system can be structured with three stations of 4 PEs that execute each one of three tasks in the first round. In the

worst case only one station contains more than one faulty node, so that at least one task is always released at the end of the first round. If needed, each of the two tasks that were undecided at the end of round 1 may be re-executed on 4 PEs in round 2. In this case each undecided task will have been executed on a total of $f_1 + f_2 + 1 = 8$ PEs by the end of round 2 and will be task diagnosable by Theorem 4.3 and Theorem 4.6.

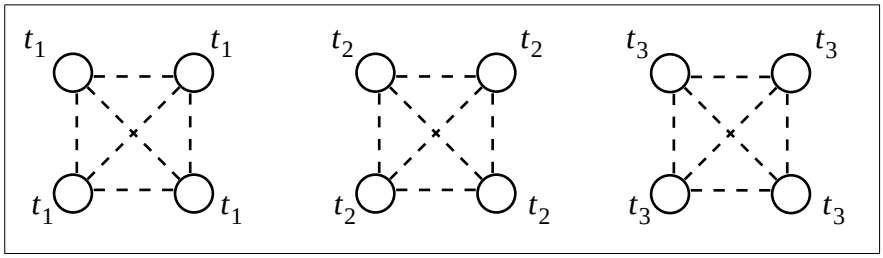


Figure 6.6 Example NMRCD system under BSC model with $f_1 = 5$, $f_2 = 2$, $n = 12$

6.4 Comparative analysis of NMRCD systems

6.4.1 Contrast of NMRCD with related techniques

The NMRCD approach corresponding to Theorem 6.3 is similar to the technique introduced by Lombardi [Lom84, Lom85]. In both cases the main part of the system is a set of k stations where each station executes a single task on $f + 1$ PEs, and partial diagnosis (called task diagnosis in this chapter) is used to release most (at least $k - 1$) of the tasks at the end of the first round. The two approaches differ, however, in the final diagnosis of the task(s) that were not released at the end of the first round. In our case, at most one single task is undecided at the end of the first round. Then the decision on the correct task result is reached by re-executing the task on another station during the second round and task diagnosing the results of this execution with those obtained for the same task on another station at the first round.

On the other hand, the system proposed by Lombardi [Lom84, Lom85] uses an auxiliary $(3f + 1)$-PE off-line control subsystem for resolving the task results that are left undecided at the end of the first round. The off-line control subsystem executes a Byzantine agreement to assert one of its PEs as fault-free. This fault-free PE is then used to diagnose the on-line PEs on which the undecided tasks were executed.

The Lombardi approach has several drawbacks with respect to our scheme. Firstly, it incurs a hardware overhead of $3f + 1$ additional PEs that becomes negligible only for large systems. Secondly the final diagnosis is adaptive, which is not well suited for a hardware implementa-

tion since all possible comparisons among the $k(f+1)$ on-line PEs and the $3f+1$ off-line PEs must be implemented either directly or through complex switching circuitry. Thirdly, if the final diagnosis is performed by comparison, the time required for releasing the last tasks cannot be less than in our approach, since it corresponds to running the Byzantine agreement algorithm plus the execution time of at least a task on one fault-free off-line PE.

This delay can be reduced if the off-line, fault-free PE exercises a diagnostic test to assess the fault status of the suspected on-line PEs. However, in this case the fault status of the results produced by the suspect on-line PEs is assessed indirectly through the determination of the fault status of the PEs that produced the results. Thus task diagnosis is replaced by processor diagnosis, and an incorrect task result can be released if the processor diagnosis did not expose the fault that produced an incorrect result. This approach can also be defeated when the faults are transient or intermittent. Such a fault could have caused an incorrect result and then be latent when the processor diagnosis is performed. In this case there would also be a risk of certifying a result that is in fact incorrect. On the other hand, the complete operation of our NMRCD system is based on task diagnosis, which allows circumventing any problem related to fault coverage of diagnostic tasks and to the existence of transient or intermittent faults.

6.4.2 Fault detection in NMRCD systems

We remark that an NMRCD system is obviously fault detectable when each task is executed on a number of PEs that is at least equal to its degree of fault tolerance, since at least one result is correct and compares to 1 with any incorrect result. With this definition, all NMRCD systems presented in this section are fault detectable. However, this raises a subtle question when tasks of different degrees of fault tolerance are executed in the system. If we take the example of Figure 6.1, there is a 5-PE system that executes task t_2 of fault tolerance degree 3 on PEs v_3, v_4, v_5 and task t_1 of fault tolerance degree 2 on v_1, v_2. Assume now that the execution of t_1 produces two incorrect results with the same value. Then two faults have occurred, but the comparison outcome $w_{1,2} = 0$ does not allow the faults to be detected. The allowed number of faulty PEs in the system has not been exceeded, but the fault tolerance of task t_2 has. Therefore if we want to ensure fault detectability in all circumstances, task t_2 should be executed on 3 PEs instead of 2.

Fault detectability ensures that at least one faulty PE is detected and also that it is identified. Once the task diagnosis is complete there is one correct result for each task. Since each task

group is connected, the correct task compares at least to another task that compares to another task and so on until an incorrect task is identified, if any. Since the incorrect task result can only be produced as a consequence of a hardware or software fault on the corresponding PE, a repair or switch-over action can be prompted on this PE.

Note, however, that the system is sequentially task diagnosable only if the testing graph is connected, as shown in Theorem 3.11 (page 57). This means that in a system where disjoint comparison testing graphs are used for fault tolerance, a few additional comparisons are required to connect the disjoint testing graphs. In the system of Figure 6.3 this can be achieved by executing each task t_i on station S_{i+1} in order to connect the k stations when a fault is detected. This requires an additional $k-1$ comparisons that may be predefined, since connecting any two PEs of two disjoint subgraphs is sufficient to satisfy the conditions defined by Theorem 3.11.

6.4.3 Reliability

In the preceding sections we have presented many new variants of NMRCD systems which complement the original structures that we have previously introduced [FCN95c, FCN95d, FCN95a, FCN94]. A common feature of all these systems is that for a given number of PEs, more job tasks per unit time can be processed than with a counterpart NMR system. This basic advantage can be traded either to increase the system throughput or the time response or both. Most of NMRCD systems have a performance advantage over NMR systems that is maximum when the system is fault-free, and no worse than NMR systems when the system is faulty. Thus this design is heavily balanced in favor of fault-free systems. This is entirely justified, for any well designed system is non-faulty most of the time.

We give now a reliability analysis [FCN95c, FCN95d, FCN95a, FCN94] which shows that in most practical situations, the throughput of a NMRCD system can be nearly 1.5 to 2 times that of the corresponding NMR system, with a similar gain in average response time. In this section, we analyze the reliability of the NMRCD system where $n = 3$ and compare it to TMR (triple modular redundancy).

In order that a TMR system produce a correct result, at least two elements must be fault-free. The reliability function of a TMR system with majority voting is defined as [Nus87]:

$$R(t) = R_d(t)[3R_1^2(t) - 2R_1^3(t)] \tag{6.1}$$

where $R_d(t)$ and $R_1(t)$ are the respective reliability functions for the voting unit and the redundant modules (assuming identically reliable modules). The NMRCD system also produces the correct results for the three tasks assigned to it provided at least two processors are fault-free. Hence the reliability $R(t)$ of the NMRCD system is the same as that of a conventional system given by equation (6.1), provided the reliability $R_d(t)$ of the TMR voting unit is the same as the reliability of the CDU in the NMRCD system. The differences will usually be small since the CDU is sufficiently simple to be built for high reliability, possibly by using triplication. Moreover, a Software Implemented Fault Tolerance (SIFT) [WLGG78] approach can be adopted for the NMRCD system by executing in each processor a small routine that performs the comparisons and the diagnosis. In this case, the CDU disappears almost entirely and the system reliability reduces to

$$R(t) = 3R_1^2(t) - 2R_1^3(t) \tag{6.2}$$

This result may seem to be surprising since the TMR system requires that only two out of three executions of the same task be correct, while the NMRCD system normally executes the same task on only two processors. However, there is no contradiction here because any single, faulty task execution is either correctly diagnosed by the NMRCD diagnosis algorithm, or at least one processor is diagnosed as fault-free, which guarantees a correct task result after executing the task on this processor in an extra round.

It can be shown by the same argument that the MTBF (Mean Time Between Failures) and the availability of the NMRCD system are the same as that of the TMR system for comparable failure rates λ, repair rates μ and fault coverage c. The fault coverage is of particular importance since a low fault coverage unduly delays the detection of a faulty processor. If we assume that all tasks have the same fault coverage even though they can expose different faults, then the processor fault detectability is the same for the two systems because each processor executes a distinct task at each round in both cases.

6.4.4 Throughput and Latency

We compare the throughput of the NMRCD system with respect to NMR for $n = 3$. It is obvious that when no faults are detected by the tasks, the throughput of the NMRCD system is 50% higher than the case of TMR, since the NMRCD system executes three tasks in two rounds, as opposed to only two tasks for the TMR scheme. This throughput advantage also translates into

lower latency—the NMRCD system produces one task result after the first round of execution and two task results after the second round of execution, while the TMR system produces only one task result at the end of each round. When a single processor is faulty and behaves as a faulty processor for the tasks that it executes, then the NMRCD system may or may not require a third round of execution depending upon the syndrome. In the worst case, the total execution time for three tasks is the same as for the two systems.

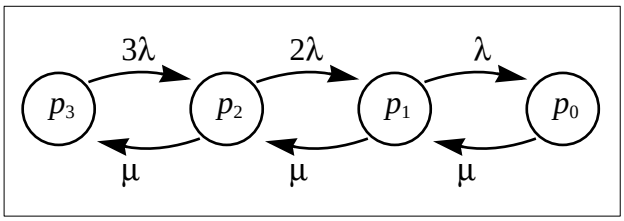


Figure 6.7 Markov diagram of a TMR system with one repair person

We seek to determine the average execution time for the NMRCD system, taking into account the failure rate and the repair rate of the replicated processors. The actual average execution time also depends upon the coverage factor *c*, but in this thesis we just establish a crude lower bound on average execution time by assuming that a third round of execution is required every time a single processor is faulty, with probability p_2. Then, if p_3 is the probability for the three processors to be fault-free, the average time *T* taken to produce three tasks results is given by

$$T = \frac{2p_3}{p_2 + p_3} + \frac{3p_2}{p_2 + p_3} \tag{6.3}$$

In this case, p_2 and p_3 are evaluated as with a simple TMR system with one repair person. Assuming p_0, p_1, p_2, p_3 are respectively the probabilities that zero, one, two and three processors are operational, the steady-state relationships among these probabilities are computed from the Markov diagram of Figure 6.7 by

$$p_0 + p_1 + p_2 + p_3 = 1 \tag{6.4}$$

$$3\lambda p_3 + \mu p_1 = (\mu + 2\lambda) p_2 \tag{6.5}$$

$$2\lambda p_2 + \mu p_0 = (\mu + \lambda) p_1 \tag{6.6}$$

$$\lambda p_1 = \mu p_0 \tag{6.7}$$

This gives

$$p_2 = \frac{3\lambda\mu^2}{\mu^3 + 3\mu^2\lambda + 6\mu\lambda^2 + 6\lambda^3} \tag{6.8}$$

$$p_3 = \frac{\mu^3}{\mu^3 + 3\mu^2\lambda + 6\mu\lambda^2 + 6\lambda^3} \tag{6.9}$$

With equation (6.3), the average time to execute three tasks becomes

$$T = \frac{2\mu^3 + 9\lambda\mu^2}{\mu^3 + 3\lambda\mu^2} + \frac{2\mu + 9\lambda}{\mu + 3\lambda} \tag{6.10}$$

This worst-case average execution time is plotted in Figure 6.8 as a function of λ/μ. It can be verified that except for very high (and somewhat unrealistic) values of λ/μ, the system takes on the average only slightly more than two time units to process three tasks. Taking, for instance, $\lambda = 1/1000$ hours and $\mu = 1/10$ hours yields only $T = 2.03$ time units, which shows a throughput advantage close to 50% when compared to TMR.

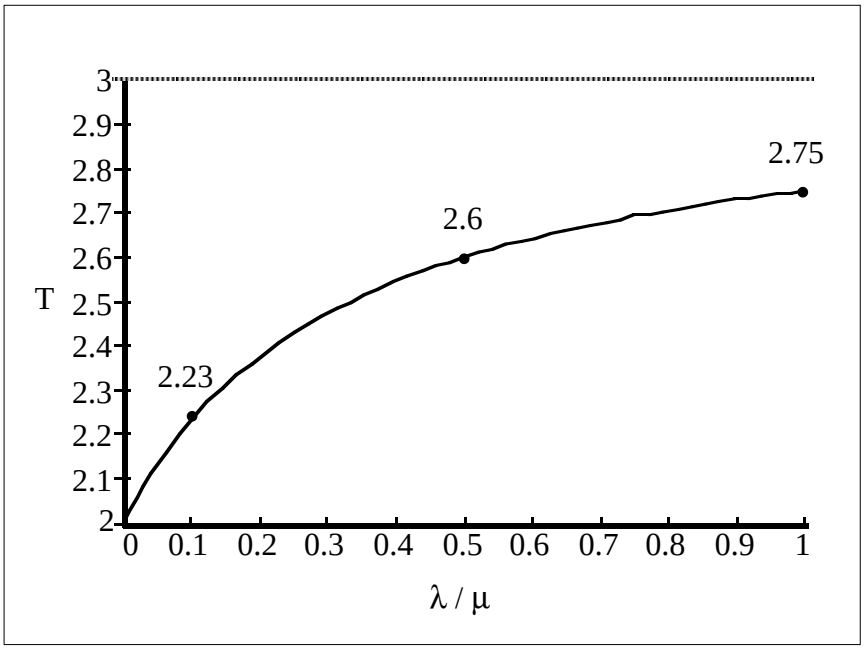


Figure 6.8 Lower bound on average execution time for a set of three tasks in NMRCD system

6.4.5 Fault tolerance

It should be stressed that this advantage is not obtained at the expense of relaxing the fault model for task execution. We still require that $f + 1$ task replicas produce the same result in order to validate this result as correct, even though this majority voting quorum may be derived by inference.

Another very important point concerning the fault model is that the validation of task results is based entirely on task diagnosis. A task is not validated indirectly because the PE on which it executed was deemed to be fault-free by a processor diagnosis procedure. As a result the NMRCD approach can tolerate transient and intermittent faults as well as permanent faults, which would not be the case if the system were relying on processor diagnosis to validate tasks.

6.4.6 Complexity

One could argue that the gains of an NMRCD system are achieved at the expense of increasing the complexity of the system. We contend that this is not the case. The systems discussed in sections 6.1-6.2 are all designed in such a way that the comparison testing graph is split into several disjoint testing subgraphs, and that re-executions, if any, take place within the existing pattern of comparison testing. As such there is no need for adaptive, comparison diagnosis, and the NMRCD system can be realized efficiently in hardware by implementing a comparator for each link in the comparison testing graph.

Since a key feature of NMRCD is to take a core NMR system and run several tasks on it instead of one, the complexity in terms of comparators and tri-state gates is about the same as that of the corresponding NMR system. This can be verified directly by considering the system of Figure 6.1. The 5-PE NMRCD system is implemented with just 3 comparators and 3 tri-state gates, and any re-execution at the second round uses the existing comparators. If we take now the system of Figure 6.3, it turns out that the NMRCD system is simpler than its NMR counterpart. We have seen with Theorem 6.3 that an NMRCD system with $k(f + 1)$ PEs requires at most $(k + 1)(f^2 + 6f)$ comparisons. Since the comparison assignment is static, the system is implemented with $(k + 1)(f^2 + 6f)$ comparators. The comparable NMR system is implemented with k stations of $2f + 1$ PEs, where each station operates independently as a single NMR system. Thus there are about twice as many PEs in the NMR stations, and this yields about four times as many comparators, since the complexity of task diagnosis is $O(n^2)$.

Even if task diagnosis has $O(n)$ complexity, as conjectured in [Xu91], the NMR system is likely to require twice as many comparators as its NMRCD rival. The diagnosis itself is slightly more complex in the case of the NMRCD system, but this concerns only bit-level computations instead of word-level computations. Thus the penalty over the NMR technique is likely to be small.

6.4.7 Scalability

Another important feature of NMRCD systems is that they scale down well, because no off-line control subsystem is required to perform auxiliary tests aimed at finalizing the diagnosis (as opposed to techniques such as Lombardi's [Lom84]). Thus an NMRCD system can be implemented either with the same number of PEs as the corresponding NMR system, with increased throughput, or with fewer PEs for the same throughput.

In the case of systems defined by Theorem 6.3, the system can be reduced to a simple two-station architecture, with a total of $2f + 2$ PEs. As such the system throughput is about twice that of a conventional NMR system for a fault tolerance capacity which is equalled.

6.4.8 Independence of tasks

The main drawback of NMRCD systems is that the computation is performed on successive groups of tasks, where the tasks within a group must be independent and have approximately the same execution time. In most cases the size of the task groups is small so that this constraint is not severe. For instance, in the case of a simple two-station system with $2f + 2$ PEs, the execution takes place by successive groups of only two tasks. Moreover most existing fault-tolerant systems [HSLJ78, WLGG78, WKF85, KWFT88] are already designed to process several tasks in parallel and to reallocate tasks to processors when faults occur.

The task independence requirement issues in NMRCD systems are addressed in more detail in [FCN95c, FCN95d, FCN95a, FCN94].

6.5 Conclusions and contributions

In this chapter we have described how to apply several of the theorems for task diagnosability defined in Chapter 5, which are in effect extensions and applications of the theorems presented in Chapter 3 for the concept of $f/(n–1)$-diagnosability. Among the applications of these theorems are the concepts of Multiple-Task N-Modular Redundancy (MTNMR) and N-Modular

Redundancy with Comparison Diagnosis (NMRCD). These two methods that allow an NMR system to achieve a better throughput and lower latency compared to primordial NMR systems with the same number of redundant elements.

We have characterized NMRCD systems in the symmetric comparison model, and we have contrasted our technique with other related techniques. A comparative analysis of NMRCD/MTNMR systems shows that the gains of the approach do not compromise the goals of fault tolerance initially present in a basic NMR scheme with the same number of redundant elements. The albeit minor drawbacks of an NMRCD/MTNMR system have been considered. When applying our methods to practical NMR systems, it is evident that the drawbacks are not necessarily problematic.

Furthermore, we consider NMRCD systems with respect to the BSC model introduced in Chapter 4. Performing NMRCD under this model (when $f_2 < f_1$) allows more faulty processors to be tolerated for the same number of processors (or requires a lower total number of processors to tolerate the same number of faulty processors). We demonstrate how the theorems of *f*/(*n*-1)-diagnosability assures a higher throughput in these systems compared to their conventional NMR counterparts.

6.6 Related publications

[FCN94] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, “Modular Redundancy with Comparison Diagnosis,” Technical Report 94/72, Swiss Federal Institute of Technology, Lausanne, October, 1994.

[FCN95a] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, “Efficient NMRCD Scheme for Fault Tolerance in Responsive Systems,” in *Proceedings of the Second International Workshop on Real-Time Computing Systems and Applications*, pp. 203-207, Tokyo, Japan, 1995.

[FCN95b] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, “Implementing Multiple Task Modular Redundancy,” in *Proceedings of the 2nd IFAC Workshop on Safety and Reliability in Emerging Control Technologies*, Daytona Beach, Florida, 1995.

[FCN95c] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, “Fault Tolerance with Multiple Task Modular Redundancy,” in *Proceedings of the Third IFAC/IFIP Workshop on Algorithms and Architectures for Real-Time Control*, Ostend, Belgium, 1995.

[FCN95d] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, “Hardware/Software Fault Tolerance with Multiple Task Modular Redundancy,” in *Proceedings of the IEEE Symposium on Computers and Communications*, pp. 171-177, Alexandria,

Egypt, 1995.

[FCN95e] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, “A Multiple-Task Approach to Efficient Fault Tolerance,” in *Proceedings of the 1995 Pacific Rim International Symposium on Fault-Tolerant Systems*, pp. 106-111, Newport Beach, California, 1995.

[FCN96] C.P. Fuhrman, S. Chutani and H.J. Nussbaumer, “Fault Tolerance with Comparison-Based Modular Redundancy,” in *Proceedings of the 14th IASTED International Conference Applied Informatics*, Innsbruck, Austria, 1996.

Chapter 7 Conclusions

This chapter summarizes the work we have presented in this dissertation and the contributions we have made. The reader can find a detailed description of the contributions at the end of each of the preceding chapters.

The goal of this thesis was to propose schemes based on comparisons in multiprocessor systems such that the inherent abundance of processing elements of these systems can be used for both reliability and efficiency, but in a way that is better than existing methods. We have achieved the goal of the thesis; we presented the comparison-based schemes of MTNMR and NMRCD in Chapter 6. These methods incorporate contributions made throughout the dissertation.

The ideas of task diagnosis presented in Chapter 5 are important for maximizing the efficiency of multiprocessor systems where comparison diagnosis is performed. The new model for comparison-diagnosis presented in Chapter 4 facilitates the diagnosis of multiprocessor systems with a large number of processors. Finally the advances introduced in Chapter 3 in the domain of $f/(n\text{-}1)$-diagnosability, which assures the theoretical lower bound on the system throughput, require fewer comparisons than previous techniques.

7.1 Summary of results

Chapter 3 presented several results aimed at exploiting the $f/(n\text{-}1)$-diagnosability concept of system-level diagnosis as a method for achieving fault tolerance. It extends the characterization of $f/(n\text{-}1)$-diagnosable systems under the symmetric comparison model. A new class of $f/(n\text{-}1)$-diagnosable structures entitled bi-corona systems is proposed, along with an algorithm for diagnosing these systems.

Chapter 4 considered models for comparison-based, system-level diagnosis. It shows how the symmetric comparison and asymmetric comparison models do not apply practically to systems with a large number of processors. Essentially using one model or the other forces the

system designer to consider all of the potentially faulty processors as behaving in the same way. Chapter 4 provided a far more flexible compromise between these two models: a new model for comparison-based, system-level diagnosis that we call the bounded symmetric comparison (BSC) model.

This new model encompasses the two of the existing models for comparison-based systems—the symmetric and asymmetric comparison models. Numerous results in this chapter show that the two well known models are in fact special cases of the BSC model. The superiority of the BSC model over the symmetric and asymmetric comparison models is that it covers the uncommon ground between the two models. This uncovered territory becomes evident when considering comparison diagnosis in systems with a large number of processors and where these processors are not dedicated to running a single task.

From a more general viewpoint, the issues of fault detectability and locatability for BSC model are considered. We obtain results in the form of theorems for the minimum number of comparisons necessary to detect and localize faults under the assumptions of the BSC model.

Chapter 5 addressed the concept of comparison diagnosis in the framework of modular redundant systems. By considering this problem from the point of view of task diagnosis as opposed to processor diagnosis, we can propose methods for improving NMR systems through the exploitation of comparison diagnosis. The key result obtained in this chapter is that of releasing task results as early as possible in order to improve the throughput of the system.

Starting with a minor contribution, Chapter 5 characterizes modular redundant, comparison-based systems including loops and chains (disjoint and continuous) for the symmetric comparison model. Furthermore it addresses the issues of imperfect comparators as well as the potential complexity added to the system by this method. It is shown that the resulting comparison-based systems are not necessarily more complex than basic NMR systems.

Chapter 6 described how to apply several of the theorems for task diagnosability defined in Chapter 5, which are in effect extensions and applications of the theorems presented in Chapter 3 for the concept of $f/(n-1)$-diagnosability. Among the applications of these theorems are the concepts of Multiple-Task N-Modular Redundancy (MTNMR) and N-Modular Redundancy with Comparison Diagnosis (NMRCD). These two methods that allow an NMR system to achieve a better throughput and lower latency compared to primordial NMR systems with the same number of redundant elements.

Furthermore, Chapter 6 demonstrates again the usefulness of the BSC model introduced in Chapter 4. Performing NMRCD under this model (when $f_2 < f_1$) allows more faulty proces-

sors to be tolerated for the same number of processors (or requires a lower total number of processors to tolerate the same number of faulty processors). We demonstrate how the theorems of $f/(n\text{-}1)$-diagnosability assure a higher throughput in these systems compared to their conventional NMR counterparts.

7.2 Contributions

Our extension to the characterization of $f/(n–1)$-diagnosable structures is a contribution to this domain, for our approach is general enough to apply to systems of any size, which as far as we know was not done previously. The bi-corona systems derived from this extended characterization require fewer comparisons and are therefore more efficient than other $f/(n–1)$-diagnosable structures.

We contribute some necessary and some sufficient conditions of one-step diagnosable systems under the BSC model. More importantly, we characterize $f/(n–1)$-diagnosable systems (Chapter 3) under our new model, which is useful for the application of this model to fault-tolerant systems with a large number of processors. Perhaps the most important aspect of our model is that it permits the diagnosis of a subdivided system on which a different task is concurrently executed on each subsystem. This is more useful in large systems where it is impractical to expect that all processors will execute the same task whose result is used for comparison diagnosis. Furthermore, these structures have a secondary advantage that the number of comparisons necessary for diagnosis is lower because of their disjointedness.

We have proposed three adaptive algorithms for $f/(n–1)$-diagnosis under the BSC model, one of which is applicable to the subdivided, multiple-task systems. That is, it can be used on the parallel, multiple-task systems for which the BSC model is ideal.

We have introduced several theorems for task diagnosability, which are in effect extensions and applications of the theorems presented in Chapter 3 for the concept of $f/(n–1)$-diagnosability. Among the applications of these theorems are the concepts of Multiple-Task N-Modular Redundancy (MTNMR) and N-Modular Redundancy with Comparison Diagnosis (NMRCD). By validating a task's early, these two methods allow a redundant system to achieve a better throughput and have lower latency in comparison with a primordial NMR configuration with the same number of redundant elements.

7.3 Future research

The contributions and approaches proposed in this dissertation suggest issues that require fur-

ther research. We present some of the possible directions for future research that one could take starting from the base of this thesis.

Self-checking systems. The bi-corona systems presented in Chapter 3 strongly relates to the work on self-checking (built-in self test or BIST) nodes in the PMC model.

Heuristics for *f*/(*n*–1)-diagnosability. As we discussed in Chapter 4, the characterization theorem for *f*/(*n*–1)-diagnosability under the BSC model requires checking the conditions of the theorem for all possible covers of consistent fault sets. It would be more feasible to propose heuristics that provide a means of quickly identifying systems which are not diagnosable for the BSC model. Such a technique might consider the topology of the graph of the system (i.e., the existence of connected components and edges) in order to determine quickly that the system cannot be always be diagnosed, since one can easily generate a cover of candidate fault sets such that the conditions of the theorem do not hold.

Extending the BSC model. An interesting idea might be to consider a different f_2 parameter for each distinct task. The justification for this might be that a *simple* task that has a binary outcome that is executed on two nodes that are faulty, then they will have produced the same result. On the other hand, a task that has *k* possible outputs has a lower chance to produce the same result on two faulty nodes, since a faulty can take $k-1$ values. Hence an f_2 parameter for each task, according to the likelihood of identical, faulty results might be interesting.

Implementation. Finally, the true validation of any theoretical proposition is a working construction of a real system that applies the theory. After such a system is implemented, the true efficiency of the techniques proposed can be measured.

Appendix A Source Code for Algorithms

A.1 Source code for Algorithm 3.1

Identification of a fault-free node in a bi-corona system for the symmetric comparison model.

```
#define MAXN 20
#define MAXE MAXN*MAXN

typedef struct edge {
  int u,v;   /* edge between two nodes */
  int i_i;   /* internal-internal pair or not */
  int comparison;
} edge_t;

typedef struct corona_t {
  int n,e;
  int lone_node[MAXN];
  int included[MAXN];
  int internal[MAXN];
  edge_t edge [MAXE];
  int result [MAXN];
  int adjacency [MAXN][MAXN];
  int comparison[MAXN][MAXN];
} corona_t;

int f,fc,p,ff_found;

init(corona_t *G)
{
  int i,j;

  for (i=0; i<MAXN; i++) {
    G->lone_node[i]=0;
    G->included[i]=0;
    G->internal[i]=0;
    G->result[i]=0;
    G->edge[i].u=0;
    G->edge[i].v=0;
    G->edge[i].comparison=0;
    for (j=0; j<MAXN; j++) {
      G->adjacency[i][j]=0;
      G->comparison[i][j]=0;
    }
  }
}

int number_of_nodes(corona_t G)
{
```

```
  int i,j;
  j=0;
  for (i=0; i<G.n; i++) {
    if (G.included[i]) j++;
  }
  printf("number_of_nodes returns %i\n",j);
  return j;
}

void validate_ff(corona_t G, int i)
{

  if (G.result[i]==0) {
    printf("Fault-free node v%i identified correctly.\n",i);
    ff_found = 1;
  }
  else {
    printf("ERROR! Faulty node v%i identified as fault-free.\n",i);
    exit();
  }
}

void readcorona(corona_t *G)
{

  int result,i,j,vi,vj,t;

  printf("Nodes...\n");

  for (i=0; i<G->n; i++) {
    result = scanf("%i %i",&t,&j);
    G->included[i] = 1;
    G->result[i] = j;
    G->internal[i] = t;
    if (t) printf("Internal ");
    else printf("External ");
    printf("node v%i result = %i\n",i,G->result[i]);
  }

  printf("Edges...\n");

  result = scanf("%i", &j);
  G->e = j;
  if (G->e>MAXE) {
    printf("Max num edges exceeded\n");
    exit();
  }

  for (i=0; i<G->e; i++) {
    result = scanf("%i %i",&vi,&vj);
    printf("Node v%i connected to v%i...        ",vi,vj);
    G->edge[i].u = vi;
    G->edge[i].v = vj;
    if (G->internal[vi] & G->internal[vj])
      G->edge[i].i_i = 1;
    else
      G->edge[i].i_i = 0;

    G->adjacency[vi][vj]=1;
    G->adjacency[vj][vi]=1;

    if (G->result[vi]==G->result[vj]) {
      G->edge[i].comparison=0;
```

```
      G->comparison[vi][vj]=0;
      G->comparison[vj][vi]=0;
    } else {
      G->edge[i].comparison=1;
      G->comparison[vi][vj]=1;
      G->comparison[vj][vi]=1;
    }
      printf("Comparison yields %i\n",G->edge[i].comparison);
  }

  /* find lone node if any */
  for (i=0; i<G->n; i++) {
    /* lone node is technically an internal node */
    if (G->internal[i]) {
      G->lone_node[i]=1;
      for (j=0; j<G->n; j++) {
        /* if adjacent node is external... */
        if (G->adjacency[i][j] && (G->internal[j]==0))
          G->lone_node[i]=0;   /* not lone node */
      }
      if (G->lone_node[i]) printf("Lone node is v%i\n",i);
    }
  }
}

void Step_1(corona_t *G)
{
  int i;

  for (i=0; i<G->e; i++) {
    if (G->edge[i].i_i==0) {   /* internal external pair */
      printf("Internal-external pair of nodes v%i and v%i\n", G->edge[i].u,
             G->edge[i].v);
      if (G->edge[i].comparison == 1) {
        /* discard the two nodes */
        printf("Discarding nodes v%i and v%i\n", G->edge[i].u, G->edge[i].v);
        G->included[G->edge[i].u]=0;
        G->included[G->edge[i].v]=0;
        fc++;
      }
    }
  }
}

void Step_3(corona_t *G)
{
  int i,j,k;

  for (i=0; i<G->n; i++) {
    /* take any remaining internal, non-lone node */
    if ((G->included[i]) && (G->internal[i]) && (G->lone_node[i]!=1)) {
      printf("Remaining node v%i...\n", i);
      for (j=0; j<G->n; j++) {
        /* compare only with remaining internal, non-lone nodes */
        if ((i!=j)&&(G->included[j])&&(G->internal[j])&&(G->lone_node[j]!=1)) {
          if (G->comparison[i][j]!=0) {
            /* discard vi and vj and their external nodes */
            printf("Discarding internal node pair v%i and v%i\n", i,j);
            G->included[i]=G->included[j]=0;
            for (k=0; k<G->n; k++) {
              if ((G->adjacency[i][k]) && (G->internal[k]==0)) {
                printf("Discarding external node v%i\n", k);
                G->included[k]=0;
```

```
              }
              if ((G->adjacency[j][k]) && (G->internal[k]==0)){
                printf("Discarding external node v%i\n", k);
                G->included[k]=0;
              }
            }
            fc += 2;
            break;
          }
        }
      }
    }
  }
}

void main()
{

  int i,j,result;
  corona_t G1,G2;

  while(1) {
    ff_found=0;

    init(&G1); init(&G2);

    /* read in nodes, adjacency and fault information */

    result = scanf("%i", &G1.n);
    if (G1.n>MAXN) {
      printf("Max num nodes exceeded for G1\n");
      exit();
    }

    printf("Reading in graph G1 with %i nodes.\n",G1.n);
    readcorona(&G1);

    result = scanf("%i", &G2.n);
    if (G2.n>MAXN) {
      printf("Max num nodes exceeded for G2\n");
      exit();
    }

    printf("Reading in graph G2 with %i nodes.\n",G2.n);
    readcorona(&G2);

    f = (G1.n + G2.n)/2;
    printf("Assuming f = %i\n",f);

    /* Do algorithm */
    printf("----------------------------------------\n");

    /* Step 0 */

    fc = 0;

    /* Step 1 */
    printf("Step 1 ...\n");
    printf("G1 ...\n");
    Step_1(&G1);
    printf("G2 ...\n");
    Step_1(&G2);
```

```
    /* Step 2 */
    printf("Step 2 ...\n");
    if (fc == f) {
      printf("STOP. The lone node in G2 is fault-free.\n");
      for (i=0; i<G2.n; i++) {
        if (G2.lone_node[i]) validate_ff(G2,i);
      }
      goto DONE;
    }
    else if (fc==f-1) {
     printf("STOP. The remaining nodes in either G1 or G2 (but not the lone node) are
\nfault-free.\n");
      printf("G1...\n");
      for (i=0; i<G1.n; i++) {
        if (G1.included[i]) validate_ff(G1,i);
      }
      printf("G2...\n");
      for (i=0; i<G2.n; i++) {
        if (G2.included[i] && !(G2.lone_node[i])) validate_ff(G2,i);
      }
      goto DONE;
    }

    /* Step 3 */
    printf("Step 3 ...\n");
    printf("G1 ...\n");
    Step_3(&G1);
    printf("G2 ...\n");
    Step_3(&G2);

    /* Step 4 */
    printf("Step 4 ...\n");
    printf("fc=%i\n",fc);
    if (fc==f) {
      printf("STOP. The lone node in G2 is fault-free.\n");
      for (i=0; i<G2.n; i++) {
        if (G2.lone_node[i]) validate_ff(G2,i);
      }
      goto DONE;
    }
    else if (number_of_nodes(G1) > f-fc) {
      printf("STOP. |V1| > f-fc\nRemaining nodes in G1 are fault-free.\n");
      for (i=0; i<G1.n; i++) {
        if (G1.included[i]) validate_ff(G1,i);
      }
      goto DONE;
    }
    else if (number_of_nodes(G2)-1 > f-fc) {
       printf("STOP. |V2|-1 > f-fc\nRemaining nodes in G2 (save the lone node) are
fault-free.\n");
      for (i=0; i<G2.n; i++) {
        if (G2.included[i] && !G2.lone_node[i]) validate_ff(G2,i);
      }
      goto DONE;
    }

    /* Step 5 */
    printf("Step 5 ...\n");
    for (i=0; i<G2.n; i++) {
      if (G2.lone_node[i]) {
        printf("Checking lone node v%i in G2...\n",i);
        for (j=0; j<G2.n; j++) {
```

```
          if ((i!=j)&&(G2.included[j])&&(G2.internal[j])) {
            printf("with v%i\n",j);
            if (G2.comparison[i][j]!=0) {
              printf("STOP. Remaining nodes in G1 are fault-free\n");
              for (i=0; i<G1.n; i++) {
                if (G1.included[i]) validate_ff(G1,i);
              }
              goto DONE;
            }
          }
        }
      }
    }
    printf("STOP. Remaining nodes in G2 are fault-free.\n");
    for (i=0; i<G2.n; i++) {
      if (G2.included[i]) validate_ff(G2,i);
    }
    printf("\n");

  DONE:
    printf("Trial %i completed\n",p++);
    if (ff_found==0) {
      printf("ERROR: No fault-free node found!\n");
      exit();
    }

  }
}
```

A.2 Source code for Algorithm 4.1

Adaptive $f/(n-1)$-diagnosis algorithm for the BSC model with $n \geq 2f_1 + 1$.

```
/* #define DEBUG */
#define MAXN 50
#define MAXE MAXN*MAXN

typedef struct graph_t {
  int n,e;
  int lone_node[MAXN];
  int included[MAXN];
  int internal[MAXN];
  int result [MAXN];
  int adjacency [MAXN][MAXN];
  int comparison[MAXN][MAXN];
  int weight[MAXN];
} graph_t;

int f1,f2,p,ff_found,F,cycle;
int L1[MAXN],L2[MAXN];
int l1,l2;

int power(int base, int n)
{
  int i, p;

  p=1;
  for(i = 1; i <= n; ++i)
    p = p * base;
  return p;
}
```

```
int compare(graph_t G, int u, int v)
{
  /* sanity check from Hakimi & Schmeichel 1984
     weight[u]==weight[v]== 2**(cycle-1) */
  if ((G.weight[u] != G.weight[v]) || (G.weight[u] != power(2,cycle-1))) {
    printf("Sanity check in compare... w(u)=%i, w(v)=%i, cycle=%i\n",
           G.weight[u],G.weight[v],cycle);
    exit();
  }
  return ((G.result[u] == G.result[v]) ? 0 : 1);
}

init(graph_t *G)
{
  int i,j;

  for (i=0; i<MAXN; i++) {
    G->weight[i]=1;
    G->lone_node[i]=0;
    G->included[i]=0;
    G->internal[i]=0;
    G->result[i]=0;
    for (j=0; j<MAXN; j++) {
      G->adjacency[i][j]=0;
      G->comparison[i][j]=0;
    }
  }
}

int number_of_nodes(graph_t G)
{
  int i,j;
  j=0;
  for (i=0; i<G.n; i++) {
    if (G.included[i]) j++;
  }
#ifdef DEBUG
  printf("number_of_nodes returns %i\n",j);
#endif
  return j;
}

void validate_ff(graph_t G, int i)
{
  int j;

  if (G.result[i]==0) {
#ifdef DEBUG
    printf("Fault-free node v%i identified correctly.\n",i);
#endif
    ff_found = 1;
  }
  else {
    printf("ERROR! Faulty node v%i id'd as fault-free, n=%i, f1=%i, f2=%i\n",
           G.n,f1,f2,i);
    for (j=0; j<G.n; j++) {
      printf("node v%i result = %i\n",j,G.result[i]);
    }
    exit();
  }
}
```

```
void readgraph(graph_t *G)
{

  int result,i,j,vi,vj,t;

#ifdef DEBUG
  printf("Nodes...\n");
#endif

  for (i=0; i<G->n; i++) {
    result = scanf("%i %i",&t,&j);
    G->included[i] = 1;
    G->result[i] = j;
    G->internal[i] = t;
#ifdef DEBUG
    if (t) printf("Internal ");
    else printf("External ");
    printf("node v%i result = %i\n",i,G->result[i]);
#endif
  }

  result = scanf("%i",&f2);
#ifdef DEBUG
  printf("f2 = %i\n",f2);
#endif

}


void main()
{

  int i,j,result,l1_n,l2_n,u,v;
  graph_t G;

  while(1) {
    ff_found=0;

    init(&G);
    for (i=0; i<G.n; i++) {
      L1[i]=L2[i]=0;
    }
    l1_n = l2_n = 0;

    /* read in nodes, adjacency and fault information */

    result = scanf("%i", &G.n);
    if (G.n>MAXN) {
      printf("Max num nodes exceeded for G\n");
      exit();
    }

#ifdef DEBUG
    printf("Reading in graph G with %i nodes.\n",G.n);
#endif
    readgraph(&G);

    f1 = G.n/2;
#ifdef DEBUG
    printf("Assuming f1 = %i\n",f1);
#endif

    /* Do algorithm */
```

```
#ifdef DEBUG
    printf("-----------------------------------------\n");
#endif

    /* Step 0 */

    F = 0;
#ifdef DEBUG
    printf("----L1-------------\n");
#endif
    for (i=0; i<G.n; i++) {
      L1[i]=i;
#ifdef DEBUG
      printf("v%i, %i\n",L1[i],G.weight[L1[i]]);
#endif
      l1_n++;
    }
    cycle =0;
    for (;;) {
      cycle++;
#ifdef DEBUG
      /* Step 1 */
      printf("Step 1 ...\n");
#endif
      if (l1_n==1) {
        printf("STOP. |L1|=1\n");
        validate_ff(G,L1[0]);
        exit(1);
        goto DONE;
      } else if (G.weight[L1[0]] > G.weight[L1[1]]) {
        printf("STOP. The first two elements of L are not equal.\n");
        validate_ff(G,L1[0]);
        exit(2);
        goto DONE;
      } else if (G.weight[L1[1]] > G.weight[L1[0]]) {
        printf("STOP. The first two elements of L are not equal.\n");
        validate_ff(G,L1[1]);
        exit(3);
        goto DONE;
      }

      /* Step 2 */
#ifdef DEBUG
      printf("Step 2 ...\n");
#endif
      do {
        u = L1[0];
        v = L1[1];
#ifdef DEBUG
        printf("Deleting nodes v%i and v%i from L1\n",u,v);
#endif
        /* delete the first two nodes of the list */
        for (i=2; i<l1_n; i++) L1[i-2] = L1[i];
        l1_n -=2;

#ifdef DEBUG
        printf("----L1-------------\n");
        for (i=0; i<l1_n; i++) {
          printf("v%i, %i\n",L1[i],G.weight[L1[i]]);
        }
#endif
        /* compare the two nodes' results */
        if (compare(G,u,v)==0) { /* comparison is 0 */
```

```
#ifdef DEBUG
          printf("Nodes v%i and v%i compared to 0\n",u,v);
#endif
          G.weight[v] *=2;

          /* put v in L2 */
#ifdef DEBUG
          printf("Putting v%i on top of L2\n",v);
#endif
          if (l2_n>0) {
            for (i=l2_n; i>0; i--)
              L2[i] = L2[i-1];
          }
          l2_n++;
          L2[0] = v;

#ifdef DEBUG
          printf("----L2--------------\n");
          for (i=0; i<l2_n; i++) {
            printf("v%i, %i\n",L2[i],G.weight[L2[i]]);
          }
#endif

          if ((G.weight[v]>f2) || (G.weight[v]>f1-F)) {
#ifdef DEBUG
            printf("STOP. The node v%i is fault-free.\n",v);
#endif
            validate_ff(G,v);
            goto DONE;
          }
        } else { /* comparison is 1 */
#ifdef DEBUG
          printf("Nodes v%i and v%i compared to 1\n",u,v);
#endif
          F += G.weight[v];
#ifdef DEBUG
          printf("F = %i\n",F);
#endif
          if (l2_n!=0) v=L2[0];
          else v=L1[0];
          if (G.weight[v]>f1-F) {
#ifdef DEBUG
            printf("STOP. w(v)>f1-F (%i>%i-%i): The node v%i is fault-free.\n",
                   G.weight[v],f1,F,v);
#endif
            validate_ff(G,v);       goto DONE;
          }
        }
      } while ((l1_n>1) && (G.weight[L1[1]] == G.weight[L1[0]]));

      /* Step 3 */
#ifdef DEBUG
      printf("Step 3 ...\n");
#endif
      /* append L1 to the end of L2 and put it as the new L1 */
      for (i=0; i<l1_n; i++)
        L2[l2_n++] = L1[i];

#ifdef DEBUG
      printf("----L1--------------\n");
#endif
      for (i=0; i<l2_n; i++) {
        L1[i]=L2[i];
```

```
#ifdef DEBUG
        printf("v%i, %i\n",L1[i],G.weight[L1[i]]);
#endif
      }
      l1_n = l2_n;

      /* L2 is now a nil list */
      l2_n = 0;

    } /* goto step 2 */

  DONE:
#ifdef DEBUG
    printf("Trial %i completed\n",p++);
#endif
    if (ff_found==0) {
      printf("ERROR: No fault-free node found! n=%i,f1=%i,f2=%i\n");
      for (i=0; i<G.n; i++) {
        printf("node v%i result = %i\n",i,G.result[i]);
      }
      exit();
    }

  }
}
```

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Index

Symbols

$\alpha_0(G)$ minimum vertex cover of G 58
° corona of two graphs 50
$f(v)$ set of subsets to which v belongs 32
$\Gamma(v)$ set of nodes adjacent to v 31
$\Gamma(X)$ set of nodes outside of X and adjacent to at least one node in X 31
G_0 0-derived graph of G 32
$H_{2r,n}$ system 125
$K_{m,n}$ complete bipartite graph 29
K_p complete graph 29
$\kappa(G)$ connectivity of G 29
$\nu(G)$ cycle rank of G 30
$\pi(G)$ cover of G 32
σ syndrome 32
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